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ORIGINAL SCIENTIFIC PAPER

Rate of Force Development Scaling Factor in Hamstring Muscles: Feasibility and Relationship to Deadlift Performance among Resistance Trained Individuals

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Abstract

The aim of this study was to examine the relationship among hamstring peak force (PF), rate of force development scaling factor (RFD-SF), countermovement jump (CMJ), and one-repetition maximum (1RM) strength in recreationally trained individuals, as well as to establish the feasibility of RFD-SF assessment for knee flexor muscles. Eighteen volunteers (12 males, 27.3±5.2 years, 6 females, 24.4±3.1 years) participated in the study. Participants performed a knee flexion maximal isometric voluntary contraction and followed a standard RFD-SF protocol. The 1RM for the deadlift was assessed to determine maximal dynamic strength, while CMJ was used to evaluate explosive movement capability. The study found no significant correlations between RFD-SF, CMJ, and 1RM. Additionally, it was observed that RFD-SF in the hamstring muscles can be effectively assessed (mean $R^2=0.92$). In addition, RFD-SF was not different between men and women, which highlights its potential as sex-independent measure of knee flexor muscle function. Future research should involve a more diverse group of athletes to further investigate the relationships among RFD-SF, strength, and explosive movements like CMJ for more comprehensive insights.

Keywords: deadlift, rapid force development, scaling factor, knee flexion, strength, countermovement jump

Introduction

Resistance training is a fundamental component of sports and exercise regimes, aimed at enhancing muscle strength, power, and local muscle endurance (Stricker et al., 2020). Its primary objectives include increased physical demands, achieve better performance outcomes, and reduce injury risks (Refalo et al., 2023; Schoenfeld et al., 2019; Stricker et al., 2020). Muscle performance assessment is vital in both scientific research and applied sports science, as well as in clinical practice (Kozinc et al., 2022; Oliveira et al., 2016). Recently, scientific research has increasingly focused on evaluating the capacity for rapid force generation, termed rapid strength (Guizelini et al., 2018; Kozinc et al., 2022; Maffiuletti et al., 2016). This

aspect of muscle function is typically measured using the rate of force development (RFD), which quantifies the rate of force increase within a specific timeframe (Aagaard et al., 2002; de Oliveira et al., 2013). RFD assessment involves analysing the force or torque curve in relation to the time of explosive muscle contractions (Aagaard et al., 2002; Methenitis et al., 2019). Notably, enhancing RFD is a key adaptation in resistance training, with larger improvements correlating to increased maximum force and speed during rapid movements (Aagaard et al., 2002; Andersen et al., 2010; Methenitis et al., 2019).

Oliveira et al. (2013) found that participants undergoing isometric resistance training for six weeks exhibited a significant 61% increase in maximum RFD, especially during the



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early phase of torque increase in the joint (de Oliveira et al., 2013). An enhancement in RFD following resistance training, as Aagaard et al. (2002) pointed out, is crucial in rapid and forceful muscle contractions where achieving maximum muscle force is limited by the short contraction time, yet rapid force generation is essential for athletic performance and daily activities (Aagaard et al., 2002). The speed of muscle contraction is influenced by factors such as the ratio of fast to slow muscle fibers, muscle cross-sectional area, viscoelastic properties, and neural factors (Andersen et al., 2010; Methenitis et al., 2019). Methenitis et al. (2019) further explored this by assessing the relationship between muscle fiber composition and RFD in well-trained individuals, identifying a strong correlation between RFD and the proportion of type II fibers, particularly type IIx, and noting that resistance training leads to significant adaptations in the utilization of hypertrophied type IIx muscle fibers (Methenitis et al., 2019). Complementary to these findings, Oliveira et al. (2016) observed improvements in early RFD in knee extensors after an eight-week maximal isokinetic training program (Oliveira et al., 2016), which was also confirmed by Blazevich et al. (2008) who reported similar RFD enhancements in knee extensors following resistance training.

The rate of force development scaling factor (RFD-SF) has recently emerged as a novel means of assessing rapid force development capacity (Bellumori et al. 2011; Smajla et al., 2021). Briefly, during rapid isometric contractions performed at varying intensities, a pronounced linear relationship is observed between the peak force (PF) achieved and the corresponding RFD. This linear relationship's slope, termed RFD-SF, is indicative of an individual's RFD capacity across a broader range of contraction intensities. The RFD's proportional scaling to PF is reflected in the linearity of the relationship, as denoted by the R^2 coefficient. A comprehensive review by Kozinc et al. (2022) highlighted the necessity for more research to elucidate the link between RFD-SF and sports performance. Further, the review concluded that RFD-SF is dependent on an individual's neuromuscular capacity, particularly the maximum motor unit firing rate, and motor control, which includes adjusting motor unit firing rates and coordinating actions across muscle groups. RFD-SF differs from maximal strength or traditional RFD assessments in that it does not necessitate maximal effort contractions, which makes it particularly suitable for clinical settings. It has been demonstrated that RFD-SF is highly sensitive to neuromuscular speed declines, a typical feature of aging and certain neurological conditions (Klass et al., 2008; Corrêa et al., 2020; Uygur et al., 2020; Uygur et al., 2022; Uygur et al., 2023). However, further research is required to ascertain if RFD-SF provides unique insights not captured by conventional measurements of peak force/torque or standard RFD, which would establish its distinct value in assessment. In particular, the literature lacks research on associations between RFD-SF and athletic performance.

Suarez et al. (2019) presented substantial evidence indicating a strong association between success in weightlifting and the strength and RFD. In other words, weightlifting performance is primarily dependent on the capacity to generate large forces, achieve high RFD, and attain maximum power during competitive lifts (Suarez et al., 2019). A key example is the deadlift, where the posterior thigh muscles – comprising the semimembranosus, semitendinosus, and biceps femoris muscles – are crucial. These muscles primarily facilitate hip exten-

sion and knee flexion (Woodley and Mercer, 2005; Koulouris and Connell, 2005). Martín-Fuentes et al. (2020) observed slightly higher activation of the semitendinosus muscle compared to the biceps femoris during deadlifts (Martín-Fuentes et al., 2020). Given the vital role of the posterior thigh muscles in performing the deadlift, and our aim to explore the association between RFD-SF and sports performance, the objective of this study was to investigate the association between higher RFD-SF in these muscles and deadlift and countermovement jump. Additionally, we aim to explore the feasibility of assessing RFD-SF in knee flexors. This study addresses the existing gap in the literature by evaluating the feasibility of RFD-SF assessment for knee flexor muscles and exploring its relationship with deadlift performance, which has not been investigated in previous research.

Methods

Participants

This study involved competitive and recreationally-active weightlifters. A total of 18 participants (12 males, 27.3 ± 5.2 years, 6 females, 24.4 ± 3.1 years), volunteered to participate in the study. Participants reported regular resistance exercise participation (training status: 4.1 ± 2.8 years) and included deadlifts in their training routine at least once a week, which was a condition for inclusion in the study. Exclusion criteria included the presence of significant injuries or illness in the six months prior to the study or the presence of pain that could affect the measurements. After presenting potential risks, benefits, and the course of the study, the subjects completed informed consent to participate in the research, in accordance with the Helsinki Declaration. Participants were also informed of the option to withdraw from the study without consequences. The study was approved by the Republic of Slovenia National Medical Ethics Committee (No. 0120-690/2017/8) and adhered to research ethics requirements.

Study Design

In this cross-sectional study, participants executed a knee flexion maximal isometric voluntary contraction and a standard RFD-SF protocol (Kozinc et al., 2022). For all subjects, only one visit was required. They were instructed to arrive well-rested, having consumed a suitable energy meal and ensuring proper hydration. Initially, participants were briefed on the procedure, risks, and benefits of the research. Following this, they completed a personal information sheet and a brief questionnaire about their injury history and training characteristics. Subsequently, each participant filled out an informed consent form to partake in the study. A warm-up session was then conducted, comprising 5 minutes of stepping onto a raised platform, dynamic stretching exercises for the entire body with 10 repetitions each (including circles for the head, shoulders, arms, hips, knees, and ankles), and 5 strengthening exercises. These exercises included 10 squats with a 4 or 6-kilogram weight lifted on toes, 10 single-leg deadlifts with a 4 or 6-kilogram weight, 10 elbow flexions in both supine and prone forearm positions with a 4 or 6-kilogram weight in each hand, and 10 single-leg bridge lifts.

Procedures

During the testing protocol, participants were seated in an isometric knee dynamometer's chair (S2P, Science to Practice Ltd., Ljubljana, Slovenia), with measurements taken at 60° of

knee flexion (full knee extension = 0°) and hips at 100°. All participants performed knee flexion with their right leg. The knee axis was aligned with the axis of the dynamometer's lever arm, and the minimally padded shank support was adjusted for each participant, approximately 2 cm above the lateral malleolus. Adequate hip and knee fixation was ensured with rigid straps over the pelvis and knee (as shown in Fig. 1.) As part of the acclimatization process, each participant executed two graded submaximal contractions at 50, 75, and 90% of their self-estimated maximal voluntary effort. After a 3-minute rest, they performed three maximal voluntary knee extensions, with a 30-second rest between each. Participants were instructed to gradually increase their torque and sustain maximal force for 3–5 seconds, which was used to determine the peak force of knee flexion. Finally, the participants undertook an RFD-SF familiarization protocol, consisting of 3 to 5 submaximal ex-

plosive contractions at 20, 40, 60 and 80 % of their maximal force recorded during maximal voluntary contraction.

After familiarization, each participant performed 15–20 explosive isometric contractions at four different submaximal levels which were selected randomly (20, 40, 60, and 80% of previously determined maximal voluntary force) (Bellumori et al., 2011; Kozinc et al., 2022). Participants were instructed to produce isometric knee flexion as quickly as possible and to relax the muscles immediately afterwards. The target level of force was presented on a computer screen in front of the participant as a horizontal line on a graph (Figure 1). Visual feedback on the amount of force the participant had generated during the pulse was also provided on the screen, while participants were instructed to apply a level of force matching (about) the red horizontal target force level during each pulse. There was a 60-s rest between two consecutive submaximal levels.



FIGURE 1. Position of the subject during the measurement of RFD-SF. 1 = a padded fixation of the pelvis; 2 = rigid straps for the knee fixation; 3 = force sensors; 4 = a monitor with visual feedback.

RFD-SF calculations were done as recommended in the literature (Kozinc et al., 2022; Mathern et al., 2019; Bellumori et al., 2011). Each pulse recorded included measurements of PF and peak RFD. To determine when the force pulses began and ended, the changes in force curves over time were calculated and filtered using a fourth-order zero-lag filter set at a frequency of 10 Hz. This specific frequency was chosen during initial testing as it closely matched the manually identified initiation and termination points. Identifying the start and end of force pulses was automated by marking the point where the force's rate of change reached 10% of its maximum value. The duration of each pulse was computed as the time gap between its termination and initiation. Pulses lasting longer than the mean plus two standard deviations of all pulse durations were excluded from the analysis.

The second set of measurements involved assessing maximum strength in the deadlift exercise. To measure the maximum strength in the deadlift, participants performed a 1RM for this exercise. They began the warm-up with an empty barbell, gradually increasing the load based on their own perception. After the initial warm-up sets, they performed only 1 repetition at subsequent loads to avoid the effects of fatigue. Participants could determine the pace of warm-up, weight in-

crements, and rest periods on their own, with vocal encouragement provided during the execution. For the 1RM test, participants were allowed 3 attempts.

Statistical Analysis

The relationships between PF and corresponding peak RFD were analysed using regression parameters, treating them as indicators of rapid force production and relaxation, respectively. The slopes of these relationships quantified the magnitude of the ability to scale RFD with the magnitude of force produced. The R^2 values from these regressions showed how consistently force development and relaxation scaled concerning peak force (Bellumori et al. 2011). Similar to prior studies (Šarabon et al.; 2020 Djordjevic and Uygur 2017), the y-intercepts of the regression lines were reported in the results, but were excluded from discussion due to their negligible impact in RFD-SF literature.

The statistical analysis was conducted using the IBM SPSS Statistics 25 software (IBM, New York, USA). After checking for normality of the data distribution with Shapiro-Wilk tests, associations were assessed with Pearson's correlation coefficient, which was interpreted as negligible (< 0.1), weak (0.1–0.4), moderate (0.4–0.7), strong (0.7–0.9) and very strong

(>0.9) (Akoglu, 2018). Differences between men and women were assessed with independent sample t-tests. Effect sizes were expressed as Cohen's d, which was interpreted as trivial (<0.2), small (0.2-0.5), medium (0.5-0.8) and large (>0.8) (Lakens, 2013). Threshold for statistical significance was set at $\alpha < 0.05$.

Results

Descriptive statistics and gender differences

The descriptive statistics for all variables are included in Table 1. The peak force and RFD seem to exhibit a good linear relationship (mean $R^2 = 0.92$; range = 0.85-0.99), demonstrating the feasibility of RFD-SF assessment in knee flexors.

Table 1. Descriptive statistics.

Outcome variable	Mean	SD	Minimum	Maximum
RFD-SF (Slope)	7.05	1.25	5.00	9.96
Intercept	23.98	21.05	-11.45	74.92
R^2	0.92	0.04	0.85	0.99
Peak Torque (Nm)	311.06	94.17	116.00	459.00
Peak Torque (Nm/kgBW)	4.00	0.88	1.79	5.56
CMJ height (m)	0.36	0.08	0.25	0.49
Deadlift 1RM (kg)	158.24	46.13	95.00	225.00
Deadlift 1RM (kg/kgBW)	2.03	0.39	1.55	2.72

SD – standard deviation; BW – bodyweight; CMJ – countermovement jump; 1RM – 1-repetition maximum

Men had larger CMJ height, as well as larger peak knee flexor torque and deadlift 1RM (all $p < 0.001$). Large differences persisted even in body-mass-normalized values. On

the other hand, RFD-SD ($p = 0.462$), y-intercept ($p = 0.216$) and R^2 ($p = 0.404$) were similar in men and women (Table 2).

Table 2. Differences between men and women

Outcome variable	Women (n = 6)		Men (n = 12)		Difference	
	Mean	SD	Mean	SD	p	ES (g)
RFD-SF (Slope)	6.76	0.74	7.21	1.47	0.462	-0.33
Intercept (Nm)	15.83	9.30	28.42	24.58	0.216	-0.58
R^2	0.93	0.02	0.92	0.04	0.404	0.46
Peak Torque (Nm)	210.33	52.53	366.00	58.31	0.000	-2.62
Peak Torque (Nm/kgBW)	3.24	0.73	4.42	0.66	0.002	-1.63
CMJ height (m)	0.28	0.03	0.40	0.06	0.000	-2.30
Deadlift 1RM (kg)	106.67	10.80	186.36	29.67	0.000	-3.02
Deadlift 1RM (kg/kgBW)	1.65	0.08	2.25	0.32	0.000	-2.16

SD – standard deviation; BW – bodyweight; CMJ – countermovement jump; 1RM – 1-repetition maximum

Correlations between RFD-SF and deadlift performance, maximal strength and CMJ

Table 3 shows the correlations between RFD-SF and deadlift performance, maximal strength and CMJ. RFD-SF and R^2 were not in statistically significant with any other outcome variable. Y-intercept was in moderate correlation

with absolute and normalized peak knee flexor torque ($r = 0.51-0.52$; $p = 0.031-0.035$). Similarly, there were no correlations when men and women were examined separately, other than very high positive correlations between y-intercept and peak knee flexor force in women ($r = 0.86-0.91$; $p = 0.012-0.028$).

Table 3. Correlations among RFD-SF, CMJ and deadlift 1RM

	Peak Torque (Nm)	Peak Torque (Nm/kg BW)	CMJ height (m)	Deadlift 1RM (kg)	Deadlift 1RM (kg/kg BW)
RFD_SF	0.01	-0.10	0.01	0.25	0.23
Intercept	0.52*	0.51*	0.15	0.32	0.25
R^2	-0.15	-0.18	-0.39	0.09	0.15

BW – bodyweight; CMJ – countermovement jump; 1RM – 1-repetition maximum

Discussion

The primary objective of this study was to determine whether a higher rate of force development scaling factor (RFD-SF) in knee flexors muscles correlates with improved performance in the deadlift and countermovement jump

(CMJ). Additionally, we aimed to assess the feasibility of measuring RFD-SF for these muscles. Our findings indicate no significant correlations between RFD-SF, CMJ, and increased one-repetition maximum (1RM) in the deadlift. Moreover, our results suggest the viability of RFD-SF assessments for knee

flexors, as evidenced by a consistently high R^2 value (mean $R^2 = 0.92$). In addition, our results are indicating it could serve as a sex-independent measure of knee flexor muscle performance.

Our study revealed a consistent linear relationship between PF and RFD, supporting the feasibility of RFD-SF assessment for knee flexors. Notably, we observed no significant differences in RFD-SF measurements between male and female participants, suggesting that RFD-SF could be a novel method for assessing muscle function irrespective of sex. However, it is important to note that our study included only six female participants, which may limit the generalizability of these findings. This observation aligns with the review by Kozinc et al. (2022), who also reported no clear sex differences in RFD-SF. Similarly, Corrêa et al. (2020) found no sex differences in grip muscle RFD-SF in young individuals, although variations were noted across different muscle groups. For instance, Bozic et al. (2013) reported higher RFD-SF in knee extensors but lower RFD-SF in knee flexors for males compared to females. Conversely, Bellumori et al. (2011) observed no significant sex differences in RFD-SF for knee extensors, elbow extensors, and index finger abductors. In summary, while gender appears to have a negligible impact on RFD-SF in certain scenarios, the overall influence remains ambiguous and may vary depending on age and specific muscle groups. Additionally, the limited data on female participants in our study hinders a comprehensive analysis. Future research should prioritize including a larger number of female participants to enable more robust analysis and validation of our study outcomes.

The absence of a correlation between RFD-SF and 1RM in our study aligns with the mixed findings in existing literature regarding the relationship between RFD and 1RM strength, especially in exercises like the deadlift and countermovement jump (CMJ). For example, McGuigan et al. (2010) and McGuigan and Winchester (2008) reported no clear association between peak RFD and 1RM strength in bench press and squat during isometric mid-thigh pulls. In contrast, other studies, such as Wang et al. (2016), have identified a moderate-to-strong correlation ($r = 0.60$ – 0.75) between RFD and 1RM strength, particularly when RFD is measured within specific time frames (90–250 ms) during the mid-thigh pull. Additionally, research indicates that resistance training can lead to parallel improvements in strength and RFD, especially when assessed during the later phase (150–250 ms) of a movement (Aagaard et al., 2002; Häkkinen et al., 1981). In sum, while some studies did not observe a direct connection between peak RFD and maximal strength, others underscore a phase-specific relationship. These findings indicate that improvements in RFD in the context of dynamic strength could be closely linked to specific time intervals or phases during a movement. This variability in findings highlights the complexity of the relationship between RFD, RFD-SF, and maximal strength.

Our findings indicated an unchanged RFD-SF with training, which could be explained by the fact that both peak strength and

RFD increased in parallel, while the scaling between the two remained the same. This means that RFD-SF, while feasible to assess in knee flexors, has limited utility in relation to weightlifting performance. In addition, recent findings, such as those by Del Vecchio et al. (2022), indicate that while maximal force can increase following short-term isometric strength training, RFD may remain relatively unchanged. This could be attributed to a combination of neural adaptations aimed at increasing RFD and musculotendinous adaptations that counteract these effects. Further studies are necessary to substantiate these findings. The lack of correlation between RFD-SF and CMJ performance in our study might be explained by the fact that none of our participants regularly perform CMJ as part of their daily routine, which hinders the reliability of the metrics related to this task. We hypothesize that including more athletes who regularly engage in jumping exercises in their training could potentially yield different results.

Strengths and Limitations

The strengths of this study include robust measurements of RFD-SF with state-of-the-art dynamometers and inclusion of highly-trained participants. However, this study also had several limitations. The small sample size may have limited the generalizability of our findings. Participants' dietary habits, which were not controlled, could have influenced the results. Moreover, not all participants were experienced with the CMJ protocol, potentially affecting the correlation analyses. Additionally, conducting all measurements on a single day might have caused participant fatigue, possibly impacted performance and underestimated the true effects on the outcomes. Future research should include a more comprehensive assessment across major lifts—squat, bench press, and deadlift—to better understand their relationship with RFD-SF and provide insights for training optimization and athletic performance enhancement. The cross-sectional design minimizes potential confounding variables, ensuring that all participants were assessed under consistent conditions to provide reliable and accurate correlations between different measures of muscle performance. However, a resistance-training-based intervention study would provide a clearer insights into RFD-SF's utility.

Conclusion

This study examined the relationship between the hamstring rate of force development scaling factor (RFD-SF), peak force, countermovement jump (CMJ), and one-repetition maximum (1RM) deadlift performance among recreationally trained individuals. While RFD-SF assessment for knee flexors was found to be feasible (mean $R^2=0.92$), no significant correlations were observed between RFD-SF, CMJ, and 1RM deadlift performance. Additionally, RFD-SF was not significantly different between men and women, indicating its potential as a sex-independent measure of knee flexor muscle function.

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Conflict of interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

The Comparison of Position-Specific Match Performance between the Group and Knockout Stage of the UEFA Champions League

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Abstract

This study aimed to compare match performance in the group and knockout stage of the UEFA Champions League (UCL) for players in different playing positions. The information on physical and technical performances was collected during all UCL matches ($n=125$) in the 2022/23 season. Players' observations ($n=1387$) were classified according to their playing positions as central defender (CD; $n=471$), fullback (FB; $n=283$), central midfielder (CM; $n=376$), winger (WM; $n=137$) and forward (FW; $n=104$). No differences in technical performance between the group and knockout stage were found for CDs, FBs, CMs, and WMs, while only FWs achieved significantly less shots on target (Cohen's d ($d=0.46$), total passes ($d=0.54$), successful passes ($d=0.49$), and more duels ($d=0.39$) in the knockout compared to the group stage. The CDs, CMs, and WMs covered greater distance at moderate ($d=0.46-0.63$) and high ($d=0.29-0.38$) speeds in the knockout compared to the group stage, while no differences in physical performance between the group and knockout stage were found for FBs and FWs. This study demonstrated that (i) the technical performance of UCL players was in general stable irrespective of the competition stage, (ii) matches played in the knockout stage were characterized by greater match intensity compared to the matches played in the group stage of UCL for CDs, CMs, and WMs. These findings may help football coaches in the decision-making process when selecting the players for specific matches and designing training programs.

Keywords: football, competition stage, playing position, running performance, technical performance

Introduction

Football (soccer) is a team sport with diverse movement patterns and fluctuating dynamics. Consequently, multiple components contribute to successful match performance, with technical, tactical, and physical performance being crucial (Andrzejewski et al., 2022). Physical performance is typically assessed by measuring various kinematic factors such as the total distance, distances covered across different speed zones, and frequency of accelerations (Modric, Versic, Sekulic, & Liposek, 2019). Shots, passes, crosses, and dribbles are commonly utilized metrics for assessing tech-

nical performance (Lorenzo-Martínez, Padrón-Cabo, Rey, & Memmert, 2021), whereas inter-player coordination and inter-team coordination are frequently analyzed to evaluate tactical performance (Memmert, Lemmink, & Sampaio, 2017).

Systematically examining match performance holds great importance as it can be used as a powerful tool (Goes et al., 2021). Specifically, match performance can be used to gain a competitive edge over the opponent, and/or to bring crucial decisions regarding the own team (Sarmiento et al., 2014). However, research investigating match performance



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shows their high variability, hampering their interpretation and application in practice (Oliva-Lozano, Muyor, Fortes, & McLaren, 2021). Empirical evidence demonstrated that myriad factors such as playing position (Modric, Versic, & Sekulic, 2021), match outcome (Chmura et al., 2018) and location (Diana et al., 2017), team formation (Aquino et al., 2020), and opponent quality (Almeida, Ferreira, & Volossovitch, 2014) may affect match performance.

Another factor shown to affect match performance in football is the competition stage (Rampinini, Coutts, Castagna, Sassi, & Impellizzeri, 2007). In general, a competitive season of professional football lasts for 9–10 months. Throughout this period, it is imperative for teams to maintain a stable (i.e., optimal) match performance (Li et al., 2023). However, research repeatedly indicated seasonal variation in match performance (Chmura et al., 2019; Li et al., 2023; Springham, Williams, Waldron, Burgess, & Newton, 2020). Briefly, Springham et al. (2020) investigated the physical performance of English Championship players and reported increased total, high-speed running, and sprinting distances in the first quarter of the season. Chmura et al. (2019) analyzed the physical and technical performance of German Bundesliga players and found that technical performance was similar throughout the season, while physical performance was seasonal dependent. Specifically, the authors reported that total distance reached its peak at two-thirds of the season and subsequently declined until the end. Li et al. (2023) investigated eight consecutive Chinese Super League seasons match performance and reported that physical performance parameters had their peak at the opening and closing matches of the season, while technical performance parameters were relatively stable throughout the season with a substantial increase in the last month.

Although previous studies provided valuable knowledge on changes in match performance according to the competition phase, it is important to note that all of them analyzed teams belonging to only one country (Chmura et al., 2019; Li et al., 2023; Springham et al., 2020). Therefore, the results were undoubtedly influenced by the geographical, cultural, historical, and social aspects of the observed countries (Sapp, Spangenburg, & Hagberg, 2018; Sarmiento et al., 2013). Furthermore, most of these studies observed sub-elite football competitions (i.e., Chinese Super League, 2nd English league) (Li et al., 2023; Springham et al., 2020). To the best of our knowledge, only Chmura et al. presented results obtained from first-division players (Chmura et al., 2019). In addition, only one study considered playing positions (Springham et al., 2020), which hampers “real world” practical applications. Finally, all previous studies were exclusively related to league competitions (Chmura et al., 2019; Li et al., 2023; Springham et al., 2020), while studies to investigate changes in match performance across different competition stages during tournaments are scarce.

The most elite and most prestigious club football competition is the UEFA Champions League (UCL) (Schokkaert & Swinnen, 2014). In the group stage of UCL, thirty-two clubs, divided into eight groups, play six games. Wins earn three points, draws earn one, and losses yield none. Two teams with the most points in each group qualify for the knockout stage, where sixteen teams compete in head-to-head elimination contests. Each pairing involves two matches, with the team scoring more goals over the two games advancing.

Such an elimination format in the knockout stage limits the team's opportunity to fix the mistakes they potentially make, resulting in a more cautious approach to the game plan compared to the group stage. Therefore, differences in match performance across competition phases may be expected.

For all aforementioned reasons, further research analyzing position-specific match performance of elite teams from multiple countries across different competition stages is warranted. The findings from such research may identify whether the match performance depends on the competition stage and provide a benchmark for elite practitioners. Therefore, this study aimed to compare match performance in the group and knockout stage of UCL for players in different playing positions. We hypothesized that both physical and technical performance will be significantly different by group stages.

Methods

Sample and design

Match performances of players were collected from all 125 UCL matches during the 2022/23 season. Of these, 96 matches were played in the group stage and 29 in the knockout stage. Two matches from the group stage were excluded initially due to poor data quality. Only data from players who completed the entire match were analyzed, with goalkeepers excluded due to the unique nature of their position. Consequently, the final dataset comprised 1387 individual match observations from 123 matches involving 437 players from 32 teams. These observations were classified automatically into five positional subsets by the data provider based on the players' tactical role in the team: central defender (CD; $n=471$), fullback (FB; $n=283$), central midfielder (CM; $n=376$), winger (WM; $n=137$) and forward (FW; $n=104$). Players' identities were anonymized following the principles of the Declaration of Helsinki to ensure confidentiality. The investigation was approved by the local university ethics board.

Procedure

Physical performance data were collected using an optical tracking system (Player & Ball Tracking System, Hawk-Eye Innovations Limited, Basingstoke, England). The system's reliability was previously assessed using the official Fédération Internationale de Football Association (FIFA) test protocol for Electronic and Performance Tracking systems (EPTS). This evaluation involved comparing the data with the Vicon system (Vicon Motion Systems, Oxford Metrics, UK) across five velocity bands (0–7 km/h, 7–15 km/h, 15–20 km/h, 20–25 km/h, and 25+ km/h). The system successfully passed this test protocol (authorization number: 1015068), indicating a high level of reliability. A detailed report is available on the official FIFA webpage. The technical performance data were registered using a WyScout® (Wyscout, Chiavari, Italy) computerized multiple-camera tracking validated analysis tool (Pappalardo et al., 2019a). The procedure of data collection has been previously described in detail (L. Pappalardo et al., 2019b).

Variables

All the physical and technical performance variables with their associated definitions are presented in Table 1.

Table 1. Match performance variables and their definitions.

Physical performance-related variables (m)	
Total distance (TD) (m)	Total distance covered in the match.
Low-intensity running (LIR) (m)	Distance covered at speed <15 km/h.
Moderate-intensity running (MIR) (m)	Distance covered at speed 15–20 km/h.
High-intensity running (HIR) (m)	Distance covered at speed >20 km/h.
Technical performance-related variables (#)	
Goals (#)	A goal scored (does not include own goals).
Assists (#)	The last action of a player from the goalscoring team, prior to the goal being scored by a teammate, or an own goal.
Shots (#)	An attempt to score a goal, made with any (legal) part of the body, either on or off target.
Shots on target (#)	An attempt to score a goal which required intervention to stop the ball going in, or resulted in a shot that would have gone in without diversion.
Passes (#)	A ball sent from one player to another.
Successful passes (#)	An intentionally played ball sent from one player to another that receives the ball.
Crosses (#)	A ball played from the offensive flanks aimed towards a teammate in the area in front of the opponent's goal.
Successful crosses (#)	Completed crosses.
Dribbles (#)	An attempt to move past an opposing player whilst trying to maintain possession of the ball.
Successful dribbles (#)	Completed dribbles.
Duels (#)	A challenge between two players to gain control of the ball, progress with the ball or change its direction.
Successful duels (#)	Completed duels.
Recoveries (#)	Any action that ends a possession of the opposition team (the last action of this possession is a loss) and starts a possession for current team.
Recoveries on opponent half (#)	Recoveries performed on opponent half.
Touches in penalty area (#)	Pass or a touch that happens in the opponent penalty area.
Interceptions (#)	An act of player actively intercepting the ball by anticipating its movement when the opponent is shooting, passing or crossing.
Clearances (#)	An action (generally a pass) when the player, while having other option, to pass or to hold the ball, is instead clearing it, either with a long pass forward without a precise target or for a throw in/corner kick, playing safe.
Sliding tackles (#)	An aggressive slide on the ground in the legs of the opposition player with a clear intention to dispossess the opponent or to clear the ball out.
Successful sliding tackles (#)	Completed sliding tackles.
Key passes (#)	Passes that immediately creates a clear goal scoring opportunity for a teammate.

Statistics

The normality of the distributions was checked by the Kolmogorov – Smirnov test and the descriptive statistics included the means $\pm$ standard deviations. Only TD showed as normally distributed (K-S $p > 0.05$); therefore, later statistical analyses were adapted accordingly. Specifically, one-way analysis of variance was used to compare differences in TD between group and knockout stages, while differences between group and knockout stages in all other variables were examined using nonparametric Mann–Whitney U test. Effect sizes (ES) were evaluated using Cohen's d , and were interpreted as follows: trivial (< 0.2), small (≥ 0.2 – 0.5), moderate (≥ 0.5 – 0.8), and large (> 0.8) (Cohen, 2013). All the analyses were performed using Statistica v.14.0.1.25 (TIBCO Software

Inc, USA), and the significance level was set to $p < 0.05$.

Results

Table 2 presents descriptive statistics and differences in CMs' physical and technical performance in the group and knockout stage of UCL. CDs covered less MIR (small ES) and HIR (small ES) in the group stage compared to the knockout stage. No differences in TD and LIR were found irrespective of the competition stages. Also, CDs' technical performance was similar both in the group and knockout stages (all trivial-to-small ES).

Table 3 presents descriptive statistics and differences in FBs' physical and technical performance in the group and knockout stage of UCL. No differences in both physical and technical performance were found irrespective of playing in

Table 2. Central defenders' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Physical performance-related variables (m)					
Total distance*	10219.4±742.3	10298.3±745.3	-1.00	0.32	-0.11 (-0.31 – 0.1)
Low-intensity running	8313.7±543.3	8121.5±839	1.66	0.1	0.31 (0.1 – 0.51)
Moderate-intensity running	1291.6±275.5	1509.9±830	-2.29	0.02	-0.46 (-0.67 – -0.25)
High-intensity running	614.1±184.5	666.9±165.8	-3.33	<0.01	-0.29 (-0.5 – -0.08)
Technical performance-related variables (#)					
Goals	0.1±0.2	0±0.2	0.36	0.72	0.11 (-0.1 – 0.32)
Assists	0±0.2	0±0	0.59	0.55	0.22 (0.01 – 0.43)
Shots	0.4±0.8	0.4±0.7	-0.39	0.7	0 (-0.21 – 0.2)
Shots on target	0.2±0.4	0.2±0.4	-0.27	0.79	-0.02 (-0.22 – 0.19)
Passes	55.9±22.3	56.1±21.2	0.02	0.98	-0.01 (-0.21 – 0.2)
Successful passes	50.3±22.2	51±20.8	-0.35	0.73	-0.03 (-0.24 – 0.18)
Crosses	0.4±1.1	0.2±0.7	0.56	0.57	0.15 (-0.06 – 0.36)
Successful crosses	0.1±0.5	0.1±0.4	0.89	0.37	0.15 (-0.06 – 0.35)
Dribbles	0.8±1.5	0.5±0.7	1.43	0.15	0.26 (0.05 – 0.47)
Successful dribbles	0.5±1	0.2±0.5	1.88	0.06	0.3 (0.09 – 0.5)
Duels	13.2±5.9	13.7±5.8	-0.69	0.49	-0.08 (-0.29 – 0.13)
Successful duels	7.7±3.9	8.2±3.9	-0.94	0.35	-0.12 (-0.33 – 0.09)
Recoveries	10.3±4.2	10.8±3.7	-1.26	0.21	-0.13 (-0.34 – 0.08)
Recoveries on opponent half	2.1±1.9	2.1±1.7	-0.58	0.57	-0.01 (-0.22 – 0.2)
Touches in penalty area	0.6±1.1	0.6±0.9	-0.2	0.84	0.04 (-0.17 – 0.25)
Interceptions	5.2±2.8	4.9±2.4	1.02	0.31	0.11 (-0.09 – 0.32)
Clearances	2.5±2.1	2.7±2	-1.37	0.17	-0.12 (-0.32 – 0.09)
Sliding tackles	0.3±0.6	0.3±0.7	0.12	0.91	-0.02 (-0.23 – 0.18)
Successful sliding tackles	0.2±0.4	0.2±0.5	-0.18	0.86	-0.03 (-0.24 – 0.18)
Key passes	0.1±0.4	0.1±0.3	0.27	0.79	0.08 (-0.13 – 0.28)

* denotes variables where T-test was calculated; MW - Mann-Whitney U test; CI – confidence interval.

Table 3. Fullbacks' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Physical performance-related variables (m)					
Total distance*	10899.3±722.8	10885.5±671.4	0.13	0.89	0.02 (-0.27 – 0.3)
Low-intensity running	8343.9±501.7	8216.6±590.7	1.62	0.1	0.24 (-0.04 – 0.53)
Moderate-intensity running	1540.5±263.6	1616.1±476.4	-0.48	0.63	-0.24 (-0.52 – 0.05)
High-intensity running	1014.9±233.2	1052.8±186.8	-1.39	0.17	-0.17 (-0.45 – 0.12)
Technical performance-related variables (#)					
Goals	0±0.2	0±0.2	0.19	0.85	0.08 (-0.21 – 0.36)
Assists	0.1±0.3	0.1±0.3	0.43	0.67	0.13 (-0.16 – 0.42)
Shots	0.7±1	0.7±1	0.36	0.72	0.04 (-0.25 – 0.32)
Shots on target	0.2±0.5	0.3±0.5	-0.34	0.73	-0.08 (-0.37 – 0.2)
Passes	48.8±17.7	47.2±18.8	0.46	0.65	0.09 (-0.19 – 0.38)
Successful passes	40.7±17.4	38.9±16.9	0.46	0.65	0.1 (-0.18 – 0.39)
Crosses	2.6±2.5	2.5±1.9	-0.37	0.71	0.04 (-0.25 – 0.32)
Successful crosses	0.8±1.1	0.8±0.9	-0.11	0.91	0.04 (-0.24 – 0.33)
Dribbles	2.6±2.7	2.5±2.5	-0.31	0.76	0.02 (-0.27 – 0.3)

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Table 3. Fullbacks' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Successful dribbles	1.3±1.7	1.5±1.6	-1.01	0.31	-0.09 (-0.38 – 0.19)
Duels	17.3±6.2	18.6±6	-1.43	0.15	-0.21 (-0.49 – 0.08)
Successful duels	8.8±3.9	9.4±3.9	-1.07	0.29	-0.16 (-0.44 – 0.13)
Recoveries	8.7±3.9	8.8±3	-0.8	0.42	-0.02 (-0.31 – 0.26)
Recoveries on opponent half	2.7±2.1	2.8±1.8	-0.82	0.41	-0.07 (-0.35 – 0.22)
Touches in penalty area	1.3±1.5	1.6±1.6	-1.44	0.15	-0.2 (-0.48 – 0.09)
Interceptions	4.8±2.6	4.8±2.3	-0.17	0.86	-0.01 (-0.3 – 0.27)
Clearances	1.7±1.6	2.3±2	-2.12	0.03	-0.35 (-0.63 – -0.06)
Sliding tackles	0.4±0.7	0.4±0.7	-0.47	0.64	-0.05 (-0.33 – 0.24)
Successful sliding tackles	0.2±0.4	0.2±0.4	-0.22	0.82	-0.03 (-0.32 – 0.25)
Key passes	0.4±0.7	0.4±0.7	0.28	0.78	0.05 (-0.24 – 0.33)

* denotes variables where T-test was calculated; MW - Mann-Whitney U test; CI – confidence interval.

the group or knockout stage (all trivial-to-small ES) for FBs.

Table 4 presents descriptive statistics and differences in CMs' physical and technical performance in the group and

knockout stage of UCL. CMs covered less MIR (medium ES) and HIR (small ES) in the group compared to the knockout stage. In addition, in the group stage, CMs performed more

Table 4. Central midfielders' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Physical performance-related variables (m)					
Total distance*	11728.2±729.9	11863.5±830.1	-1.56	0.12	-0.18 (-0.4 – 0.05)
Low-intensity running	8899.3±484.8	8634.1±864.5	2.68	0.01	0.43 (0.2 – 0.66)
Moderate-intensity running	1977±405.1	2295.8±876.7	-3.55	<0.01	-0.55 (-0.78 – -0.32)
High-intensity running	851.9±249.4	933.6±194.2	-3.5	<0.01	-0.35 (-0.57 – -0.12)
Technical performance-related variables (#)					
Goals	0.1±0.3	0.1±0.3	0.1	0.92	0.02 (-0.2 – 0.25)
Assists	0.1±0.3	0±0.2	0.45	0.65	0.13 (-0.09 – 0.35)
Shots	1±1.3	1.1±1.2	-0.88	0.38	-0.04 (-0.27 – 0.18)
Shots on target	0.3±0.6	0.3±0.6	0	1	0.03 (-0.2 – 0.25)
Passes	50.3±19.7	50.5±18	-0.28	0.78	-0.01 (-0.23 – 0.22)
Successful passes	44.2±19.6	45.1±17.6	-0.63	0.53	-0.05 (-0.27 – 0.18)
Crosses	0.8±1.3	0.9±1.5	-0.92	0.36	-0.07 (-0.3 – 0.15)
Successful crosses	0.3±0.6	0.3±0.7	-0.62	0.53	-0.12 (-0.35 – 0.1)
Dribbles	1.6±1.7	2±2.1	-1.38	0.17	-0.21 (-0.43 – 0.02)
Successful dribbles	0.9±1.2	1.2±1.4	-1.23	0.22	-0.19 (-0.41 – 0.04)
Duels	16.6±6	17.2±6.4	-0.94	0.35	-0.11 (-0.34 – 0.11)
Successful duels	8.4±3.5	8.6±3.9	-0.21	0.83	-0.06 (-0.28 – 0.17)
Recoveries	8.7±3.9	8.3±4	0.83	0.41	0.09 (-0.13 – 0.32)
Recoveries on opponent half	3.5±2.3	3.4±2.3	0.15	0.88	0.02 (-0.2 – 0.25)
Touches in penalty area	1.2±1.7	1.2±1.5	-0.32	0.75	0 (-0.22 – 0.22)
Interceptions	4.4±2.7	4±2.5	1.17	0.24	0.16 (-0.06 – 0.39)
Clearances	1±1.4	1.1±1.4	-0.7	0.48	-0.08 (-0.3 – 0.15)
Sliding tackles	0.5±0.8	0.6±1	-0.08	0.94	-0.07 (-0.3 – 0.15)
Successful sliding tackles	0.2±0.5	0.3±0.8	-0.16	0.87	-0.12 (-0.35 – 0.1)
Key passes	0.3±0.6	0.3±0.6	-0.43	0.67	-0.02 (-0.24 – 0.21)

* denotes variables where T-test was calculated; MW - Mann-Whitney U test; CI – confidence interval.

LIR (small ES) compared to the knockout stage. CMs' technical performance was similar both in the group and knockout stages (all trivial-to-small ES).

Table 5 presents descriptive statistics and differences in WMs' physical and technical performance in the group and knockout stage of UCL. WMs covered less MIR (medium ES) and HIR (small ES) in the group stage compared to the

knockout stage. Also, in the group stage, WMs performed more LIR (small ES) compared to the knockout stage. In addition, WMs in the group stage had less successful dribbles (small ES) than in the knockout stage, while no differences in all other technical performance variables were found irrespective of the competition stages (all trivial-to-small ES).

Table 5. Wide midfielders' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Physical performance-related variables (m)					
Total distance*	11126.6±933.8	11093.9±1050.2	0.18	0.85	0.03 (-0.33 – 0.39)
Low-intensity running	8411.6±611	7903.8±1042.3	2.86	<0.01	0.66 (0.29 – 1.02)
Moderate-intensity running	1678.8±380.4	2062.9±941.9	-2.12	0.03	-0.63 (-0.99 – -0.26)
High-intensity running	1036.2±231.8	1127.2±260.2	-2.06	0.04	-0.38 (-0.74 – -0.01)
Technical performance-related variables (#)					
Goals	0.3±0.5	0.2±0.5	1.2	0.23	0.23 (-0.13 – 0.59)
Assists	0.2±0.5	0.1±0.4	0.45	0.65	0.18 (-0.18 – 0.53)
Shots	2.1±2	1.6±1.5	1.03	0.3	0.25 (-0.11 – 0.61)
Shots on target	0.9±1.2	0.7±0.9	0.68	0.5	0.19 (-0.17 – 0.55)
Passes	38.8±17.5	38.5±12.6	-0.48	0.63	0.02 (-0.34 – 0.38)
Successful passes	32.4±16.5	31.7±11.5	-0.37	0.71	0.04 (-0.32 – 0.4)
Crosses	2±2.5	2.2±1.6	-1.39	0.16	-0.08 (-0.43 – 0.28)
Successful crosses	0.7±1	0.6±0.8	0.43	0.67	0.14 (-0.22 – 0.5)
Dribbles	4.3±3.8	5.2±3.8	-1.73	0.08	-0.25 (-0.6 – 0.12)
Successful dribbles	2.4±2.3	3±2.3	-2.12	0.03	-0.29 (-0.65 – 0.07)
Duels	19.8±6.6	21.6±6.7	-1.59	0.11	-0.27 (-0.63 – 0.09)
Successful duels	9.1±3.8	9±3.6	0	1	0.01 (-0.35 – 0.37)
Recoveries	5.1±3.3	4.8±2.7	0.17	0.87	0.1 (-0.26 – 0.46)
Recoveries on opponent half	2.3±1.5	2.9±2	-1.41	0.16	-0.32 (-0.68 – 0.05)
Touches in penalty area	3.3±3.1	3.5±2.4	-1.48	0.14	-0.1 (-0.45 – 0.26)
Interceptions	2.7±2.5	2.7±1.8	-1.01	0.31	-0.02 (-0.38 – 0.34)
Clearances	0.7±1.3	0.7±0.9	-0.79	0.43	0.04 (-0.31 – 0.4)
Sliding tackles	0.2±0.4	0.4±0.8	-1.05	0.29	-0.43 (-0.79 – -0.06)
Successful sliding tackles	0.1±0.3	0.2±0.5	-0.77	0.44	-0.26 (-0.62 – 0.1)
Key passes	0.8±1	0.7±0.7	-0.08	0.94	0.11 (-0.25 – 0.47)

* denotes variables where T-test was calculated; MW - Mann-Whitney U test; CI – confidence interval.

Table 6 presents descriptive statistics and differences in FWs' physical and technical performance in the group and knockout stage of UCL. No differences in physical perfor-

mance were found irrespective of playing in group or knockout stage (all trivial-to-small ES) for FWs. However, FWs in the group stage executed more shots on target (small ES), passes

Table 6. Forwards' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Physical performance-related variables (m)					
Total distance*	10586.4±1108.2	10549±1233.7	0.15	0.88	0.03 (-0.39 – 0.46)
Low-intensity running	8294.2±725.1	8037.7±868.5	1.33	0.18	0.33 (-0.09 – 0.76)
Moderate-intensity running	1431.8±388.4	1583.8±814.1	-0.4	0.69	-0.28 (-0.7 – 0.15)
High-intensity running	860.4±262.6	927.5±244.9	-1.09	0.27	-0.26 (-0.68 – 0.17)

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Table 6. Forwards' match performance in different competition stages of UEFA Champions League.

	Group stage	Knockout stage	Analysis of variance	p	Effect size
	Mean±SD	Mean±SD	T/MW-test		Cohen's d (95%CI)
Technical performance-related variables (#)					
Goals	0.5±0.7	0.2±0.5	1.78	0.08	0.45 (0.02 – 0.88)
Assists	0.2±0.5	0.1±0.3	0.76	0.45	0.3 (-0.12 – 0.73)
Shots	2.7±2	2.3±1.3	0.6	0.55	0.23 (-0.2 – 0.65)
Shots on target	1.3±1.1	0.8±0.9	2	0.05	0.46 (0.03 – 0.88)
Passes	30.6±15.3	22.9±12.2	2.49	0.01	0.54 (0.1 – 0.96)
Successful passes	24.7±14	18.3±10.9	2.34	0.02	0.49 (0.06 – 0.91)
Crosses	1.1±1.4	0.7±1	1.33	0.18	0.33 (-0.1 – 0.75)
Successful crosses	0.3±0.5	0.2±0.6	0.41	0.68	0.06 (-0.36 – 0.48)
Dribbles	2.6±2.7	2.6±2.6	0	1	0.02 (-0.41 – 0.44)
Successful dribbles	1.4±1.7	1.1±1.3	0.99	0.32	0.22 (-0.2 – 0.64)
Duels	18.8±9	22.2±7.1	-2.46	0.01	-0.39 (-0.82 – 0.04)
Successful duels	7.7±4	8.4±4	-1.11	0.27	-0.19 (-0.61 – 0.24)
Recoveries	3.7±3.3	2.9±1.7	0.34	0.73	0.27 (-0.15 – 0.7)
Recoveries on opponent half	2.1±1.5	1.8±1.3	0.49	0.62	0.18 (-0.25 – 0.6)
Touches in penalty area	4.2±3	3.8±2.1	0.31	0.75	0.14 (-0.28 – 0.57)
Interceptions	2±2.3	1.9±1.5	-0.33	0.74	0.06 (-0.36 – 0.48)
Clearances	0.5±1	0.8±1	-1.23	0.22	-0.24 (-0.66 – 0.19)
Sliding tackles	0.3±0.9	0.2±0.4	0.03	0.97	0.16 (-0.26 – 0.59)
Successful sliding tackles	0.1±0.5	0.1±0.3	0.08	0.93	0.11 (-0.31 – 0.53)
Key passes	0.7±1	0.4±0.5	0.83	0.41	0.32 (-0.11 – 0.75)

* denotes variables where T-test was calculated; MW - Mann-Whitney U test; CI – confidence interval.

(medium ES), and successful passes (small ES) compared to the knockout stage. Also, less duels (small ES) were performed by FWs in the group compared to the knockout stage. All other technical performance variables for FWs were similar irrespective of the competition stages (all trivial-to-small ES).

Discussion

This study aimed to compare match performance in the group and knockout stage of UCL for players in different playing positions. The main results were that (i) the technical performance of all players was consistent across both stages of the competition, except for FWs who achieved less shots on target, total and successful passes, and more duels in the knockout compared to the group stage, (ii) CDs, CMs, and WMs covered significantly more MIR and HIR in the knockout compared to the group stage, while FBs' and FWs' physical performance were similar irrespective the competition stage. Considering evident position-specific changes, the hypothesis that both physical and technical performance will be significantly different by group stages was partially accepted.

Representing the overall amount of space covered throughout the match, TD is one the most common variables used to assess physical performance in football (Modric et al., 2019). Previous research investigating TD through the various competition stages reported diverse results. Briefly, Chmura et al. showed that the average TD had the highest value at two-thirds of the season in the German Bundesliga (Chmura et al., 2019), while Li et al. indicated that average TD in the Chinese Super League reached its peak in the initial and fi-

nal fixtures (Li et al., 2023). In contrast, our results showed no significant difference in TD irrespective of the competition phase of UCL, indicating similar overall running efforts in the group and knockout stage of most elite football players in all playing positions. The possible explanation for such inconsistent findings compared to previous studies (Chmura et al., 2019; Li et al., 2023) may be a different concept of football competition analyzed (De Albuquerque Freire et al., 2022). While previous studies analyzed match performance in league type of competition (Chmura et al., 2019; Li et al., 2023), the current study analyzed tournament type of competition. In such competitions, the later (i.e., knockout) stage consists of higher-quality teams compared to the earlier (i.e., group) stage which consists of lower-quality teams. However, in UCL, both greater and lower-quality teams typically demonstrate similar TD (Modric et al., 2023), possibly explaining their similar TD in both group and knockout stages.

Arguably some of the crucial running performance indicators in football are MIR and HIR because of its relationship with match outcome (Chmura et al., 2018) and situational efficacy (Modric et al., 2019), respectively. The research investigating these metrics across various competition stages remained equivocal. For example, Chmura et al. indicated the lowest MIR and HIR values during the initial and final fixtures (Chmura et al., 2019). However, Li et al. reported gradually increased MIR and HIR in the final periods of the Chinese Super League (Li et al., 2023). Despite the different concepts of football competitions, to some extent, our results can be comparable to the results from the study of Li et al.

who highlighted increased physical performance in later competition stages. Specifically, our results showed significantly greater MIR and HIR in the knockout (i.e., the later stage of the season) compared to the group (i.e., earlier stage of the season) for CDs, CMs, and WMs (all small to medium ES). Although it did not reach statistical significance, our descriptive parameters show a similar trend among FWs and FBs as well, suggesting heightened match intensity in the knockout stage for all players. Due to the higher standard of teams competing in the UCL knockout stage (i.e., as discussed previously), it is possible that players performed more attacking actions in the knockout than in the group stage. Since attacking actions are related to intensive efforts (Andrzejewski, Chmura, Konefał, Kowalczyk, & Chmura, 2017; Di Salvo, Gregson, Atkinson, Tordoff, & Drust, 2009; Modric et al., 2023), it may explain greater physical performance at moderate and higher speeds (i.e., MIR and HIR).

Irrespective of causality, it is evident that achieving the football match-play level required in the knockout stage of UCL might not be possible without a high level of physical performance at moderate and higher speeds (Mackenzie & Cushion, 2013). However, for competitive success in football, players' technical performance is decisive (Moalla et al., 2018; Modric, Versic, & Jelcic, 2022). As teams promoted to the knockout stage can be characterized as more successful compared to those not promoted from the group stage, it is expected that players from teams playing in the knockout stage will achieve increased technical performance. However, our results showed no significant differences in technical performance irrespective of playing either in the group or knockout stage for almost all players. Specifically, all CDs', FBs', CMs', and WMs' technical performance variables were similar both in the group and knockout stage of UCL. Such findings are generally in line with previous studies which demonstrated weak differences in technical performance in various competition stages (Chmura et al., 2019; Li et al., 2023). Therefore it seems that the technical performance of top-elite football players (i.e., those playing in UCL) was at a high level irrespective of the competition stage, indicating the importance of players' technical proficiency throughout the whole season.

However, it is noteworthy that the technical performance of top-elite FWs was dependent on the competition stage. Specifically, our results showed that FWs had more duels and less shots on target, total, and successful passes in the knockout stage. This may be a consequence of their playing against higher-standard defenders in the knockout stage. In general, defenders' main match duty is to prevent the opponent's attacks on their goal (Modric et al., 2019). To complete this efficiently, defenders keep close to the opposing players, most often to the FWs', disabling them from receiving the ball and performing passes and/or shots. Due to the increased standard of teams competing in the UCL knockout stage (i.e., as discussed previously), it is possible that defenders from the knockout stage were more efficient in these duties. As a consequence, FWs in knockout stage matches likely needed to

be more active to achieve their good positioning, which may explain their increased number of duels. For the same reason (i.e., playing against higher-standard defenders), FWs in the knockout stage had less opportunities to receive the ball than in the group stage matches, resulting in their decreased shots on target, total, and successful passes.

Several limitations should be noted when interpreting the findings of this study. Firstly, the influence of contextual factors that can affect match performance (Jerkovic, Modric, & Versic, 2022), such as the quality of the opponent, match outcome, and location, were not considered. However, as previous research indicated the weak effect of these factors on match performance in UCL (Modric et al., 2023), the influence on match performance in the current study may be negligible. Secondly, match observations for group and knockout stage matches were unequal in their sample sizes, and such discrepancy likely influenced the findings. Thirdly, for methodological reasons, we included only players who played a whole match, which reduced the number of observations and may have affected match performance. Further studies with longitudinal designs that control for contextual factors are necessary to confirm the present results. Finally, for a more detailed understanding of position-specific match performance in different competition stages, there should be some attempt to perform a deeper analysis of playing positions; for example, splitting the CMs into the defensive, central, and offensive midfielders.

This research suggests two main practical applications. Firstly, the results revealed that the technical performance of UCL players was in general similar irrespective of the competition stage. As UCL is the highest-level football competition consisting of the world-best football players whose technical performance is certainly on a high level (Modric et al., 2023), such findings suggest that the technical proficiency of UCL players was on the highest possible level both in the group and knockout stages. Therefore, participation in UCL, even in the group stage, requires technical high-skilled players. This information may be of great help to the football coaches in the decision-making process when selecting the players for specific matches during the whole competitive season. Secondly, our results indicated that CDs, CMs, and WMs covered more MIR and HIR in the knockout stage than in the group stage of UCL. Considering their similar TD irrespective of the competition stage but greater distance covered at higher speeds, such findings suggest that for CDs, CMs, and WMs matches played in the knockout stage were characterized by greater match intensity compared to the matches played in the group stage of UCL. Therefore, physical conditioning programs preceding matches of the knockout stage for these players should be designed to enable them to respond to these needs. In addition, considering their increased HIR, which may be associated with greater fatigue, football coaches should consider extended periods of recovery or supplementary practices (e.g., ice submersions, massage, etc.) following matches of the knockout stage.

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Conflict of Interest

The authors report no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Relationship between Rectus Femoris Muscle Architecture and Isokinetic Peak Knee Extension Torque in Physically Active Men

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Abstract

The aim of this study was to examine the relationship between rectus femoris muscle architecture and isokinetic concentric peak knee extension torque at various velocities. Twenty physically active men (age = 21.1 ± 1.41 years, weight = 69.9 ± 5.7 kg, height = 176.4 ± 7.07 cm) had their isokinetic concentric knee extensor (Biodex 4 Pro, Biodex Medical Inc, Shirley, USA) strength assessed at 60°/sec, 180°/sec and 300°/sec. Pennation angle, muscle thickness, and fascicle length of the rectus femoris was measured using real-time B-mode ultrasound (Logiq P5, GE Healthcare, UK). Pennation angle had a significant positive correlation to peak torque at 60°/sec ($r=0.731$, $p=0.001$), 180°/sec ($r=0.802$, $p=0.001$) and 300°/sec ($r=0.685$, $p=0.001$). There was a significant positive correlation between muscle thickness and peak torque at 60°/sec ($r=0.718$, $p=0.001$), 180°/sec ($r=0.749$, $p=0.001$) and 300°/sec ($r=0.722$, $p=0.001$). However, there was no significant correlation between fascicle length and the isokinetic peak torque values ($p>0.05$). In addition, pennation angle and muscle thickness were found to be significant contributors for predicting isokinetic knee extension torque ($R^2=0.47-0.64$; $p<0.01$). Pennation angle and muscle thickness best predicted peak knee extension torque at 180°/sec (explained variance = 64% and 56%, respectively). This study suggests that pennation angle and muscle thickness can be used to predict isokinetic knee extension torque in physically active men.

Keywords: football, isokinetic, ultrasound, skeletal muscle

Introduction

The measurement of muscular strength is of clinical significance given its relationship to morbidity rates in males (Ruiz et al., 2008) and injury risk identification (Duhig et al., 2019; Van Dyk et al., 2016; Zinke, Warnke, Gäbler, & Granacher, 2019). Isokinetic dynamometry is the gold standard in strength assessment and provides reliable measures of joint torque (Brown & Greig, 2024; Verdijk, Van Loon, Meijer, & Savelberg, 2009). However, maximal isokinetic strength assessment can be challenging. It requires a skilled technician, familiarization period and is expensive. One

alternative to the aforementioned challenges is the use of surrogate measure such as ultrasonography, which can be used to assess muscle function. Ultrasound examination is a simple, non-invasive solution for measuring the architectural features of muscles, but it also requires an experienced technician (Chauhan, Hamzeh, & Cuesta-Vargas, 2013).

Muscle architecture consists of pennation angle, muscle thickness and fascicle length (Coratella et al., 2020) and plays a pivotal role in muscle performance. The architectural characteristics have an effect on force generation and speed of contraction of the muscle-tendon unit (De Boer



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et al., 2008; Lieber, 2010; Nuell et al., 2021). Increasing or decreasing in parallel sarcomeres leads to greater or lesser pennation angles which affects the physiological cross-sectional area and in turn increases or decreases force generation (Cunnane et al., 2023; Dias et al., 2016). The fascicle length directly affects the force-velocity performance and the force transferred to tendons (Abe, Fukashiro, Harada, & Kawamoto, 2001). The relationship between ultrasound measures of quadriceps muscle thickness and pennation angle is well established (Abe, Kojima, & Stager, 2014; Cadore et al., 2012; Fukumoto et al., 2011; Strasser, Draskovits, Praschak, Quittan, & Graf, 2013; Watanabe et al., 2013). Raj, Bird and Shield (2017) have shown that quadriceps muscle thickness is a robust predictor of knee extensor strength in older adults, however, there is limited research examining the relationship between muscle architecture and strength at isokinetic velocities. The role muscle architecture plays in force production is recognized, although further exploration on its influence on joint velocity is warranted.

The aim of this study is to examine the relationship between rectus femoris architecture and knee extension torque at various velocities. We hypothesized that (i) muscle architecture would have a significant positive correlation with knee extension torque, and (ii) muscle architecture can be used as an independent contributor to peak torque prediction models.

Methods

Experimental Design

In this study, we used a cross-sectional study design to examine whether there is a relationship between rectus femoris muscle architecture and peak torque during a concentric isokinetic knee extension at various velocities. Each participant visited the clinic on three separate days at approximately the same time. During the first visit, the participants were familiarized with the whole isokinetic dynamometer protocol before the actual testing days. The participants' anthropometric measurements were taken on the same day. The second visit occurred at least 48 hours later to avoid morphological acute changes that may have occurred after the familiarization session. At the second visit, a 2D real-time B-mode ultrasound was performed by a musculoskeletal physician with more than 10 years of experience in musculoskeletal ultrasound analysis to determine the pennation angle, muscle thickness and fascicle length of the dominant leg of the rectus femoris muscle. During the third visit, isokinetic dynamometer strength assessments were performed at 60°/sec, 180°/sec and 300°/sec.

Participants

Twenty healthy and physically active men took part in this study (age = 21.1±1.4 yrs, weight = 69.9±5.7 kg, height = 176.4±7.07 cm). Sixty percent of the participants played football (n=12), 20% volleyball (n=4) and 20% basketball (n=4). Participants' training status (experience and training volumes) are presented in Table 1. The inclusion criterion was that they had been exercising for ≥5 years, while the exclusion criteria for participants included current or previous musculoskeletal injuries and a history of lower limb surgery. Sample size was determined by a priori power analysis using G-Power (3.1.9.3). The effect size was based on previous

studies that reported a coefficient of determination between the combined rectus femoris and vastus intermedius thickness and maximal isometric knee extension at 60°/sec ($R^2 = 0.60$) (Raj et al., 2017). An effect size of 0.77 was considered a reasonable and conservative point for determining the sample size. The type I error (α) was 0.05, and power ($1-\beta$) 0.80 with a two-tail bivariate normal model. The model indicated a minimum total sample size of 10 participants, but considering a miss rate, we decided to have a total of 20 participants. All participants were informed of the benefits and risks of the investigation prior to signing an institutionally approved informed consent document to participate in the study. This study was approved in advance by Research Ethics Committee of Inonu University (approval number: 2017/61). Each participant voluntarily provided written informed consent before participating.

Body Composition Measurements

All measurement procedures were performed with minimal clothing (e.g. shorts, underwear, no shoes and with jewellery removed). Participants' height was measured with a portable stadiometer (Seca Ltd., Bonn, Germany) with an accuracy of 0.1 cm, with the head in the Frankfort plane, while the body was upright and weight evenly distributed across both legs. Body mass and body fat percentage were measured using a body composition scale with a capacity of 270 kg and a sensitivity of 100 g (Tanita SC-330S, Amsterdam, The Netherlands).

Muscle Architecture Measurements

Architecture was assessed using 2D real-time B-mode ultrasound device (Logiq P5, GE Healthcare, UK) with a 4.5 cm linear array transducer (frequency 10-MHz and depth 5 cm). Imaging was conducted supine, with the rested leg supported in passive extension. Rectus femoris pennation angle, muscle thickness and fascicle length were measured with the probe placed transversely and longitudinally. The transducer was placed with adequate use of contact gel and minimal pressure to avoid excessive compression of the muscle at 50% of the distance from the superior border of the patella and great trochanter. Pennation angle was determined as the angle between the muscle fascicles and the deep aponeurosis. Muscle thickness was determined as the distance between the superficial and deep aponeurosis. Fascicle length across the deep and superficial aponeurosis was estimated from isolated muscle thickness and angle using the following equation:

$$\text{fascicle length} = \text{muscle thickness} \cdot \sin \alpha \cdot 1$$

where α is the pennation angle of the muscle as determined by ultrasound (Kawakami, Abe, Kuno, & Fukunaga, 1995). All measurements were carried out by the same experienced sonographer who was blinded to participants. Three ultrasound images were taken from each trial in order to increase the reliability and the average of the three values for each variable was used for statistical analysis (Kwah, Pinto, Diong, & Herbert, 2013). The images were assessed using software in-built in the Ultrasound Device, real time. Intrarater reliability for muscle architecture was evaluated using intra-class correlation coefficient (ICC), coefficient of variance (CV), and standard error of mean (SEM). The ICCs were 0.997, 0.999 and 0.989 for pennation angle, muscle thickness and fascicle length, respectively. The mean CVs

and SEMs were 1.26 % and 0.50° and 0.42% and 0.07 cm and 1.33% and 0.19 cm for pennation angle, muscle thickness and fascicle length, respectively.

Isokinetic Concentric Peak Torque Assessment

Participants were provided detailed briefing before data collection session. Approximately 48 hours before the test, participants were informed about the isokinetic dynamometer and a familiarization trial conducted. Concentric isokinetic peak torque test of the dominant leg (based on kicking preference) was measured with the use of a dynamometer (Biodex 4 Pro, Biodex Medical Inc, Shirley, USA). The control strap was secured over the chest and abdominal area to minimize the motion of the upper body and to avoid compensatory movement (Feiring, Ellenbecker, & Derscheid, 1990). The lever range of motion was set between 0° to 100° flexion. Prior to the testing protocol, a warm-up consisting of four repetitions of 50% maximal effort was performed at each speed (300°/sec, 180°/sec, 60°/sec) (Mota et al., 2015). After warming up, participants performed four repetitions of concentric knee extension tests at different angular speeds of the dominant leg across 100 degrees knee extension (300°/sec, 180°/sec, and 60°/sec, respectively). The order of angular velocities was randomized for each participant, who performed four contractions at each speed with a 30-s rest between contractions and a 1-min rest between speeds (Raj et al., 2017). Torque values from the trials were automatically corrected for the effects of gravity by the Biodex Advantage Software program (ver-

sion 4.0, Biodex Medical Inc, Shirley, USA). The highest peak torque obtained at the end of the repetitions was recorded in Newton meters (Nm).

Statistical Analysis

Statistical analyses were performed using GraphPad Prism version 7.0 (GraphPad Software, USA). All data are presented as mean $\pm$ (SD: standard deviation). The homogeneity of the study data was determined by skewness and kurtosis values (between 1.5 and -1.5) and the Shapiro Wilks test used as sample size was less than 50. The correlation between muscle architecture and isokinetic performance was evaluated by means of the Pearson product-moment correlation coefficient with 95% confidence interval. Correlation was interpreted as follows: an r between 0 and 0.3, was considered small; 0.31–0.49, moderate; 0.5–0.69, large; 0.7–0.89, very large; and 0.9–1, near perfect for relationship prediction (Hopkins, 2013). Linear regression analysis was used to determine whether the muscle architecture values predicted the isokinetic performance scores. The alpha level was accepted at $p < 0.05$.

Results

Participants' demographic, training status, architectural characteristics and isokinetic peak torque values are shown in Table 1 as mean values $\pm$ standard deviations (min-max). Data were normally distributed. The isokinetic measures showed a negative trend of increasing torque as the angular velocity decreased.

Table 1. Descriptive statistics (mean $\pm$ SD; min-max) for the physical characteristics, training status, muscle architecture characteristics and isokinetic peak torque values of participants (n=20)

Physical characteristics				
Age (year)	Mass (kg)	Height (cm)	BFP (%)	BMI (kg/m2)
21.1±1.3 (19-23)	69.9±6.3 (60.6-80)	174.9±7.2 (1.61-1.86)	9.3±2.1 (5.56-12.5)	22.7±1.6 (19.56-25.31)
Training status				
Training experience (year)	Training frequency (session/week)		Training frequency (hours/week)	
7.3±1.8 (5-11)	4.3±0.9 (3-6)		5.8±1.6 (3-8.5)	
Muscle architecture characteristics				
Muscle thickness (cm)	Pennation angle (°)		Fascicle length (cm)	
3.20±0.4 (2.6- 3.8)	16.6±2.3 (13.3-21.2)		11.06±0.9 (9.9-13.2)	
Isokinetic peak torque values				
300°/sec (Nm)	180°/sec (Nm)		60°/sec (Nm)	
125.1± 25.5 (85-164.3)	154.7±30.3 (100-215)		202.9±40.6 (110-260.1)	

Note n: number of participants; BFP: body fat percentage; BMI: body mass index

Figure 1 shows a very large positive linear correlation between muscle thickness and pennation angle ($r=0.801$; 95% CI=0.584 to 0.922, $p < 0.01$) and a large negative correlation between fascicle length and pennation angle ($r=-0.544$; 95% CI=-0.772 to -0.276, $p=0.013$). There was no significant relationship between muscle thickness and fascicle length ($r=0.058$; 95% CI=-0.333 to 0.453, $p=0.807$).

Figure 2 shows large and very large correlation between

the pennation angle and 60°/sec ($r=0.731$; 95% CI=0.426 to 0.886), 180°/sec ($r=0.802$; 95% CI=0.557 to 0.918), 300°/sec ($r=0.685$; 95% CI=0.348 to 0.865). Very large correlations were found between muscle thickness and 60°/sec ($r=0.718$; 95% CI=0.403 to 0.880), 180°/sec ($r=0.749$; 95% CI=0.458 to 0.894), 300°/sec ($r=0.722$; 95% CI=0.410 to 0.882). However, no correlation was found between fascicle length and isokinetic peak torque values ($p > 0.05$).

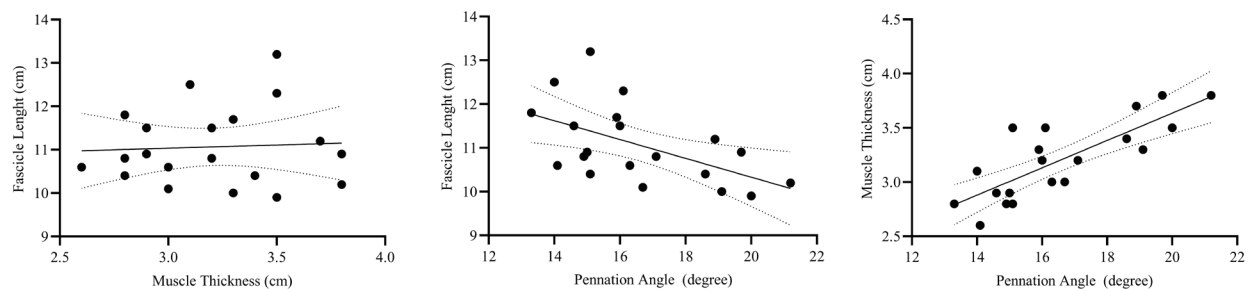


FIGURE 1. Correlation between the architectural values of the rectus femoris with linear regression line with 95% CI.

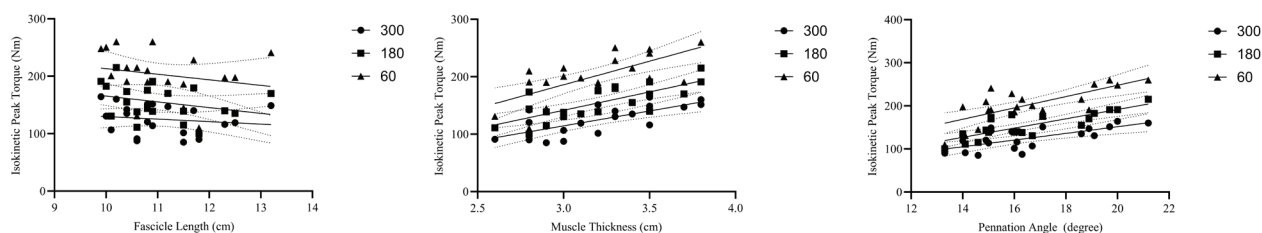


FIGURE 2. Illustrates the correlation between the isokinetic testing scores and muscle architecture. Individual data points are presented as well as linear regression line with 95% CI.

Table 2 shows that pennation angle and muscle thickness could predict peak torque ($p < 0.01$), but fascicle length could not predict peak torque ($p > 0.05$). Rectus femoris pennation angle was a significant independent contributor to the models predicting the isokinetic knee extensor torque at 60°/sec, 180°/sec, 300°/sec ($R^2 = 0.53, 0.64$ and 0.47 respectively). In addition,

rectus femoris muscle thickness was a significant independent contributor to the models predicting isokinetic knee extensor torque at 60°/s, 180°/s and 300°/s ($R^2 = 0.51, 0.56$ and 0.52 , respectively). Peak torque at 180°/s was best predicted when using pennation angle and muscle thickness as a single predictor with 95% CI.

Table 2. Regression constants and coefficients for relationships between muscle architecture and isokinetic strength

Dependent variables	Regression equation	R ²	Adj-R ²	p value
PT at 60°/sec (Nm)	-15.183 + 13.150 (pennation angle)	0.53	0.51	0.001*
	-59.955 + 82.015 (muscle thickness)	0.51	0.49	0.001*
	310.279 + (-9.704) (fascicle length)	0.05	0.21	0.367
PT at 180°/sec (Nm)	-23.667 + 10.757 (pennation angle)	0.64	0.62	0.001*
	-49.881 + 63.844 (muscle thickness)	0.56	0.54	0.001*
	265.591 + (-10.018) (fascicle length)	0.09	0.04	0.207
PT at 300°/sec (Nm)	-3.252 + 7.740 (pennation angle)	0.47	0.44	0.001*
	-40.796 + 51.767 (muscle thickness)	0.52	0.49	0.001*
	173.785 + (-4.398) (fascicle length)	0.02	-0.03	0.517

Note PT: Peak knee extension torque; R²: variance explained; Adj-R²: Adjusted R²; * denotes significance $p < 0.05$

Discussion

The findings from this study suggest rectus femoris pennation angle and muscle thickness are associated with peak torque across all three velocities, and muscle architecture can be used to predict isokinetic knee extensor torque in physically active men.

There is growing evidence to support the use of muscle architecture to predict strength (Abe et al., 2015; Moreau, Simpson, Teeffey, & Damiano, 2010; Strasser, et al., 2013). Previous study by Raj et al. (2017) showed that combined rectus femoris and vastus intermedius thickness was a significant ($p < 0.05$) independent contributor to predict isometric and isokinetic knee extensor torque at 60°/s ($R^2 = 0.63$), 120°/sec ($R^2 = 0.68$), 240°/sec ($R^2 = 0.65$) and 360°/sec ($R^2 = 0.66$).

Treize et al. (2016) found that vastus lateralis, vastus intermedius and rectus femoris architecture and neuromuscular variables showed the greatest effect on maximum knee extension torque ($R^2 = 0.72$). Housh, Housh, Johnson and Chu (1993) reported significant ($p < 0.05$) correlations between the isokinetic peak torque for arm flexor and muscle cross sectional area (forearm) at isometric ($r = 0.73$), 60°/sec ($r = 0.85$), and 120°/sec ($r = 0.76$). Our findings further support the relationship between muscle architecture and function as they showed significant correlations between peak torque at all angular velocities ($r = 0.685-0.802$; 47-64% explained variance). In addition, muscle architecture was a significant independent contributor to the models predicting isokinetic knee extensor torque at 60°/sec, 180°/sec, and 300°/sec ($p < 0.01$).

Both type I and II fibers can be maximally activated at lower speeds, while the slow-twitch type I fibers initially remain passive at higher angular velocities (Iodice et al., 2020; Kannus, 1994). Therefore, it is to be expected that the torque is greater at lower speeds. It has been mentioned that all three muscle fibers type (such as Type I, IIa and IIb) contribute to force generation at low angular velocities and there is a transition towards Type IIb as the angular velocity increases (Kocahan et al., 2017). Also, it has been suggested that larger pennation angles may be associated with slower contraction velocities, and that increasing the pennation angle can negatively affect a muscle's ability to produce maximum contraction velocity (Narici, 1999; Wakeling et al., 2020). Therefore, training at low and moderate angular velocities (at 30°/sec, 60°/sec or 180°/sec, etc.) may be preferred for increasing pennation angle and muscle thickness. Conversely, lesser pennation angles increase muscle shortening velocity, as longer fibers shorten at greater absolute speeds than a similar muscle with shorter fibers (Cormie, McGuigan, & Newton, 2011). Gür, Gransberg, van Dyke, Knutsson and Larsson (2003) analyzed muscle fiber types via biopsy with participants performing isokinetic maximal contraction at 30°/sec and 240°/sec. Significant correlations were observed between both peak torque at 240°/sec and the relative type II fiber area ($r=0.46$; $p<0.05$). Furthermore, Bartolomei et al. (2001) found the existence of positive correlations between pennation angle and muscle thickness of the vastus lateralis and sprint agility test in field hockey players ($r=0.62$; $p=0.006$ and $r=0.51$; $p=0.032$, respectively). Therefore, isokinetic exercise programs performed at high angular velocities may be optimal for improving explosive power performance.

Although the muscle architecture contributes significantly to the differences in the torque, other factors also have an effect on torque performance. Factors such as microRNA3 expression and motor unit action potential amplitudes at the cellular level (Mitchell et al., 2018). Therefore, in our study, it is not surprising that only 47–64% ($R^2=0.47-0.64$) of the variability in the torque at different angular velocities of movement is accounted for by muscle architecture.

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Conflict of Interest

The author declares that there is no conflict of interest.

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The finding the rectus femoris fascicle length is not related to isokinetic strength does not support our hypothesis of a positive relationship between muscle architecture and strength. However, a previous study found that vastus lateralis fascicle length is related to time to peak isokinetic torque, vertical jump height and timed up and go performance, but not peak torque (Raj et al., 2017). Also, Abe et al. (2001) observed that sprinters have longer vastus lateralis and gastrocnemius medialis fascicle lengths than endurance runners. These studies imply that muscle fascicle lengths may influence muscle contraction speeds, but not necessarily muscle strength.

The study emphasizes the importance of ultrasound for athletes, which offers a more accessible and faster examination option than other imaging techniques (such as MRI or CT). In addition, our results may provide important insights into the relative importance of architectural variables for torque in isokinetic activities. However, this study is not without limitations. First, this paper conducted the research with physically active men, which may result in low generalizability in other groups (such as patient, sedentary or gender differences). Second, we only examined one fascicle of interest at a given point and this may not be truly representative of how the whole muscle performs (Charles, Kissane, Hoehfurner, & Bates, 2022). Finally, the design of the study was cross-sectional, which does not allow the identification of which angular velocity can modify muscle architecture characteristics in the long-term. We suggest future studies focus on evaluating how isokinetic exercise can alter rectus femoris architecture. Despite these limitations muscle architecture can be used to reliably predict strength in young males.

Conclusion

Muscle architecture parameters are strongly correlated with isokinetic peak torque. This study shows the validity of the analysis of muscle architecture to predict isokinetic knee extensor torque in physically active men. We suggest the use of ultrasound to understand strength capabilities, as it is an easier and more accessible method in comparison to the isokinetic test.

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ORIGINAL SCIENTIFIC PAPER

Positional Differences in the Efficacy of Critical End-of-Game Possessions in EuroLeague Basketball

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Abstract

In close basketball games where the final score is within a 5-point difference, end-of-game possessions are very important. However, previous studies have not examined the efficiency of various play types' possessions regarding player positions. Therefore, the study aimed to identify play type actions during end-of-game possessions across player positions (guard, forward, and big) that directly influence the possession's outcome. The analysis evaluated 1009 possessions from 100 EuroLeague games from the 2022-2023 and 2023-2024 regular seasons using Sport Scout observation software (SportScout Group, Thessaloniki, Greece). The variables observed included isolation (ISO), pick and roll ball handler (PnRBH), pick and roll screener (PnRSC), off-ball screens (OBS), transition (TR), cuts (CUT), spot-up shooting (SUP), post-ups (PUP), handoffs (HO), inbound plays (IN), and putbacks (PB). Possession efficiency was evaluated based on points per possession (PPP) and the possession's outcome (positive or negative). Chi-square (χ^2) tests compared categorical variables, while the Kruskal-Wallis test assessed PPP differences across player positions. Additionally, the chi-square automatic interaction detector (CHAID) decision tree model was used to classify data and make meaningful predictions for the possession play types. The findings revealed that player positions significantly influenced possession efficiency, with noticeable variations in possession distribution. Decision tree analysis underscored the impact of possession play types on outcomes across player positions. In conclusion, the study highlights the predominant role of guards in end-of-game possessions, relying heavily on isolation plays but achieving higher efficiency with teamwork-oriented strategies. Forwards demonstrate effectiveness in offensive rebounding situations and off-ball movement, while big players exhibit efficiency in proximity to the basket.

Keywords: basketball, play type, observation analysis, crunch time

Introduction

Observational analysis in basketball entails systematically examining various aspects of the game to gain insights into player performance and team dynamics. This analytical approach is essential for coaches, analysts, and researchers aiming to deepen their understanding of the sport. Evaluating critical performance indicators of players and teams is essential in sports analysis (Sarlis & Tjortjis, 2020).

The offensive strategies employed by teams are essential

to their success (Fichman & O'Brien, 2019). Teams adopt various offensive play types tailored to the abilities of their players (Chen, Zhang & Xu, 2023). In basketball, a play type is a specific pattern or strategy a team utilizes during offensive possessions to create scoring opportunities. While a team typically incorporates multiple play types into its offensive playbook, not all may suit every player. Examples of play types include isolation (ISO), pick and roll (PnR), off-ball screens (OBS), transition (TR), cuts (CUT), spot-up



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shooting (SUP), post-ups (PUP), handoffs (HO), inbound plays (IN), and putbacks (PB) (Božović, 2021; Christmann, Akamphuber, Müllenbach & Güllich, 2018).

Classifying actions within play types and analyzing their relationship to the outcome of these actions allows coaches to develop and improve offenses based on team strengths and individual player abilities. Research into team offensive play types underscores the importance of PnR effectiveness for overall team success (Marmarinos, Apostolidis, Kostopoulos & Apostolidis, 2016), the efficacy of ISO actions in EuroLeague play (Matulaitis & Bietkis, 2021), and the effectiveness of shorter-duration actions (Christmann et al., 2018). Akinci (2023) discovered that playoff teams prominently used OBS possessions, while PnR ball handlers (PnRBH), and SUP possessions and points were used by highly skilled players.

Traditionally, basketball player positions have been classified into three main categories based on their similarities: guards, forwards, and centers (Sampaio, Ibañez Godoy, Gómez Ruano, Lorenzo Calvo, & Ortega Toro, 2008). The analysis of players' technical-tactical performance has been studied for several years (Remmert & Chau, 2019; Vaquera, García-Tormo, Gómez Ruano, & Morante, 2016). According to Sampaio, Janeira, Ibañez, and Lorenzo (2006), evaluating players' game performance requires reference to specific normative data pertaining to their playing position. In recent years, researchers have increasingly employed dynamic and sophisticated analytical techniques such as decision trees (DT) to identify patterns in player behavior and performance (Morgan, Williams & Barnes, 2013).

In competitive basketball, match outcomes often depend on end-of-game possessions, with strategic maneuvers and tactical decisions of players under intense pressure playing a critical role (Bar-Eli & Tractinsky, 2000). During these critical moments, the ability of the players to rise to the occasion becomes the defining factor. However, few studies have explored the impact of different play types on game performance during end-of-game possessions at the player level in close EuroLeague contests.

The existing literature primarily categorizes players into playing positions like ball handlers, wings and bigs, exploring offensive structures, player roles, and possession styles in basketball. Still, it does not extensively analyze how these roles influence performance in clutch situations, with player play type being a relatively underexplored area, with limited research on this topic (Chen et al., 2023). While some studies touch upon the play types in critical end-of-game situations (Christmann et al., 2018; Foteinakis, Pavlidou & Stavropoulos, 2024), revealing that more cooperative play types were more efficient in crunch time situations and longer-lasting possessions were less effective, there is a lack of comparative analysis between different player positions in the context of critical end-of-game possessions.

Understanding how guards, forwards, and bigs perform in high-pressure situations and how their roles influence team success in crunch time across different play type possessions is essential for optimizing performance in late-game scenarios.

Addressing these research gaps can contribute to a better understanding of positional differences in the efficacy of critical end-of-game possessions and inform coaches, players and analysts in developing more effective game strategies.

Therefore, this study aimed to identify the play type actions during end-of-game possessions across player positions (guard, forward, and big) that directly influence the outcome of each possession.

Methods

Sample, Data Collection, and Observation Instrument

The study sample comprised 100 games from the 2022–2023 and 2023–2024 regular seasons of the EuroLeague, observed via the publicly accessible EuroLeague TV (<http://tv.euroleague.net/>). Matches were selected based on a criterion of ending with a score differential within five points to focus on games where the outcome remained uncertain in the final 2 min. Categorized as miscellaneous play types, 233 possessions were excluded from analysis, resulting in 1009 possessions recorded within the last 120 seconds of the fourth quarter and overtime. The mean score difference at the games' conclusion was 3.17 ± 1.44 points (mean $\pm$ SD).

The sample encompassed all EuroLeague teams from the specified regular seasons. As the study was purely observational, conducted in a natural setting without any experimentation, and the data collected were publicly televised and accessible, obtaining informed consent from the competitors was not deemed necessary (American Psychological Association Ethical Principles of Psychologists and Code of Conduct, 2002).

The Sport Scout observation analysis software (Sport Scout STA Ver. 3.2, SportScout Group, Thessaloniki, Greece), facilitated game analysis, with results recorded on a Microsoft Excel spreadsheet (Microsoft Inc., Washington USA).

Procedure and Variables

The games were analyzed by a seasoned professional basketball coach and Sports Sciences graduate with over 20 years of experience in basketball performance analysis. The observation protocol was designed to document the offensive play types used by the teams during the games. Based on previous research (Božović, 2021; Christmann et al., 2018), 11 play-type possessions were selected, with definitions provided in Table 1. Player positions were categorized into three levels: guard, forward, and big. To assess the efficiency of end-of-game possessions play types, the following variables were analyzed: points per possession (PPP) and possession outcome (whether a 2-point or 3-point shot was made, fouls without and with bonus free throws, goal fouls, and turnovers, including offensive fouls).

A possession was deemed positive if the team scored a two-point (2p) or three-point (3p) shot, obtained a basket and foul for 2p or 3p (2pF, 3pF), or received a foul without bonus free throws (FO) or with bonus free throws (FT1, FT2, FT3). Conversely, a possession was considered negative if the team missed a shot (–3p, –2p), committed any type of turnover (TO), or had a shot blocked (BL).

The observer reanalyzed five randomly selected games after a 4-week interval to validate the data. The results from the initial observation were compared with those from the second, yielding Cohen's Kappa intra-observer correlation coefficients ranging from 0.81 to 0.96, indicating excellent agreement rates (Altman, 1991).

Statistical Analysis

This study utilized SPSS 29 software (IBM Corp, Armonk,

Table 1. Definition and description of the examined play types and the offensive outcome of the possession.

Variables	Definition and description
Offensive Outcome	
Positive	Possession finished with a made Field Goal for 2 or 3 points (2P, 3P), a basket and foul for 2-p or 3p (2PF, 3PF), and when the offense received a foul with no bonus Free Throws (FO) or a foul with bonus Free Throws (FT1, FT2, FT3).
Negative	Possession finished with a missed shot (-3P, -2P), made a turnover of any kind (TO, including offensive fouls) and had a blocked shot (BL).
PPP	The average number of points that a team scores during a possession.
Offensive Play Types	
Pick and Roll Ball-Handler (PnRBH)	Possession finished by the ball-handler in the pick-and-roll. Includes off-dribble shots or direct shots before dribbling off the screen, pull-ups, and floaters, by that player. Includes also possessions where the ball handler rejects the screen (dribbles away from the screen).
Pick and Roll Screener (PnRSC)	Possession play type where the ball-screener receives the ball. These are the slips, rolls, and pops from screeners in the pick-and-roll.
Transition (TR)	Transition possession before the defense is set within 5-8 seconds.
Spot Up (SUP)	Possession finished with a catch and shot or catch and drive. They can be catching and shooting, attacking a close-out by dribbling into a pull-up, dribbling into a floater, or driving to the rim.
Isolation (ISO)	Possession play type where the player is in a 1vs1 situation trying to size up and create space from his defender.
Hand-Offs (HO)	Possession play type where the screener with or without dribbling hands over the ball to a player cutting towards him, acting as an immediate screener creating space for his teammate.
Cuts (CUT)	Possession play type, in which the player without a screen cuts out or toward the ball to receive it. Includes backdoor cuts, flash cuts, and spacing when a player is getting open near the basket.
Putbacks (PB)	Possession play type where a tip in or quick shot happens after offensive rebound.
Post Ups (PUP)	Possession with the player receiving the ball with his back to the basket in proximity or inside the paint area.
Off Ball Screen (OBS)	Possession is generated by a player running off a screen, whether it be a pin-down, flare screen, elevator screen, or any other of the plethora of screen variations before they receive the ball. This includes curls and fades.
Miscellaneous (M)	Possession that does not fit in the above categories. Includes but is not limited to possessions such as: Players being fouled in the backcourt, errant passes out of bounds, possessions where the player dribbles into a pull-up 3-point shot in the halfcourt, inbounds passes that go directly out of bounds, technical fouls, etc.
Inbound Play (IN)	Possession that is generated from the sideline or the baseline and creates a spatial advantage for the player. For this study, the inbound play types were recorded only when it was concluded within 4 seconds after the inbound pass.

NY, USA) for statistical analysis (Beddo & Kreuter, 2004). Contingency table analysis, employing a chi-square (χ^2) distribution, was conducted for group comparisons among categorical variables (play type, player position, and offensive outcome). Interpretation of chi-square contingency-table test results was facilitated through post hoc and planned comparison procedures (Garcia-Perez & Nunez-Anton, 2003). Furthermore, a Kruskal-Wallis test was employed to assess the variance in PPP across player positions. To evaluate the relative significance of play type and player position in determining possession effectiveness and to construct an efficient end-of-game play type model, we developed a chi-square automatic interaction detector (CHAID) decision tree model. The model delineated parent nodes from no fewer than 100 cases and subsidiary nodes from no fewer than 50 cases. Classification trees, known for their capacity to model non-linear phenomena and make insightful predictions, were utilized to analyze the data (Marques & Ighalo, 2022). A value of $p < 0.05$ was applied to determine statistical significance.

Results

According to the results (Table 2), a significant effect of the position of the player was observed regarding the frequency distribution of the different play types ($\chi^2=519.276$, $df=20$, $p < 0.001$). Guards used statistically more significant ISO play type possessions (31.2%, $p < 0.001$), followed by PnRBH (23.9%, $p < 0.001$), and SUP (16%) possessions. On the other hand, the predominant play-type possessions for the forward position were SUP (28.3%, $p < 0.001$), followed by ISO (22.4%) possessions. PnRSC (29.8%, $p < 0.001$), PB (16.2%, $p < 0.001$), and PUP (12.6%, $p < 0.001$) were the most frequent play-type possessions between the big position. Overall, guards finished 59.4% of the total end-of-game possessions, while forwards 21.7% and bigs 18.9%.

Concerning the effectiveness (positive or negative) of the possession between the positions of the players, CHAID analysis was used (Figure 1). The end-of-game possessions play type and the different positions of the players were the independent variables included in the model. The root of this tree contains all 1009 ob-

Table 2. The percentages of the possessions play-type regarding the position of the player.

Play Types	Position of the Player			
	Guard	Forward	Big	Total
Transition	13.9%**	10.3%	6.9%**	11.8%
PnR Ball-Handler	23.9%**	6.9%**	-	15.7%
PnR Screener	-	2.7%*	29.8%**	6.2%
Isolation	31.2%**	22.4%	9.9%**	25.3%
Spot Up	16%	28.3%**	9.4%**	17.4%
Post Up	0.7%**	5.5%	12.6%**	4%
Putback	0.5%**	4.6%	16.2%**	4.4%
Hand-Off	1.2%	0.5%	1%	1%
Off Ball Screen	3.7%*	2.3%	0.5%*	2.8%
Inbound	8.2%	10.5%	3.7%*	7.8%
Cut	0.8%**	5.9%*	9.9%**	3.7%
Total	59.4%**	21.7%	18.9%	100%

Note Bottom row (total) percentages are row-wise: Total distribution of play types between the position of the players. Right column (total) percentages are column-wise: total distribution of play-types. Percentages within cells are column-wise: distribution of play types within the position of the player. * $p < 0.05$; ** $p < .001$.

servations in this dataset. 506 (50.1%) possessions had a positive outcome. The model successfully classified 294 of 506 positive outcomes of possession (58.1%) and 287 of the 503 negative possession outcomes (57.1%). The big position (Node 2) was found to have 62.3% possessions with a positive outcome, which is a statistically more significant result ($\chi^2=17.838$, $df=1$, $p<0.001$) compared with guard and forward positions (47.3% positive possessions, Node 1). Regarding the play type of the possessions, guard and forward positions were found to be statistically more effective ($\chi^2=19.315$, $df=2$, $p<0.05$), when the end-of-game possessions were PUP, PnRBH, TR, HO, PB, and CUT (54.9% positive outcome, Node 3),

while possession play type SUP (65.2% negative outcome, Node 5) and ISO, PnRSC, OBS and IN possession play types were less effective (54% negative outcome, Node 4). Concerning the position of the big players, the results revealed that play-type possessions of TR, HO, PB, CUT, and IN ($\chi^2=14.857$, $df=1$, $p<0.05$) were statistically more effective (80.3% positive outcome, Node 7) compared with PUP, PnRBH, ISO, PnRSC, SUP and OBS play type possessions (51.7% positive outcome, Node 6).

Regarding the efficiency of the possession, the Kruskal-Wallis test indicated that there was no significant difference in PPP across the positions of the players ($\chi^2=5.708$, $df=2$,

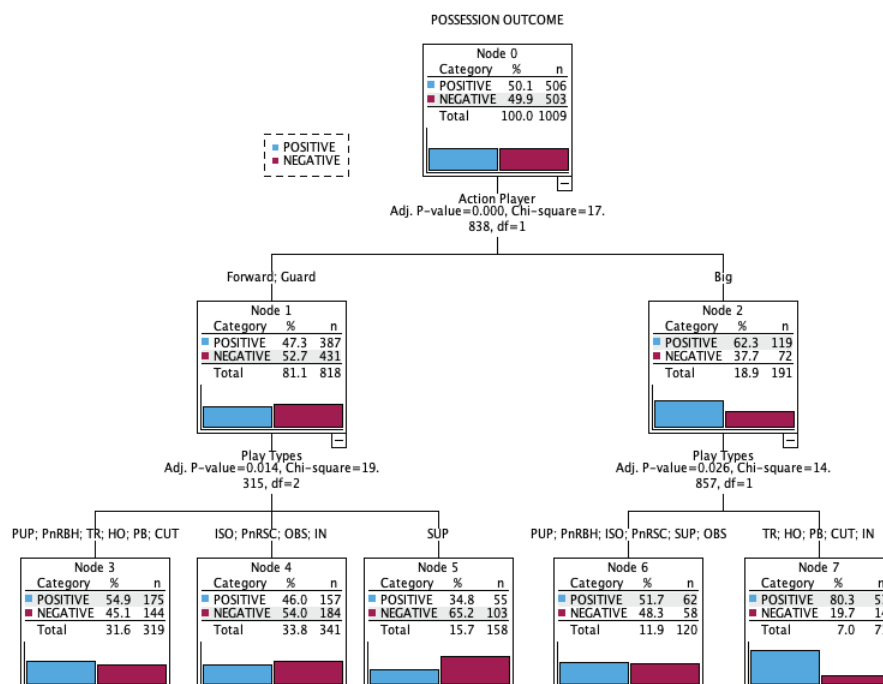


FIGURE 1. CHAID tree describing the frequency effectiveness (%) of the classification and predictive end of game play type possessions.

Note PUP - post-ups; PnRBH - pick and roll ball handler; TR - transition; HO - handoffs; PB - putbacks; CUT - cuts; ISO - isolation; PnRSC - pick and roll screener; OBS - off-ball screens; IN - inbound plays; SUP - spot-up shooting

$p=0.058$) with a mean rank PPP score of 498.37 for guards, 477.30 for forwards, and 539.37 for bigs. As depicted in Table 2, guard's most effective play-type possession was TR (1.31 PPP), followed by CUT (1.20 PPP) and PnRBH (1.09 PPP).

Forwards, on the other hand, produced 2.5 PPP from OBS possessions, 1.55 PPP from CUT, and 1.20 PPP from PUP. IN (1.83 PPP), TR (1.67 PPP), and CUT (1.59 PPP) produced higher-scoring possessions for the big playing position.

Table 3. Points Per Possession (PPP) between the playing position of the players regarding the play type of the possession.

Play Types	Efficiency (Points Per Possession, PPP)			
	Position of the Player			
	Guard	Forward	Big	Total
Transition	1.30	.88	1.67	1.25
PnR Ball-Handler	1.11	.42	-	1.05
PnR Screener	-	.33	1.09	1.02
Isolation	.96	0.98	1.16	.98
Spot Up	.95	.87	1.22	.95
Post Up	.25	1.33	.70	.85
Putback	2	.80	1.35	1.27
Hand-Off	2.2	0	1	1.80
Off Ball Screen	.68	2.6	-	1
Inbound	1.02	1.0	1.83	1.08
Cut	1.20	1.54	1.74	1.59
Total	1.05	.97	1.22	1.07

Note Bottom row (total) is row-wise: Total PPP between the position of the players. The right column (total) is column-wise: total PPP of all positions for the different play types. PPP within cells is column-wise: PPP of the play types within the position of the player.

Discussion

The assessment of players' positions and performance is a continual focus for researchers (Babae et al., 2021; Chen et al., 2023; Rangel, Ugrinowitsch, & Lamas, 2019; Sampaio et al., 2006; Wang & Zheng, 2022). For researchers, delving into specific roles can refine basketball performance analysis at a micro level (from teams to individuals), offering practitioners more tailored insights (Zhai, Guo, Zhang, Li & Liu, 2021). Variations in playing positions have been shown to impact ball possession, highlighting the importance of coaches and players understanding positional demands (Te Wierike, Huijgen, Jonker, Elferink-Gemser & Visscher, 2018). However, exploration of possession play types across playing positions remains limited, particularly in the final minutes of games.

Our findings revealed that guards accounted for 59.4% of total possessions in the last 2 min of the fourth quarter and overtime, with ISO play type actions being the most prevalent (31.2% of total possessions), followed by PnRBH actions (23.9%) and SUP (16%). This trend can be attributed to the central role of guards in ball distribution, ball handling, controlling game rhythm, passing, and organizing offensive tactics (Fewell, Armbruster, Ingraham, Petersen & Waters, 2012). As highlighted by Courel-Ibáñez, McRobert, Toro, and Vélez (2017), guards primarily bear the responsibility for ball handling and shooting, which are pivotal aspects of the offense. Ball handlers have a crucial role within the team, with a significant portion of set offense initiated through ball screens (Vaquera et al., 2016). The prominence of PnRBH possessions (second most frequent) aligns with findings from previous studies (Akinci, 2023; Demenius, 2020).

Forwards accounted for 21.7% of the total end-of-game possessions, with the most frequent play type actions being SUP (28.3%) and ISO (22.4%). Forwards are responsible for shooting from long and medium distances and play a crucial

role during fast break situations, either shooting or passing (Courel-Ibáñez et al., 2017). However, players in the big position completed 18.9% of the total possessions, more frequently utilizing PnRSC actions (29.8%), followed by PB (16.2%). Previous studies on player classification consistently illustrate the comparatively lower offensive role of players in the big position (Chen et al., 2023). As noted by Çene (2018) in his study, players in the center position with specific skills, such as long-range shooting or high post-up efficiency, are rare on most teams, which may explain the lower percentage contribution of the big position in end-of-game possessions.

In terms of the effectiveness of possession play types, although ISO and SUP were among the most common actions for guards in this study, their efficiency did not rank highest among the play types, consistent with previous findings (Zukolo, Dizdar, Selmanović & Vidranski, 2019). Matulaitis and Bietkis (2021) noted in their research that ISO possessions at the end of the ball possessions tend to be inefficient. Similarly, Christmann et al. (2018) found in their examination of endgame possessions of close NBA matches that ISO possessions were the least effective, attributed to their static nature and prolonged duration, similar to our results. Conversely, TR (1.30 PPP) and CUT (1.20 PPP) play type actions emerged as the most efficient among guards in end-of-game possessions. These offensive strategies prioritize teamwork and have demonstrated efficacy in various studies (Christmann et al., 2018; Foteinakis et al., 2024; Matulaitis & Bietkis, 2021; Zukolo et al., 2019). In particular, TR possessions have been noted for their high efficiency (Matulaitis & Bietkis, 2021), prompting coaches to strategize for increased opportunities for TR possessions (Christmann et al., 2018). PnRBH play type actions were also highly effective, yielding 1.11 PPP, consistent with findings from other studies (Christmann et al., 2018; Foteinakis et al., 2024). Overall, guards exhibited an efficiency

of 1.05 PPP across all end-of-game possessions in this study.

In the forward playing position, OBS (2.6 PPP), CUT (1.54 PPP), and PUP (1.33 PPP) play type possessions were the most effective at the end of the game. Stavropoulos, Papadopoulou, & Kolias (2021), in their assessment of OBS effectiveness on the weak side, highlighted its dependency on various factors, such as defensive tactics, player attributes, the game score, finishing moves, and screen types. In our study, OBS yielded a notably high 2.6 PPP, primarily due to the limited number of possessions recorded for the forward position ($n=5$). Although CUT may not be as prevalent as other offensive strategies, it still holds statistical significance in influencing game outcomes (Akinci, 2023). PUP possessions, as emphasized by Akinci (2023), play a significant role in EuroLeague playoff performances. Zukolo et al. (2019) stated that winning teams demonstrated greater efficiency in possessions with back-to-the-basket plays than defeated teams. Matulaitis and Bietkis (2021) found that winning teams exhibited increased frequency and efficiency in low post-offense. In our study, the forward position recorded a PPP efficiency of 0.97.

Although the big playing position recorded the lowest percentage of end-of-game possessions in our study, their possessions exhibited the highest efficiency among the three playing position groups (1.22 PPP). The significance of the big playing position in basketball success is well-established (Chen et al., 2023). The lower frequency of end-of-game possessions for big players can be attributed to their reliance on teammate support for execution. However, their efficiency in specific play types is notable, with IN (1.83 PPP), CUT (1.74 PPP), and TR (1.67 PPP) possessions aligning with the characteristics of the big playing position, where plays near the basket tend to have higher success rates. Additionally, PB possessions yielded 1.35 PPP for big players. Chen et al. (2023) identified cuts, low post offense, and free throws as defining aspects of the offensive role of big players, highlighting the importance of off-ball movement and offensive rebounding. Ozmen (2016) found that EuroLeague teams with a single additional offensive rebound have a 6.3% higher chance of winning. Conversely, Lutz (2012) suggested in his study that big players do not typically positively affect game outcomes. This finding contrasts with our results regarding the efficiency of the big playing position.

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Conflict of Interest

The authors declare that there is no conflict of interest.

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The decision tree analysis revealed that positive possession outcomes were associated with the use of TR, HO, PB, IN, and CUT play type possessions for the big playing position. Conversely, the most effective play types for guards and forwards combined included PUP, PnRBH, TR, HO, PB, and CUT possessions. These findings underscore the importance of coaches balancing inside-outside gameplay among guard, forward, and big playing positions and recruiting players who can collaborate effectively and work as a team during end-of-game possessions.

This study offers novel insights into the efficiency of end-of-game possessions across playing positions, providing a fresh perspective on possession play types in close games. This nuanced approach enhances our understanding of how different player roles contribute to team success, addressing the gap in research on positional differences in late-game scenarios. By advancing our understanding of how player positions across different play type possessions influence team success, enriches the existing body of knowledge in the field. On the contrary, this research focused on offensive contributions and overlooked the defensive aspect of the game, limiting the study's comprehensive understanding of positional efficacy in different play types in late-game scenarios. Future research endeavors, incorporating larger game samples and advanced statistical methodologies like cluster analysis or machine learning, hold promise for further elucidating the roles of playing positions in possession efficacy at the end of a game.

Understanding positional differences in end-of-game possessions regarding the different play types, allows coaches to make strategic adjustments, lineup changes and play-calling tailored to the players strengths and weaknesses. Additionally, coaches can foster a supportive team culture, emphasizing in team collaboration and chemistry, which has been proved critical during end-of-game possessions. Gaining insights from the positional differences can also serve to guide player development and inform scouting reports against opposing teams by analyzing tendencies and vulnerabilities during critical possessions. By implementing these practical implications, teams can optimize their game performance during critical end-of-game possessions, ultimately leading to improved success in close contested games.

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ORIGINAL SCIENTIFIC PAPER

Physical Literacy Levels in the Croatian Adult Population; Gender Differences and Associations with Participation in Organized Physical Activity

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Abstract

Physical literacy (PL) is the foundation for lifelong participation in physical activity. Most previous research concentrated on PL examinations in younger groups, while such research is scarce in the adult population. This study aimed to determine the PL levels of the Croatian adult population. Also, the aim was to investigate whether individuals with longer participation in organized physical activity have higher PL levels and do males and females differ in PL levels. The sample comprised 561 individuals (321 females, 240 males) aged 27.65 ± 12.13 years. PL levels were assessed by the Croatian version of the Perceived Physical Activity Questionnaire (PPLQ-Cro). Demographic characteristics included age, gender, and years of involvement in organized physical activity. Gender differences were calculated using the Mann-Whitney U test and effect sizes. Spearman correlation coefficients were used to determine associations between study variables. Males had significantly higher PL levels compared to females (83.10 ± 16.17 for males and 74.27 ± 20.07 for females, $Z = -6.88$, $p = 0.001$). Years of organized physical activity were associated with PL levels ($R = 0.48$, $p = 0.001$) in the total sample as well in males ($R = 0.55$, $p < 0.001$) and females ($R = 0.24$, $p < 0.01$) separately. The finding that years of organized physical activity are associated with total PL supports the theory that PL is the cornerstone for participating in lifetime physical activity. The results of this study emphasize the importance of introducing the concept of PL in numerous institutions and organizations connected with organized physical activity.

Keywords: physical exercise, health behaviors, lifestyle medicine, well-being, older individuals

Introduction

Physical activity (PA), defined as any motion generated by skeletal muscles resulting in the expenditure of energy, plays a pivotal role in promoting overall health and well-being (Caspersen et al., 1985; Peralta et al., 2021). The benefits of embracing a physically active lifestyle extend beyond individual well-being to encompass broader public health objectives, including weight management, the prevention of chronic diseases such as cardiovascular disease, type II diabetes, and cancer, as well as the enhancement of bone and muscle strength. Indeed, a broad review study reported that PA is the most effective prevention strategy for more than 25 chronic medical illnesses, with a 20-

30% risk reduction range (Rhodes, Janssen, Bredin, Warburton, & Bauman, 2017). However, there is a global problem of low PA levels in children and adolescents, and this problem tracks into adulthood (Telama et al., 2014; Vukelja, Milanovic, & Salaj, 2022).

According to the World Health Organization, adults should strive to engage in 150–300 minutes of moderate-intensity exercise, 75–150 minutes of vigorous-intensity PA, or a combination of both each week to optimize their physical health (DiPietro et al., 2020). However, despite these recommendations, a significant proportion of the global adult population fails to meet these guidelines. In 2016, 28% of adults aged 18 and older worldwide



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did not meet the recommended activity threshold, with 23% of men and 32% of women being inactive, which stands for more than 1.4 billion adults not being sufficiently active (Guthold et al., 2018). Therefore, as the problem of increased inactivity still has a rising trend, researchers have been investigating the concepts and determinants of this issue. The concept which emerged as very important for maintaining an optimal PA through life is called physical literacy (PL).

PL is most commonly defined as “the motivation, confidence, physical competence, knowledge and understanding to value and take responsibility for participation in lifelong physical activity” (Edwards et al., 2017). Individuals with greater PL are anticipated to achieve health benefits more readily and sustain PA throughout their lives (Whitehead, 2013). The International Physical Literacy Association (IPLA) outlines PL as comprising various elements including fitness level, daily habits, motivation, confidence, and knowledge (Shearer et al., 2018). These components collectively form a comprehensive framework that fosters lifelong engagement and enjoyment in PA (Longmuir & Tremblay, 2016). PL can be seen as a factor influencing health through a series of interconnected pathways: heightened PL fosters increased engagement in PA, which in turn prompts beneficial physiological, social, and psychosocial adjustments, ultimately leading to enhancements in physical, mental, and social well-being. This pathway is expected to persist and evolve throughout the lifespan, from early childhood to old age (Cairney, Clark, Dudley, & Kriellars, 2019).

The importance of organized PA on the PL level has been repeatedly reported in previous studies. Namely, a study on Croatian adolescents reported that individuals involved in out-of-school sports had better PL compared with those not involved in sports (Sunda et al., 2022). Also, this was indirectly shown in a study where children and adolescents who participated in sports had a greater likelihood of participating in sports as adults (Perkins, Jacobs, Barber, & Eccles, 2004). Moreover, as the importance of enabling a greater number of people to participate in sports and improve their PL, the Physical Activity and Sport Participation framework has been developed for stakeholders to follow for creating a policy model toward being physically active across the lifespan (Westerbeek & Eime, 2021). However, studies directly investigating the relationship between organized PA and PL are generally lacking.

Most studies examining the relationship between PL and PA have been conducted on younger individuals, while research on the adult population is lacking. However, this population is equally important because PL shapes lifelong habits and activities (Whitehead, 2013). Also, developing PL in adulthood is extremely important as lifelong participation in PA leads to lower risks of chronic illness and decreased odds of falls. Even though they are scarce, there are several studies which investigated PL in the adult population (Boldovskaia et al., 2023; Ryom et al., 2022). A study which gathered information regarding self-reported assessment of PL concluded that the existing instruments cover only affective and physical domains of PL, while there is no holistic assessment of PL (Ryom et al., 2022). Nevertheless, a recent study on the German and Austrian populations constructed the holistic PL questionnaire which covered all PL domains, which makes this instrument currently the best choice for assessing PL of adults (Holler, Carl, Poppel, & Jaunig, 2023).

Previous studies reported that there are gender differences in PA participation, with boys being more active than girls, which

could imply that males will have higher PL levels compared to females (Brown et al., 2020; Saffer et al., 2013). Therefore, as previous research noted the gender differences in both PA participation and PL levels, the aim of this study was to establish gender differences in the PL levels and associations with participation in organized PA. We hypothesized that males will have higher PA participation and higher PL levels compared to females. The results from this research would have practical implication for implementing public health recommendations and guidelines connected with PA participation.

Materials and methods

Participants

This research included 561 adult individuals (321 females, 240 males) from Croatia aged 27.65 ± 12.13 years. The study's inclusion criteria were: (a) Croatian citizenship, (b) age 18 or older, (c) ability to comprehend and speak Croatian, and (d) granting informed consent to participate. Exclusion criteria included: (a) having a medical or psychological condition that might impair one's ability to understand and answer survey questions, and (b) having previously taken the survey. The procedures were conducted according to the Declaration of Helsinki and were approved by the Ethical Board of the University of Split, Faculty of Kinesiology (ref.no 2181-205-02-05-23-029, Approved 6/12/2023).

Variables and procedures

The research included demographic information (age, gender), participation in organized physical activity and PL levels evaluated by the translated version of the Perceived Physical Literacy Questionnaire for adults (PPLQ-Cro).

The PPLQ has recently been developed and was used for holistically assessing PL levels (Holler et al., 2023). This study used the PPLQ-Cro version which has previously been translated into the Croatian language, culturally adapted and checked for psychometric properties (study in the publishing process). The Perceived Physical Literacy Questionnaire (PPLQ) assesses perceived PL levels and its six domains in the general adult population following the IPLA definition of PL (Shearer et al., 2018). Originally conducted in German and in Austria, it's also available in English (Holler et al., 2023). It's grounded in established PL theories, with 24 items organized into six domains: Physical competence, Understanding, Motivation, Confidence, Knowledge, and Physical activity behavior. Responses for the first four domains use a 6-point Likert scale (5 to 0), while Knowledge uses closed response categories and Physical activity behavior uses open response categories. Precisely, PA behavior is assessed using the International Physical Activity Questionnaire Short Form (IPAQ-SF). Also, Metabolic equivalents of task (METs) representing the ratio of the rate of energy expended during PA to the rate of energy expended at rest were calculated from the IPAQ-SF to provide detailed insight into PA levels. Precisely METs in the moderate-to-vigorous intensity PA were calculated. The PA behavior domain score is based on a theory-driven nonlinear saturation function, with 300 minutes of MVPA per week equating to a score of 100. Domain scores range from 0 to 100, with higher values indicating stronger competency. The total PL score is a composite of the six domain scores, each weighted at 16.67%. The composite score ranges from 0 to 100, with higher scores indicating higher PL levels. The questionnaire demonstrates sufficient internal consistency for all domains ($0.70 \leq \omega \leq 0.90$) and acceptable model fit. The

answers to the questionnaire were collected using the online platform SurveyMonkey which was sent through social media to all interested participants.

Statistical analysis

The normality of the distribution was checked by the Kolmogorov-Smirnov test for normality, which allowed to choose the parametric or non-parametric tests in further statistical calculations. Due to highly skewed data, variables were log-transformed to decrease the non-uniformity of error. Analysis were conducted on log-transformed data but the results in the figures and tables are presented as actual, non-log-transformed values.

Mann-Whitney-U test was used to determine the differences between males and females in study variables. Additionally, effect sizes (ES) were calculated to further check for gender differences, and they were interpreted as: <0.2 = very small; $0.21-0.49$ = small; $0.50-0.79$ = moderate; ≥ 0.8 = large; ≥ 1.2 = very large (Cohen, 2013). The correlations between the study variables

were evaluated by Spearman Rank Order correlation coefficients, gender-stratified. Additionally, values of correlation coefficients were presented into magnitude-based descriptors as follows: very low correlation ($0.00-0.20$), low correlation ($0.21-0.40$), moderate correlation ($0.41-0.60$), high correlation ($0.61-0.80$), and very high correlation ($0.81-1.00$) (Schober, Boer, and Schwarte, 2018). Statistical analyses were conducted using Statistica ver.13 (Tibco, Palo Alto, California) with a p-value of 0.05.

Results

According to IPAQ-SF activity classification, 76.44% of participants were categorized as highly active 18.2% as moderately active, and 5.36 % with low activity. Moreover, males accumulated a mean of 4258.61 METmin/week, while females accumulated 3085.23 METmin/week ($Z=-5.50$, $p<0.001$).

Descriptive statistics (means and standard deviations) and differences between males and females in the study's variables are presented in Figure 1. The most significant differences between

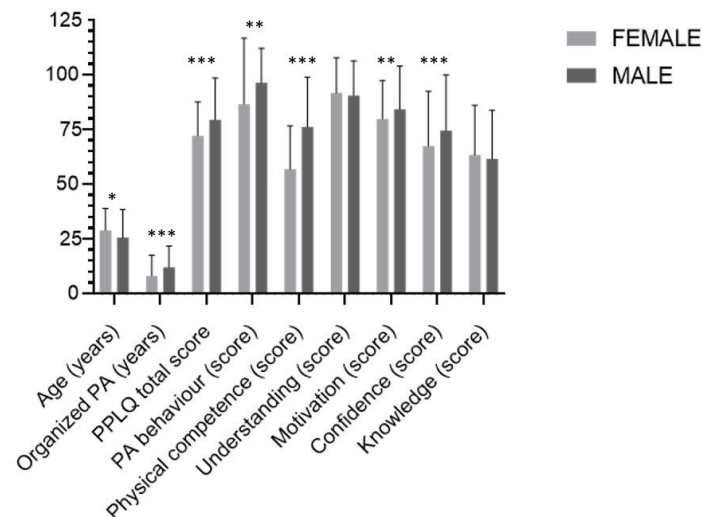


FIGURE 1. Gender differences in physical literacy scores and organized physical activity. Note * denotes $p<0.05$; ** denotes $p<0.01$, *** denotes $p<0.001$, PPLQ – Perceived Physical Literacy Questionnaire, PA- Physical activity.

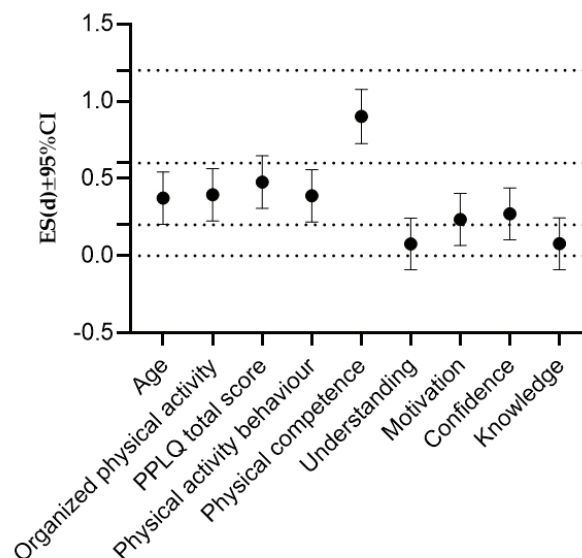


FIGURE 2. Gender differences in physical literacy and physical activity are expressed through effect sizes and confidence intervals. Note PA – physical activity.

males and females were noted in years of organized PA, PPLQ total score, physical competence, and confidence, with males having higher scores than females.

Gender differences expressed through effect sizes are shown in Figure 2. Small differences (ES) were noted for age, years of organized physical activity, physical literacy total score, physical activity behavior, motivation and confidence, while physical competence had moderate ES.

The gender-stratified correlation coefficients and magnitude of correlations are presented in Figure 3. In the total sample, years of organized PA were correlated to PL total score ($p<0.001$), Physical activity behaviour ($p<0.001$), physical

competence ($p<0.001$), Understanding ($p<0.001$), Motivation ($p<0.001$), Confidence ($p<0.001$), Knowledge ($p<0.01$), and Moderate-to-vigorous PA ($p<0.001$). Among males, years of organized PA were correlated to PL total score ($p<0.001$), Physical activity behaviour ($p<0.001$), physical competence ($p<0.01$), Understanding ($p<0.01$), Motivation ($p<0.05$), Confidence ($p<0.001$), Knowledge ($p<0.01$), and Moderate-to-vigorous PA ($p<0.001$). Moreover, in females, years in organized PA were correlated to PL total score ($p<0.001$), Physical activity behaviour ($p<0.001$), physical competence ($p<0.001$), Understanding ($p<0.001$), Motivation ($p<0.01$), and Moderate-to-vigorous PA ($p<0.01$).

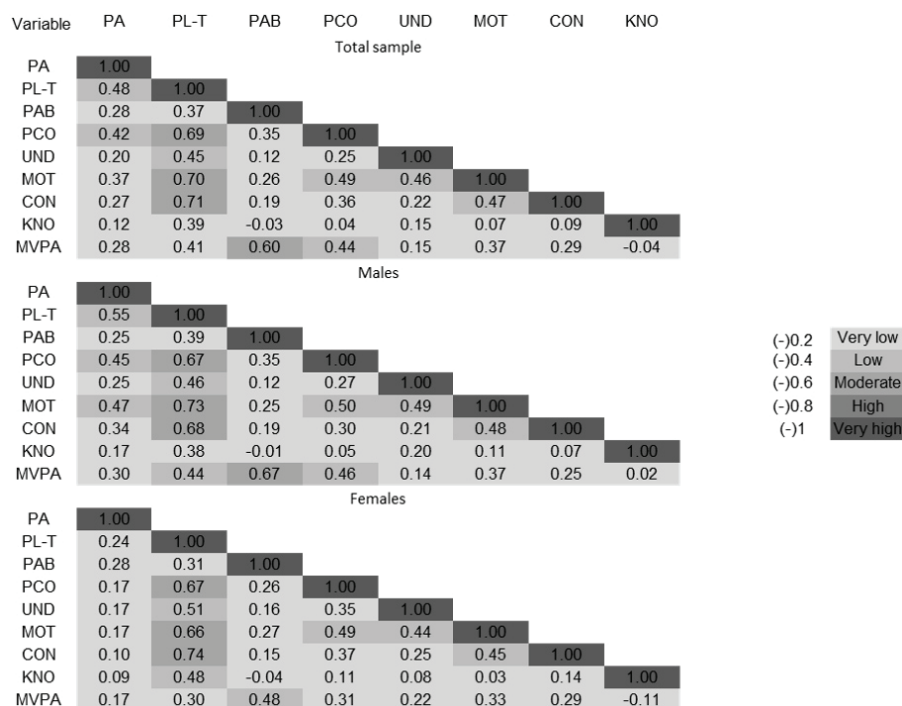


FIGURE 3. Gender-stratified correlation coefficients and magnitude of correlations between study variables.

Note PA – Years of organized physical activity, PL-T – Total physical literacy score, PAB – physical activity behavior, PCO – Physical competence, UND – Understanding, MOT – Motivation, CON – Confidence, KNO – Knowledge, MVPA – Moderate-to-vigorous physical activity.

Discussion

This research aimed to investigate gender differences in PL levels and associations between PL and participation in organized PA. The most interesting findings from this study are that (i) males had higher participation rates and PL scores compared to females and (ii) participation in organized PA was associated with PL levels.

Gender differences in participation in organized physical activity and physical literacy levels

The most significant differences between males and females were noted in years of organized PA, PPLQ total score, physical competence, and confidence, with males having higher scores than females. The reason for these findings can be found in the following explanation. Precisely, males are generally more active than females, which is noted from childhood till adolescence (Guthold et al., 2018). The reason for gender differences in childhood and adolescence can be found in the type of activity. Notably, boys are more involved in vigorous-intensity PA such as team sports (e.g., soccer, basketball, handball) while girls are more involved in low-to-moderate intensity PA

such as dance, artistic gymnastics, ballet, artistic swimming (Frömel, Grofik, Šafar, & Mitaš, 2022). However, the gender differences in adults can be influenced by some other factors.

In the past, men were seen as the foundation responsible for contributing to their families and ensuring life's essentials. Even today, due to social norms and accepted roles, men are often more active than women. This is supported by numerous studies that have identified competitiveness and the desire for greater strength as key motivators for men's participation in PA (Edwards et al., 2017; Molanorouzi, Khoo, & Morris, 2015). Considering the importance of social support in promoting PA, it becomes clear how interactions and support from the environment can strongly influence individuals' motivation and engagement in PA. Research has shown that males are often in a more favorable position when it comes to receiving support, whether from peers, parents, or the broader community, compared to females. This support, coming from various sources, not only provides emotional backing but is also perceived as a means that positively impacts individuals' well-being. Through these positive interactions and support, individuals feel encouraged to be active, further promoting

their participation in PA. Additionally, social support can provide practical assistance, such as engaging in sports activities together or supporting goals related to PA (Smith, Moyle, & Burton, 2023). Through these mechanisms, individuals feel connected to others and feel part of a community that supports their commitment to a healthy lifestyle. Therefore, understanding the importance of social support can be crucial for promoting PA and a healthy lifestyle.

A lack of confidence is certainly one factor contributing to lower PA among girls compared to boys. According to a recent study conducted in the UK on girls, one-third stated that they did not participate in PA due to low confidence in their physical abilities, while another third avoided PA due to feeling scrutinized and judged by others regarding their bodies (Cowley et al., 2021). When we consider opportunities and resources, it can be noticed that many areas offer a limited number of sports activities mostly targeted towards males, while options for females are lacking. For example, males can participate in recreational soccer, basketball, handball, and combat sports, while for females the offer is mainly limited to some dance and light fitness classes. These lower opportunities for females are most likely one of the decisive factors for lower participation in organized PA.

Also, as males are more involved in sports activities, it is logical that they also have higher PL levels as organized PA is generally associated with increased PL (Belanger et al., 2018). Access to sports facilities, coaching, and organized sports teams all have an impact on PL levels. In certain areas, females may have fewer possibilities to participate in organized sports or leisure activities than males. Furthermore, societal norms or family dynamics may favour boys' engagement in sports over girls' (Westerbeek & Eime, 2021). All of these factors can influence PL levels and can be responsible for gender differences found in this research.

Associations between participation in organized physical activity and physical literacy levels

The second most interesting finding of this study is that years of organized PA were correlated to PL total score. However, the magnitude of correlations was higher in males than in females, and certain PL domains were not significantly correlated with PA among females, which is in line with the first result of this study. The result that years of organized PA were correlated to PL total score and subdomains physical competence, Understanding, Motivation, Confidence, and Knowledge are consistent with numerous previous studies regarding associations between PL and PA. Precisely, it was found that children who met PA guidelines had higher PL scores, particularly in the domains of Physical Competence and Motivation and Confidence (Belanger et al., 2018). Similarly, it was reported that higher PL was linked to better health indicators, with the relationship between PL and aerobic fitness being influenced by moderate-to-vigorous PA (Caldwell et al., 2020). Noteworthy, one study demonstrated a correlation between

PL and self-perceived fitness in children and adolescents, with higher self-perceived fitness being associated with greater PL (Pastor-Cisneros et al., 2021). These findings collectively suggest that participation in organized PA can enhance PL levels, leading to improved health and fitness outcomes.

In addition, persons who have been active in organized physical exercise for a longer period of time are more likely to have participated in many various activities. Organized PA often exposes participants to a range of motions and exercises, which broadens their movement repertoire and improves overall PA. For example, a person who participates in numerous sports would acquire a broader range of movement abilities than one who exclusively participates in one activity (Mackenzie-Stewart et al., 2023). Furthermore, many organized programs incorporate assessment and feedback tools to track progress and offer suggestions for improvement. This feedback loop is critical for growing PL because it helps people recognize their own strengths and areas for improvement, resulting in better-informed decisions.

Overall, participation in organized PA provides a structured and supportive environment for individuals to develop and enhance their PL, which encompasses not only physical skills but also knowledge, motivation, confidence, and enjoyment in being physically active.

Limitations and strengths

One of the main limitations of this study comes from the sample included in the research. Specifically, we included mainly active participants. Therefore, the results of this study should be considered with caution and future studies should consist of a heterogeneous sample regarding PA involvement. Moreover, this study did not include objectively measured PA or physical fitness measures which could add value to this research. On the other side, the main strength of this research is that this is among the first studies investigating PL levels in Croatia and the entire Southeastern Europe and can be used as the direction for future similar studies.

Conclusions

The purpose of this study was to investigate the gender differences in PL levels as well as the relationship between PL and involvement in organized PA. The most noteworthy findings from this study are that males had greater participation rates and PL scores than females, and participation in organized PA was connected with PL levels. The finding that years of organized PA are associated with total PL supports the theory that PL is the cornerstone for lifetime participation in PA. The results of this study emphasize the importance of introducing the concept of PL in numerous institutions and organizations connected with organized PA. Also, special attention should be paid to increasing the offer of female organized PA which should also increase their PL levels. Future studies should investigate more heterogeneous samples regarding PA participation to gain detailed insight into herein studied topic.

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Conflicts of interest

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ORIGINAL SCIENTIFIC PAPER

Motor Competencies among Athletic Population of 9- and 10- Year-Old Children

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Abstract

The aim of this study was to assess the basic motor competencies of primary school children in relation to their participation in sports. The research involved 86 children aged 9 and 10 (46 boys, 40 girls), comprising a group of 53 athletes (30 boys, 23 girls) and 33 non-athletes (16 boys, 17 girls). Using the MOBAK 3–4 test battery, the study compared motor competence data between these groups. Results indicated that sports-playing children demonstrated motor skills comparable to non-athletes in terms of object-movement (OM) abilities. Specifically, significant differences were observed in OM skills (mean =4.52), notably in the throwing test (mean =0.94) favoring the sporting population ($p<0.05$). While in other OM tests there was no difference between the groups. In the self-movement (SM) area, there was no difference between the athletic and non-athletic group. However, the overall score for SM (mean =3.43) demonstrated significance in favor of sports-playing children. The total score (mean =7.96) significantly favored the sports-playing group ($p<0.05$). The findings confirm that children engaged in regular sports activities exhibit higher levels of physical literacy, as reflected in their measurable motor competencies. This underscores the importance of incorporating motor competence diagnostics into school practices to enhance physical education initiatives.

Keywords: *primary school children, motor development, MOBAK test*

Introduction

Children's general motor predispositions, which shape their motor competencies, significantly influence their motor literacy (Mačura, Hubinák, & Krška 2021). Motor competencies and physical activity affect overall movement literacy (Mačura et al., 2021). Childhood, particularly during the primary school years, is a critical phase for motor skill development. However, there has been a considerable decline in children's motor competencies over recent decades (Masaryková, 2021; Ružbarská, 2018). Furthermore, for some children, physical education classes are the primary context for motor skill experiences. Masaryková (2021) states that innovative perspectives on motor competencies, differing from those in Slovakia, have long been conceptualized in the European context. These innovative approaches in motor skills assessment aim to redefine how children's achievements are evaluated. Traditionally, German-speaking

countries and Slovakia have prioritized the evaluation of motor abilities, while English-speaking countries have focused predominantly on motor skills assessment (Masaryková, 2021). Recognizing these different assessment methodologies, experts were forced to think about creating a unified framework for assessing skills and abilities simultaneously, leading to the emergence of the concept of motor competence, which encompasses both components. Thus, motor competence can be understood as a set of motor abilities and motor skills (Masaryková, 2021).

Mačura et al. (2021) elaborate on the concept of motor competence, identifying it as an intermediary stage between motor abilities and skills. They characterize motor competencies as inherent capacities for motor performance developed through specific situational demands or functional expressions translated into movement. The authors emphasize that motor competencies, alongside relational



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dynamics and knowledge, significantly influence movement literacy. These competencies extend beyond sports, encompassing daily life activities and various environmental contexts, serving as preventive measures against lifestyle-related illnesses (Valková, 2010). Hermann and Seelig (2014) define basic motor competencies, highlighting their non-observable nature while enabling children to execute specific tasks and develop them over time. They assert that competencies are regulatory mechanisms for motor skills, such as determining a child's strength. From a theoretical perspective, motor competencies can be considered as the result of motor skills and abilities. As proposed by Whitehead (2019), the acquisition of motor competencies contributes to achieving movement literacy, spanning from infancy to the acquisition of specialized physical education skills in various activities.

The extent of basic motor competencies (BMC) among primary school students varies owing to internal and external factors. Several studies have investigated BMC levels (Herrmann, 2018; Herrmann, Heim, & Seelig, 2019; Quitério et al., 2018; Scheuer, Bund & Herrmann, 2019; Strotmeyer, Kehne & Herrmann, 2020; Tumynaitė, 2016). While these investigations reached differing conclusions favoring the sport-

ing population, some also noted a satisfactory BMC level in the general population of children. For instance, in Finland, organized sports participation correlated with enhanced levels of SM and OM (Niemistö, Finni, Cantell, Korhonen & Sääkslahti, 2019). The aim of this study was to examine the basic motor competencies of primary school children relative to their sports participation.

Methods
Participants

The assessment was conducted in a group of primary school children engaged in sports activities (n=86, boys =46, girls =40). The study comprised an athletic group (n=53), with an average age of ($x=9.03\pm0.73$ years), actively participating in physical education classes, and a non-athletic group (n=33) with an average age of ($x=8.90\pm0.67$ years). Gender distribution is outlined in Table 1. Testing occurred during regular physical education sessions in the gymnasium, with each participant's parent providing voluntary written informed consent beforehand. The research adhered to the principles of the Declaration of Helsinki and was approved by the Ethical Commission at the University of Presov (ECUP 102023PO, 2023).

Table 1. Results of the number of participants relative to gender

	TOTAL		BOYS		GIRLS	
	N	%	N	%	N	%
Athletic population	53	61.63	30	56.60	23	43.40
Non-athletic population	33	38.37	16	48.48	17	51.52

Note: N- sample size %- Percentage

Procedure

A team of three trained test leaders administered the MOBAK 3–4 test (Herrmann, 2018). During the testing, participating school classes were divided into small groups of four to six children. Guided by one of the test leaders, these groups completed all MOBAK 3–4 test battery items at designated stations. Before each test item, the leader explained the motor task and provided a demonstration. Practice attempts were not allowed. Children completed all test items in accordance with the MOBAK 3–4 protocol. The purpose of the test battery is to convince children that they can perform natural movement activities (Machura et al., 2021).

Comprising two areas, the self-movement (SM) and object-movement (OM), the MOBAK 3–4 test battery is a suitable diagnostic tool for evaluating exercise competence in 9–10-year-olds. The SM section includes tasks such as balancing, rolling, jumping, and running, while the OM section assesses throw, catch, bounce, and dribble abilities. In the throwing and catching tests, each child had six attempts, with point scores allocated as follows: 0 to 2 hits =0 points, 3 to 4 hits =1 point, and 5 to 6 hits =2 points. For other test items, each child had two attempts, with scoring

criteria as follows: both attempts failed =0 points, one attempt passed =1 point, and both attempts passed =2 points (Herrmann, 2018).

Statistical analyses

The observed data were analyzed using TIBCO Statistica™ 14.0.0. (Santa Clara, USA, 2020). Descriptive statistics such as mean, standard deviation, and percentage were used to characterize the data. Evaluation of the obtained data involved analysis, synthesis, and a factual assessment of the research results. The Shapiro–Wilk normality test was employed to assess the normality of data distribution. Differences between the observed groups were assessed using the non-parametric Mann–Whitney U-test. A statistical significance level of 0.05 was applied for all analyses.

Results

Table 2 presents the measured values of individual anthropometric indicators for the observed groups of primary school-aged children. Our sample from the athletic population indicates a relatively high participation rate (62%) in extracurricular physical activities among children.

Table 2. Descriptive indicators of anthropometric characteristics

	N	Age ($x\pm SD$)	Body height ($cm\pm SD$)	Body weight ($kg\pm SD$)	BMI (kg/m^2)
Athletic population	53	9.03±0.73	138.93±5.79	34.40±6.94	17.79±2.81
Non-athletic population	33	8.90±0.67	139.06±7.97	34.73±10.00	17.59±3.42
Mann-Whitney U-test	86	0.45	0.72	0.64	0.51

Note: N-sample size, SD - standard deviation, BMI- Body mass index, statistical significance *p<0.05, x- average.

Table 3. Result of Mann-Whitney U test of motor competencies areas

Object-movement	Athletic population [x±SD]	Non-athletic population [x±SD]	U	Z	p-value
Throwing	0.94±0.71	0.30±0.55	451.00	-3.75	0.00*
Catching	0.77±0.72	0.54±0.66	724.50	-1.32	0.18
Bouncing	1.52±0.72	1.45±0.75	827.50	-0.41	0.67
Dribbling	1.28±0.76	1.42±0.75	783.50	0.80	0.42
Self-movement	Athletic population [x±SD]	Non-athletic population [x±SD]	U	Z	p-value
Balancing	0.43±0.63	0.30±0.58	776.00	-0.87	0.38
Rolling	1.22±0.93	0.87±0.89	696.50	-1.58	0.11
Jumping	0.52±0.79	0.24±0.61	712.00	-1.44	0.15
Running	1.24±0.80	1.21±0.73	842.50	-0.28	0.77

Note: N-sample size, SD - standard deviation, U- value of Mann-Whitney U-test, Z- approximation, statistical significance *p<0.05.

Table 3 presents data on the basic motor skills of the observed groups across both tested areas. Notably, in the OM area, the throwing test exhibited the most significant differences, statistically favoring the sporting population of children at a significance level of $p<0.05$. Additionally, improved results were observed in the catching and bouncing test items among the sports population. However, it was surprising to find that children from the general population outperformed the sports population in the dribbling test, a result that was evaluated negatively.

In the SM area, significantly higher performance values were evident across all test items in favor of the sporting pop-

ulation. Although the largest discrepancy in measured values was noted in the SM tests, statistical confirmation was not achieved.

Table 4 presents the total score of basic motor competencies measured among groups of primary school children under observation. Analysis of Table 4 data led to the conclusion that while the monitored group achieved significant values in basic motor competencies, it was insufficient to indicate the effectiveness or significant positive impact of free-time and interest-based movement activities on the basic motor competencies of primary school-aged children.

Table 4. Result of Mann-Whitney U test of basic motor competencies – Total score

	Athletic population [x±SD]	Non-athletic population [x±SD]	U	Z	p-value
Object-movement	4.52±1.81	3.72±1.50	665.00	-1.85	0.06
Self-movement	3.43±1.89	2.64±1.67	651.00	-1.98	0.04*
Total score	7.96±2.99	6.36±2.61	628.50	-2.18	0.02*

Note: N- sample size, SD - standard deviation, U- value of Mann-Whitney U-test, Z- approximation, statistical significance *p<0.05.

Discussion

Whiting et al. (2021) conducted a Europe-wide analysis, revealing a 44.1% membership rate of children aged 6–9 in sports clubs. The difference in basic anthropometric indicators among the monitored groups is negligible.

Masaryková (2021) noted that it is possible to assess the relationship between basic anthropometric characteristics and motor tests, demonstrating negative correlations with children's body weight. Comparisons with Slovak studies such as Halmová and Šimonek (2020), Štetinová and Leütterová (2021), and Leütterová and Štetinová (2021) reveal similarities in measured values. In contrast, compared to the study by Mačura et al. (2021), the groups monitored in our study exhibited a significantly lower body mass index (BMI) indicator level, nearly 0.44 kg/m² lower. Studies by Štetinová and Leütterová (2021), and Leütterová and Štetinová (2021) found BMI levels for the general population of 9–10-year-old boys to be 16–18 kg/m² and for girls 16.69–17.72 kg/m², corroborating the similarity of values obtained in our study. However, noteworthy differences emerge when comparing data with Masaryková (2021), who reports an average BMI value in Slovak children at 16.34 kg/m². Our results align more closely with the BMI values observed in European countries, averaging 17.49 kg/m².

Understanding the level of basic motor competencies is crucial for enhancing the efficiency of the educational process.

Therefore, utilizing a tool for extension is imperative in pedagogical practice. The primary objective of this study was to contribute to understanding the current level of basic motor competencies among primary school children. From a diagnostic perspective, it was pertinent to analyze the percentage of children who did not reach the boundary value of 3 points in individual competence areas of the MOBAK test, indicating the need for educational motor support (Herrmann, 2018; Ružbarská, 2023).

Based on our results, 15% of athletic and non-athletic populations demonstrated motor deficits in the OM domain. At the same time, higher values were observed in SM, with 34% of athletic and 48% of non-athletic populations exhibiting motor deficits. These findings align with the study by Ružbarská and Boržíková (2023), where approximately 20% of girls and 11% of boys showed motor deficits in OM. Moreover, a cross-sectional international study by Wälti et al. (2023) reported that over 25% of children aged 8–10 required educational motor support in OM and over 20% in SM. Interestingly, our research revealed a degree of motor insufficiency even among the athletic population, particularly in the SM component. In terms of OM competencies, both athletic and non-athletic populations showed levels comparable to those found in similar studies. For instance, the non-athletic population in our study demonstrated levels similar to Chilean children (mean =3.71; Carcamo-Oyarzun & Herrmann, 2020). The athlet-

ic population outperformed German children (mean =4.03; Herrmann, Heim, & Seelig, 2019) but fell slightly behind Swiss children (mean =4.67; Herrmann, Gerlach, & Seelig, 2015) and Portuguese children (mean =4.90; Quitério et al., 2018). The athletic population's OM scores were comparable to those in the study by Šiška et al. (2024) (mean =4.46). In the SM component, both the non-athletic and athletic populations achieved slightly lower levels than Chilean children (mean =4.72; Carcamo-Oyarzun & Herrmann, 2020), German children (mean =4.48; Herrmann, Heim, & Seelig, 2019), Portuguese children (mean =4.70; Quitério et al., 2018), Swiss children (mean =5.48; Herrmann, Gerlach, & Seelig, 2015), and Slovak children (mean =5.07; Šiška et al., 2024). Higher values in the SM area were also observed in Slovak children by Ružbarská and Boržíková (2023) (mean =5.25) and Wälti et al. (2023; mean =5.57). A negative evaluation is evident upon comparing our results with the study conducted by Šiška et al. (2024). It was initially assumed that the sporting population under observation would demonstrate higher motor competence values compared to the average Slovak general population of primary education children. However, when comparing the total scores with the study conducted by Mačura et al. (2021), our monitored sports population exhibited significantly higher scores, surpassing by 0.53 points, which was evaluated positively. Conversely, in the case of the non-athletic population, the total scores differed by 0.27 points in favor of Mačura et al. (2021), who monitored motor skills in 3rd and 4th-grade primary education children within a sample of 481.

Wälti et al. (2023) examined the influence of exogenous factors on motor competence. They found that extracurricular physical activities were significant predictors in the SM field, with ball sports being significant positive determinants of physical competence. Children participating in extracurricular physical activity demonstrated a higher level of basic physical competence in at least one area. Our research also confirmed the significant influence of extracurricular physical activities on SM, with the athletic population showing a statistically significantly higher level. These findings align with those of Drenowatz and Greier (2019), Schembri, Quinto, Aiello, Pignato, and Sgrò (2019), and Niemistö et al. (2020). Additionally, Madrona, Martínez and Faraco (2021) found higher average ranges in laterality, postural tonic control, and spatial orientation among children engaged in extracurricular physical activities.

Conclusions

Based on the obtained results, several key findings emerge that could inform pedagogical practices. It would be partic-

ularly beneficial to elucidate the basic motor competencies of primary school-aged children in the athletic population because this area has attracted increasing attention in recent studies. Our research focused on analyzing the motor skills of primary school-aged children using the MOBAK 3–4 test battery. The findings reveal that children engaged in sports demonstrate motor skills comparable to those of non-athletic counterparts in the OM domain. Notably, a significant value was observed in the OM area (mean =4.52). Among the individual test items in the OM domain, the throwing test (mean =0.94) exhibited significance ($p < 0.05$) in favor of the sporting population.

The SM area analysis and comparison of its test items revealed higher average values favoring the sporting population, although without statistical significance confirmation. However, in the overall score of the SM area (mean =3.43), statistical significance favoring the sporting population was confirmed when compared to the sets. Additionally, the total score (mean =7.96) in favor of the sporting population was statistically significant at $p < 0.05$. These results confirm that children engaged in regular sports activities attain a statistically higher level of physical literacy, as evidenced by measurable motor competencies. Based on the obtained results, it is evident that this study has certain limitations. Generalizing the findings is challenging owing to the low prevalence of children in the observed athletic and non-athletic populations. Additionally, it would be beneficial to specify the type and extent of leisure activities performed by children, particularly in the context of monitoring those involved in sports. It is essential to determine whether these activities are purely recreational or aimed at enhancing mobility performance under the guidance of qualified trainers. Children in the athletic population engage in various leisure activities that may impact their overall performance in the target exercise areas. However, these effects were not consistently confirmed across all test item indicators. This inconsistency can be attributed to irregular attendance and scheduling conflicts in sporting activities.

A strength of this study is that the used instrument for the assessment is well established and economical.

This study addresses the growing demand in scientific research for expertise in educational practice aimed at stimulating basic motor competencies. Furthermore, it underscores the necessity of establishing diagnostics for these competencies within the school setting. Effective educational support in physical education is crucial to providing children with a solid foundation for healthy motor development and lifelong physical education.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

The Impact of Motivation on Decision-Making of Futsal Goalkeepers in Indonesia: An Analysis of Self-Confidence as Mediator

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Abstract

This study aims to explore the relationship between motivation and decision-making of futsal goalkeepers through self-confidence variable. It is a quantitative correlational research using structural equating modeling (SEM) approach with Smart-PLS analysis. The sample was 150 elite goalkeepers (103 men and 47 women), aged 23.4 ± 7.8 , had 7.24 ± 5.6 years of career in competition, and had 9.78 ± 2.2 years of training experience. A questionnaire on the Google form was used as research instrument. It was arranged and adjusted to the conditions in real context by modifying the results of previous research through the (focus group discussion (FGD)). The results of this research show that the instrument is declared valid based on the Outer Loading value (>0.7) and average variance extracted (AVE; >0.5), then the reliability test is based on Cronbach's Alpha (>0.7) and Composite Reliable (>0.7). The Goodness of Fit (GOF) in this research model is Standardized Root Mean Square (SRMR; <0.10 or <0.08), showed 0.073. Path Coefficients (P-Values) showed the effect of motivation on self-confidence at 0.001, the effect of motivation on decision-making at 0.045, the effect of self-confidence on decision-making at 0.000, and motivation on decision-making through self-confidence at 0.002. It conclusion, the proper training method, coach support, team and environment must constantly be optimized to increase motivation. Furthermore, participating in trial matches and competitions at the regional and provincial levels is an effective initial method to develop and train self-confidence. Ultimately, a goalkeepers increasing skill at decision-making will lead to becoming an elite athlete.

Keywords: futsal, athlete performance, competition, sports psychology

Introduction

Futsal is a type of sport that has the character of a fast and dynamic variant of football. From the relatively small field size, there are few opportunities for errors, so it requires quick and precise decision-making skills, especially from a goalkeeper (Windoro et al., 2019). Goalkeeper is a player the team can trust to repel attacks from the opposing team. As the last guard in team defense, futsal goalkeepers face situations requiring instant decisions to anticipate opponent attacks and protect the goal (Szwarc & Oszmaniec, 2021). Futsal goalkeeper decision-making involves a quick evaluation of various factors, such as the direction of the opponent's

kick, the position of the teammate, and the overall game situation (Medina et al., 2020; Méndez-Domínguez et al., 2021). Ideally, a goalkeeper should have a high technical and tactical ability to improve his skills in repelling the ball and preventing situations that lead to goals (Szwarc & Oszmaniec, 2021).

Decision-making is crucial for athletes in all sports to determine the following strategy and movement. However, in team sports, decision-making is the most complicated thing compared to individual and team sports (Araújo et al., 2023; Nagarabetta, 2023). Decision-making is one of the factors resulting from the performance of futsal athletes, so athletes



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must undergo a lengthy training process to improve performance (Esteves et al., 2022). The factors that influence the decision-making of futsal goalkeepers are an essential focus of research to understand the dynamics of this game in more depth. From the initial observation, it can be concluded that motivation plays a crucial role in decision-making ability and performance improvement for futsal athletes in all positions (Choi, 2023; Leibo et al., 2021; Rico-Lara et al., 2023; Sanmiguel-Rodríguez & Giraldez, 2021).

As an initial observation, the Indonesia Futsal Federation has noted that the participation of early childhood and adolescents in futsal coaching has increased in recent decades (Dwi Agestin et al., 2023; Rivaldi et al., 2022; Supiati et al., 2021). Furthermore, enthusiasts who chose the goalkeeper position also showed improvement. The initial survey of prospective athletes who choose the goalkeeper position explains the initial selection process that is easy compared to other positions, wants to be recognized by everyone after becoming part of the club, makes parents proud, increases self-esteem and searches for identity. The initial survey results refer to an athlete's motivational factors (Duclos Bastías et al., 2021). Motivation is an encouragement for athletes to achieve success in achieving achievements (Katanic et al., 2022; Posavac et al., 2023). Thus, the motivation that will continue to increase can be a driving force for goalkeepers to choose specific strategies in dealing with diverse game situations through the right training process and long experience (Barrero, 2023).

Initial observations are also addressed to futsal goalkeepers who have high competition experience. The initial observations showed that goalkeepers with high experience did not show problems, and there was no decrease in motivation (Csáki et al., 2017). However, further investigation through unstructured interviews will be conducted by adopting a research questionnaire on futsal goalkeeper decision-making from Windoro et al. (2019). The results showed doubts about cutting the ball during corner kicks, passing the ball short to the defender, assisting the attack when the score was behind, and dispelling the arrival of a tight-bottom ball (Castillo-Rodríguez et al., 2022). The following result was that the goalkeepers said they were worried, nervous, and unsure of the match's outcome before the match (Mouloud, 2019). Then, as the match progresses, sometimes it takes time to quickly adapt, easily intimidated and unsure of winning when seeing a better and solid opponent play (Madsen et al., 2022). Based on the initial interview results, the goalkeeper's confidence was affected (Álvarez-Kurogi et al., 2019). The interview answers are based on factors showing that optimists' self-confidence variable is not worrisome and can adapt (Supriyanto et al., 2022). Previous studies have reported that confidence can affect the skills of futsal athletes (Febrinaldo et al., 2023; Hidayati et al., 2023).

The relationship among motivation, decision-making, and confidence of futsal goalkeepers is an exciting research subject, especially in efforts to improve individual and team performance. Until now, no scientific publications have analyzed the impact of motivation and confidence in decision-making on futsal goalkeepers. This study aims to explore the relationship between motivation and decision-making of futsal goalkeepers through self-confidence variable. By understanding how motivation influences the decision-making process through self-confidence, it is expected that new strategies can be found in the training and coaching of futsal goalkeepers that can improve the quality of team play.

Method

Procedures

This correlational quantitative study investigates the influence between variables. The analysis of this study used a structural equating modeling (SEM) approach because it involved a mediator variable, self-confidence. The independent variable is motivation, and the dependent variable is the decision-making. The advantage of SEM research is that it can comprehensively analyze each variable based on direct and indirect influence. This research was conducted for three months, from January to March 2024, with a sequence of identification based on problem phenomena, formulating research objectives, data collection, and preparation of scientific reports in this article.

Participants

The population of this study was futsal athletes in Indonesia, but the sample selection used purposive sampling. Thus, the goalkeeper criteria in this study were still active as athletes, and had career in a club that competes in the Indonesian Pro Futsal League and the Indonesian National Sports Week (PON) in 2024. PON is a multi-event sports championship held every four years in Indonesia, one of which is futsal. The sample was 150 elite goalkeepers (103 men and 47 women), aged 23.4 ± 7.8 , had 7.24 ± 5.6 years of career in competition, and had 9.78 ± 2.2 years of training experience.

Measurements

The research instrument was adopted from the results of previous studies. The motivation variable was adopted from the Sport Motivation Scale (SMS-6), which consisted of six dimensions, namely amotivation, external regulation, introjected regulation, identified regulation, integrated regulation, and intrinsic motivation (Kawabata & Mallett, 2013; Mallett et al., 2007;). The self-confidence variable consists of five dimensions, namely Optimistic, Independent, Sportsman-like, Not Worrisome, Self-Adaptable (Amir, 2015; Arif & Sujarwo, 2021). The decision-making variable adopted from the results of the study by Windoro et al. (2019) consists of five dimensions, namely reading the movement of the ball (Y1), regulating the tempo of the game (Y2), keeping the ball away from the circle and goal line (Y3), angling for the opponent (Y4), and blocking the shot (Y5). After the instrument preparation material had been determined, a focus group discussion (FGD) was carried out to determine valid items. This FGD was attended by lecturers with a minimum doctor qualification who was an expert in futsal coaching and a lecturer in sports psychology. Futsal coaches with AFC level 1 licenses were also involved. The specified scale was in the form of a Likert scale of 1 to 5.

Statistical Analysis

The analysis of this study used the help of Smart-PLS (Hair et al., 2019). There were two model approaches, namely the first-order model used for the motivation variable (X) and self-confidence (M), and the second-order model used for decision-making (Y). The decision-making variable (y) consisted of five dimensions, namely reading the movement of the ball (Y1), setting the tempo of the game (Y2), keeping the ball away from the circle and goal line (Y3), angling for the opponent (Y4), blocking the shot (Y5). Then there are two methods for reporting results. The first is in the PLS Algorithm

menu which aims to analyze convergent validity values, the outer loading value must be >0.7 and the average variance Extracted (AVE) value >0.5 . Then the reliability test is based on Cronbach's Alpha (>0.7) and Composite Reliability (>0.7). Next, report the goodness of fit (GoF) value to see the goodness of this research model. Second, in the bootstrapping

menu, the results of this menu will show direct and indirect effects based on P Values (<0.05).

Results

The following is a report using the Smart-PLS software with the PLS Algorithm menu according to Figure 1.

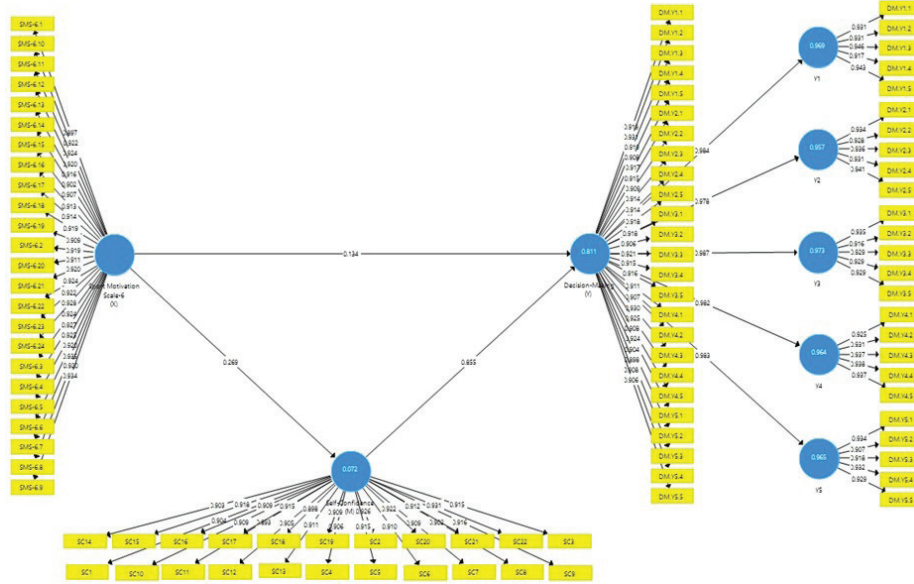


FIGURE 1. Analysis Results on PLS Algorithm Menu

Validity and Reliability

The convergent validity results based on the outer loading value for each item in the instrument show >0.7 (Figure 1), then the average value for each variable shows an AVE value >0.5 (table 1).

Furthermore, the Cronbach's Alpha value for each variable shows results >0.7 and the Composite Reliability value for each variable also shows results >0.7 . Thus, the research instrument shows valid and reliable results.

Table 1. Construct Validity and Reliability Result

	Cronbach's Alpha	rho_A	Composite Reliability	Average Variance Extracted (AVE)
Decision-Making (Y)	0.992	0.992	0.992	0.836
Self-Confidence (M)	0.990	0.990	0.991	0.830
Motivation (X)	0.992	0.993	0.992	0.844
Reading The Movement of The Ball (Y1)	0.963	0.963	0.971	0.871
Regulating The Tempo of The Game (Y2)	0.963	0.963	0.972	0.873
Keeping The Ball Away from The Circle and Goal Line (Y3)	0.959	0.959	0.969	0.860
Angling for The Pponent (Y4)	0.963	0.963	0.971	0.871
Blocking The Shot (Y5)	0.957	0.957	0.967	0.854

Fit Model

The fit model test was carried out to see suitability of this research model. The criteria for the fit research model are

RMS Theta (Root Mean Square Theta) Value is <0.102 , SRMR (Standardized Root Mean Square) Value is <0.10 or <0.08 , and NFI Value is >0.9 .

Table 2. Goodness of Fit (GoF)

	Saturated Model	Estimated Model
SRMR (Standart Root Mean Residual)	0.073	0.073
d_ULS	5.138	5.215
d_G	5.867	5.867
Chi-Square	2316.884	2316.884
NFI (Normed Fit Index)	0.613	0.613
RMS Theta (Root Mean Square Residual Covariance)	0.131	

Based on the results in Table 3, the RMS Theta (Root Mean Square Theta) value is $0.131 > 0.102$, and the NFI value is $0.613 < 0.9$. So, based on the two assessments of the model, it does not meet the criteria for the fit model.

However, based on the SRMR (Standardized Root Mean Square) value, the value is $0.073 < 0.10$ or < 0.08 , so the model

is fit. Thus, it can be said that the model fits the data.

Path Coefficient

Path analysis in this study used the bootstrapping menu shown in the following Figure 2. The analysis will be explained based on the table.

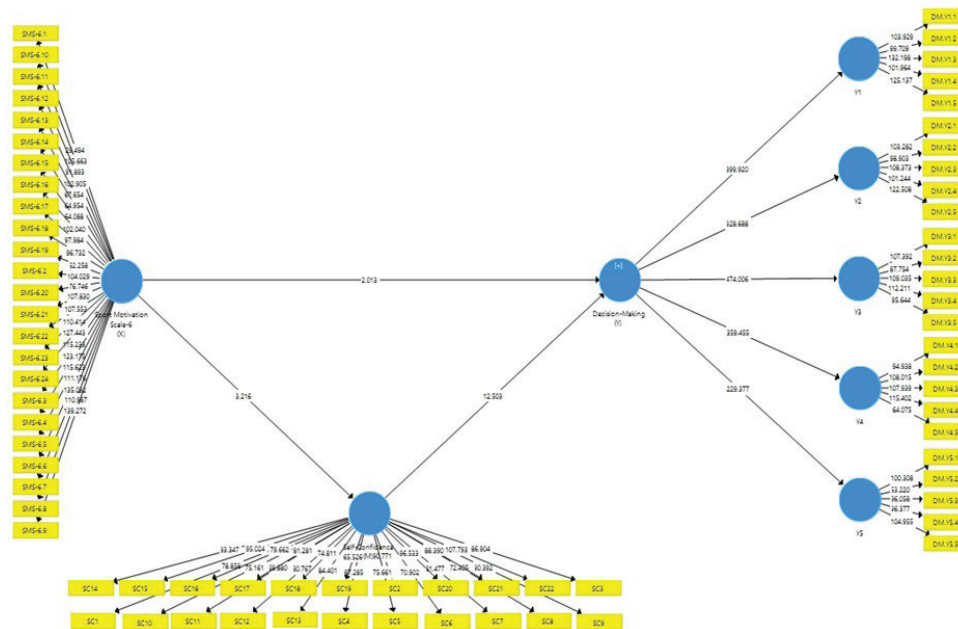


FIGURE 2. Path Coefficient Results of Bootstrapping

The Smart-PLS model in this research combines the first-order model and the second-order model. Based on Table 4, the second-order model is used on the dependent variable, namely decision-making. The decision-making variable consisted of five dimensions, namely reading the movement of the

ball (Y1), setting the tempo of the game (Y2), keeping the ball away from the circle and goal line (Y3), angling for the opponent (Y4), blocking the shot (Y5). From the five dimensions, it shows that the P-Values are 0.000, meaning that the five dimensions in decision-making are stated to be interconnected.

Table 3. Results of Path Coefficient Analysis

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
Decision-Making (Y) -> Y1	0.984	0.984	0.002	399.920	0.000
Decision-Making (Y) -> Y2	0.978	0.979	0.003	328.688	0.000
Decision-Making (Y) -> Y3	0.987	0.987	0.002	474.006	0.000
Decision-Making (Y) -> Y4	0.982	0.982	0.003	359.455	0.000
Decision-Making (Y) -> Y5	0.983	0.983	0.004	229.377	0.000
Self-Confidence (M) -> Decision-Making (Y)	0.855	0.852	0.068	12.503	0.000
Motivation (X) -> Decision-Making (Y)	0.134	0.138	0.066	2.013	0.045
Motivation (X) -> Self-Confidence (M)	0.269	0.273	0.084	3.216	0.001
Motivation (X) -> Self-Confidence (M) -> Decision-Making (Y)	0.230	0.233	0.075	3.074	0.002

Furthermore, the results of the path analysis are: 1) the effect of motivation has a positive and significant effect on self-confidence, namely P-Values 0.001; 2) the effect of motivation has a positive and significant effect on decision-making, namely P-Values 0.045; 3) the effect of self-confidence has a positive and significant effect on decision-making, namely P-Values 0.000; 4) the effect of motivation on decision-making through self-confidence, namely P-Values 0.002.

Discussion

Based on the findings of this research, motivation influences decision making either directly or through self-confidence. These results show that the higher the level of motivation a futsal goalkeeper has, the better his ability to make decisions while on the field. The results of this research are in accordance with the results of previous research which explains that a goalkeeper's motivation can encourage them to look for the best solution in different situations when playing futsal (Alves

et al., 2022; Choi, 2023). Other findings also show that motivation influences self-confidence. In the next stage, this self-confidence influences decision making among futsal goalkeepers. These results explain that the motivation to become professional athletes, they tend to have greater self-confidence in their ability to achieve achievements and overcome challenges, of course by not letting the ball enter the goal. Analysis of this performance, it has been found in previous research that goalkeepers who have high experience will choose to make the decision to move early during a penalty and have a greater chance of successfully blocking the ball (Noël et al., 2021). Meanwhile, experience has a positive relationship with the level of motivation, so that the more competitions an athlete participates in, the more motivation will increase (Hussein, 2023). In the discussion section of the research results of Noël et al. (2021), the goalkeeper who moves at the start is due to having strong self-confidence, which will ultimately affect the goalkeeper's performance, namely the chance of successfully blocking the ball is very high.

Then, the level of self-confidence is also directly caused by the athlete's experience in participating in competitions. This statement was found in this research that motivation directly influences the self-confidence of futsal goalkeepers in various ways, such as increasing training time at school or university and looking for a team for match trials. This explanation is supported by previous research. One method for increasing achievement in futsal is to encourage athletes to increase their training time outside of their training schedule (Permadi et al., 2023; Syaifuddin et al., 2023). The more training and competition time they participate in, the more confident the goalkeeper will be in his own ability to make decisions quickly and correctly (Iusepolsky et al., 2022). The training method involves modifying a small side game of 3 versus 3 involving the goalkeeper (Alex et al., 2024). Thus, achievement motivation can increase decision-making in futsal goalkeepers directly and also through increased confidence (Brendon, 2022).

Previous research studies have proven that motivation can have an impact on athlete performance (Juita et al., 2024; Prayoga et al., 2024; Wibowo et al., 2024). Motivation factors towards decision-making of futsal goalkeepers must also be trained in their training programs, such as setting challenging but realistically achievable goals to increase motivation. For example, they aim to catch the ball's direction to improve reaction ability in certain situations on the field. Recognition of achievement and progress in technical and physical quality can be a strong driver to increase a goalkeeper's motivation (Bastias et al., 2021). It can be in the form of compliments from coaches and teammates, prizes from the organization or even awards in competitions. A goalkeeper should also be taught to be open to suggestions. Having an open attitude towards learning and developing and seeing failure as an oppor-

tunity to learn and grow can increase achievement motivation, indirectly affecting their skills. Another study shows that support from coaches, teammates, family support, and other close people can increase motivation by providing encouragement, support, or praise (Mandan et al., 2024).

Motivation and trust are interrelated psychological elements (Ita et al., 2022; Levine et al., 2022; Park & Jeon, 2023). The factor that increases self-confidence is when they have successful experience in training or previous matches. Each small success can reinforce the belief that they can perform the task as goalkeepers. Therefore, coaches must consider holding trials against teams whose playing qualities are lower than the team being coached, with the aim of producing a win to increase mental motivation and self-confidence before taking part in a long competition (Almeida et al., 2019). The coach must also teach how to form a realistic understanding of each goalkeeper's strengths and weaknesses and focus on mastering the necessary skills. Training models using positive reinforcement techniques, such as imagery, visualization exercises, or positive self-affirmations, can help increase self-confidence (Rozi et al., 2023). The training model has been proven to improve the skills of a goalkeeper, which will affect decision-making during matches.

By paying attention to the right training program such as challenging games, then environmental factors such as team support, coaches, organizations can increase the motivation and self-confidence of a futsal goalkeeper. The limitation of this research is the time used in the data collection process because this research is purely a type of quantitative research. Combining quantitative methods as primary data and qualitative methods using interviews to strengthen research results would be better.

Conclusion

This study reveals that motivation plays a vital role in mental health and impacts the ability of futsal goalkeepers. These results indicate that motivation affects the self-confidence of a goalkeeper, which in turn will positively affect decision-making. Motivation must be built and developed when the athlete chooses to be in the goalkeeper position. The conclusion is that the proper training method, coach support, team and environment must constantly be optimized to increase motivation. Furthermore, participating in trial matches and competitions at the regional and provincial levels is an effective initial method to develop and train self-confidence. Thus, suitable training methods, positive environmental support, and match experience that are often followed will affect the decision-making of futsal goalkeepers. In the end, a goalkeeper's better and improved skills will lead to becoming an elite athlete who will play in the professional futsal league and at the international level.

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Conflict of Interest

The researchers have no conflicts both with the researchers and the results of other studies.

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ORIGINAL SCIENTIFIC PAPER

Analyzing the Associations between Physical Literacy, Physical Activity Levels, and Sedentary Behavior: Cross-sectional Study in Preadolescent Children

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Abstract

Physical literacy (PL) is considered an important determinant of physical activity level (PAL), but studies have rarely examined the associations between PL and PAL in preadolescents. This study aimed to evaluate associations between PL, PAL and sedentary behavior (SB) in preadolescents from southeastern Europe. The participants were 9- to 11-year-old children from Croatia and Bosnia and Herzegovina (n=333, 52% girls) who were tested for PL, PAL and SB. The PLAYself questionnaire was used for the evaluation of PL, while PAL was estimated by data provided throughout the Physical Activity Questionnaire for Children (PAQ-C). Pearson's correlations and multiple regressions were calculated to identify the associations between study variables. The results revealed a significant correlation between PL and PAL in the total sample (16% of the common variance) among boys (10% of the common variance) and girls (25% of the common variance), which was additionally confirmed by multiple regression calculated between the PL subdomains and PAL ($R^2=0.23, 0.15$, and 0.31 for total sample, boys and girls, respectively). The SB was not significantly correlated with PL ($<2\%$ of the common variance; $p>0.05$) or with PAL ($<1\%$ of the common variance; $p>0.05$). The results indicated a stronger association between PL and PAL in girls than in boys, indicating the possible influence of the types of physically demanding activities children are involved in on the studied associations. To explore causality more specifically, intervention studies are warranted.

Keywords: prepubescent children, questionnaire, regression, correlation

Introduction

There is strong evidence that physical activity (PA) has multiple benefits for children and adolescents, improving physical and mental health, sleep quality, brain development, bone health, and social, psychological, and cognitive health (Alvarez-Pitti et al., 2020). Furthermore, PA is recognized as an important factor in the prevention of overweight

and obesity (Ariza et al., 2019). However, despite the current knowledge on the benefits of PA, there is an increasing level of sedentary behavior characterized by low physical activity levels (PAL). Unfortunately, the majority of the young population is not physically active, and global reports indicate that $>80\%$ of children and adolescents do not have sufficient PAL (Guthold, Stevens, Riley, & Bull, 2020; Kuna, Duvnjak,



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& Sklempe Kokić, 2023). Of particular concern is that recent reviews of scientific papers emphasize that a progressive decline in moderate to severe PAL occurs even before adolescence, beginning between early and middle childhood, that is, between the ages of six and nine (Farooq et al., 2020; Vukelja, Milanovic, & Salaj, 2022). Therefore, it is necessary to investigate methods to increase the level of PAL in children and determine the factors associated with it. As one of the novel solutions, the concept of physical literacy (PL), which is considered a key factor that leads to an increase in the PAL was proposed (Caldwell et al., 2020; Geets-Kesić, Maras, & Gilić, 2023).

Physical literacy (PL) is defined as “the motivation, confidence, physical competence, knowledge, and understanding to value and take responsibility for engagement in physical activities for life (Whitehead, 2013). The concept of PL encompasses the development of skills, knowledge and attitudes that empower individuals to lead a physically active lifestyle throughout their lives and, as such, emphasizes the importance of encouraging lifelong engagement in PA. Although different concepts of PL and tools for assessing the level of PL have been developed around the world, there is no consensus on which is the most appropriate (Longmuir & Tremblay, 2016). However, the most popular and commonly used tools for assessing PL are the Canadian Assessment of Physical Literacy (CAPL) and the Physical Literacy Assessment of Youth (PLAY; de Dieu & Zhou, 2021). Recent studies have shown better reliability and validity of the PLAY questionnaire in the evaluation of PL in preadolescents (Vuletic et al., 2023). This is important to highlight because, from the aspect of the development of lifelong participation in PAs, preadolescence is a crucial period.

More precisely, the PL recognizes several subdomains, including (1) the physical domain (which includes motor skills and physical capacity), (2) the affective domain (which includes motivation and confidence) and (3) the cognitive domain (which includes knowledge and understanding). These subdomains are interrelated and are critical for supporting participation in PA. Therefore, encouraging the development of PL in children from an early age is important for a lifelong active and healthy lifestyle. However, although it can be assumed that PL and PA are related, studies have rarely investigated their associations in children and adolescents, and the results are generally inconsistent. In brief, Canadian studies have shown a certain positive association between PL and PAL in children aged 8 to 12 years (Belanger et al., 2018; Stearns, Wohlers, McHugh, Kuzik, & Spence, 2019). Similar results were reported in another Canadian study of children aged 7-14 years (Bremer et al., 2020). Moreover, a study performed in China on adolescents aged 12 to 18 years evidenced a low (although significant) correlation between PL and PAL (Choi, Sum, Leung, & Ng, 2018). Finally, a recent Croatian study revealed a significant correlation between PL and PAL in girls aged 9-11 years but not among boys of the same age (Rajkovic Vuletic et al., 2024).

From the previous literature overview, it is clear that there is increased interest in the association between PL and PAL in children (Belanger et al., 2018; Bremer et al., 2020; Choi et al., 2018; Stearns et al., 2019). However, the majority of previous studies used various instruments for PL assessments that focused on motor competence (e.g., the Canadian Agility and Movement Skill Assessment, PLAYfun) and cog-

nitive domains (knowledge and understanding), while there are few studies that have used the PLAYself to assess the cognitive and affective domains of PL in younger children (Stearns et al., 2019). Furthermore, studies performed thus far included samples of participants of relatively wide age spans, which could influence the reported findings simply because of the known decrease in PAL with the age of the children (Bremer et al., 2020; Choi et al., 2018). Therefore, the aim of this study was to investigate the association between PL levels assessed with PLAYself and PAL in an international sample of children aged 9-to-11 years from south-eastern Europe. Additionally, we evaluated an association between sedentary behavior and PAL in preadolescents.

Materials and methods

Participants

This study involved preadolescent children (n=333; 170 girls) from Croatia and Bosnia and Herzegovina. All participants were in good health, attended the 3rd or 4th grade of elementary school (4th and 5th in Bosnia and Herzegovina), and regularly attended physical education (PE) classes. During the research, all participants were aged between 9 and 11 years. Children who were ill or had an injury affecting their movement in the two weeks before the examination were not included in the study. The research team informed the parents or guardians about the research's objectives and procedures and obtained their consent for their children's participation. This research was approved by the Ethics Committee of the Faculty of Kinesiology, University of Split.

Variables and procedures

In addition to age (years), sex (male or female), we observed PL, PAL and sedentary behavior (SB). The PL was assessed using the PLAYself questionnaire, which is a component of the PLAY tools (PLAYfun, PLAYbasic, PLAYself, PLAYparent, PLAYcoach, and PLAYinventory). This questionnaire is used by children and adolescents for self-assessment of their PL level and has four subdomains: (i) Environment, which assesses the degree of confidence of movement in different environments (e.g., activities in the gym, in and on water, and on snow and ice), including questions such as “How good are you at doing sports and activities in the gym?” or “How good are you at doing sports and activities in and on the water?” (ii) Self-reports of PL, which assess a number of affective and cognitive segments (e.g., motivation, confidence, and self-esteem) related to PL that determine an individual's self-efficacy and ability to participate in PA, including questions such as “It does not take me long to learn new skills, sports or activities” or “I think I can take part in any sport/PA that I choose”. (iii) The relative ranking of literacy with its subdomains of literacy, numeracy, and physical literacy assesses literacy in different environments (e.g., school, home, and friends) and evaluates the importance that an individual places on each of the mentioned literacies. (iv) The level of fitness is assessed by the following question: “My state of fitness is good enough to allow me to participate in all the activities I choose”. This subdomain is excluded from the final PLAYself score. The overall result of the questionnaire is calculated by summing the points from the first three subdomains and dividing by the total number of questions (27 in total). The maximum possible self-concept score for PL is 100, and it is represented

by PLAY self points. The Croatian version of the PLAYself questionnaire has recently been shown to be feasible, valid, and reliable in Croatian children aged 9 to 11 years (Vuletic et al., 2023).

PAL was evaluated using the “Physical Activity Questionnaire for Older Children” (PAQ-C) (Crocker, Bailey, Faulkner, Kowalski, & McGrath, 1997). The PAQ-C-Croatian version of the questionnaire for assessing the level of PA was constructed for children of similar age (from 8 to 14 years) to assess the overall level of arithmetical PA (Samaržija & Mišigoj-Duraković, 2013). The questionnaire comprises 9 questions that are evaluated on a 5-point scale, for example, “In the last 7 days, what did you mostly do during a long break (except eating a snack)?” and “In the last 7 days, during your physical education (PE) classes, how often were you very active (playing hard, running, jumping, throwing)?”. The total score for PA was determined by calculating the mean of the responses provided. A maximum PAQ-C score of 5 represented high activity, while a score of 1 represented very low or no activity.

The level of SB was evaluated by a questionnaire consisting of questions about the time children spent in six different groups of behaviors in hours and minutes per day: (1) watching television, (2) playing computer games, (3) browsing the internet, (4) homework and studying, (5) listening to music, and (6) reading (Štefan, Horvatin, & Baić, 2019).

Statistical analysis

The normality of the variables was tested with the Kolmogorov–Smirnov test. For normally distributed vari-

ables, means and standard deviations were reported, while counts and percentages were reported for categorical and ordinal variables.

Differences between boys and girls were evidenced by *t*-test for independent samples.

The associations between study variables were first established by calculating the Pearson’s product moment correlation coefficients. In the next phase, multiple regressions were performed with PAL as the criterion and different subdomains/facets of PL as predictors. Generally, multiple regressions were calculated between facets of PL and PAL, due to established correlations (please see previous phase, and Results for more details). The coefficient of the multiple correlation (*R*), coefficients of determination (*R*²), standardized (β) and nonstandardized (*B*) regression coefficients are reported.

Statistica ver 13.5 (Tibco Inc., Palo Alto, CA, USA) was used for all calculations, and a *p*-level of 95% was applied.

Results

The descriptive statistics and differences between sexes in the study variables are presented in Table 1. Boys and girls were of the same age, and no significant differences were found between boys and girls for the PL subdomains or total PL score. However, boys had greater PAL (*t* test = 2.55, *p* < 0.01) and less sedentary time than girls did (*t* test = 3.37, *p* < 0.001).

The Pearson’s product moment correlation coefficients between the study variables calculated for the total sample and for girls and boys are presented in Table 2. No significant correlation between SB and PL or PAL was detected. PAL

Table 1. Descriptive statistics and differences between boys and girls

	Girls (n = 170)		Boys (n = 163)		t test	
	Mean	Std.Dev.	Mean	Std.Dev.	t value	p-level
Age (years)	9.99	0.66	9.95	0.74	0.61	0.54
PL-1 (score)	361.13	120.70	353.35	113.98	0.61	0.55
PL-2 (score)	891.88	163.51	915.95	146.41	-1.42	0.16
PL-3 (score)	228.83	64.68	222.52	71.61	0.84	0.40
PL-4 (score)	236.17	63.20	224.68	75.61	1.50	0.13
PL-5 (score)	253.54	60.42	259.76	52.15	-1.01	0.31
PLAYself (score)	73.02	12.27	73.19	11.03	-0.14	0.89
SB (score)	19.04	5.98	21.62	7.86	-3.37	0.001
PAQ-C (score)	3.02	0.63	3.21	0.73	-2.55	0.01

Note. PL-1 – environment subdomain of the PLAYself questionnaire; PL-2 – self-description subdomain of the PLAYself questionnaire; PL-3 – literacy subdomain of the PLAYself questionnaire; PL-4 – numeracy subdomain of the PLAYself questionnaire; PL-5 – physical literacy subdomain of the PLAYself questionnaire; PLAYself – total score for the PLAYself questionnaire; SB – sedentary behavior; PAQ-C – physical activity level obtained by the PAQ-C questionnaire

was significantly correlated with the first two subdomains of the PL in the total sample (<17% of the common variance), in girls (<25% of the common variance), and in boys (<20% of the common variance). Additionally, the correlation between PAL and the PL total score reached statistical significance for the total sample (16% of the common variance), girls (25% of the common variance), and boys (10% of the common variance).

The results of the multiple regression calculations for the PAQ-C as criterion variables with the PL subdomains/facets and PL-total score as predictors are presented in Table 3.

Predictors explained a significant proportion of the variance in the criterion when regressions were calculated for the total sample and separately for boys and girls (23%, 15%, and 31% of the explained variance, respectively). In the total sample, significant partial predictors were the first two facets of PL (β =0.31 and 0.24, respectively). The first facet of the PL was a significant partial predictor for boys (β =0.28), while the first two facets were significantly partially associated with the criterion for girls (β =0.38 and 0.23, respectively). In general, the positive association between specific PL facets and PAL should be highlighted.

Table 2. Correlation between study variables for the total sample of participants and stratified by sex (* indicates coefficients significant at $p < 0.05$)

		PL-1	PL-2	PL-3	PL-4	PL-5	PLAYself	SB
PL-1	Total	-						
	Girls	-						
	Boys	-						
PL-2	Total	0.50*	-					
	Girls	0.46*	-					
	Boys	0.51*	-					
PL-3	Total	0.11*	0.22*	-				
	Girls	0.11	0.25*	-				
	Boys	0.06	0.11	-				
PL-4	Total	0.11	0.19*	0.69*	-			
	Girls	0.16*	0.26*	0.69*	-			
	Boys	0.01	0.05	0.67*	-			
PL-5	Total	0.21*	0.33*	0.49*	0.50*	-		
	Girls	0.17*	0.32*	0.55*	0.63*	-		
	Boys	0.19*	0.23*	0.40*	0.38*	-		
PLAYself	Total	0.69*	0.81*	0.59*	0.58*	0.62*	-	
	Girls	0.67*	0.82*	0.59*	0.63*	0.63*	-	
	Boys	0.69*	0.77*	0.56*	0.51*	0.55*	-	
SB	Total	0.00	0.06	0.04	0.00	0.09	0.06	-
	Girls	0.02	0.05	0.02	0.12	0.01	0.06	-
	Boys	-0.04	-0.03	0.02	-0.09	0.08	-0.04	-
PAQ-C	Total	0.43*	0.40*	0.07	0.10	0.19*	0.42*	0.06
	Girls	0.50*	0.43*	0.14	0.20*	0.20*	0.50*	0.08
	Boys	0.35*	0.29*	-0.03	-0.01	0.06	0.28*	-0.05

Note. PL-1 – environment subdomain of the PLAYself questionnaire; PL-2 – self-description subdomain of the PLAYself questionnaire; PL-3 – literacy subdomain of the PLAYself questionnaire; PL-4 – numeracy subdomain of the PLAYself questionnaire; PL-5 – physical literacy subdomain of the PLAYself questionnaire; PLAYself – total score for the PLAYself questionnaire; SB – sedentary behavior; PAQ-C – physical activity level obtained by the PAQ-C questionnaire

Table 3. Multiple regression calculation for physical activity level as a criterion variable for the total sample and stratified by sex (* indicates coefficients significant at $p < 0.05$)

	Total sample		Boys		Girls	
	β	B	β	B	β	B
Intercept		1.41*		2.02*		1.31*
PL-1	0.31*	0.00*	0.28*	0.00*	0.38*	0.00*
PL-2	0.24*	0.00*	0.15	0.00	0.23	0.00*
PL-3	-0.07	0.00	-0.09	0.00	-0.03	0.00
PL-4	0.04	0.00	0.04	0.00	0.08	0.00
PL-5	0.05	0.00	-0.01	0.00	0.03	0.00
R	0.48		0.38		0.55	
R ²	0.23		0.15		0.31	
p	0.001		0.001		0.001	

Note. PL-1 – environment subdomain of the PLAYself questionnaire; PL-2 – self-description subdomain of the PLAYself questionnaire; PL-3 – literacy subdomain of the PLAYself questionnaire; PL-4 – numeracy subdomain of the PLAYself questionnaire; PL-5 – physical literacy subdomain of the PLAYself questionnaire; R – coefficient of multiple correlation; R² – coefficient of determination; p – level of significance; β – standardized regression coefficient; B – nonstandardized regression coefficient

Discussion

The few most important findings of this study are discussed in the following text. First, despite the significant correlation between PL and PAL in the total sample of participants, these

associations were weak. Second, the correlation between PL and PAL was greater among girls than among boys. Third, SB was not significantly associated with PL in the studied preadolescent children.

Low correlation between PL and PAL in the total sample

To the best of our knowledge, few studies have investigated the correlations between PL and PAL in preadolescent children. Additionally, the results are generally inconsistent. In brief, when studying a large sample of Canadian children, Belanger et al. reported that children who met PA guidelines had greater PL, confirming a positive correlation between PL and PAL (Belanger et al., 2018). In another Canadian study, the PL of preadolescent children was also significantly associated with PAL, as evaluated through moderate-to-vigorous PAL (Caldwell et al., 2020). However, in both previously presented investigations, the sample of participants was “one”, not dividing boys from girls or younger from older children (although the authors included gender/sex as covariate in the calculated regression models). Moreover, a very recent Croatian study performed with preadolescents revealed a positive correlation between PAL and PL (i) in girls but not in boys and (ii) in older (4th graders) and not younger (3rd graders) children (Rajkovic Vuletic et al., 2024). Therefore, our results on the generally poor (although significant) association between PL and PAL when preadolescents were observed as a unique sample are in line with previous reports. There are several possible explanations for such findings, which we will discuss in the following text.

First, the self-evaluation of the PL is based on one's perception of (personal) physical abilities and physical competencies in comparison to those of other people (i.e., peers, friends, schoolmates) (Vuletic et al., 2023). Therefore, the children observed in this study compared themselves to others who were participating in activities where physical competencies were present (i.e., sport training, PE, and free play). Naturally, boys compared themselves with boys, while girls compared themselves with peer girls. When translated to the PL scale, there was no significant difference between boys and girls in the PL subscale. However, it is well known that the PLs of boys and girls are not similar, even in preadolescence. Overall, this approach results in specific bias in the scoring of the PL and translates to the calculation of the correlation between the PL and PAL when calculated for the total sample of participants (not dividing boys and girls).

Second, despite previous findings, there is a certain possibility that in this period of life, PL and PAL are not as associated as they appear later in life. Specifically, although there is a global consensus that PL is an important determinant of PAL in life and that “physically literate” individuals will have greater PAL, the association between PL and PAL is expected to be stronger in adulthood. Namely, at younger ages children are physically active irrespective of their physical literacy simply because (i) young children are frequently engaged in free play and (ii) most of them participate in PE (Veitch, Salmon, & Ball, 2010). In other words, in this period of life, PA is not determined and conditioned by “physical competence”, as is the case in later life. Taken together, these facts ecologically decrease the correlation between PL and PAL in preadolescence.

Differences in correlations between PAL and PL for boys and girls

Interestingly, sex-stratified analyses of the correlations between PL and PAL are not common. This is particularly important considering that studies thus far have confirmed (i) nonsignificant gender/sex differences in PL between genders during childhood and adolescence and (ii) gender/sex differences in magnitude of correlations between PL and

PAL (Rajkovic Vuletic et al., 2024; Vuletic et al., 2023). For example, research on preschool children revealed that the PL is predictive of PAL in girls but not in boys (Cairney et al., 2018). Similarly, the authors found no significant correlation between PL and PAL in preadolescent boys, but a significant correlation between PL and PAL was found in same-age girls from Croatia (Rajkovic Vuletic et al., 2024). Although our results are not entirely consistent with those of previous reports (i.e., we found a significant correlation for both boys and girls), our findings support those of previous studies. Namely, the correlation we have evidenced for girls is almost twice as large as the correlation evidenced for boys.

The first issue that deserves attention in explaining sex-specific associations between PL and PAL is free-play. Free-play (i.e., participation in nonstructured activities) is a significant source of PA in preadolescence (Rajkovic Vuletic et al., 2024). Additionally, in the region where this study was conducted, boys are regularly involved in physically demanding nonstructured activities, while girls participate in physically less demanding games (Maric et al., 2020). Moreover, although it is an important source of PA, participation in nonstructured physically demanding activities is not as strongly associated with PL, as is the case for structured activities (PE and organized sport participation), where PL is specifically accentuated and developed (Sunda et al., 2022). On the other hand, the PAL of girls is more a result of participation in structured activities, such as PE and organized sports, where coaches and PE teachers specifically emphasize PL. As a result, girls who are physically active are more likely to be “physically literate”.

Another explanation for the finding that PL and PAL are more strongly associated in girls than in boys is based on practical/professional knowledge of the authors who are long-term professionals in PE and sport. In brief, the authors share the opinion that participation in any kind of physically demanding activity in girls is accompanied by a certain profound interest in the activity itself. In other words, girls rarely participate in any form of PA if they are not “interested in it”. Meanwhile, boys are frequently engaged in physically demanding activities (only) emotionally to compete and win. This is supported in studies where authors described the motives for being physically active between genders/sexes. In brief, boys are often motivated by competition, social recognition, and challenges, while girls are more likely to be motivated by intrinsic factors such as fun and being with friends (Portela-Pino, Lopez-Castedo, Martinez-Patino, Valverde-Esteve, & Dominguez-Alonso, 2019). Consequently, girls' participation in PA (i.e., PE, sport) actually leads to better PL than is the case for boys, resulting in a greater correlation between PAL and PL among girls.

Lack of correlation between SB and PL in preadolescence

Although sedentary behavior is often considered a habit that is directly opposed to PA, the association between PAL and SB has not been frequently explored. Most of the related studies have examined problems in late adolescence and young adulthood, and the related research has yielded mixed results. In brief, when authors summarized studies performed with children and adolescents, Pearson et al. reported a small negative association between sedentary behavior and PA in children and adults (Pearson, Braithwaite, Biddle, van Sluijs, & Atkin, 2014). In contrast, a more recent study revealed a

weak but positive correlation between SB and PA in undergraduate students (de Andrade Leão, dos Santos, & da Silva, 2018). Therefore, our finding of a nonsignificant association between sedentary time and PAL is not surprising.

Indeed, SB is a certain counterpoint of physically active behavior. It is logical to consider that people who have greater SB are less physically active. For that reason, “sedentarism” (high SB) and “physical inactivity” (i.e., low PAL) are frequently used interchangeably as synonyms (Forero, Morales, & Forero, 2023). Naturally, sedentary people spend more time in sedentary activities (i.e., watching TV, reading, playing video games, reading), which results in less time available to be physically active. Additionally, there is no doubt that sedentarism negatively influences physical capacity by decreasing muscular and functional capacities (i.e., fitness), which is another reason why people with high SB have lower PAL, simply because proper fitness allows one to be physically active at first place. As a result, a negative correlation between PAL and SB could be (naturally) expected.

On the other hand, modern life is often saturated with more or less mandatory sedentary activities. If we focus solely on children, at the very beginning of the mandatory school education (starting from six or seven years, depending on country), they are obligated to dedicate 5-6 hours of sedentarism during direct or indirect schooling (Tremblay et al., 2011). If we add the necessity of transport to different activities (including sports), SB increases to 7-8 hours daily. Furthermore, children often socialize through sedentary activities (mostly screen time), resulting in up to 9-10 hours of SB in regular circumstances. It is also common for parentally guided children to be involved in other forms of education (i.e., language, music), which additionally increases SB. Notably, children who have “increased SB” due to music or foreign-language education are often the same children who are engaged in out-of-school sport activities. While PAL exponentially increases as a result of sport participation lasting (for example) six hours per week, the duration of SB increases very much because of the other activities that are regular in preadolescence. Taken together, these results revealed small and nonsignificant correlations not only between SB and PAL but also between SB and PL. Therefore, a clear distinction

should be made between “sedentary behavior” and “sedentary lifestyle (sedentarism)”.

Limitations and strengths

The most important limitation of this study is that PAL was evaluated by questionnaire and not directly measured. Therefore, further analyses including direct measurements of the PAL at this age are needed. Next, this is a cross-sectional analysis, and consequently, causality cannot be explicitly identified. In future intervention studies, clear cause-effect relationships should therefore be evaluated.

This is one of the first studies in which the associations between PAL, SB and PL were evaluated via a sex-specific approach. This approach allowed specific and clear interpretations of the identified associations and provided directions for further studies in the field. Additionally, while PAL, PL and SB should be specifically investigated in different regions and under different sociocultural circumstances, this is one of the first investigations on a problem performed in southeastern Europe (e.g., the territory of former Yugoslavia).

Conclusion

The results clearly supported the necessity of a sex-stratified approach for evaluating associations between PL and PAL in preadolescence. Namely, the evaluation of the PL is based on comparisons with peers and friends, which is in this age group that is defined by sex/gender.

The greater correlation between PL and PAL in girls than in boys is probably influenced by differences in PA participation between boys and girls. While boys in preadolescence frequently participate in nonstructured activities, girls' PA is almost exclusively defined by participation in structured PA. Consequently, PA among girls is naturally followed by higher PL to a greater extent than among boys.

Although this is one of the first studies in which the correlation between SB and PAL was evaluated in preadolescents, it seems that SB and PAL are poorly correlated in this period of life. Most likely, the time spent in schooling and other educational activities increases sedentary time for all children, despite their PAL, leading to a lack of correlation between PAL and SB.

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Effects of Post-Activation Performance Enhancement in Kayak Sprint Competition within Same Day

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Abstract

The purpose of this study was to determine the post-activation performance enhancement in kayak sprint competitive simulation 2 times (S1 and S2) within the same day. This study was a crossover design, ten kayak sprint paddles national team performed different activations. All participants will be randomized crossover trials, get three interventions consisting of the Resistance program (2 x 4 x 80%1RM x Bench press and Bench row), Maximum sprint paddle (Ergometer all-out 2 x 10-sec x rest interval 2 min) and Control group (CON; Individual activity condition) followed 3-minute test (3MT). Maximum power, power average, stroke per minute, total distance, percentage of heart rate, and blood lactate concentration were recorded during the simulation competition. The main effect of the analysis showed that maximum power of the Resistance program was higher than the Control group in S1. The average power and total distance of the Resistance program in S2 were lower than in S1. Blood lactate concentration of the Resistance program and Maximum sprint paddle was higher than the Control group after the intervention, before the 3MT period, and remained rise until after 3MT, especially in S2. Also, blood lactate concentration before the intervention of the Resistance program in S2 was higher than in S1. The preconditioning by using the resistance or maximum sprint paddle improves maximum power only the first competition of the day. It seems that the resistance program leads to accumulated blood lactate concentration, and may decrease performance in the second competition of the day.

Keywords: *paddles sport, stimulate muscle, simulation competition, maximum voluntary contraction, Strategies warm-up*

Introduction

Preparing physically for performance in sports that use speed, power, and anaerobic energy is critical. The coordination of the upper body muscles is essential for athletic performance in Kayak sprints (Michael, Rooney, & Smith, 2008). A warm-up strategy is a way to help activate the muscles to be ready for a competitive day. Post-activation performance enhancement (PAPE) methods are preconditioning widely used (Prieske, Behrens, Chaabene, Granacher, & Maffiuletti, 2020). The physiological mechanism, the post-activation performance enhancement, had increased muscle temperature, dynamic fluid

within the muscle fibre, and motivation from muscle activation (Blazevich & Babault, 2019; Boullousa, 2021). The intervention method had a maximum voluntary contraction (MVC) performance enhancement and effected a rise in the rate of force development (RFD) after activation for several minutes (7-15 min) (Blazevich & Babault, 2019; Prieske et al., 2020).

According to theories, short-term high-intensity exertion should follow a low-intensity warm-up (Blazevich & Babault, 2019; Boullousa, 2021). Even after an intermittent high-intensity warm-up, blood lactate concentration levels remain high in the body (Khamros, Peepathum, Senakham, Sriramatr, &



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Phongsri, 2023). Dynamic movements with maximum effort activate motor units (Tillin & Bishop, 2009), causing the neural drive to increase the rate of force development. Besides, dynamic resistance can improve movement ability for a short time (Young, Haff, Newton, Gabbett, & Sheppard, 2015) when slow rhythm movement stimulates and increases acceleration better than fast movement. However, the muscles should not be stimulated with heavy load resistance, which may lead to accumulated fatigue. The resistance load should be used with a sub-maximum load (Chen et al., 2023; Farup & Sørensen, 2010). The resistance method methodology should be applied to the specific muscle group of each type.

Pull and press exercises movement for the upper body muscle group are used for stimulation in sports where the upper musculature is primarily used, consistent with kayak sprint (Romagnoli et al., 2022). Stimulation of muscle upper body parts will effectively enhance kayaking performance (Khamros et al., 2023; McKean & Burkett, 2014). It is still being determined whether a preconditioning method can improve performance in athletes competing more than twice daily. Since competition occurs more than once daily, the recovery time is relatively short; such competition may impair the athlete's performance, although the recovery strategy has been reinforced. Athletes must have a proper strategy for competition preparation, such as implementing a pre-competition warm-up or incorporating a re-warm-up. It is recommended that specific guidelines regarding training methods for athletes be followed. The technique involves using the intensity load for a short duration of two minutes by method is adopting a Short Active Warm Up pattern, gradually increasing the intensity from moderate to near-maximal levels (Silva, Neiva, Marques, Izquierdo, & Marinho, 2018) or the resistance activities to activate the muscles have also been beneficial for athletes (Farup & Sørensen, 2010; Khamros et al., 2023; Young et al., 2015). However, it is important to be mindful of the consequences if too much stimulation affects anaerobic energy systems and muscle contraction, especially in kayak paddling, by Bishop, Bonetti, and Spencer (2003) have emphasised the need to balance muscle activation and over-stimulation when using these methods. The strategies for warm-up were essential for preparing before a competition again (Swinton, Symon, Maughan, Burgess, & Dolan, 2022).

When athletes compete many times in one day, there may be an accumulation of fatigue that affects and impairs physical performance. In addition to the recovery strategies used in many previous studies, implementing a strategic activation to prepare for the next competition is no less critical. There is little current evidence in the literature to suggest the effects of post-activation performance enhancement used more than once in competitions. Studies tend to focus only on the immediate impacts of various methods. For instance, a study conducted by Soh, Aziz, and Lee (2020) on a specific acute phase stimulation method

for kayaks indicated no positive effects on athletes. However, in contrast, a study by Khamros et al. (2023) that used a PAPE theoretical load stimulation method yielded positive results on athlete performance. However, neither study specified post-activation performance when applied twice within the same day.

Applying the principles of sports science using the PAPE model to help stimulate increased performance in two competitions on the same day. This study aimed to determine whether post-activation performance enhancement, the methodology for preconditioning before the competition kayak sprints twice daily, can enhance performance. The outcome of this study may help explain the application of PAPE to increase the performance levels of athletes in competition.

Method

Participant

The participants were ten male kayakers who had attended international kayak sprint competitions for more than three years (Age 26.20 ± 3.70 years, Height 178 ± 4.48 cm, Weight 73.10 ± 5.51 kg) and had no injuries to joints or muscles within one year, as shown in Table 1. The participants will be randomized crossover trials, all get experimental treatments in three interventions by a Latin square design experimental plan and have 48 hours to washout. The study included three interventions groups: a resistance programme (RSP), a maximum sprint paddle (MSP), and a control group (self-activation). The resistance programme used heavy load ($2 \times 4 \times 80\%1RM$) in the bench press following the bench row exercise. The maximum sprint paddle involves kayakers paddling maximum effort on the ergometer (2×10 second rest intervals for 2 minutes). The control group usually warmed up by themselves. It continued to the 3-minute test to assess the intervention difference per performance of the simulated kayak sprint competitive two times within one day (The first simulated: S1 and the second simulated: S2). Before attending the experiment, the participant did not take dietary supplements for seven days, avoided drinks containing caffeine and alcohol, and performed physical activities at a moderate or vigorous level for at least 48 hours.

Ethics

Following the Declaration of Helsinki, this study was approved for ethics in human committee research by The Strategic Wisdom and Research Institute, Srinakharinwirot University, number SWUEC-G210/2565E. Each participant voluntarily provided written informed consent before participating.

Experiment in laboratory

The experiment occurred in the Sports Science Laboratory from 6 to 9 a.m. every morning. Participants partook in five sessions, with a control temperature range of $22-25^\circ\text{C}$.

The first session was a body composition assessment, in-

Table 1. Descriptive physical characteristics of participant

Physical Characteristics	Mean $\pm$ SD
Age (years)	26.20 $\pm$ 3.70
Weight (kg)	73.10 $\pm$ 5.51
Height (cm)	178 $\pm$ 4.48
VO2Max (ml/kg/min)	53.45 $\pm$ 6.10
1 RM (Bench Press)	111.46 $\pm$ 11.09
(kg) (Bench Row)	109.33 $\pm$ 8.10

cluding weight, height, and resting heart rate. The next test followed them. The one repetition maximum test (1RM) was used following Haff and Triplett's (2016) protocol for bench press and bench row exercises. In the second session, the participants went to the laboratory for the maximal oxygen consumption test (VO₂max test). The VO₂max test used an increment step test applied to the protocol from Santos (2012) by using a portable gas analyser (PNOE, ENDO Medical, Palo Alto, CA, USA). The participants wore a portable gas analyser on a kayak ergometer (Kayak Ergometer, WEBA Sport und med. Artikel GmbH, Liesneckgasse, Wien, Austria) starting at 100 watts and increasing to 30 watts every 2 minutes until the end of the test. At the end of the test, researchers determined criteria 2 from 4 specifications, such as the VO₂ plateau <150 ml/min, the respiratory exchange ratio (RER) is not less than 1.15, and maximum heart rate and exhaustion cannot continue testing. The sample was to become familiar with the protocol (Intervention programme and equipment for testing) by trying all the processes. After both experiments in the laboratory, the subject rested 48 hours before the next test.

In the third to fifth sessions in the laboratory, the sample was randomised to one of the intervention programmes in each experiment to simulate a kayak sprint two times within one day. The sample measured blood lactate concentration before warm-up (BFW) in capillary samples from the earlobe and then started a general warm-up at 50–60% MHR combined with dynamic and static stretching for 10 minutes. After the general warm-up, the sample rested for 5 minutes following the preconditioning. Blood lactate was immediately measured after an intervention (AIV), then rested for 7–10 minutes to fully stimulate the physiological mechanisms. Then, the blood lactate concentration was checked before assessing the 3-minute paddling test (BF3). The 3-minute paddling test assessed physical performance after the physiological mechanism of intervention. After the 3-minute paddling test, the sample was tested for blood lactate concentration after resting for 6 minutes (AF3). After the first simulation, the sample will have 2 hours of rest and continue to the second simulation using the same protocol as the first simulation. All protocol processes were completed, and the sample had a 72-hour rest period and got to the crossover trial of another intervention.

Intervention

The resistance programme (RSP) uses bench press and bench row exercises to determine repetition and load for intervention at 2 sets x 4 repetitions at 80% of 1RM and 2-minute rest interval between exercises by using an alternating exercise

protocol suggested by Khamros et al. (2023) adapt from Boullousa (2021). The maximum sprint paddle (MSP) was the maximum paddling effort on the kayak ergometer 2 set x 10 seconds and rest interval 2 minutes (Harat et al., 2020). Finally, the control group will exercise 10 minutes to precondition themselves for this intervention.

The 3-minute paddle test

The maximum effort is paddling on an ergometer for three minutes. This method is designed for critical power testing, and anaerobic work capacity is the main object of the test (Bergstrom et al., 2012). The protocol for the 3-minute test started without setting intensity in the first 1-2 minutes, then the sample paddled at maximum effort within 3 minutes. During the test, the sample was not monitored on the ergometer to avoid strategies of the paddle. The researcher controlled the testing, monitored, and asked for cooperation with the sample. During the test, the maximum power (MxP), average power (AvP), stroke per minute (SPM), total distance (TD) and Percentage decrement of maximum power and average power were recorded by using Weba Science software. The heart rate (PHR) percentage was recorded and controlled throughout the paddle (Polar H10, Kempele, Finland). Blood lactate concentration ([La-]b) (Lactate Plus Meter, Nova Biomedical, MA, USA) was measured in capillary samples obtained from the earlobe.

Statistical Analysis

The statistical analysis is represented as mean $\pm$ SD by using SPSS version 26 (IBM, Chicago, Illinois, United States of America) Normality of distribution data used for the Shapiro-Wilk test. Maximum power, average power, stroke per minute, and total distance and percentage of heart rate variables used One-way MANOVA. Blood lactate was measured using two-way repeated-measures MANOVA analysis each period to compare the interaction between intervention x periods. The Bonferroni test determined which measures differed significantly. All determined level of significance was set at $p \leq 0.05$. Further, the percentage of power decrement was compared between the first simulation (S1) and the second simulation (S2) for a comping percentage of power decreased (maximum power and average power) by the formula $[(S1 - S2) / S1] \times 100$.

Results

The main result of two-way repeated-measures MANOVA was that the interaction (Time x Trial) was $p=0.013$, the affected trial had a significant $p=0.047$, and time had a considerable $p=0.002$, respectively. The RSP and MSP had maximum power

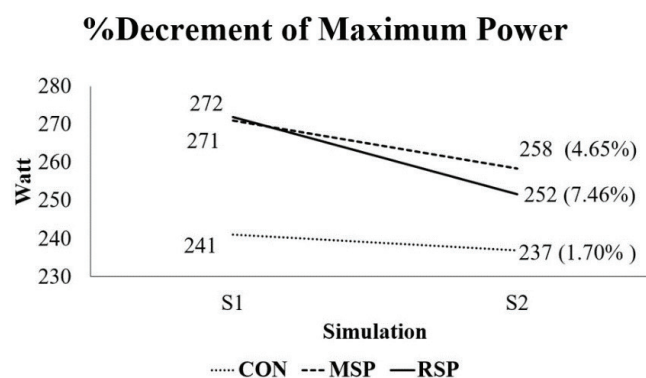


FIGURE 1 Percentage decrement of Maximum Power, Con - Control group, MSP - Maximum sprint paddle RSP - Resistance program

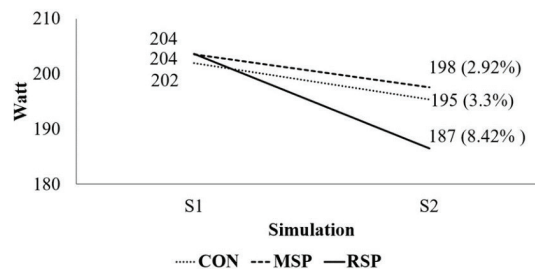
%Decrement of Average Power

FIGURE 2 Percentage decrement of Average Power, Con - Control group, MSP - Maximum sprint paddle RSP - Resistance program

er higher than CON in S1 $p=0.017$ (RSP: 272.00 ± 26.84 , MSP: 271.00 ± 27.26 , and CON: 241.00 ± 16.02), shown in Table 2. The average power in MSP and RSP found that S2 less than S1 all had significant $p \leq 0.05$, same as total distance that S2 lower than S1 all had significant $p \leq 0.05$ for all data as shown in Tables 2. The percentage of power decrement in RSP was higher than MSP and CON (7.46% for MxP and AvP 8.42%).

Conversely, MSP can maintain the average power of paddles in S2, like S1 (AvP=2.92%) all shown in figures 1 and 2.

The percentage of heart rate was not significant for all periods. The blood lactate analysis in S1 revealed a substantial difference in AIV. RSP and MSP showed a considerable difference compared to CON (RSP vs CON $p=0.024$ and MSP vs CON $p=0.001$). Additionally, AF3 period in

Table 2 The analysis compared the variable between intervention in each simulation competitive (A) and compared between simulation within group (B)

Intervention		MxP	AvP	SPM	TD	PHR	MANOVA (p)			
		Mean±SD	Mean±SD	Mean±SD	Mean±SD	Mean±SD				
A	Simulation 1	CON	241.00± 16.02	202.00± 13.71	101.78± 4.55	633.33± 20.02	89.33± 2.60	.016*		
		MSP	271.00± 27.26	203.56± 13.01	105.00± 3.28	642.20± 12.04	92.00± 2.12			
		RSP	272.00± 26.84	203.67± 13.09	102.00± 2.36	635.22± 10.44	89.89± 2.85			
		p	.017*	.957	.131	.371	.083			
	Simulation 2	CON	236.88± 32.84	195.33± 15.00	104.55± 3.67	626.66± 15.81	90.44± 2.78	0.517		
		MSP	258.40± 32.21	197.60± 29.75	104.90± 3.17	635.00± 30.64	92.50± 3.37			
		RSP	251.70± 21.43	186.50± 14.81	103.50± 3.92	623.00± 21.10	91.30± 3.52			
		p	0.280	0.480	0.669	0.517	0.398			
	B	CON	S1	241.00± 16.02	202.00± 13.71	101.78± 4.54	633.33± 20.02	89.33± 2.60	0.388	0.013*
			S2	236.88± 32.84	195.33± 15.00	104.55± 3.67	626.66± 15.81	90.44± 2.78		
p			0.753	0.202	0.110	0.262	0.070			
MSP		S1	271.00± 27.26	203.56± 13.01	105.00± 3.28	642.22± 12.04	92.00± 2.12	0.626		
		S2	258.40± 32.21	197.6± 29.75	104.90± 3.17	635.00± 30.64	92.50± 3.37			
		p	0.069	0.041*	0.803	0.044*	1.00			
RSP		S1	272.00± 26.84	203.67± 13.09	102.00± 2.36	635.22± 10.44	89.89± 2.85	0.114		
		S2	251.70± 21.43	186.50± 14.81	103.05± 3.92	623.00± 21.10	91.30± 3.52			
		p	0.036*	0.001*	0.754	0.012*	0.073			

Note MxP - Maximum power, AvP - Power average, SPM: Stroke per minute, TD - Total distance, PHR - percentage of heart rate; * $p \leq 0.05$

MSP, and RSP were significantly higher than CON with $p \leq 0.05$. In BF3, MSP had a different significance from RSP and CON ($p=0.015$), and RSP higher than CON had a significant $p=0.013$. Blood lactate in S2 and MSP difference CON in AIV and BF3 was significant ($p=0.021$ and $p=0.006$, respectively). In the AF3 period, MSP and RSP differed significantly (MSP vs CON $p=0.026$ and RSP vs CON $p=0.001$). Compared within the group, RSP in S2 higher than S1 had significant $p=0.009$ (RSP: S1= 1.13 ± 0.18 and S2= 1.45 ± 0.29).

Discussion

From this study, the resistance method tended to reduce performance in the second simulation but not decrease physical performance in other preconditions. By the results that occur after the first simulation, The Maximum voluntary contraction (MVC) after preconditioning using the resistance or maximum paddling effort plays a vital role in improving peak power performance for kayak sprint athletes (Borba et al., 2017; Brink et al., 2021; Chen et al., 2023; Krzysztofik et al., 2021). The maximum effort in a short time for activation produces a level of neural drive, which occurs after the mobilization of the muscle motor units during short, high-velocity contractions. The mechanism is the release of rate-coding potential, where rate-coding plays a significant role in muscle power and speed, leading to muscle contraction that results in maximum voluntary contraction (MVC) (Heckman & Enoka, 2012). As the motor unit recruitment increases, the rate coding becomes more efficient. It affects the activation of more muscle fibres, thus making the contraction of the muscles with maximum force more effective (Heckman & Enoka, 2012). The working process is constantly changing during movement. Over a while, the coding rate of neuronal discharge continues to decrease (Enoka & Duchateau, 2017). This results in an increase in the rate of force development (Aagaard, Simonsen, Andersen, Magnusson, & Dyhre-Poulsen, 2002; Tillin & Bishop, 2009).

Resistance exercises to activate muscles will cause muscle stiffness after exertion against external resistance (Dankel & Razzano, 2020). Downregulation of rate coding results from motor neuron and muscle remodeling (Taylor, Amann, Duchateau, Meeusen, & Rice, 2016). This causes the effect of activation that has previously mobilized the motor units to work. There is a decrease in performance. The mechanism will cause an increase in force. An increase in the intramuscular fluid has been associated with increased muscle stiffness and changes in muscle temperature caused by fluid circulation within the muscle. Although the initial stimulation produced positive results for the athletes performance the effects of resistance activation cause muscles to remain under tension (TUT) and continue into a state of ischemia, leading to more fatigue compared to other preconditions (Scott, 2006). the average power performance decreased, and the fatigue was cumulative from the first simulation, causing a change in kinematics and decreasing velocity during the second simulation's race.

Preconditions affected physical performance, so in the second simulation, the resistance method decreased the percentage of maximum power and the average percentage of paddling power to more than the maximum paddling exertion or self-warmup. Additional research has indicated

that the disparities in performance observed between the initial and subsequent simulations of the resistance condition could be influenced by various other factors such as body balance, alertness, rest intervals, duration of warm-up sessions, and environmental conditions during the day's competition (Behrens et al., 2023; Jeffrey, Kristin, Nick, Harrison, & Tim, 2020). These factors affected the physical ability to perform high-intensity activities. However, the duration of the competition may not affect the competitive performance of athletes who are well-trained and continuously training (Kusumoto, Ta, Brown, & Mulcahey, 2021).

After preconditioning blood lactate increase, it is caused by high levels of energy expenditure from the resistance condition increasing lactate content (Lawson, Vann, Schoenfeld, & Haun, 2022; Lintmeijer, Hofmijster, Schulte Fishedick, Zijlstra, & Van Soest, 2018) and the maximum padding condition (Sang-Yong, Duk-Mook, & Jin-Seok, 2018) muscles remain in time under tension (TUT) and used anaerobic energy for metabolism. It causes a sudden spike in blood lactate (Gentil, Oliveira, & Bottaro, 2006; Lopes et al., 2018; Scott, 2006). Blood lactate remains in the muscles before subsequent breakdown (Sang-Yong et al., 2018). During rest between competitions, the athletes will have active and passive rest parallel to recover and prepare for their next competition.

During the second simulation, the blood lactate concentration was higher than the first, especially when using the resistance condition method. This high blood lactate concentration can lead to fatigue after a high-intensity competition, combined with accumulated resistance load and decreased muscle function for several hours. Therefore, blood lactate concentration can serve as an indication of the occurrence of this condition (Manojlović & Erčulj, 2019). However, blood lactate determinations of fatigue may not accurately assess the extent of fatigue. A high blood lactate may be related to the highest muscle oxidative capacity during resting competition. The oxidative ability of the energy system can contribute to the better breakdown of blood lactate and has been associated with delayed fatigue during high-intensity exercise. This was evident from the fact that the maximum paddling condition was like blood lactate concentration, but no physical performance was decreased.

Applying PAPE can increase maximum power in the first competition. The resistance model should not be used in subsequent competitions within the same day as it will reduce the athlete's performance. In contrast, paddling at maximum speed seemed more effective in maintaining performance than the resistance method in the second competition of the day. The accumulation of blood lactate from both preconditions could not explain the fatigue and performance degradation caused by the activation load. Coach or trainers should adopt the best methods to enhance their athletes' competitive performance. This study is an initial study of the application of the PAPE method to sports where two events are played within one day, showing that although it has a positive effect on the first event of the day, it may reduce it. Athlete's performance in the second competition of the day. As for applying the methods in this study to other sports, there may not be much consistency. In the next study, studies should be conducted in other sports. There are different competition periods.

Conclusion

This study describes an approach for implementing a pre-conditioning method for two daily bouts. The introduction of a maximum-speed paddle style of stimulation is likely to improve some aspects of the athlete's performance and is appropriate only for first-time competition. However, when used twice within the same day, stimulating methods may increase their effects on other physiological functions. Coach or trainers should adopt the best methods to enhance their athletes' competitive performance.

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Conflicts of interest.

The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

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ORIGINAL SCIENTIFIC PAPER

A Point Difference Analysis Established by Winners and Losers in International Badminton Doubles Matches

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Abstract

The aim of this study is to analyze the point difference established in three different phases between winners and losers in international men's and women's badminton doubles matches. Analyzing 183 matches from the 2021 TotalEnergies World Championship, each match was divided into three phases: Phase 1 (0-7 points), Phase 2 (8-14 points), and Phase 3 (15-21 points). Maximum Point Difference (MPD) and Maximum Consecutive Points (MCP) were examined to assess player performance. Kruskal-Wallis H and Mann-Whitney U statistical tests were conducted. Significant differences between winners and losers were found in all phases, with the difference increasing significantly over phases with medium to large effect sizes. The first seven points significantly increase a player's chances of winning. 72% (men's category) and 75% (women's category) of players won, who was ahead in the point difference in the first phase of the game. In the men's category, 67.4% won who were ahead in point difference in all three phases. Whereas in the women's category, it is 70.45%. Winners in both categories had significantly ($p < 0.05$) higher consecutive points than the losers. In the second phase, the winners averaged 5 to 6 points over the losers in both categories. The winners kept increasing their point difference in every phase. On the other hand, in losers, the point difference decreased throughout the game. These findings underscore the importance of early lead acquisition and suggest potential strategies and tactics to enhance players' winning probabilities.

Keywords: badminton, match analysis, consecutive point, match results

Introduction

The sport of badminton stands out as one of the fastest games played globally, particularly in the doubles format, requiring an immense level of fitness, skill, and energy (Butterworth et al., 2012). In addition, it requires a quick series of high-intensity movements (Abian-Vicen et al., 2013). Badminton sport entails five disciplines: women's singles, men's singles, women's doubles, men's doubles, and mixed doubles. As a result of gender differences, each of these disciplines has its playing requirements (Gawin et al., 2015). Abian-Vicen et al. (2013) conducted a comprehensive investigation

into the temporal and notational patterns of elite badminton matches involving both male and female players. Their study revealed notable gender differences among variables like shots per rally, rally time, rest time, & match duration, with male players displaying significantly larger values. The association between technical skills, stroke play, and player performance was analyzed in international badminton matches. There is a significant difference in the percentage of shots played and the type of shots used by female and male athletes (Lee et al., 2005; Zhang et al., 2013).

The Badminton World Federation (BWF) annually hosts a



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series of international tournaments covering all five aforementioned disciplines. Within these tournaments, several leagues or knockout games adopt a three-set format, each consisting of 21 points. This format shift occurred in 2006, transitioning from the previous three sets with 15 points each (Hsin-Lian et al., 2018). This change subsequently altered the physical, technical, and physiological aspects of the sport, highlighting the need for increased scientific investigation (Ming et al., 2008; Phomsoupha & Laffaye, 2015).

While the application of data analytics to assess performance in various sports is growing, research on the utilization of point differences to analyze badminton matches remains limited.

Studies in other sports such as soccer, tennis, and squash have explored the implications of point differences on match outcomes, competitiveness, and predictive models (Baert & Amez, 2018; Karlis & Ntzoufras, 2009). However, certain studies have concentrated on analyzing the impact of goal difference by examining the variations in match running performance pre & post scoring a goal (Buchheit et al., 2018). In racquet sports like Tennis and Squash, researchers examined point difference analysis between players to determine their competitiveness, performance level, and rating. In addition, a statistical model was proposed to assess the probability of winning based on the point difference (Ley et al., 2018; Marcus, 2001; Pfeiffer et al., 2010). Existing research on point difference in badminton is scarce; two of these studies have examined men's and women's singles matches, showing similar findings (Bagchi et al., 2022; Barreira et al., 2016). However, there remains a gap in understanding point difference in badminton, particularly within the context of doubles matches, calling for further investigation.

The current literature on the badminton doubles discipline covers: technical and tactical analysis, a comparison of doubles and singles badminton based on physiological demands, a comparison of men's and women's doubles matches based on timing factors and competition analysis between singles and doubles discipline (Abián-Vicén et al., 2018; Alcock & Cable, 2009; Gawin et al., 2015; Pérez-Turpin et al., 2020; Zhang et al., 2013).

The initial moments of a game hold significant importance in predicting its outcome. A player's likelihood of winning increases as they establish a lead in the match (Barreira et al.,

2016). Consequently, studying the point difference becomes crucial for anticipating the game's conclusion. This analysis can aid coaches and players in formulating pregame strategies and in-game tactics. Thus, the primary aim of this study is to examine the point difference established in three different phases between winners and losers in international men's and women's badminton doubles matches.

Methods

Participants

Data from men's and women's doubles matches played during 2021 TotalEnergies BWF World Championships held in Spain from 12th to 19th December were collected to conduct this study. This data was publicly available after the Badminton World Federation (BWF) championship and was collected from tournament software (www.tournamentsoftware.com). This championship draw included three rounds, quarterfinals, semi-finals, and finals. A total of 183 games (men's – 95 and women's – 88) were analyzed from the 2021 TotalEnergies World Championship matches. In each game, we recorded maximum point differences (MPD) and maximum consecutive points (MCP) separately for winners and losers in each phase.

Data Collection

Data from the 183 games were meticulously collected and organized in Microsoft Excel (Microsoft Corporation, Redmond, WA, USA) spreadsheets. Each game was divided into three phases to analyze the point difference: phase 1, phase 2, and phase 3. These phases, namely phase 1, phase 2, and phase 3, were defined based on point ranges: phase 1 encompassed point differences between 0 and 7, phase 2 consist of 8 to 14 points and Phase 3 is of 15 to 21 or above points. Based on a study conducted by Barreira et al. (2016), a point difference is a maximal difference between the scores of the winner and loser established in each phase of the game. It is calculated by subtracting the maximum score attained from the minimum score conceded by a player during the phase. In cases where neither the winner nor the loser attained a lead during a particular phase, a value of zero was attributed. To gain a clearer insight into the maximum point difference at every stage of the game, we constructed a sample line graph (as illustrated in Figure 1).

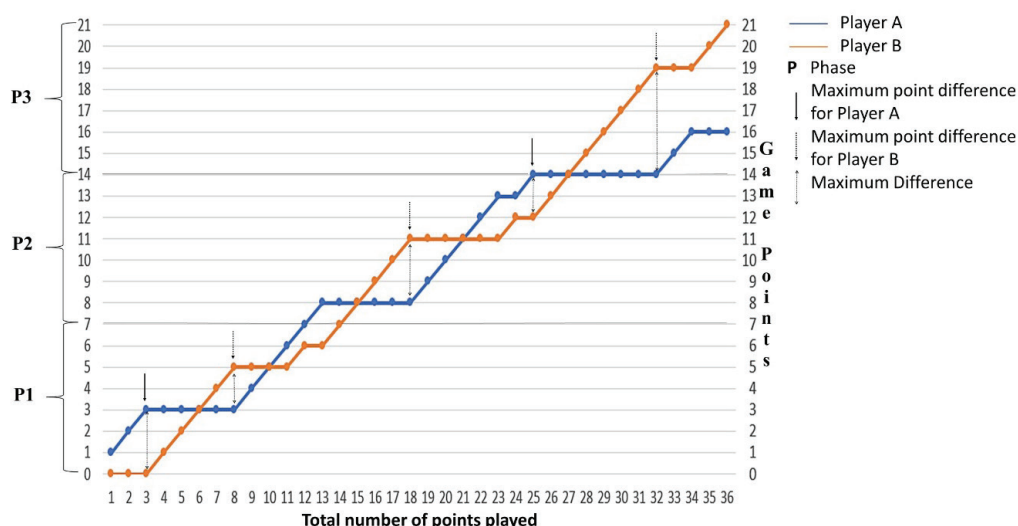


FIGURE 1. Illustrates the maximum point difference established in each phase between winners and losers of the match

Statistical Analysis

The data collected were described using descriptive statistics such as average, median, standard deviation, quartiles, minimums, and maximums. The distribution of the data was supported by using Skewness, Kurtosis and their standard error values were calculated. The maximum point difference (MPD) and maximum consecutive points (MCP) between winners and losers were compared in each phase using the Mann-Whitney U test. Utilizing the Kruskal-Wallis H test, we conducted a comparative analysis of the point differences between winners and losers across the three stages. Effect size measurements were employed to determine the magnitude of the observed point difference.

es. Statistical analyses were performed using SPSS software (version 24, IBM, New York, USA), maintaining a significance level of 0.05 for all tests.

Results

The findings of this study suggest that in both men's and women's matches, winners (Men's Category - MC: 5.03 ± 3.13 , Women's Category - WC: 5.77 ± 3.44) are ahead of losers (MC: 0.72 ± 1.34 , WC: 0.63 ± 1.20) throughout all the phases of the game. In both categories, winners exhibit a lower level of variation around the mean in point difference and demonstrate a higher frequency of scoring consecutive points compared to losers (refer to Table 1).

Table 1. Descriptive Statistics for Phase-wise Maximum Point Difference and Maximum Consecutive Points (MCP) of Winners and Losers in Men's and Women's Categories

	Gender	Match Outcome	Mean	Md	SD	CV	Skewness	Std. Error of Skewness	Kurtosis	Std. Error of Kurtosis
Overall	Men	Win	5.03	5	3.1	0.6	0.79	0.14	0.6	0.29
		Loss	0.72	0	1.3	1.8	3.18	0.14	15.61	0.29
	Women	Win	5.77	5	3.4	0.6	0.7	0.15	0.16	0.3
		Loss	0.63	0	1.2	1.9	2.07	0.15	3.87	0.3
Phase - 1	Men	Win	2.91	3	1.5	0.5	0.05	0.25	-0.37	0.49
		Loss	1.12	1	1.2	1	0.87	0.25	-0.13	0.49
	Women	Win	3.28	3	1.7	0.5	0.17	0.26	-0.27	0.51
		Loss	1.11	1	1.3	1.2	1.15	0.26	0.73	0.51
	Men	Win	5.15	5	2.7	0.5	0.01	0.25	0.16	0.49
		Loss	0.65	0	1.5	2.3	3.46	0.25	16.1	0.49
Phase - 2	Women	Win	5.78	5	2.8	0.5	0.06	0.26	-0.5	0.51
		Loss	0.49	0	1.3	2.6	2.79	0.26	7.42	0.51
	Men	Win	7.03	7	3.4	0.5	0.45	0.25	-0.3	0.49
		Loss	0.38	0	1.3	3.3	5.55	0.25	38.4	0.49
	Women	Win	8.25	8	3.5	0.4	0.28	0.26	-0.4	0.51
		Loss	0.3	0	0.8	2.8	2.94	0.26	7.89	0.51
Phase - 3	Men	Win	4.73	4	1.8	0.4	0.79	0.25	0.92	0.49
		Loss	3.16	3	1.4	0.4	1.04	0.25	5.42	0.49
	Women	Win	5.83	6	1.9	0.3	0.83	0.26	0.7	0.51
		Loss	3.27	3	1.3	0.4	1.52	0.26	3.96	0.51

Note: MCP - Maximum Consecutive Points; Mean – Arithmetic mean; Md - Median; SD – Standard Deviation; CV - The Coefficient of Variation

Table 2 shows that 72% (MC) and 75% (WC) of players won the match who were ahead in point difference during the first phase of the game. Similarly, 86% (MC) and 89% (WC) of players who were not ahead in point difference anytime during the first phase lost the game. The analysis of the players who lost the game shows that 76% (MC) and 83% (WC) of the athletes were not ahead on the scoreboard at any time during the second phase, and this frequency got increased to 84% (MC) and 86% (WC) in the final phase of the game.

For players who lost the game, 35% (MC) and 40% (WC) were not ahead during any phase of the game, and 20% (MC) and 18% (WC) opened the highest point difference than the opponent in the first phase of the game. Our analysis revealed that players needed to maintain a lead of more than one point in the second and third phases to secure victory. Specifically, in men's doubles matches, the average point difference be-

tween winners and losers increased from five points in the second phase to seven points in the third phase. Similarly, in women's doubles, this metric increased to eight points in the third phase from six in phase two.

The analysis revealed significant trends in match outcomes based on point differences across various phases of the game. During the first phase, a considerable percentage of players won the match when leading in point difference: 72% in the men's category (MC) and 75% in the women's category (WC). Conversely, a substantial proportion of players who trailed in point difference during this phase ended up losing the match: 86% in MC and 89% in WC.

Further analysis of players who ultimately lost the game unveiled notable patterns. In the second phase, the majority of players who were defeated did not hold a lead on the scoreboard at any point: 76% in MC and 83% in WC. This trend

intensified in the final phase, with 84% in MC and 86% in WC failing to take the lead at any time.

For those players who ultimately lost, a significant proportion failed to secure a lead throughout any phase of the game: 35% in MC and 40% in WC. Additionally, a noteworthy finding was that 20% (MC) and 18% (WC) of losing players achieved the highest point difference compared to their opponents during the first phase.

Further analysis revealed that a lead of more than one point

in the second and third phases significantly increased the likelihood of winning the game. Specifically, in men's doubles, winners maintained an average lead of five points over their opponents in the second phase, which expanded to seven points in the third phase. Similarly, in women's doubles, the average lead increased from six points in the second phase to eight points in the third phase. These findings highlight the critical role of point differences in determining match outcomes and suggest strategies for achieving success in doubles matches.

Table 2. Analysis of Match Outcomes Based on Point Differences in Different Phases of the Game

Metric	MC	WC
Players won when ahead in point difference during first phase	72%	75%
Players lost when not ahead in point difference during first phase	86%	89%
Players lost when not ahead at any time during the second phase	76%	83%
Players lost when not ahead at any time during the final phase	84%	86%
Players lost and were not ahead during any phase of the game	35%	40%
Losing players who had the highest point difference in the first phase	20%	18%
Average point difference between winners and losers (second phase)	5 points	6 points
Average point difference between winners and losers (third phase)	7 points	8 points

For further analysis, Kruskal Wallis H and Mann Whitney U tests were used as the data failed to satisfy the assumptions of normal distribution. The Kruskal-Wallis H test was employed to assess the phase-wise MPD, while the comparison of MCP difference between Winners and Losers was conducted

using the Mann Whitney U test.

As shown in Figure 2, in all three phases, the maximum point differences are significantly ($p < 0.05$) higher in winners than losers.

In Table 3, the effect size was computed for each of the

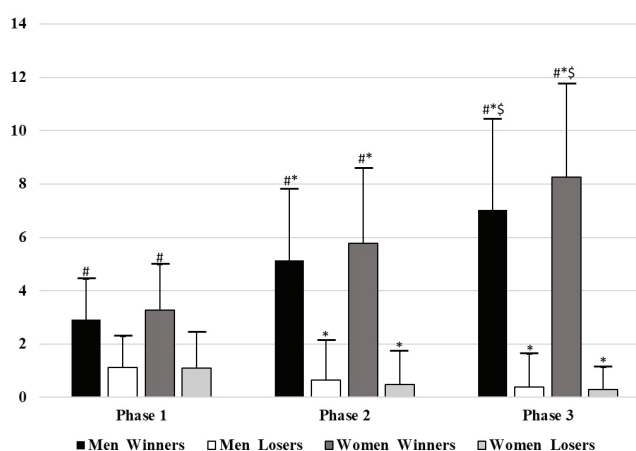


FIGURE 2. Illustrates the mean and standard deviation (SD) of the maximum point difference between the winners and losers of men's and women's doubles matches across each phase of the games throughout the entire championship. Notably, # denotes a significant difference in each phase compared to the loser, * represents a significant difference compared to phase one, and \$ represents a significant difference compared to phase two

phases. In all three phases, the magnitude of the difference was significantly large for both categories (MC: 0.56, 0.76, 0.87 and WC: 0.6, 0.81, 0.9). Therefore, it has shown a very high point difference between winners and losers with a large magnitude and kept on increasing throughout the phase.

While comparing the point differences for winners and losers in each phase separately, there were significant ($p < 0.05$) differences in all the phases of winners and losers for both categories. In both categories, the point difference in winners kept increasing from Phase 1 to Phase 3. Whereas in losers, this difference got reduced. Post hoc test, followed by the effect size for each significant difference, was calculated.

In both categories, while comparing the overall point dif-

ference between all three phases of winners and losers separately, both showed a large (Winners - MC: 0.3 and WC: 0.35, Losers - MC: 0.15 and WC: 0.14) effect size. In the case of winners, it showed a significant difference ($p < 0.05$) across all the comparisons, with a large effect size in phases 1 & 2 (MC : 0.68 and WC: 0.67), a moderate effect size in phases 2 & 3 (MC: 0.37 and WC: 0.48), and again a large effect size in phase 1 & 3 (MC: 0.88 and WC: 0.97). Whereas losers in both the categories showed a significant difference in phases 1 & 2 and phases 1 & 3 with moderate (MC: 0.45 and WC: 0.49) and large (MC: 0.63 and WC: 0.59) effect sizes, respectively. In the case of phases 2 & 3, losers of both categories failed to show significant ($p > 0.05$) differences in point differences.

Table 3. Statistical Comparison across different phases in Men's Doubles and Women's Doubles

		Men Doubles		Women Doubles	
		Mean±SD	p-value [r] Magnitude	Mean±SD	p-value [r] Magnitude
Phase 1	Winners	2.91±1.52	0.000 [0.56]L	3.28±1.69	0.000 [0.6]L
	Losers	1.12±1.16		1.11±1.33	
Phase 2	Winners	5.15±2.67	0.000 [0.76]L	5.78±2.84	0.000 [0.81]L
	Losers	0.65±1.5		0.49±1.25	
Phase 3	Winners	7.03±3.39	0.000 [0.87]L	8.25±3.53	0.000 [0.9]L
	Losers	0.38±1.25		0.3±0.83	
MCP	Winners	4.7±1.8	0.000 [0.48]M	5.8±1.9	0.000 [0.67]L
	Losers	3.2±1.4		3.3±1.3	
Winners (Overall)	Phase 1	2.91±1.52	0.000 [0.3]L#	3.28±1.69	0.000 [0.35]L#
	Phase 2	5.15±2.67		5.78±2.84	
	Phase 3	7.03±3.39		8.25±3.53	
Losers (Overall)	Phase 1	1.12±1.16	0.000 [0.15]L#	1.11±1.33	0.000 [0.14]L#
	Phase 2	0.65±1.5		0.49±1.25	
	Phase 3	0.38±1.25		0.3±0.83	
Winners (post hoc)	Phase 1	2.91±1.52	0.000 [0.68]L	3.28±1.69	0.000 [0.67]L
	Phase 2	5.15±2.67		5.78±2.84	
	Phase 2	5.15±2.67	0.000 [0.37]M	5.78±2.84	0.000 [0.48]M
	Phase 3	7.03±3.39		8.25±3.53	
	Phase 3	7.03±3.39		8.25±3.53	
Losers (post hoc)	Phase 1	1.12±1.16	0.000 [0.45]M	1.11±1.33	0.000 [0.49]M
	Phase 2	0.65±1.5		0.49±1.25	
	Phase 2	0.65±1.5	0.118 [0.16]S	0.49±1.25	0.469 [0.08]T
	Phase 3	0.38±1.25		0.3±0.83	
	Phase 3	0.38±1.25		0.3±0.83	

Note: r – Correlation Coefficient ($Z/\sqrt{n}$) Effect Size; # – Eta Squared (η^2); T – Trivial Effect Size; S – Small Effect Size; M – Moderate Effect Size; L – Large Effect Size

Figure 3 shows a significant difference in maximum consecutive points between winners and losers with an effect size ($r = Z/\sqrt{n}$) of 0.48 in the men's category and 0.67 in

the women's category. Therefore, the match is more likely to be won by the winners because they have more consecutive points.

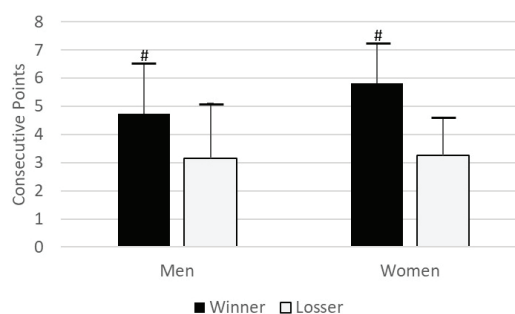


FIGURE 3. Mean and SD of maximum consecutive points between winner and loser during all the games in the entire championship. Note- # represents a significant difference (<0.05) in each phase in relation to the loser.

Discussion

The research findings have demonstrated that winners in both categories (Men and Women Doubles) consistently exhibited notably wider point differences than losers throughout

various phases of the game. The players need to score consecutive points and keep their opponents' point differences from increasing to win the game. Winners of both categories showed the highest point difference in the game's final phase. In com-

parison, losers showed up in the first phase. However, the winners of men's doubles were, on average, five points ahead of their opponents in phase two, which increased to seven in the third phase. Similarly, in women's doubles, this metric increased to eight points in the third phase from six in phase two.

In badminton, irrespective of the format, the outcome of the match can be predicted based on the performance in the initial phase of the game. Taking the lead during the first phase significantly enhances the likelihood of winning the game. The game's later stages demanded more endurance, strategic decisions, and high concentration to sustain the performance (Gómez et al., 2021). Hence, it reduces the chances of losing if the players try to take the lead at least once in the first phase of the game. It has been established in several studies that the first stage of a match is crucial to the outcome of the game (Duarte et al., 2012; García-Rubio et al., 2015). It is essential to understand the scoring performance dynamics to develop better match strategies (Gómez et al., 2021).

Bagchi et al. (2022) and Barreira et al. (2016) have studied the point difference analysis between the winners and losers among male and female badminton players, respectively; it has been determined that the style of playing badminton and the selection of shots differ quite a bit between men and women. Real-time played, percentage of time played, shots per rally, rally time, strokes per point, and work-to-rest ratio were significantly higher, and the number of shorts played per second was lower in the women's doubles category compared to men's doubles. Hence, in women's doubles, the matches were longer with greater real-time, while men's doubles were more intense (Abián-Vicén et al., 2018; Gawin et al., 2015; Torres-Luque et al., 2019). One study disclosed that in badminton, as the number of unforced errors per rally increases, there is a greater likelihood of losing a match. The chances of winning a game increase when you make more winning shots than your opponent (Cabello Manrique & González-Badillo, 2003). Alder and Broadbent (2017) after extensive research in anticipatory behaviour in badminton during competition, found that women's doubles exhibit a higher percentage of anticipatory behaviors compared to men's doubles. Conversely, men's doubles display a notably longer average response time compared to women's doubles.

It has been reported that expert badminton athletes grasp game situations more readily than novices (Blomqvist et al., 2000). The findings imply that success in a badminton match depends on knowing the opponent's strategy and quick analysis, which should be done primarily in the opening stages of the game. 86% of players in men's doubles lost the game who could not get ahead in the scoreboard anytime during the first phase, and this metric rose to 89% in the women's doubles category. Therefore, to enhance the likelihood of winning the game, it's crucial to lead on the scoreboard during the initial phase. In men's doubles, 20% of losing teams have established the largest point difference over their opponents in the first phase of the game. Similarly, in women's, this metric got decreased to 18%. It is not about only open-

ing highest in phase one but also trying to maintain it in later stages. Chances are, a player won't win the game if they're not leading on the scoreboard during the first phase.

Research on similar lines can be done for other sports as more literature is needed to analyze the sports using point difference analysis. Based on such data, a mathematical prediction model can be developed for predicting the outcome of a match. Based on the study's analysis, winning chances can be enhanced, and the result can be predicted. Coaches and players can use it to analyze the opponent before a game to devise strategies for victory. Additionally, this will help coaches and players adjust tactics during matches based on point differences.

Limitations and Future Directions

The research offers valuable insights into the dynamics of point differences in international-level badminton tournaments, specifically in men's and women's doubles matches. This focus helps us understand how point differences influence match outcomes in these competitive formats. However, the study's findings cannot be extrapolated to other levels of badminton tournaments. Additionally, the study did not include mixed doubles matches, leaving that part of the competitive format unexplored. The study uses data from matches played in 2021, giving us a snapshot of that year but not a full picture of long-term trends. Analyzing data of several years (eg. Last 5 years) could provide a more comprehensive view and make the study's conclusions even stronger. Future studies can focus on the aforementioned limitations.

Conclusions

This study aimed to analyze the point difference established at different phases of the game by the men's and women's doubles players in international badminton matches. The game was divided into three phases (P1 – 0 to 7 points, P2 – 8 to 14 points and P3 – 15 to 21 points). The results depicted that the winners in both categories consistently held higher point differences than losers across all phases underscores the importance of gaining and maintaining leads throughout the match. Furthermore, the notable increase in point differences as matches progress highlights the momentum-building aspect of maintaining an advantage. We also found that winners of men's and women's doubles were, on average, 5 to 6 points ahead of their opponents in phase two and 7 to 8 points ahead in phase 3. The significance of this study lies in its contribution to understanding the dynamics of point differentials in men's and women's doubles badminton matches at the international level. By dissecting the game into distinct phases and analyzing the point differences established by players, the research sheds light on critical patterns and trends crucial for strategizing and improving performance. Overall, this study enriches the understanding of competitive dynamics in badminton doubles matches, offering valuable implications for training, tactics, and performance enhancement in the sport.

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Conflict of interest:

The authors declare that there are no conflict of interest:

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ORIGINAL SCIENTIFIC PAPER

Using the Relative Handgrip Strength in Identification of the Under-Aged of Female Gender Candidates Exposed at Risk in Developing of Sarcopenic Obesity

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Abstract

Detecting children and minors who are at risk of developing childhood sarcopenic obesity frequently necessitates the use of specialized equipment and costly testing procedures. Finding more affordable and efficient techniques would be extremely advantageous, particularly if they are applicable in various field environments. The aim of this research was to investigate if a comprehensive evaluation of hand grip strength relative to BMI could distinguish young female candidates who may be at risk of developing second-class sarcopenic obesity. The study involved 535 female participants chosen at random from 9 elementary schools in the Skopje region of the Republic N. Macedonia. Several anthropometric measures, body composition measures, and absolute and relative handgrip strength were utilized to achieve the research goals based on the definition of sarcopenic obesity. Bioelectrical impedance was used to determine the percentage of body fat and skeletal muscle mass, with the skeletal muscle mass to fat ratio being calculated. Considering the relative handgrip strength in predicting a sarcopenic obesity class II in female candidates the area under the curve was 0.816. Relative handgrip strength (RHGS kg/kg/m²) is a useful indicator for identifying sarcopenic obesity in young females and can be valuable for guiding health interventions.

Keywords: bioelectrical impedance, muscle mass, ratio of skeletal muscle mass

Introduction

Continuous monitoring of health and fitness throughout one's life is crucial, particularly for early detection and treatment of diseases (Henriksson et al., 2019; Ortega et al., 2008). Health examinations that are thorough can be costly and necessitate specific equipment and skilled personnel that are typically found only in medical facilities. Alternatively, field fitness tests are cost-effective, convenient, and enable the assessment of the physical attributes of a large number of individuals quickly (Marques et al., 2021).

For instance, the handgrip test is a key component in various international and national physical fitness assessment tests

(Bianco et al., 2015; Marques et al., 2021), as it provides a quick way to evaluate overall muscle strength (Wind et al., 2010), and is linked to several medical conditions across different age brackets. More precisely, inadequate handgrip strength is linked to higher metabolic risk in children and (Cohen et al., 2014), other cardio metabolic risk factors in adults and the elderly (Peterson et al., 2016), and is also connected to various aspects of physical fitness (Matsudo et al., 2015).

Rosenberg (1989) explains that sarcopenia develops due to a decline in muscle mass and strength related to aging, typically appearing in later stages of aging. Although sarcopenia is typically seen in the elderly population, new research suggests



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that children and adolescents can also experience this condition (Biolo et al., 2014). Despite the fact that children and adolescents may have limited muscle mass, the rise in obesity, which has become a global epidemic, is a significant problem (Katanic et al., 2023; Lobstein et al., 2004).

It is unclear in children and adolescents whether the absence of muscle mass leads to obesity or if it is the other way around. However, previous studies suggest that obesity may play a role in the development of sarcopenia, leading to the condition known as “sarcopenic obesity” (Cauley, 2015).

Sarcopenic obesity occurs when there is an imbalance between the fat-free mass and fat mass in the body (Cauley, 2015; Stenholm et al., 2008). Sarcopenic obesity is characterized by an imbalance between muscle mass and adipose tissue, which is commonly seen in children rather than the progressive loss of muscle mass typically seen in the elderly with sarcopenia (Fielding et al., 2011; Morley et al., 2014). Children and adolescents who do not look obese on the outside may still have less muscle mass compared to their peers, possibly due to having a high amount of body fat, giving the illusion of a normal or healthy appearance. This creates challenges in identifying children who may have sarcopenic obesity. Having a diagnostic tool to identify children and adolescents with sarcopenic obesity is extremely important. Neglecting to treat any diseases in children that can impact their future health is detrimental (Gontarev et al., 2020; Steffl et al., 2017).

Due to the varying rates at which children mature and develop, it may be more suitable to use relative measures, as they can be more easily applied to both genders in different developmental stages, rather than absolute measures derived from fitness assessments. Hence, McCarthy and colleagues (2014) indicates that skeletal muscle mass together with the body fat component, i.e. calculating their ratio (MFR), can serve as an indicator of metabolic risk in children. Park et al. (2013) were the first to propose the ratio between muscle mass and fat components to determine the association between muscle mass and metabolic syndrome, but McCarthy et al. (2014) took an additional step and proposed a method to calculate limit values in children using the body mass index together with the ratio of skeletal muscle mass and the fat component. Kim and colleagues (2016), utilized the McCarthy method to detect sarcopenic obesity in children. Regrettably, accurately determining skeletal muscle mass and fat component ratio depends on costly body composition measurements, posing a challenge when assessing a large group of individuals. Nevertheless, in clinical settings, assessing muscle strength rather than skeletal muscle mass is commonly utilized for diagnosing sarcopenia. Cruz-Jentoft et al. (2010) suggested the handgrip dynamometry test as a diagnostic indicator for sarcopenia. Handgrip strength is a quick and easy method to measure in field research, in comparison to techniques like dual energy X-ray absorptiometry and bioelectrical impedance. Considering the developmental changes in children's maturational and body size, it is recommended that handgrip strength be measured in relative terms. It is suggested to use the handgrip-to-BMI ratio as a way to measure handgrip strength relative to body size, by dividing the handgrip dynamometry test result by the body mass index. The grip strength relative to body mass index (BMI), also known as the grip-to-BMI ratio, is suggested as a method to identify sarcopenia in the elderly population (McLean et al., 2014).

While the correlation between a lower fat-free component and a higher grip-to-BMI ratio (McLean et al., 2014) has been

confirmed in older individuals, there is limited research on this association in adolescents in scientific literature.

The aim of this study is to explore the connection between skeletal muscle mass ratio and fat component (MFR) and relative handgrip strength. The goal is to determine if relative handgrip strength can be used as a substitute for skeletal muscle mass ratio and fat component (MFR) in identifying under-aged females who may be at risk of sarcopenic obesity

Methods

Participants

The study involved 535 female participants selected randomly from various primary schools in the Skopje region. Their average age was 12.47 (± 1.1) years old. The research involved students whose parents consented to participate and who were physically and mentally healthy, and regularly attended physical education classes. Candidates of this type were dealt with in compliance with the Helsinki Declaration (revision of Edinburgh 2000). The protocols were approved by the Ethics Committee (Number 549, 10.05.2021) at the Ss. Cyril and Methodius University of Skopje.

Anthropometric measurements

Measurements of the anthropometric characteristics were conducted following the standard procedures of the International Biological Program (Weiner & Lourie, 1969), Anthropometric Standardization Reference Manual (Lohman, Roche, & Martorell, 1988) and Weiner-Lurie (1981). The weight was measured using a Tanita BC-418MA scale (Tanita Corporation, Tokyo, Japan), to the nearest 0.1 kg, with the participant wearing underwear and no shoes. Height was measured without shoes using Martin's anthropometry tool, to the nearest 0.1 cm, in the Frankfort horizontal plane. Body mass index is determined by dividing body weight in kilograms by the square of height in meters [$BMI = \text{weight (kg)} / \text{height (m)}^2$].

Components of body composition

Body mass and estimated body composition were assessed using the Tanita BC-418MA single-frequency (50 Hz) segmental Body Composition Analyzer (Tanita Corporation, Tokyo, Japan) with adjustments made for light indoor clothing. The subject was asked to stand barefoot on the analyzer and hold a pair of handgrips, one in each hand, as part of the measurement procedure. It took approximately 30 seconds per subject to complete the BIA component of the measurements. The BIA monitor used in this study provided individual measures of body fat mass (FM) and fat-free mass (FFM), as well as predicted skeletal muscle mass (SMM) in the limbs and trunk. The sum of SMM in the four limbs was also calculated as appendicular skeletal muscle mass (SMMa).

Measurements of segmental electrical properties such as resistance, reactance, and impedance were utilized to assess skeletal muscle mass in the limbs. The predictive equations were used to determine appendicular lean soft tissue, which comprises mostly of skeletal muscle. The impedance device used in this study was tested against DXA in various populations of children and adults, and was found to be superior to previous BIA methods (Pietrobelli et al. 2004).

The Tanita BC418MA model was validated in pediatric population against DXA and air-displacement plethysmography (BodPod Life Measurement, Inc, Concord, CA, USA). Once again, the results showed a strong correlation with the DXA, and there

was no significant difference in mean values. The latest research found that the daily variation in limb SMM was less than 1%, which is consistent with previous findings (Pietrobelli et al., 2004).

Handgrip strength

We utilized the Takei TKK 5101 dynamometer from Takei Scientific Instruments Co. Tokyo, Japan. Handgrip strength was measured for both hands using a dynamometer with a range of 1-100 kg. The participant holds the dynamometer in the hand that is being tested, with the arm bent at a 90-degree angle and the elbow next to the body. Once prepared, the candidate (participant) exerts maximum isometric force by squeezing the dynamometer for approximately 3 seconds. However, the subject must not engage in any other body movements and should be strongly motivated to exert their fullest effort. In this case, the highest value obtained from either the right or left hand was chosen as the maximum handgrip strength value. The grip-to BMI ratio was determined by dividing maximal handgrip strength by the BMI.

Sarcopenia risk diagnostics

Female participants were divided into quintiles of the BMI z-score, and the mean and standard deviation of MFR were calculated for each quintile. Cut-off values were determined based on the mean and standard deviation of the MFR for the third BMI quintile. The mean value minus 2 standard deviations of the MFR for the third BMI quintile was used as the cutoff value, and the percentage of sarcopenic obesity individuals was analyzed (Kim et al., 2016; McCarthy et al., 2014).

Statistics and Data Analysis

ROC analyses were conducted to compare the predictive accuracy and determine the best cut-off points (Akobeng, 2007). Pearson product-moment correlations were utilized to examine the relationships between the MFR and grip strength with BMI. The sensitivity and specificity measures based on grip-to-BMI ratio cut-points were used to plot the ROC curves. A diagnostic test with an AUC value of 1 is completely accurate, while another test with a value of 0.5 has no ability to dis-

tinguish between different groups (Krzanowski & Hand 2009; Pintea & Moldovan, 2009). The values that resulted in the highest AUC from the ROC curves were indicative of optimal sensitivity and specificity. Using a binary logistic regression model adjusted for age, the probability of developing sarcopenic obesity based on MFR was estimated in individuals at risk of sarcopenic obesity according to the cutoff values of grip-to-BMI. Similarly, a binary logistic regression model adjusted for age was used to calculate the probability of developing sarcopenic obesity based on MFR in subjects considered at risk of sarcopenic obesity using cutoff values in grip-to-BMI. Effect sizes were presented as odds ratios (OR), i.e., exponentiated estimates. All analyses were conducted using the Statistical Package for Social Sciences software (SPSS, v. 26.0 for Windows; SPSS Inc., Chicago, IL, USA) and MedCalc (Version 19.1.3).

Results

In Table 1, the essential descriptive statistical parameters of anthropometric measurements for body mass, both absolute and relative values obtained from the handgrip dynamometry test are presented for the candidates mentioned above.

Table 2 displays the MFR ratio based on BMI z scores quintiles in girls, showing the relationship between skeletal muscle mass and fat component. After reviewing table 2, it is evident that the cut-off value for sarcopenia obesity in girls was 0.860, calculated as the average of the MFR minus 2 standard deviations for the 3rd quintile of BMI. Furthermore, 11.1% of girls fell below this cut-off value. There were no girls with sarcopenia obesity in the first quintile compared to those without it. The second quintile has 0.0%, the third quintile has 0.0%, the fourth quintile has 5.9%, and the fifth quintile has 50.0%. The cutoff values for sarcopenia grade I were determined by calculating the arithmetic mean of the MFR - 1SD for the 3rd quintile of the BMI. In girls, the cutoff value was found to be 1.130, resulting in a sarcopenia grade I proportion of 30.3%. The percentage of girls with grade I sarcopenia compared to those without it was 0.0% in the first quintile, 4.1% in the second quintile, 10.3% in the third quintile, 48.5% in the fourth quintile, and 90.6% in the fifth quintile.

Table 1. Descriptive statistics of anthropometric measurements, components of body composition and handgrip strength in girls

	Mean	SD	Min	Max
Age	12.47	1.13	11.00	14.00
TV (cm)	156.26	8.16	116.30	177.00
TT (kg)	51.30	11.36	27.90	98.10
BMI (kg/m ²)	20.80	3.78	13.40	34.30
BMI z	0.00	1.00	-1.85	3.45
BFP (%)	24.99	7.37	6.20	45.90
BFM (kg)	13.44	6.58	1.99	38.19
SMM (%)	33.18	2.40	23.60	39.70
SMM (kg)	16.88	3.18	9.18	28.35
FFM (kg)	37.85	5.97	24.18	68.15
MFR (kg/kg)	1.51	0.68	0.60	5.81
HGS (kg)	22.53	5.25	8.20	40.15
RHGS (kg/kg/m ²)	1.10	0.25	0.36	1.92

TV = weight; TT= height; BMI = body mass index; BMI z= Z-Score of body mass index; BFP%= body fat percentage; BFM= body fat mass; SMM%= percentage of skeletal muscle mass; SMM = skeletal muscle mass; FFM = fat-free mass; MFR= ratio of skeletal muscle mass and fat component; HGS = handgrip strength; RHGS = relative handgrip strength (grip-to BMI ratio).

Table 2. Skeletal muscle mass ratio and fat component (MFR) shown through the quintiles of the body mass index (BMI) z values in girls

BMI	BMIZ		MFR		Sarcopenia I degree		Sarcopenia II degree	
	Mean	SD	Mean	SD	f	%	f	%
Q1	-1.15	0.25	2.34	0.86	0	0.0%	0	0.0%
Q2	-0.65	0.12	1.71	0.40	4	4.1%	0	0.0%
Q3	-0.17	0.12	1.40	0.27	12	10.3%	0	0.0%
Q4	0.41	0.19	1.16	0.19	49	48.5%	6	5.9%
Q5	1.58	0.64	0.90	0.19	96	90.6%	53	50.0%
Total			1.51	0.68	161	30.3%	59	11.1%

The class I MFR cut-off for girls (mean - 1SD of MFR for the 3rd BMI quintile) to determine sarcopenia = 1.130. The class II MFR cut-off for girls (mean - 2SDs of MFR for the 3rd BMI quintile) to determine sarcopenia = 0.860

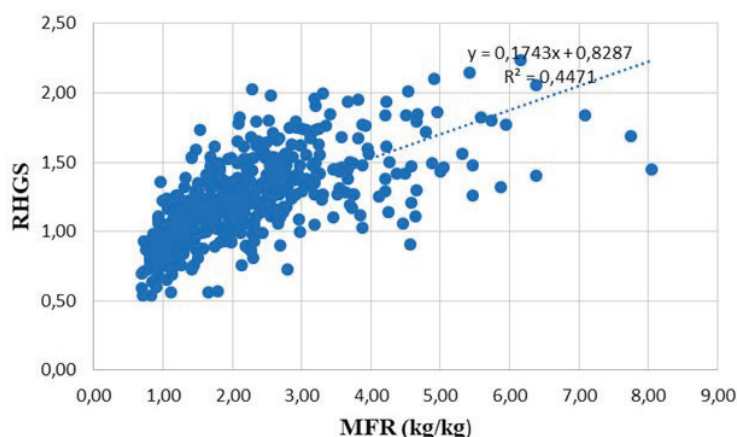
Table 3 and graph 1 demonstrate Pearson's correlation coefficients between the relative values from the relative handgrip dynamometry test (kg/kg/m^2) and the MFR. Upon reviewing the table and graph, it is evident that there is a significantly positive correlation between the relative handgrip

dynamometry test results (kg/kg/m^2) and the MFR in girls ($r=0.349$, $p<0.001$). Intercorelation between ratio of skeletal muscle mass and fat component, as well as relative handgrip strength, was analyzed in female candidates and presented in Table 3.

Table 3. Pearson's correlation between MFR and grip-to-BMI

	MFR (kg/kg)
RHGS (kg/kg/m ²)	.349*

*Correlation is significant at the level of $p < 0.001$ (2-tailed).



GRAPH 1. Scatter plot between ratio of skeletal muscle mass and fat component, relative handgrip strength in female candidates

Table 4. Determination of the optimal cut-off value of relative handgrip strength in predicting sarcopenic obesity grade II in female candidates

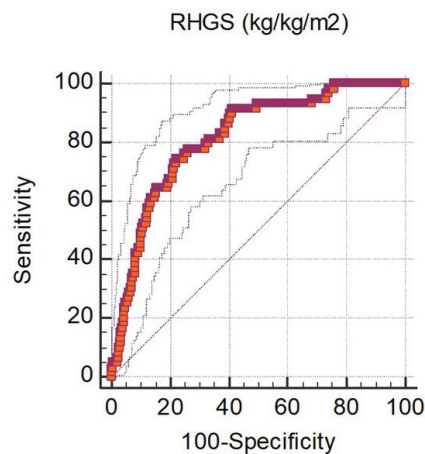
Area under the ROC curve (AUC)	0.816
Standard Error ^a	0.027
95% Confidence interval ^b	0.781 to 0.848
z statistic	11.566
Significance level P (Area=0.5)	<0.001
Youden index J	0.526
Cut-off point	≤0.950
Sensitivity	74.58
Specificity	78.01
Sample size	532
Positive group ^a	59 (11.09%)
Negative group ^b	473 (88.91%)

* $p<0.001$.

The research found that determining the best cut-off for relative handgrip strength to predict grade II sarcopenic obesity in female candidates resulted in an area under the ROC curve with a value of 0.816 (95% CI 0.781–0.848), with a sen-

sitivity of 74.58% and a specificity of 78.01%.

A logistic regression analysis was used to assess the relationship between relative hand grip strength and the risk of sarcopenic obesity grade II in female candidates. This analysis aimed



GRAPH 2. Receiver-operating characteristic curve on the relative handgrip strength (grip-to-BMI ratio) in girls ($p < 0.001$)

to distinguish between candidates at risk and those not at risk. The results are shown in table 5. Upon reviewing the table, it is evident that the adjusted probability quotient for age (OR 95% CI) is 19.09 (9.43–38.66, $p < 0.001$). In female candidates, it indi-

cates that those who exceed the cutoff value (meeting the criterion reference standard) are 19 times less likely to have grade II sarcopenic obesity, based on the average MFR - 2SD (two standard deviations below the mean) for the 3rd quintile of BMI.

Table 5. Relative handgrip strength in discriminating female respondents who are at risk, compared to those who are not at risk of grade II sarcopenic obesity

Explanatory variables	Unstandardized coefficient (B)	Wald statistics	p	OR (95% CI)
Years	.713	42.29	0.000	2.04 (1.64–2.53)
Sarcopenia class II				
MFR = >0.860	2.95	67.14	0.000	19.09 (9.43–38.66)
MFR = ≤ 0.860 (Ref)				1.000

*Dependent variable: The relative strength of the handgrip (Cut-off point ≤ 0.950 kg/kg/m²)

Discussion

The findings of this research demonstrated that by using relative values from the handgrip dynamometry test (ratio of SMM and BFM), it is possible to identify female underage individuals who are at risk of sarcopenic obesity.

In this study, the prevalence of sarcopenic obesity II among girls was 11.1%, which was higher than the 3.8% found in the study by Kim et al (2016) on Korean girls. Nevertheless, in this study, when the authors defined sarcopenia class I as 1 standard deviation below the mean MFR for the 3rd BMI quintile, the prevalence was 24.3% in girls, as compared to 30.3% in the previous study (Kim et al., 2016). McCarthy and colleagues (2014) found that 9.8% of girls were at risk for class II sarcopenic obesity. According to Steffl et al's research in 2017, the prevalence of sarcopenic obesity risk class II was 9.3% for girls and 7.2% for boys. Other studies suggest that the cutoff point for relative handgrip strength (grip-to-BMI ratio) in girls aged 4 to 9 is estimated to be .680 kg/kg/m², while for boys it is .721 kg/kg/m². In a study conducted by Gontarev, Jakimovski, and Georgiev (2019), the prevalence of risk of sarcopenic obesity class II in younger school-aged boys and girls was found to be 5.9% and 9.2%, respectively. The researchers also found that the cutoff values for relative grip strength (specifically, the ratio of grip to BMI) in this group of subjects were 0.658 kg/kg/

m² and 0.669 kg/kg/m² for boys and girls, respectively.

The findings of the study show that relative handgrip strength is highly effective (AUC=0.82) in determining the presence of sarcopenic obesity grade II in female participants. The criterion reference standard for evaluating grade II sarcopenic obesity in female candidates aged 11 to 14 years old is set at .950 kg/kg/m². This indicates that if girls score at or below this result on the handgrip dynamometry test divided by the body mass index, they are highly likely to have grade II sarcopenic obesity. Steffl et al found that the AUC in girls aged 10–14 years was 0.79 (95% CI 0.688–0.890, $p < 0.001$). In females aged 11 to 14 years, the cut-off point (the criterion reference standard) for identifying grade II sarcopenic obesity is .920 kg/kg/m² (Steffl et al., 2017).

While sarcopenia has typically been linked to muscle mass loss in older adults, recent studies have found that sedentary children and adolescents can also develop sarcopenia (Kim et al., 2009). Unlike elderly candidates who develop sarcopenia due to degenerative processes, children and adolescents are at risk of developing sarcopenia for entirely different reasons. Obesity is likely a significant factor in the development of sarcopenia in children and adolescents due to the widespread prevalence of obesity at an epidemiological level globally (Coles, 2016; Lobstein, Baur & Uauy, 2004; Peulić et al., 2024).

This is primarily caused by insufficient physical activity and poor nutrition. Female adolescents may experience an increased risk of higher body fat percentage and lower muscle mass percentage due to physiological changes in body composition, decreased insulin sensitivity, hormonal changes, and psychological adjustments (Todd et al., 2015).

In children and adolescents, low muscle mass is often indicated by the MFR, which measures skeletal muscle mass and fat content (Kim, Hong, & Kim, 2009). Unfortunately, the calculation of MFR relies on anthropometric measurements and body composition assessments that necessitate the utilization of specialized equipment like DXA, BIA, computed tomography, or magnetic resonance imaging. Alternatively, measuring grip strength is a cost-effective and easily applicable method. When the handgrip dynamometry test results are adjusted for relative values by dividing the absolute values by the body mass index (grip-to-BMI ratio), it is possible to differentiate between children and adolescents who are at risk of sarcopenic obesity and those who are not at risk. Children with lower relative handgrip strength are at a higher risk of being diagnosed with sarcopenic obesity, as determined by the MFR which looks at skeletal muscle mass and fat component. The standard interpretation of the AUC grip-to-BMI indicates that it accurately estimates sarcopenia.

One of the main limitations is that the sample used in the

study did not capture the diversity of under-aged female candidates from the Republic of North Macedonia as it did not include candidates from all eight planning regions. Nevertheless, the sample size was adequate and accurately represented. Determining precise cutoff values is challenging because increasing sensitivity will inevitably lead to a reduction in specificity.

Conclusion

The set criteria refer to standard limits that take into account the relative handgrip strength, providing a simple and cost-effective way to identify young females who may be at risk of second-degree sarcopenic obesity, making it a useful field method. While the methods presented in this study may not definitively diagnose sarcopenic obesity in younger individuals, they can still be a cost-effective and useful way to detect those who might be at risk and in need of further medical evaluation, dietary changes, or physical exercise programs.

The findings from this study enable countries with comparable economic, ethnic, and social backgrounds to utilize these boundary values. Physical education teachers and public health professionals can also utilize the findings from the research. Incorporating a healthy diet and regular physical activity should be a key aspect of overall health policy and daily routines, targeting not just individuals but also families and the entire population.

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Conflict of Interest

The authors state that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Is There a Meaningful Association Between Physical Literacy and Health Literacy? A Cross-Sectional Study in Older Adolescents

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Abstract

Health literacy (HL) is hypothesized to be associated with physical literacy (PL), but studies have rarely examined this association in adolescents. This study aimed to investigate the possible relationships between PL and physical activity levels (PAL), with HL in high-school adolescents. The participants in this cross-sectional study were high school students ($n=204$; 100 females; 16 to 19 years of age). Variables included evaluation of the HL, PL, and PAL. The Physical Activity Questionnaire for Adolescents was used for analysis of the PAL. PL was assessed by the Canadian Assessment of Physical Literacy, and the Physical Literacy Assessment of Youth (PLAYself). HL was evaluated using the European Health Literacy Survey Questionnaire. Univariate and multivariate correlations were calculated for the total sample and stratified by gender. Boys had greater PAL than girls (t test $=6.76$, $p<0.001$), but no significant gender differences were found in PL and HL. The results showed no significant association between PL and HL in boys. PL was significantly associated with HL in girls (17% of the explained variance, $p<0.001$), with PLAYself being a significant partial regressor ($\text{Beta}=0.38$, $p<0.001$). Due to the established gender-specific associations between HL and PL, in developing and implementing educational strategies aimed at improving PL and HL in adolescents, a gender-specific approach is warranted.

Keywords: *physical activity levels, physical literacy assessment, health literacy survey, adolescents, physical education*

Introduction

Health literacy (HL) is the ability to obtain, read, understand, and use healthcare information to make appropriate health decisions and follow instructions for treatment (Schulenkorf, Krah, Dadaczynski, & Okan, 2021). It involves a range of skills, including (i) reading and comprehension (i.e., understanding health-related materials such as prescription labels, appointment slips, medical brochures, doctor instructions, and consent forms); (ii) numeracy (i.e., performing tasks requiring math skills, such as measuring medications, understanding nutrition labels, and interpreting blood sugar levels or other test results); (iii) communication (i.e., effectively communicating with healthcare providers, expressing needs, asking questions, and understanding responses); (iv)

decision-making (i.e., evaluating options for treatment and prevention, understanding risks and benefits, and making informed decisions about health care); and (v) navigation (i.e., understanding how to access healthcare services, follow the healthcare system's administrative procedures, and utilize available resources) (Geets-Kesić, Maras, & Gilić, 2023). Today, it is generally accepted that HL is essential because it impacts individuals' ability to manage their health, navigate the healthcare system, prevent and manage disease, and adhere to treatment plans. Low health literacy can lead to poorer health outcomes, higher rates of hospitalization, less frequent use of preventive services, and overall higher healthcare costs. Improving health literacy can empower individuals to take control of their health, enhance the quality of care, and pro-



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more better health outcomes (Francis-Oliviero et al., 2023).

The importance of HL is especially highlighted in adolescents since improving HL in this period of life can lead to healthier, more informed individuals who are better prepared to manage their health throughout their lives (Francis-Oliviero et al., 2023). Therefore, schools, healthcare providers, and community programs play a vital role in fostering HL among young people. Indeed, studies have shown that higher HL in adolescents is associated with better health-promoting behaviors, body mass index, oral health, osteoporosis prevention, self-efficacy, self-care, hair and skincare, breakfast consumption, and nutrition behaviors (Olyani & Peyman, 2021). However, there are gaps in the general understanding of HL in this age group, including the lack of prevalence data and developmental frameworks (Okan, Pinheiro, Zamora, & Bauer, 2015). Factors such as general knowledge, infectious disease knowledge, health ideas, and school performance have been identified as predictors of better health literacy in adolescents, while the role of media-based HL in preventing noncommunicable diseases in adolescents has also been highlighted (Hidayatullaili, Musthofa, & Margawati, 2023). One of the insufficiently described areas is the association between HL and physical literacy.

In most common terms, physical literacy (PL) is the ability to move with competence and confidence in a wide variety of physical activities in multiple environments that benefit the healthy development of the whole person (Geets-Kesić et al., 2023; Sunda et al., 2022). It involves a combination of physical, cognitive, and emotional components. Similar to HL, the PL also involves several aspects/subdomains, including (i) physical competence (i.e., developing fundamental movement skills and patterns such as running, jumping, throwing, catching, and balancing), (ii) confidence and motivation (i.e., having self-assurance and enthusiasm to participate in physical activities), (iii) knowledge and understanding (i.e., understanding the principles of movement and the health benefits of physical activity and knowledge about how the body moves, the importance of fitness, and how to engage in different physical activities safely and effectively), and (iv) engagement in physical activities (i.e., being regularly active in a variety of physical pursuits). In general, PL is considered important because it lays the foundation for a healthy, active lifestyle that contributes to overall physical, mental, and social well-being (Carl et al., 2023). Not surprisingly, HL and PL are considered to be interconnected concepts.

For example, while HL provides individuals with knowledge about the benefits of physical activity, the risks of sedentary behavior, and the importance of maintaining a healthy lifestyle, this knowledge can motivate individuals to develop and maintain PL. On the other hand, through regular participation in physical activities, PL reinforces the practical application of HL since engaging in physical activities can provide experiential learning opportunities that enhance the understanding of health concepts. Furthermore, individuals with high HL are more likely to make informed decisions regarding their physical activities, such as choosing appropriate exercises, understanding the significance of warm-ups and cool-downs, and recognizing the need for rest and recovery. On the other hand, PL fosters a positive attitude toward physical activity, encouraging individuals to integrate movement into their daily routines, which in turn promotes better health outcomes, which are often associated with greater HL (Kesić, Savicević, Perić, Gilic, & Zenić, 2022).

According to the previous literature overview, it is clear that HL and PL are both important concepts in modern life, and especially in adolescence when life-long habits and behaviors are formed. Additionally, the relationships between HL and PL and between HL and physical literacy are synergistic. Therefore, it could be expected that HL and PL are intercorrelated, but there is an evident lack of studies that have specifically investigated the associations that may exist between HL and PL in adolescence. One of the rare investigations of such examined the correlations between HL and PL in older adolescents (medical school students) and reported a relatively poor correlation between HL and PL and theorized that the established level of correlation could be characteristic of the study sample (adolescents who were directly interested and specifically educated in health/medicine areas), indicating the necessity of further research in “nonselected” samples of participants (Kesić et al., 2022). Therefore, the aim of this study was to evaluate the associations between PL and HL in older adolescents. Since PL generally develops earlier in life than does HL, in this study, we used HL as a criterion variable and applied a gender-specific approach to study possible associations between PL and physical activity (predictors) and between PL and HL (criterion). We hypothesized that PL would be significantly associated with HL irrespective of gender.

Materials and methods

Participants and study design

The participants in this cross-sectional study were high school students from Split, Croatia ($n=204$; 100 females; all 16 to 19 years of age). All participants were in good health and regularly participated in physical education classes. Since the study involved analysis of physical activity levels (PAL), in this report, we did not include those participants who reported illness and/or injury that did not allow them to be regularly physically active during the last week (please see later on study variables, specifically analysis of PAL). Participants voluntarily participated in the investigation after being informed about the risks and benefits of the study and after their parents (or themselves if they were older than 18 years) signed informed consent for the study participation. The study was approved by the Ethical Board of the Faculty of Kinesiology (EBO: 2181-205-02-01-21-0011; date of approval, 23-10-2021). The study was performed in early 2024.

Variables and measurement

Apart from gender (male, female, none) and age (in years), the variables in this study included evaluation of the HL, PL, and PAL. All data were collected online through the Survey Monkey platform (SurveyMonkey Inc., San Mateo, CA) provided by the Faculty of Kinesiology, University of Split, Croatia during the physical education classes taught by first two authors of the paper.

The European Health Literacy Survey Questionnaire (HLS-EU-Q) was used to evaluate HL (Sorensen et al., 2012). The HLS-EU-Q comprises 47 questions indicating an individual's ability to access, understand, appraise, and apply health-related information. The general index of HL was constructed using a 4-point Likert scale, with responses ranging from very difficult – 1 to very easy – 4. The score was calculated using the following formula: $\text{index} = (\text{mean} - 1) \times (50/3)$. The HL scale (from 0 to 50) was used, where 0 represented the lowest score and 50 represented the highest score. Four levels of HL

were defined: inadequate (from 0 to 25), problematic (26–33), sufficient (34–42), and excellent (43–50). The Croatian version of the HLS-EU-Q was checked for reliability and validity, and the results are presented elsewhere (Kesić et al., 2022).

To evaluate the PL, we applied two widely used questionnaires, namely, (i) the second/shorter version of the Canadian Assessment of Physical Literacy (CAPL-2) and (ii) the Physical Literacy Assessment of Youth (PLAYself). Croatian versions of both tools were studied for reliability and validity in this age group, and the results are presented previously (Sunda et al., 2022). The CAPL-2 knowledge and understanding questionnaire included 12 questions aimed at evaluating the participants' knowledge of the necessity and importance of daily PA, the problem of sedentarism, the importance and definition of cardiorespiratory fitness and muscular strength, the concept of fitness and its impact on a physically active lifestyle, ways of improving one's motor skills, and the components of overall fitness. Each correct answer was assigned 1 point, while each incorrect answer received a score of 0, resulting in a total score ranging from 0 to 12. The PLAYself questionnaire establishes the perceived level of PL and includes four subsections: (i) environment, assessing the degree of movement confidence in different environments (i.e., gym, water, snow); (ii) affective and cognitive aspects related to PL that determine individuals' self-efficacy with respect to their participation in PA; (iii) relative ranking of literacy, numeracy, and physical literacy in different settings, including school, home, and social life with friends, examining how much an individual values each literacy type; and (iv) fitness. The final score consists of the subtotals from the first three subsections divided by the number of questions (27 in total). The maximum PLAYself score is 100, representing a high self-perceived PL.

The Physical Activity Questionnaire for Adolescents (PAQ-A) was used for analysis of the PAL. This questionnaire consists of nine items asking participants to complete a seven-day recall. The theoretical score ranged from 0 (minimal) to 5 (maximal physical activity level). The first 8 items were scored on a 5-point scale and included questions on different types of physical activity (i.e., activity during sports, physical

education, active transportation, and free play); the ninth item did not contribute to the overall score, and it was only used for the selection of participants whose results should not be observed as reliable and valid (e.g., evidencing participants who suffered a sort of injury/illness and whose physical activity was therefore reduced).

Statistics

Data were checked for normality of the distributions by the Kolmogorov-Smirnov test, and descriptive statistics included calculations of the frequencies (for gender), and means and standard deviations (for remaining variables).

Differences between boys and girls in study variables were established by t-tests for independent samples, which consequently allowed us to identify possible associations between gender and study variables (note that two participants defined gender as "none", which did not allow us to calculate statistics for this subsample). To determine the associations between PL, PAL and HL, Pearson's correlations were calculated. Next, multiple regression calculations were applied to evaluate possible multivariate associations between measures of PL and PAL (observed as predictors) and HL (observed as criterion).

Statistica ver. 14.0 (Tibco Inc., Palo Alto, CA) was used for all calculations, and a p value of 0.05 was applied.

Results

Differences in the study variables between boys and girls are presented in Figure 1. In brief, boys achieved greater PAL than girls did (2.79 ± 0.73 vs. 2.18 ± 0.65 ; t test = 6.76, $p < 0.001$) (Figure 1A). Further, total PL scores of boys and girls did not differ according to the PLAYself (65.11 ± 13.21 and 63.7 ± 11.23 for boys and girls, respectively; t test = 1.02, $p = 0.30$) (Figure 1B) or CAPL-2 (7.91 ± 2.12 and 8.11 ± 1.93 for boys and girls, respectively; t test = 1.60, $p = 0.10$) (Figure 1C). Finally, no significant difference was found in HL as obtained by HLS-EU-Q (31.62 ± 9.29 and 33.52 ± 7.44 for boys and girls, respectively; t test = 1.66, $p = 0.10$) (Figure 1D). However, with the average HLS-EU-Q score of 32.11 ± 6.72 HL level of the studied adolescents is "problematic".

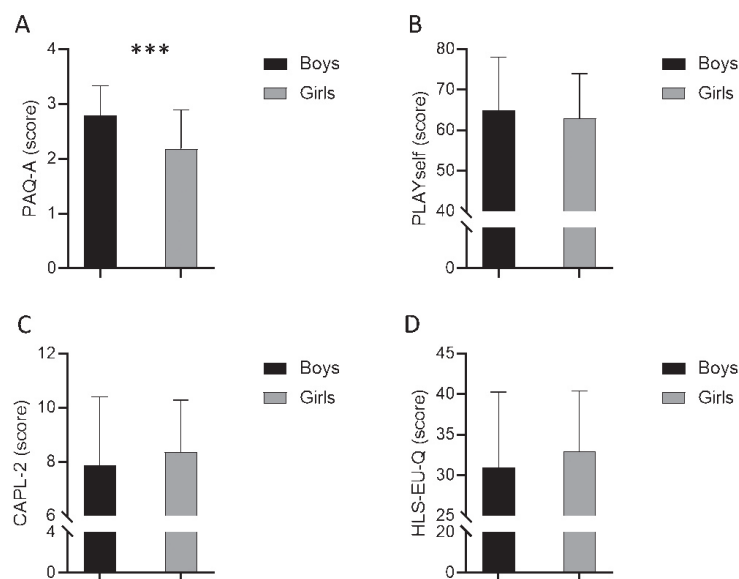


FIGURE 1. Descriptive statistics (results are presented as Means ± Standard Deviations), and t-test differences between genders in PAQ-A (Figure 1A), PLAYself (Figure 1B), CAPL-2 (Figure 1C), and HLS-EU-Q (Figure 1D), *** indicates t-test significance of $p < 0.001$.

Table 1 presents univariate correlations between study variables for the total sample and stratified by gender. The PLAYself score was positively correlated with the PAL obtained by the PAQ-A (Pearson's $R=0.48$, $p<0.001$) and the HL score (Pearson's $R=0.19$, $p<0.01$). When gender was stratified, no

significant correlation between HL and PL measures was detected among boys, while PLAYself was significantly correlated with HL among girls (Pearson's $R=0.40$, $p<0.001$). PLAYself scores were significantly correlated with PAL in boys (Pearson's $R=0.52$, $p<0.001$) and in girls (Pearson's $R=0.45$, $p<0.001$).

Table 1. Pearson's correlation coefficients between study variables

	Total (n = 204)			Boys (n = 104)			Girls (n = 100)		
	PLAYself	CAPL-2	PAQ-A	PLAYself	CAPL-2	PAQ-A	PLAYself	CAPL-2	PAQ-A
PLAYself	-			-			-		
CAPL-2	-0.02	-		0.01	-		-0.05	-	
PAQ-A	0.48***	-0.04	-	0.52***	0.09	-	0.45***	-0.10	-
HLS-EU-Q	0.19**	0.06	0.08	0.07	0.11	0.10	0.40***	-0.06	0.22**

Legend: PAQ-A – Physical Activity Questionnaire for Adolescents, PLAYself – Physical Literacy Assessment of Youth, CAPL-2 – Canadian Assessment of Physical Literacy, HLS-EU-Q – European Health Literacy Survey Questionnaire, *** $p < 0.001$, ** $p < 0.01$

Multiple regression calculations between PAL and PL scores observed as predictors and the HL criterion are presented in Table 2. In the total sample, predictors explained 5% of the criterion variance ($p<0.05$), and the PLAYself score was identified as the only significant predictor (Beta = 0.21, $p<0.01$).

When multiple regression was calculated for boys, no significant association was found between predictors and criterion (<2% of the explained variance). In girls, predictors explained 17% of the common variance ($p<0.001$), with PLAYself being a significant partial regressor of HL (Beta = 0.38, $p<0.001$).

Table 2. Multiple regression results for criterion - health literacy (obtained by HLS-EU-Q)

	Total			Boys			Girls		
	Beta	b	p	Beta	b	p	Beta	b	p
Intercept		21.90	0.001		24.63	0.00		17.42	0.001
PLAYSELF	0.21	0.14	0.01	0.04	0.03	0.74	0.38	0.26	0.001
CAPL-2	0.06	0.23	0.37	0.10	0.36	0.32	-0.04	-0.16	0.65
PAQ-A	-0.02	-0.23	0.81	0.06	0.85	0.59	0.04	0.54	0.70
R	0.21			0.14			0.40		
Rsqr	0.04			0.01			0.16		
p	0.03			0.58			0.001		

Legend: PAQ-A – Physical Activity Questionnaire for Adolescents, PLAYself – Physical Literacy Assessment of Youth, CAPL-2 – Canadian Assessment of Physical Literacy, HLS-EU-Q – European Health Literacy Survey Questionnaire, R – coefficient of multiple correlation, Rsqr – coefficient of determination, p – level of significance

Discussion

This study aimed to investigate the possible relationships of PL with HL in high school adolescents. With regard to the study aim, we may highlight several important findings. First, boys and girls did not differ in HL or total PL score, but boys had greater PAL than girls. Second, we found no significant correlation between PL (and PAL) and HL in boys, while PL was significantly associated with HL in girls. Therefore, our initial study hypothesis may be partially accepted.

No significant difference in health literacy between boys and girls

Studies have investigated gender differences in HL, and our findings of nonsignificant differences between boys and girls are not in accordance with the majority of previous studies in which authors mostly noted higher HL in girls. In brief, studies have shown that girls generally have higher HL levels, particularly in areas such as numeracy, use, communication, and self-efficacy (Karimi et al., 2019). Additionally, girls are more likely to seek health information from school, parents, and medical personnel (Brown, Teufel, & Birch, 2007).

However, girls also report more smoking and less exer-

cise (Cohen, Brownell, & Felix, 1990), which consequently could decrease the total HL score (note that in this study, we observed total HL as the final score). These differences may be influenced by factors such as age, with junior high school being a critical period for the development of health habits (Cohen et al., 1990). Additionally, in previous studies, the authors reported that social disparities, including income and education, contribute to differences in health literacy (Fleary & Ettienne, 2019). However, in our study, such differences between genders are not expected due to the similar socioeconomic statuses of boys and girls.

In explaining the lack of significant differences in HL between boys and girls, we can highlight several factors, both inherent and environmental. The first factor that should be highlighted is the fact that all the students we assessed had equal access to education. Therefore, irrespective of their gender and possible interests, boys and girls received the same school health-related programs. It means that both genders were exposed to the same information, skills, and resources related to health. The second factor is parental influence. It is regularly confirmed that parental health literacy and behavior signifi-

cantly influence children's understanding of health (de Buhr & Tannen, 2020). From their own experience, authors can witness that parents equally share their health knowledge and practices with their children, regardless of gender, and parents typically model health behaviors without gender bias, which develop similar levels of health literacy for both boys and girls.

Third, public health campaigns and initiatives in Croatia are generally designed to be inclusive, targeting broad audiences without gender discrimination (Geets-Kesić et al., 2023; Kesic et al., 2022). Additionally, many general- and community-based health programs and resources are equally accessible to both boys and girls. Also, the widespread availability of health information through digital media, including websites, social media, and online forums, offers equal access to health information for both genders. This democratization of information helps level the playing field in all areas, including the terms of HL.

In addition to the previously explained similar opportunities, it must be highlighted that boys and girls of adolescent age follow the same cognitive development (Choudhury, Blakemore, & Charman, 2006). It logically includes the ability to understand and process all information (including health information), which follows a similar trajectory for boys and girls. Both genders develop critical thinking and comprehension skills at comparable rates, contributing to similar levels of health literacy. However, while individual learning styles may vary, there is no clear evidence to suggest that boys and girls differ significantly in their ability to learn health-related information. In other words, it is clear that educational strategies designed to cater to diverse learning styles benefit all students, irrespective of gender.

Lack of correlation between physical literacy and health literacy in boys

As said previously, studies rarely examined the associations between PL and HL in high school adolescents. On the basis of some previous studies in which similar associations were examined (i.e., associations between PL and PAL), we decided to examine this association using the gender stratified approach. (Sunda et al., 2022). Evidently, this was the right decision due to the evident gender-specific correlation. In brief, the correlations between PL and HL did not reach statistical significance among boys, while the correlations were significant among girls of the same age. Several reasons could explain such findings.

The first factor that could contribute to gender-specific associations is related to personal interest and motivation toward physical activities (i.e., sports). More precisely, adolescent boys are generally more inclined toward physical activities, such as sports and games, than girls. These activities develop PL but do not necessarily involve learning about health topics such as nutrition, disease prevention, and personal health management. Therefore, the PL of boys increases as a result of sport participation, but it is not followed by HL. Indeed, studies in Croatia and the region regularly confirmed greater participation of adolescent boys in sports, which directly supports previous considerations (Gilic, Zenic, Separovic, Jurcev Savicevic, & Sekulic, 2021).

The lack of correlation between PL and HL in boys is aggravated by the fact that some adolescent boys may perceive health information as less relevant or interesting compared to physical activities. This perception can lead to lower en-

gagement with health education and a weaker development of HL (Gray, Klein, Noyce, Sesselberg, & Cantrill, 2005). Additionally, the authors can see that schools and communities often offer a wide range of extracurricular activities that focus on sports and physical fitness but fewer opportunities that specifically address HL. By all means, HL programs do not receive the same level of attention or resources as physical activities and sports, and this imbalance almost certainly results in higher PL and increased PAL without a corresponding increase in HL.

Boys might be more focused on immediate, tangible activities such as sports rather than abstract concepts related to HL (Choudhury et al., 2006). Adolescence is a critical period for identity formation, and many boys might develop identities centered around physical competence and athleticism. This identity formation process might not prioritize health knowledge and literacy. This is additionally supported by social and cultural influences. Media portrayals and role models for boys often emphasize physical prowess and athletic success rather than health knowledge. This cultural emphasis can lead to a stronger development of physical skills without a corresponding focus on HL. Altogether, this can result in a discrepancy between boys' PL and HL.

Finally, health education is generally taught through theoretical subjects, while physical education is thought to be a mostly practical subject, with abilities and skills being the most important and relevant topics (Forte, Pugliese, Ambretti, & D'Anna, 2023). Despite recent efforts to equalize course topics, health education emphasizes knowledge about nutrition, disease prevention, and general health practices, while physical education focuses on developing physical skills and promoting physical activity (Hrg, Lončar, Zelanto, Novak, & Podnar, 2023). There is no doubt that such separation results in developing skills and knowledge in one area without a corresponding increase in the other.

Positive correlation between physical- and health-literacy in girls

Apart from previously described factors that can contribute to the lack of correlation between HL and PL in boys, some additional explanations can be offered to explain the positive correlation between HL and PL in girls. First, and probably the most important explanation can be found in the fact that adolescent girls who have a high level of HL are more likely to understand the importance of physical activity for their overall health (Rutkauskaitė & Kuusinen, 2019). This understanding can translate into increased participation in physical activities, enhancing individuals' PL. While this is a general consideration, its background also deserves attention.

As discussed previously, boys are more interested in sports because of play and fun, which consequently increases their PL with little or no influence on HL. Meanwhile, girls' motives for being physically active are different, and those girls who are educated about how physical activity can improve their health and well-being are more likely to be interested in being physically active (Litt, Iannotti, & Wang, 2011). Additionally, girls are generally more interested in psychological well-being and therefore are better informed about the mental health benefits of physical activity, such as stress relief and improved mood. Therefore, a substantial number of girls are physically active while seeking the psychological benefits of such practices (Litt et al., 2011).

Finally, we cannot neglect certain biological factors that

could contribute to the correlation between PL and HL in girls. Namely, physical development during adolescence can affect interest in and participation in physical activities. More specifically, girls may experience changes in their bodies that affect their comfort and interest in certain types of activities. Logically, they will be more oriented toward activities that are more efficient in modifying body composition, which is logically followed by their interest in some topics directly connected to HL (i.e., nutrition, metabolism) (Niven, Fawkner, Knowles, Henretty, & Stephenson, 2009). Therefore, it is no surprise that this will be directly translated to a simultaneous increase in PL and HL in girls. Taken together, these factors probably contributed to finding that HL and PL are positively correlated in girls.

Limitations and strengths of the study

The first limitation of the study is its cross-sectional design, which did not allow clear interpretation of the causality between the study variables. However, it is generally known that PL develops beginning in early childhood, even nonformally, which allows us to make some contextual interpretations of these relationships. Future studies should explore this problem through prospective or intervention approaches to evaluate cause-effect relationships more specifically. Additionally, this study included adolescents from one urban center. Therefore, the generalizability of the findings is limited to similar samples of participants.

The main strength of the study is the use of a gender-specific approach, which allowed us to identify the context of the studied associations more clearly. Additionally, the fact that the authors were able to observe the problem from a re-

al-world professional perspective (note that first two authors are teachers in schools where the sample was drawn from, and therefore are well informed about the situation and context of the research), is a significant strength of the study.

Conclusion

The lack of significant differences in HL between boys and girls can be attributed to equal access to health education, similar parental influences, inclusive public health campaigns, and widespread digital information. This equality suggests that HL initiatives and educational programs are equally effective in reaching both genders, promoting a uniform level of HL among studied adolescents from Croatia.

The positive correlation between HL and PL in girls and the lack of a significant correlation in boys can be attributed to several factors, with basic motives for physical activity (and sports) likely being the most important. Namely, boys participate in physical activity because of fun, play and competition, which logically increases their PL but not their HL. On the other hand, the health benefits of physical activity are more important motives for being active among girls, which simultaneously increases their PL and HL.

Both PL and HL are important concepts and should be developed, especially during adolescence. Therefore, specific efforts are needed to design specific educational programs aimed at the development of PL and HL in adolescents. However, our results suggest that in developing and implementing educational strategies aimed at improving PL and consequently HL in this age group, a gender-specific approach should be applied.

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Conflict of interest:

The authors declare that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Evaluation of Prevalence, Location and Pattern of Musculoskeletal Pain and Discomfort among Dancers

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Abstract

Professional dancers are contemplated as athletes as it is involving repetitive and rhythmic movement. These strenuous activities are the negative stressors and reason for overuse injuries leading to discomfort and pain. This pain and discomfort results in some major issues in the future and it can affect their professional performance or career. This study aimed to exploring the prevalence, common region involved and relation of course of pain in different regions of the body in dancers. A total of 110 dancers, both male and female, participated in this cross-sectional study using the Nordic MSD questionnaire tool. The mean and standard deviation for age (years), height (cm), weight (kg), BMI (kg/cm²), experience (years), and weekly practice hours were calculated as follows: for women, 21.87±4.47, 165.57±9.89, 60.98±17.85, 22.20±5.93, 8.05±5.47, and 8.98±6.93, and for men, 21.83±6.59, 173.73±7.34, 70.41±11.43, 23.23±3.24, 7.84±5.55, and 9.36±6.95. The result findings revealed, 30.8% participants had pain in the low back region that in last 7 days, followed by shoulder 27.3% and ankle 25.5%. In last 12 months 51% participants reported pain, discomfort and numbness in the lower back region followed by ankle/ foot and knee (31%). In response towards the restriction in the ADL, lower back pain cases (30%) were highest followed by knee (23.6%). The results also revealed that neck pain is strongly and positively associated with upper back pain ($r=.601$) and moderately related to shoulder pain ($r=.467$). The study concludes that the lower back region is more prevalent area for pain following the ankle and knee among dancers.

Keywords: *biomechanics, dancing, foot, ankle, injury, prevalence, epidemiology*

Introduction

Dancers are considered as athletes, using artistic statement with athletic skills performing a series of rhythmic body movements to the beats of music (Costa et al., 2016). Dance is an art involving motor activities with expression to interact with society (Aweto et al., 2014). In the past, dancing was considered to have cultural links, but nowadays it is also opted as a profession and forms the statement of social style rather than cultural links. Professional dancing requires hard training and more practice hours. The movement patterns in dancing involve transitions from one position to another that can be challenging and strenuous at the same time. This places a high physical and

physiological demand on all the body's musculoskeletal systems (Motta-Valencia, 2006). This makes them more susceptible to musculoskeletal injuries, pain and discomfort, affecting their performance level and career (Russell, 2013). Myriads factors involve in different dancing activities, placing dancers at the risk of injuries resulting in pain and discomfort (Campoy et al., 2011). The factors are biomechanical faults, and mal-alignment caused over time due to different body movements, lack of flexibility and strength because of non-involvement in exercise regime before dancing (Huang et al., 2022).

Evidences underscore the pivotal role of biomechanical analysis in dance, emphasizing its significance in understand-



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ing movement patterns, skill enhancement, and injury prevention (Koutedakis et al., 2008; Sadhu et al., 2021). While existing research has delved into aspects such as prevalence rates ranging from 26% to 84%, commonly injured lower limb body parts, and preventive measures (Anand Prakash et al., 2023; Gopi & Shah, 2022; Uršej & Zaletel, 2020; Vassallo et al., 2019), there remains a notable gap in tracking the development and progression of pain over time. Moreover, the interconnected biomechanical impacts have not received adequate exploration. Comprehensive data on the overall pain patterns experienced by dancers, including which body parts are affected together, are lacking. Therefore, our study aims to fill this gap by investigating the pain patterns across different anatomical sites. By doing so, we seek to inform the implementation of appropriate preventive measures in advance. Our study encompasses the determination of the incidence rate of musculoskeletal pain, identification of the most common regions for experiencing pain among dancers, and analysis of sex differences within the sample. Through these undertakings, we aim to contribute to the enhancement of dancer well-being and performance.

Material and methods

Study design and participants

A cross-sectional study with a convenient sampling method was conducted from September 2023 to December 2023 among the dancer population of India. The study was conducted by using both online and offline platforms. The participants from different parts of India was approached, and a questionnaire was shared with them via the google form and personally. All form of male and female dancers within the age group of 18-45 years, willing to participate voluntarily and ready to provide signed consent, without any diagnosed psychological issues and resident of India were included in this study. This cross-sectional study was carried out according to established criteria included in Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) (Cuschieri, 2019).

Ethical consideration

Ethical approval was acquired on 20 April 2023 from the Genibandhu Independent Ethics Committee under Ref-ECG005/2023. The study's objectives were mentioned and explained to the participants before filling out the questionnaire. The signed consent was obtained from the respondents before filling out the questionnaire. The moral principles of ethics were considered as per the Helsinki declaration, and privacy were maintained.

Data collection tool

The Nordic MSD Questionnaire was used for to analysing the pain and discomfort. This questionnaire has three sections with questions of pain or discomfort in the last seven days and other two sections having questions about pain, discomfort and problems faced in performing household or other activities in the last 12 months (Crawford, 2007).

Procedure

After obtaining the ethical approval a well-structured data collection form was constructed including three sections: Section 1: It includes the title of the study with all the objectives and procedure disclosed followed by the consent form for the participant. Section 2: This section includes the general information related questions of the

participants (name, age, height, weight, BMI, dance form, dance type, dance practicing hours per week/day, number of years practicing dance). Section 3: This section includes the Nordic musculoskeletal questionnaire for pain and discomfort assessment.

The platform used for this study was both online and offline. After considering inclusion and exclusion criteria, the dancers were approached individually telephonically and on a one-on-one basis. A total of 110 responses were received. The participant responses were evaluated properly for any missing numbers and data analyses were carried out by using software IBM SPSS Statistics 22 (IBM Corp. Released 2013. IBM SPSS Statistics for Windows, Version 22.0. Armonk, NY: IBM Corp) and MS Excel 2016 (Microsoft Excel 2016 is developed by Microsoft Corporation, headquartered in Redmond, Washington, USA.) (Abbott, 2011).

Statistical analysis

The data's normality was tested by using Shapiro-Wills and data was normally distributed. Descriptive statistics were used to calculate the samples' mean $\pm$ SD and median (range); the confidence interval was 95%. Independent t test was used to evaluate any gender difference ($p < 0.05$). Pearson's correlation coefficient was utilized to detect the relationships among the variables.

Results

Descriptive statistical analysis was used to define the variables and characters. Frequency distribution tables were used to explain the results more clearly.

Socio-demographic and clinical information

A total of 110 dancers (71 females and 39 males) participated in the study, with the mean and standard deviation for various attributes such as age, height, weight, BMI, experience, and weekly practice hours were calculated as for females 21.87 ± 4.47 , 165.57 ± 9.89 , 60.98 ± 17.85 , 22.20 ± 5.93 , 8.05 ± 5.47 , and 8.98 ± 6.93 , and for male 21.83 ± 4.59 , 165.89 ± 9.53 , 61.26 ± 18.19 , 22.19 ± 5.96 , 7.84 ± 5.55 , and 9.36 ± 6.95 respectively. In our investigation, 45% (49) of the dancers practiced traditional dance form, while 27% (30) were engaged in Non-traditional dance, and 28% (31) participated in both forms. Subsequent analysis revealed that 23% (25) dancers practiced all types of dance forms, 24% (26) focused on Western type, 8% (9) specialized in Bharatanatyam classical dance, 15% (17) in Bhangra dance, 10% (11), 6% (7) in Kathak dance, 5% (6) in Kuchipudi dance, 4% (4) in Rajasthani folk, and 5% (5) in Bihu folk dance.

Nordic musculoskeletal assessment questionnaire

In our investigation, subsequent to inquiring about the pain status over the previous seven-day period as depicted in Table 1, it was observed that the highest number of cases reported was related to low back pain (30.8%), followed by shoulder (27.3%) and ankle (25.5%). In light of the limitations in the performance of daily activities, as depicted in Table 2, instances of lower back pain (30%) were predominant, followed by knee (23.6%) and then neck (21.8%). The findings of our study demonstrate that, during the past 12 months, as illustrated in Table 3, 51% of the participants encountered pain, discomfort, and numbness in the lower back area, followed by the ankle/ foot and knee regions (31%).

Gender difference

On comparing the mean difference between the genders, there was no significant difference between male and female

except pain/discomfort in the neck region in last seven days (0.017) and Hip pain/discomfort in last 12 months (0.044) as illustrated in Table 1,2, and 3.

Table 1. Pain/discomfort in a particular body region at any time during the last seven days

Region	Total (110)	Males (39)	Females (71)	P value
Neck	22 (20%)	3	19	.017
Upper back	26 (23.64%)	7	19	.302
Lower back	34 (30.90%)	10	24	.380
Shoulder	30 (27.27%)	9	21	.469
Elbow	11 (10%)	4	7	.948
Wrist/hand	15 (13.64%)	6	9	.695
Hip	11 (10%)	5	6	.469
Knee	18 (16.36%)	6	12	.839
Ankle/foot	28 (25.45%)	7	21	.184

Table 2. Prevented from doing your normal work due to pain in a particular body region for the last 12 months

Region	Total	Males	Females	P value
Neck	24 (21.82%)	6	18	.230
Upper back	18 (16.36%)	7	11	.742
Lower back	33 (30%)	15	18	.154
Shoulder	18 (16.36%)	7	11	.742
Elbow	15 (13.64%)	5	10	.855
Wrist/hand	13 (11.82%)	4	9	.710
Hip	10 (9.09%)	6	4	.090
Knee	26 (23.64%)	8	18	.572
Ankle/foot	21 (19.09%)	8	13	.781

Table 3. Trouble (such as pain, ache, discomfort, numbness) in particular body region in last 12 months

Region	Total	Males	Females	P value
Neck	23 (20.91%)	7	16	.620
Upper back	26 (23.64%)	11	15	.408
Lower back	56 (50.91%)	21	35	.651
Shoulder	24 (21.82%)	9	15	.815
Elbow	13 (11.82%)	4	9	.710
Wrist/hand	12 (10.91%)	5	7	.637
Hip	20 (18.18%)	11	9	.044
Knee	35 (31.82%)	11	24	.551
Ankle/foot	35 (31.82%)	12	23	.863

Correlation between the socio-demographic characteristics and pain/discomfort history of different regions

To assess the relationship between gender, age, height, weight, BMI, dancing experience, and daily practice hours, Pearson's correlation statistical test was used (refer to Table 4). The findings demonstrated a significant inverse correlation between gender and neck pain experienced within the last seven days (-.228), as well as a significant positive correlation with hip pain or discomfort encountered within the past 12 months (.193). Age exhibited a positive correlation with neck pain experienced within the past seven days (.259), neck pain experienced within the past 12 months (.215), and

wrist pain that hindered work within the past 12 months. Furthermore, age displayed a negative correlation with wrist pain experienced within the last seven days (-.190). Height manifested both positive and negative correlations with neck pain experienced within the past seven days and hip pain experienced within the past 12 months (-.238, .245). Practice hours were significantly and positively associated with lower back pain experienced within the past seven days and 12 months, as well as hip pain experienced within the past 12 months (.318, .279, .285). Additionally, practice hours exhibited a negative association with pain in the elbow and ankle that impeded work (-.198, -.190).

Table 4. Correlation between socio-demographic characteristics and pain/discomfort history of different regions

Variable	Hours	Years	BMI	Height	Weight	Age	Gender
Neck1	-.061	.029	.063	-.238*	-.038	.259**	-.228*
Neck2	-.075	.076	.053	-.186	-.023	.173	-.115
Neck3	-.001	.048	.068	-.149	.010	.215*	-.048
Upperback1	-.141	-.016	.031	-.176	-.043	.083	-.099
Upperback2	-.010	-.098	.045	.008	.050	.134	.032
Upperback3	.005	-.119	.041	-.021	.032	.112	.080
Lowerback1	-.012	.114	.099	-.127	.045	.125	-.084
Lowerback2	.318**	.104	-.066	.089	-.021	.063	.137
Lowerback3	.279**	.090	-.155	.065	-.113	-.024	.044
Shoulder1	.025	.006	-.112	-.168	-.164	.045	-.070
Shoulder2	-.088	.091	.135	-.070	.094	.079	.032
Shoulder3	.129	.008	-.106	-.062	-.122	-.109	.023
Elbow1	-.175	.053	-.028	.084	.000	.064	.006
Elbow2	-.198*	.089	.045	-.019	.033	.005	-.018
Elbow3	-.093	.157	.042	.043	.051	.099	-.036
Wrist1	-.172	.021	.002	-.012	-.003	.190*	.038
Wrist2	-.207*	.028	.050	-.071	.021	.245*	-.036
Wrist3	.098	-.024	.086	-.053	.056	.141	.045
Hip1	-.082	-.125	-.038	-.015	-.042	.057	.070
Hip2	.161	.003	.025	.079	.052	.080	.162
Hip3	.285**	-.071	-.102	.245**	.008	.013	.193*
Knee1	.023	.127	.043	.095	.071	.173	-.020
Knee2	.005	.046	.012	-.033	-.003	.011	-.054
Knee3	.180	.111	-.069	.029	-.053	-.086	-.057
Ankle1	-.056	-.032	-.018	-.110	-.052	.007	-.128
Ankle2	-.190*	.030	.074	-.035	.053	.154	.027
Ankle3	-.004	-.031	-.086	.032	-.065	.055	-.017

Note: 1= Trouble in a particular body region at any time during the last 7 days, 2= Prevented from doing your normal work due to pain in a particular body region last 12 months, 3= Trouble (such as pain, ache, discomfort, numbness) in particular body region in last 12 months.

Correlation between pain/discomfort history of different regions

Our study findings demonstrate a strong association between various factors, revealing patterns of pain/discomfort. The heat map correlation table (Figure 1) and Figure 2 visually represent these associations through color coding. At a significance level of $P \leq 0.01$, we observed a positive correlation among variables related to neck pain. Individuals experiencing neck pain in the past seven days were likely to have also experienced it in the past 12 months ($r=.341$) and reported activity limitations due to pain ($r=.412$). Dancers with recent neck pain also frequently reported upper back pain, showing a moderate association ($r=.524, .332, .364$), as well as shoulder pain ($r=.408, .270$). Moreover, dancers reporting upper back pain in the past year displayed a strong association with activity restrictions due to upper back pain ($r=.506, .322$). Those with recent lower back pain showed

a significant and positive association with lower back pain in the past 12 months and activity limitations due to pain. Furthermore, dancers with activity restrictions due to low back pain exhibited positive and significant relationships with hip ($r=.267, .311, .369$), knee ($r=.287, .280$), and ankle ($r=.280$) pain.

Discussion

In our study, 110 dancers participated, in these participants only 17.27% ($n=19$) participants did not report any pain or discomfort but 82.73% ($n=91$) reported pain at least at one site of the body. These results agree with the two studies conducted in Nigeria and New Zealand among professional dancers and pre-professional dancers. This study concluded the results with a prevalence rate approximately 86%. (Anulika Aweto, 2014; Lee et al., 2017) Similar findings were found in

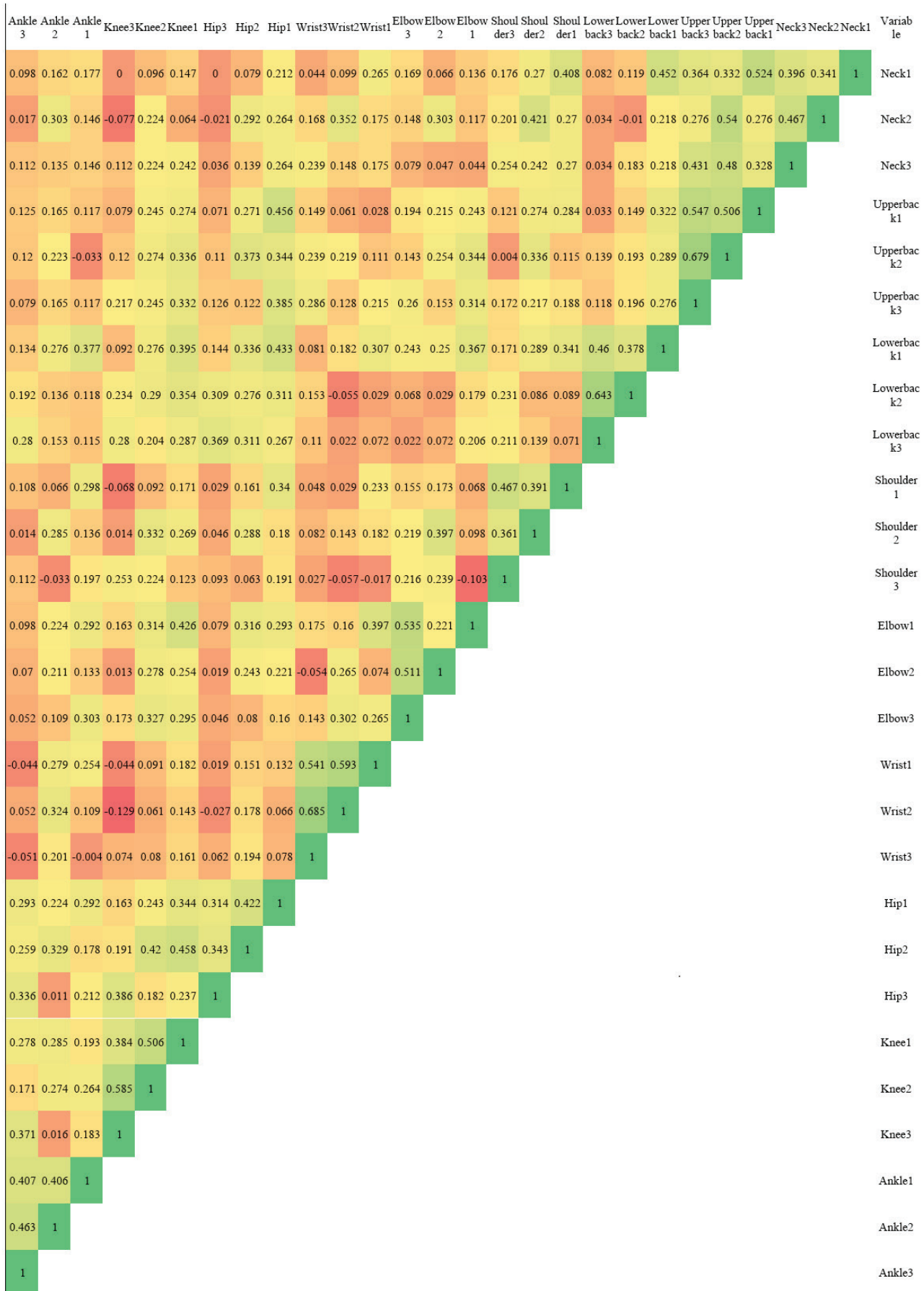


FIGURE 1. Heat map Correlation table between the different regions of the body

Note: 1= Trouble in a particular body region at any time during the last 7 days, 2= Prevented from doing your normal work due to pain in a particular body region last 12 months, 3= Trouble (such as pain, ache, discomfort, numbness) in particular body region in the last 12 months

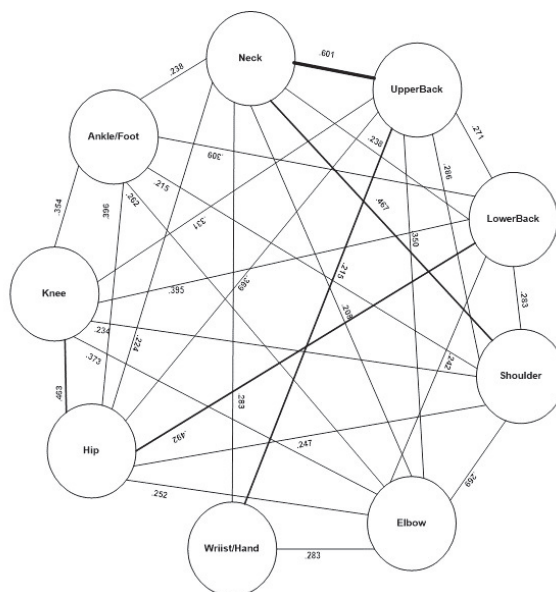


FIGURE 2. Correlation diagram representing association between the different regions of the body with respect to pain and discomfort (association is represented by thickness of lines; thickest line determines strongest correlation)

one of the studies conducted in India, stating the 81% of pain prevalence among the dancers (Nair et al., 2018). A study conducted in South Korea among Korean break dancers reported 95.2% of injury prevalence (Cho et al., 2009) and Ruanne et al. reported an injury rate ranging from 67% to 95% pre-professional dancers (Lai et al., 2006). This indicates there is a higher incidence of pain/discomfort among the dancers irrespective of the dance form they are practicing. Every form has a having high injury risk.

The findings of this study are supported by a retrospective cross-sectional study conducted in Germany stating that the lumbar spine is the most affected area (Adam et al., 2004). In our study, in all the three sections, the lower back site was the most affected, area followed by the ankle and knee. Most of the studies related to dancer's injury/pain have reported similar findings. The lower limbs are more affected as compared to the upper limbs, which could be due to the repetitive stamping of limbs on to the ground. The ground reaction force offered by the surface will eventually have an impact on the part, which is in contact. So prevention strategies should be adapted to prevent these issues.

The mean difference between males and females with respect to pain/ discomfort was not significant. This implies both the sexes have equivalent pain and discomfort rates. A similar retrospective study was conducted by Jarneja et al. (2019) with different result findings stating that females had significantly higher incidence rates as compared to males (Premelč et al., 2019).

To estimate the pain pattern relationship between the different anatomical sites, Pearson's correlation coefficient analysis was done, it revealed that neck pain is strongly and positively related to the upper back ($r=.601$) and moderately to the shoulder ($r=.467$). Lower back pain is moderately related to hip, knee and ankle pain ($r=.492$, $.395$, $.309$). Hip pain was associated with knee and ankle pain ($r=.463$, $.396$) and knee pain was associated with ankle ($r=.354$). The study concludes that the pain in the proximal anatomical sites was strongly and significantly related to the pain in the distal anatomical sites. This can give us an idea about the pain pattern and we will be able to stop the course of the pain to further sites. This is the

first study to find the correlation of pain/discomfort in different anatomical sites. So there is no literature to support it but one study was conducted among the European population to find the correlation between back and neck as per the working condition and the results supported our study by stating the facts that there was a moderate to strong correlation between the prevalence of back and neck pain (Rizzello et al., 2019).

Thus, the result findings of our study revealed that the lower back is the most common region to get affected in dancer's populations so we should strengthen our core musculature to prevent this pain and our study also shows that pain is not caused randomly in any part. The pain always follows a pattern. If we stop the pattern there is a strong probability that it will not transfer to other parts. The limitation of our study is the small sample size. This study should be done by taking a large sample size so that predictive analysis can be used to prevent the further distribution of the pain.

Conclusion

The findings from the study indicate a significant prevalence of pain among participants, particularly in the lower back region. Specifically, 30.8% of participants reported experiencing lower back pain in the past 7 days, and over the past 12 months, 51% of participants reported pain, discomfort, or numbness in the lower back, followed by 31% experiencing similar issues in the ankle/foot and knee.

When considering the impact of pain on activities of daily living (ADL), lower back pain was the most restrictive, affecting 30% of participants, with knee pain affecting 23.6%. Additionally, the study found a strong positive correlation between neck pain and upper back pain ($r=.601$), and a moderate correlation between neck pain and shoulder pain ($r=.467$). It also revealed that Lower back pain is moderately related to the hip ($r=.492$). Hip pain was associated moderately with the knee pain ($r=.463$). In conclusion, the study highlights that the lower back is the most prevalent area for pain among dancers, followed by the ankle and knee. This underscores the need for targeted interventions to address and mitigate pain in these regions to improve dancers' health and performance.

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Conflict of interest

The authors declare no conflict of interest

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ORIGINAL SCIENTIFIC PAPER

Relationships between Physical Fitness Attributes, Food Habits and Nutritional Knowledge in Youth Basketball Players

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Abstract

Understanding food habits and nutritional knowledge in youth athletes could contribute to optimal recovery and improving performance in basketball. Therefore, the aim of this study was to assess the relationships between physical fitness, with food habits and nutritional knowledge in basketballers. Twenty-three healthy youth players to the same elite basketball academy participated in this study. Basketballers performed a basketball fitness test battery including jumping tests [countermovement jump (CMJ) and drop jump (DJ)]; 5, 10 and 20 m-linear sprints; change of direction (COD) tests [Lane Agility Drill test and 505 test], and repeated change-of-direction (RCOD) sprint and completed a Turconi questionnaire based on eating habits and nutritional knowledge. While no significant ($p > 0.05$) correlations were found between physical fitness attributes and total score obtained in sections C and H in dietary questionnaire, significant correlations ($r = 0.606 - 0.971$; large - nearly perfect; $p < 0.001 - 0.003$) were found among jumping, COD, RCOD and LSST tests. These findings suggest that physical performance of youth basketball players is not influenced by their level of dietary behaviors and nutritional knowledge.

Keywords: associations, nutrition, conditioning, dietary behavior, team sport

Introduction

Basketball is an intermittent team-sport involving repeated transitions between offence and defense and frequent changes in movement, which require performing a great number of high-intensity actions such jumps and short-term and high intensity actions during match-play (Castillo et al., 2021). Specifically, basketball players cover 200 m at speeds exceeding $24.1 \text{ km} \cdot \text{h}^{-1}$ per minute, reaching peak speeds of over $19 \text{ km} \cdot \text{h}^{-1}$. In addition, approximately 16 accelerations, 16 decelerations and 0.1 jumps occur per minute during match-play (Vazquez-Guerrero et al., 2019). Since these high-intensity

actions are strongly associated with successful on-court performance (Mancha-Triguero, García-Rubio, Gamonales, & Ibáñez, 2021), adequate physical conditioning is required.

To optimize the training process, it is necessary to consider the specific characteristics of each population (Latzel et al., 2018). During adolescence (i.e., 12–14 years old), young basketball players undergo dramatic body changes, such as sudden increases in height and body mass, as well as alterations in motor control (Faigenbaum et al., 2009), so special attention must be focused on these groups to understand the motor abilities. Several laboratory and field test are available



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to assess the fitness of young basketball players (Drinkwater, Pyne, & McKenna, 2008). When selecting the test battery to use, relevant factors must be considered, such as the specific physical capacity to be assessed, the time required to conduct the test, and the necessary material and their cost (Morrison et al., 2022). Therefore, it is essential to understand the relationships between the different tests to carry out a comprehensive and efficient assessment of young basketball players in a short time and with limited resources. However, to date, no studies have been conducted with this purpose in young basketball players.

Nevertheless, there are certain variables that indirectly could influence in physical fitness, which are related to healthy lifestyle habits (Sánchez-Díaz, Yanci, Raya-González, Scanlan, & Castillo, 2021). Among them, the food habits of basketballers are considered relevant parameters for contributing to optimal recovery, performance, and injury risk in young athletes (Jeukendrup & Cronin, 2010), positively affecting to psychological variables such as self-concept and self-efficacy (Zaccagni, Rinaldo, Bramanti, Mongillo, & Gualdi-Russo, 2020). However, several factors may contribute to poor food habits in young athletes, so education strategies for improvement are necessary. In this regard, nutritional knowledge has been considered a key modifiable determinant of food habits (Trakman, Forsyth, Devlin, & Belski, 2016). Despite the presumed negative impact of food habits have on the physical fitness of young basketball players, and previous research have confirmed the association between nutrition knowledge and physical fitness (Nikolaidis & Theodoropoulou, 2014). Therefore, future research that explores the relationships between food habits and nutritional knowledge, with physical fitness attributes will enable practitioners to individualize the strategies used to optimize the performance of young basketball players.

To overcome the limitations and gaps in the literature, we assessed physical fitness attributes, food habits and nutritional knowledge during the in-season period. Specifically, the main aim of this study was to assess the relationships between physical fitness, with food habits and nutritional knowledge in U-14 basketball players. We hypothesized that physical fitness performance will be related with the scores obtained on the nutritional knowledge questionnaire.

Methods

Study design

An observational study design employing a cross-sectional approach was implemented. The participants underwent anthropometric measurements, performed a basketball fitness test battery, and completed a questionnaire on eating habits and nutritional knowledge. These assessments were conducted sequentially within a single testing session during the in-season phase (October) of a competitive season. The test battery included jumping tests [countermovement jump (CMJ) and drop jump (DJ)]; 5, 10 and 20 m-linear sprints; change of direction (COD) tests [Lane Agility Drill test and 505 test], and repeated change-of-direction (RCOD) sprint on an indoor basketball court (15–18 °C, 60–70% relative humidity). Five min of passive, standing recoveries were administered between tests. A month before the start of the study, players were familiarized with the physical tests by performing them during training. Prior to starting the physical test, participants performed a standardized ≈20-minutes

warm-up protocol, including running at a moderate intensity, jumps, dynamic stretching exercises and 20-m running bouts (Sánchez-Díaz et al., 2021). All tests were conducted between 4:00 and 7:00 pm three days after the official match. Players were advised to refrain from physical activity for two days prior to the test and were advised to arrive at the test to ensure adequate hydration and nutritional status. Fitness tests were conducted in groups of 3–4 players to ensure consistent recovery times between tasks for all players can be managed throughout the test.

Participants

The subjects were 23 healthy players (13.80±0.66 years; height, 167.45±9.35 cm; weight, 55.86±10.71 kg); body fat composition, 15.37±7.40%; muscle composition, 37.22±3.80%; body mass index, 20.04±2.21 kg·m⁻²) to the same elite basketball academy (i.e., Spanish Asociación de Clubes de Baloncesto [ACB] League) being members of teams competing in the first Spanish basketball Under 14 division. Players met the following inclusion criteria: a) completed all fitness assessments and the questionnaires, b) had exercised for than four consecutive weeks in the two months preceding the test, and c) were not taking any drugs, medications or ergogenic supplements that could alter performance. All players completed on-field team training three times a week, including pre-match training and physical training. Each session usually lasting 75 to 90 minutes, and players participated in an official game a week during the test period.

Before providing written informed consent, players and their legal guardians were informed of the procedures, potential risks, and benefits of participating in this research. This research was performed in accordance with the 2013 Declaration of the World Medical Association of Helsinki and was approved by the University Ethics Committee on July 30, 2020 (code: FUI1-P007).

Physical fitness tests

Jumping tests

A photocell system (Optojump, Microgate™, Bolzano, Italy) was used to measure the lower limb performance. Players performed two trials each of the CMJ and DJ with a 45-s of passive, standing recovery, and the highest jump was used for subsequent analysis in each test (Mancha-Triguero et al., 2021; Sánchez-Díaz et al., 2021). For CMJ, players performed unilateral and bilateral jumps and were instructed to perform a downward movement followed by a complete, explosive extension of the lower limb, maintaining their hands on their hips while jumping as high as possible (Heishman et al., 2020). For DJ trials, players were instructed to step from a wooden box (30 cm high) and immediately following ground contact, jump for maximal height as quickly as possible. The between-trial intraclass correlation coefficients (ICCs) for jump height attained during both tests was 0.91.

Change-of-direction (COD) speed tests

To assess COD speed, we used the Lane Agility Drill test and the 505 COD test according to the protocols described by Castillo et al. (Castillo et al., 2021). Players completed two trials of each test with 90 s of passive, standing rest between trials. On both test a photocell system (Microgate™ Polifemo, Bolzano, Italy) was used and the fastest trial was registered for further analysis in each test. The ICCs were 0.88 for the Lane

Agility Drill test and 0.74 for the 505 COD test in the current sample of players.

Repeated change-of-direction (RCOD) sprint test

Players performed a single attempt of the RCOD sprint test consisting of 5 × 30-m shuttle sprints (15 m + 15 m) interspersed with 30 s of passive recovery between each sprint, according to the protocol described by Castillo et al. (2021). A single pair of photoelectric cells (Microgate™ Polifemo, Bolzano, Italy) were placed at the start line to record performance time (s) during each shuttle. The sum of all shuttle sprint times (total performance time) was calculated and the value obtained was used for subsequent analysis.

Linear straight sprint test (LSST)

Linear speed was assessed by means two trials of a 20-m linear sprint using four pairs of photoelectric cells (Microgate™ Polifemo, Bolzano, Italy) to record sprint split times at 5 m, 10 m and 20 m. Players started the sprint test half a meter behind the first timing gate on their own volition. The fastest time (s) for each split (irrespective of the trial) was used for subsequent analysis. The ICCs were higher than 0.87 for all sprint distances.

Questionnaires

The questionnaires were conducted following a typical week in which players maintained their normal daily routines involving attendance at school on five days, three on-court training sessions, and participation in an official match during the weekend.

Dietary questionnaire

The food habits and nutritional knowledge of players were evaluated through the utilization of a dietary questionnaire (Turconi et al., 2003). Originally comprising ten sections but only two sections (i.e., C and H) were considered pertinent to this study (Turconi et al., 2003). In addition, slight modifications were made to each section, incorporating a 'non-reported' option for every item. This allowed players to indicate

instances where they did not feel obliged, lacked comfort, or possessed incomplete information regarding specific quantities when responding to a question. The two sections from the dietary questionnaire used included:

Section C pertains to the examination of breakfast constituents, the frequency of daily meals, the consumption of fruits and vegetables, as well as the intake of soft drinks and alcoholic beverages. This section comprises 14 questions, but we used 13 items to capture the perceived frequency of consumption. The maximum attainable score for this section was 52.

Section H delves into nutritional knowledge and encompasses 11 questions. Each question revolves around crucial nutritional aspects, such as the functions of specific macronutrients and micronutrients, as well as the significance of nutrition. For each question, players are provided with four answer choices, of which only one is correct. A point is awarded for each accurate response, while no points are granted for incorrect answers. The maximum total score achievable in this section was 11.

Statistical analysis

Data are presented as mean ± standard deviations (SD) for quantitative variables or frequencies and percentages for qualitative variables. After performing Shapiro-Wilk test, the Spearman correlation test was used to assess the correlation between physical fitness attributes, and the total scores in sections C and H in the dietary questionnaire for each group of males and females. The following scale of magnitudes was used to interpret the correlation coefficients: <0.1, trivial; 0.1 to 0.3, small; 0.3 to 0.5, moderate; 0.5 to 0.7, large; 0.7 to 0.9, very large; >0.9, nearly perfect (Hopkins, Marshall, Batterham, & Hanin, 2009). Data analysis was carried out using the JASP 0.16.3.0 software (University of Amsterdam, Amsterdam, Netherlands). Statistical significance was established at $p < 0.05$.

Results

Tables 1, 2 and 3 shows the descriptive results of physical fitness attributes, food habits and nutrition knowledge in youth basketballers, respectively.

Table 1. Descriptive results of physical fitness attributes in youth basketballers.

Variables	Mean	Standard deviation	Minimum	Maximum
Jumping test				
CMJ (cm)	27.67	6.93	18.30	49.10
CMJr (cm)	13.62	4.11	6.50	21.80
CMJl (cm)	14.17	4.29	6.90	24.20
DJ (cm)	27.00	6.34	17.10	41.70
Change of Direction Ability (CODA) test				
Lane Agility Drill (s)	14.29	0.82	12.80	16.45
505-COD (s)	2.63	0.17	2.20	2.93
Repeated change-of-direction (RCOD) sprint test				
RCODtotal (s)	33.23	1.75	29.16	37.02
Linear Sprint Straight Test (LSST)				
5 m sprint (s)	1.19	0.28	0.99	2.40
10 m sprint (s)	2.00	0.14	1.72	2.33
20 m sprint (s)	3.56	0.26	3.00	4.14

ES: effect size; CI: confidence limit; CMJ: countermovement jump; CMJr: countermovement jump with right leg; CMJl: countermovement jump with left leg; DJ: drop jump.

Table 2. Descriptive results of food habits (%) in youth basketballers.

Items	Frequency consumption in percentage (%)				
	Never	Sometimes	Often	Always	NR
C1 Breakfast	0	0	4.76	95.24	0
C2 Beverage breakfast	Tea	Juice	Chocolate	Milk	NR
	5	0	5.00	90.00	0
C3 Eat breakfast	Cheese	Pizza	Bread	Fruit	NR
	0	0	90.48	9.52	0
	Never	Sometimes	Often	Always	NR
C4 Fruit	14.29	14.29	28.57	42.86	0
C5 Vegetables	14.29	42.86	28.57	14.29	0
C6 Cake	33.33	42.86	19.05	4.76	0
C8 Three meals	0	0	9.52	90.48	0
C9 Diet	Monotony	Different on weekend	Different sometimes	Different all days	NR
	4.76	4.76	14.29	76.19	0
C10 Diet based on	Different all days	Carbohydrate	Lipids	Protean	NR
	76.47	5.88	5.88	11.77	0
C11 Snacks	Sweets	Fried	Bread	Fruit	NR
	10.00	15.00	20.00	55.00	0
C12 Beverages	Juice	Wine, beer	Refresh	Water	NR
	0	0	4.76	90.48	4.76
	Never	Sometimes	Often	Always	NR
C13 Milk	0	0	9.52	90.48	0
C14 Water	0	9.52	9.52	80.95	0
C Total	41.48 ± 2.96				

Note: All questions were obtained from section C in the Turconi questionnaire; total score is presented as mean ± standard deviation; NR = not reported.

Table 3. Descriptive results of nutrition knowledge (%) in youth basketballers.

	All	
	Correct (%)	Incorrect (%)
H1 Carbohydrates	60	40
H2 Fibre	50	50
H3 Fat	35	65
H4 Protein	85	15
H5 Calories	45	55
H6 Energy	50	50
H7 Vitamins and minerals	45	55
H8 Balanced diet	65	35
H9 Daily energy expenditure	65	35
H10 Biological foods	65	35
H11 Transgenic foods	40	60
H Total	6.05 ± 2.16	

Note: All questions were obtained from section H in the Turconi questionnaire; total score is presented as mean ± standard deviation for the number of correctly answered questions.

Table 4 presents the relationships among physical fitness attributes, with food habits, and nutrition knowledge in youth basketballers. While no significant ($p>0.05$) correlations were found between physical fitness attributes and total score ob-

tained in sections C and H in dietary questionnaire, significant correlations ($r=0.606$ – 0.971 ; large - nearly perfect; $p<0.001$ – 0.003) were found among jumping, COD, RCOD and LSST tests.

Table 4. Relationships between physical fitness attributes, and total scores in sections C and H in the dietary questionnaire for youth basketballers.

		CMJ	CMJr	CMJI	DJ	Lane Agility Drill	505- COD	RCODtotal	5 m sprint	10 m sprint	20 m sprint
CMJr	Spearman's rho	0.840									
	p-value	< 0.001									
CMJI	Spearman's rho	0.878	0.818								
	p-value	< 0.001	< 0.001								
DJ	Spearman's rho	0.875	0.754	0.890							
	p-value	< 0.001	< 0.001	< 0.001							
Lane Agility Drill	Spearman's rho	-0.702	-0.633	-0.740	-0.775						
	p-value	< 0.001	0.002	< 0.001	< 0.001						
505-COD	Spearman's rho	-0.623	-0.730	-0.636	-0.630	0.782					
	p-value	0.002	< 0.001	0.001	0.002	< 0.001					
RCODtotal	Spearman's rho	-0.724	-0.719	-0.745	-0.787	0.785	0.717				
	p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001				
5 m sprint	Spearman's rho	-0.840	-0.828	-0.734	-0.766	0.606	0.627	0.756			
	p-value	< 0.001	< 0.001	< 0.001	< 0.001	0.003	0.002	< 0.001			
10 m sprint	Spearman's rho	-0.808	-0.810	-0.790	-0.805	0.720	0.681	0.836	0.928		
	p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001		
20 m sprint	Spearman's rho	-0.779	-0.739	-0.779	-0.799	0.754	0.696	0.804	0.871	0.971	
	p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	
C Total	Spearman's rho	-0.121	-0.158	0.091	0.120	0.111	0.102	-0.030	0.132	0.066	0.035
	p-value	0.623	0.518	0.711	0.624	0.650	0.678	0.901	0.589	0.789	0.887
H total	Spearman's rho	-0.148	-0.051	-0.163	-0.269	0.292	0.304	-0.102	-0.051	0.003	0.020
	p-value	0.557	0.840	0.517	0.281	0.239	0.220	0.687	0.841	0.992	0.936

Note: Correlation established at $p < 0.05$

Discussion

The main aim of this study was to assess the relationships between physical fitness, nutritional knowledge, and nutritional habits in U-14 basketball players. We hypothesized that physical fitness performance will be related with the scores obtained on the nutritional knowledge questionnaire. However, no significant correlations were found between physical fitness attributes, food habits and nutritional knowledge since the different physical fitness attributes only correlated significantly between them.

The current findings show significant negative correlations among jumping performance and COD, RCOD and LSST tests. These findings are somewhat expected. Previous studies with adult basketball players reporting significant negative correlations between CMJ performance and repeated sprint ability (RSA) total time ($r = -0.74$; $p < 0.01$) (Stojanovic, Ostojic, Calleja-González, Milosevic, & Mikic, 2012) and COD performance ($r = -0.594$; $p < 0.01$) (Alemdaroğlu, 2012). Likewise,

youth male basketball players showed significant negative correlations between both bilateral and unilateral CMJ height and COD performance ($r = -0.777$ – -0.656 ; $p < 0.05$) (Pérez-Ifrán et al., 2022). On the other hand, the current relationships are similar to previous findings with youth basketball players indicating that COD performance is positively related to RSA total time and RSA best time ($r = 0.720$ – 0.805 ; $p < 0.05$) (Pérez-Ifrán et al., 2022) and to 10- ($r = 0.53$; $p < 0.05$) (Scanlan et al., 2021) and 15-m sprint times ($r = 0.814$; $p < 0.05$) (Pérez-Ifrán et al., 2022). The relationships between jumps performed in the vertical axis and short-distance sprinting, COD and RCOD performed in the horizontal axis is due to the similarity of biomechanical and neuromuscular prerequisites and energy systems involved in the acceleration and the application of force in different axes. However, the magnitude of the relationships varies depending on the test characteristics (e.g., unilateral vs bilateral actions, sprint distance, the number of turns and turning angles associated with each COD test) or individual

performance (e.g., jumping mechanics, running mechanics – forward vs. lateral-shuffles –, adjustments in stride pattern to decelerate and reaccelerate, COD technique – sharp vs. round –) (Suarez-Arrones et al., 2020). Therefore, youth basketball strength and conditioning coaches should consider these results when select tests to monitor physical performance of players depending on their physical fitness attributes. This information may help them to develop the most efficient, cost-effective, and practical testing procedure.

A balanced and appropriate diet in youth athletes is especially important to cope with physiological maturation needs and their athletic (Bergeron et al., 2015). In fact, the sports nutrition practices of U-18 Spanish basketball players have been reported to be deficient and inadequate, which may interfere with their health and performance (Escribano-Ott, Mielgo-Ayuso, & Calleja-González, 2022). In this regard, nutrition knowledge is a key factor for youth basketball players development, being highly relevant to establish correct nutritional habits, because consolidates the rationales to improve dietary behavior and grant the required competencies to make adequate food choices (Vázquez-Espino, Rodas-Font, & Farran-Codina, 2022). However, no significant relation was found between nutrition knowledge and physical fitness attributes in youth basketball players. Measurement of nutrition knowledge is challenging due to limitations in the adequacy, validity and precision of the tools used, which this could affect the reliability of the results. Thus, the absence of relationships could be explained by the lack of connectedness between the theoretical and declarative aspects inquired in the nutrition knowledge questionnaire and the objective fitness data obtained with the physical fitness tests.

A lower level of nutritional knowledge may contribute to poor nutritional habits (e.g., food choices and dietary behavior) (Vázquez-Espino et al., 2022). However, association between nutritional knowledge and food habits has previously been reported either as weak or lacking (Spronk, Kullen, Burdon, & O'Connor, 2014). Nutritional knowledge is only one factor that influences food habits, and other factors may be more important. Cultural factors, appetite, taste predilections or attitude towards nutrition have been reported to motivate food choices and dietary behavior in both general population and athletes (Birkenhead & Slater, 2015). Other related factors are changes in body composition or aesthetics (Byrne & McLean, 2002; Sundgot-Borgen & Torstveit, 2004) and peer pressure and teammates' choices. The complex interaction between these factors and the lack of connectedness between the theoretical and declarative aspects inquired in the food habits questions and the fitness attributes measured by the physi-

cal fitness tests may explain the absence of relationships. In addition, youth athletic performance not only relies on optimal eating patterns, but also depends on coaching styles and physical conditioning programs (Bergeron et al., 2015) and genetic variation (Guth & Roth, 2013). Thus, the relationships between food habits and physical fitness tests may also be influenced by other training- or genetic-related factors that were not measured in this study.

Some limitations should be acknowledged. Firstly, participants recruited were U-14 players of the same elite basketball academy, so the generalization of the findings is restricted. Thus, further studies involving a larger sample and using a wider range of age categories and levels of play (i.e., regional, international) are required. Secondly, sample include both males and female basketball players, so gender-related differences in physical fitness may influence the relationships obtained. Sánchez-Díaz et al. (2021) reported that U-14 male players outperformed female basketball players in jump, sprint, CODA and RCOD tests, and Mancha-Triguero et al. (2021) reported higher performance in jumping tests among U-14 male players compared with their female counterparts. Additionally, it is plausible that gender-related differences in nutritional knowledge influenced the magnitude of relationships, since female senior athletes who were involved in different sports, including basketball, reported greater nutritional knowledge compared to senior male athletes (Ali, Al-Siyabi, Waly, & Kilani, 2015). Thus, future studies should analyze whether the relationships between physical fitness, food habits and nutritional knowledge are influenced by the players' gender. Thirdly, the maturity status performance was not measured in the current study. This information could help practitioners to have a better understanding of the relationships between different physical fitness attributes in youth basketball players. Accordingly, the maturity status should also be measured in future studies. Finally, potential social desirability and self-reporting biases may be considered when interpreting the self-reported data collected with the questionnaire (Althubaiti, 2016), so the current findings should be analyzed with caution.

Conclusion

The findings of the current study indicate that no significant relationships exist between physical fitness attributes, food habits and nutritional knowledge in youth basketball players. The different physical fitness attributes only were significantly related between them. This suggest that physical performance of youth basketball players is not influenced by their level of dietary behaviors and nutritional knowledge.

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Conflicts of interest

The authors certify that they have no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Adaptation of the Oslo Sports Injury and Health Problems Questionnaire to the Bahasa Version and Its Application for Athletes Surveillance

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Abstract

The Oslo Sports Trauma Research Center (OSTRC) questionnaire on injuries and health problems is one of the recommended methods for monitoring athletes' health problems, but until now it has not been available and validated in a Bahasa version. This study aims to develop the OSTRC Questionnaire into Bahasa and assess its application in adolescent athlete surveillance. The research process is divided into two stages; The adaptation and validity assessment according to ISPOR rules involving 42 adolescent athletes for 2 weeks and the second stage is 8 weeks of surveillance involving 46 adolescent athletes. Results of the analysis found that the Bahasa version of the OSLO Questionnaire for shoulder, knee, ankle, and health problems has good validity and reliability with Cronbach- α 0.905, 0.940, 0.933, and 0.840 respectively. The intraclass correlation coefficient for health problems was 0.905, the shoulder region was 0.94, the knee region was 0.933 and the ankle region was 0.840, with all p-values <0.001. The sensitivity of the health problems questionnaire reached 97.6% and specificity 99.4%, while the sensitivity of the injury questionnaire reached 100% and specificity reached 99.4%. The Bahasa version of OSLO Sports Trauma Research Center Injury and Health Problems is valid and reliable and shows high sensitivity and specificity to detect and separate athletes with injuries or health problems.

Keywords: injury surveillance, health surveillance, injury, OSTRC Bahasa Version, adolescent athlete

Introduction

Theisen et al. (2014) said in the last decade, the number of adolescent athletes has grown rapidly. Several studies (Patel et al., 2017; Theisen et al., 2014) revealed that the incidence of sports injuries at the age of 12-19 years reached 1.3-3.0/1000 hours of activity for racket sports, 2.0-3.8/1000 hours for individual sports, and 4.6-6.5/1000 hours for team sports activities, with 40% of injuries experienced being overuse injuries. Other studies (Carragher et al., 2019; Theisen et al., 2014) also mentions that a total 31.1% of injuries cause time loss of 7-28 days and 68.9% take 7 days to recover from illness. Injuries and illness cause impaired athlete development and Application (Carragher

et al., 2019; Theisen et al., 2014). The International Olympic Committee (IOC) recommends carrying out regular monitoring of health problems to reduce the risk of athlete illness and injury (Engebretsen et al., 2014). One method the IOC recommends is the Oslo Sports Trauma Research Center (OSTRC) questionnaire as a self-reported tool to monitor athletes' health problems (Engebretsen et al., 2014). This questionnaire has 4 questions that can assess participation, modification, performance, and symptoms experienced by athletes when there are health problems (Clarsen et al., 2014). OSTRC questionnaire can also be used for Para-athletes (Hirschmüller et al., 2021). This ability has made many countries adapt and develop surveillance systems using



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the OSTRC injury and health questionnaire, including French, German, Danish, Japanese, Thai, and Persian. (Beaudart et al., 2023; Hirschmüller et al., 2017; Jorgensen et al., 2016; Kaewkul et al., 2021; Mashimo et al., 2021; Mirkarimpour et al., 2018). Until now in Indonesia, there is no self-reported questionnaire for monitoring athletes' health condition, so this study aims to adapt and validated the OSTRC questionnaire to the Bahasa version and assess its application in monitoring the health problems of adolescent athletes in Indonesia.

Methods

Procedures

The OSTRC questionnaire was developed by Clarsen et al, consists of four questions with a total score range of 0 to 100, with details of questions 1 and 4 having 4 answer choices with answer scores A=0, B=8, C=17, D=25, and questions 2 and 3 have answer scores of A=0, B=6, C=13, D=19 and E=25 (Clarsen et al., 2013). In 2020 the OSTRC questionnaire underwent improvement in reporting methods, namely; question 1 is used to assess participation, question 2 to assess modification, question 3 to assess Application, and question 4 to assess pain, with each answer score for the 4 questions being A=0, B=8, C=17, D=25 (Clarsen et al., 2020), so the answer score range is between 0-100, this score is also called the severity score. For this study, the latest update of the 2020 OSTRC questionnaire was used to be adapted into a Bahasa version. This study is divided into 2 stages, namely; (1) the adaptation and validity stage and (2) the surveillance stage. The adaptation process stage is carried out based on the International Society for Pharmacoeconomics and Outcomes Research (ISPOR) guidelines for Translation and Cultural Adaptation (TCA) (Wild et al., 2005). The adaptation and translation process were carried out in five stages, namely; Stage 1; forward translation: using 2 independent translators (T1 and T2) who translate the English version of OSTRC into Bahasa. Stage 2; reconciliation: The translated texts (T1 and T2) in the Bahasa version were then combined by the research team into a Bahasa version of the manuscript, which was then assessed for suitability of meaning and meaning in the language by the 3rd translator who is a Bahasa expert and became the initial version of the OSTRC questionnaire in Bahasa Indonesia (T123). Stage 3; back translation review and harmonization: The language version of the OSTRC questionnaire (T123) was again translated into English by the research team and reassessed for the suitability of the language version, the original version, and the back-translated version by the 4th independent translator (T1234). Stage 5: Cognitive debriefing until proofreading is carried out by the 5th translator, to assess the suitability of the meaning and meaning with the original text, so that the Bahasa version of the final questionnaire text is produced. This questionnaire will then be tested for validity and reliability on adolescent athletes, which will be done by filling out the questionnaire twice over 7 days. In this stage, the injury questionnaire was divided into three regions; shoulder, knee, and ankle.

The surveillance stage was carried out for 8 weeks on adolescent athletes at the Cibubur Youth Athlete Sports Center. Every 7 days researcher will send a questionnaire via a messaging application to the athlete. In this stage, athletes are free to report the location and number of injuries they have experienced.

Participants

Sample size calculations for the validity stage were calculated based on the ratio of subjects to items to be validated (Anthoine et al., 2014; Osborne & Costello, 2004). For this study, a ratio of

10:1 was used, so the injury and health questionnaire required a minimum of 40 subjects. For the surveillance stage, sample size calculation was calculated using the equation formula from Haldun Akoglu's study (Akoglu, 2022), with a significance level of 0.05 and test power of 0.90, so the surveillance stage required a minimum of 19 subjects, and in this stage, the researcher used the population at Cibubur Youth Athlete Sports Center, with a total of 50 athletes. The subject inclusion criteria for this research are: (1) Athletes aged 13-18 years, (2) Have ever participated in a sports club or sports school, (3) Have participated in training at a sports club or sports school. Subject exclusion criteria for the validity stage are: (1) Did not fill out the questionnaire completely, (2) Did not fill out the questionnaire at the specified time, and (3) Refused to confirm the condition. Meanwhile, the exclusion criteria at the surveillance stage are: (1) Not filling out the questionnaire for 2 consecutive weeks, (2) Refusing to confirm the condition. A total of 42 athletes took part in the validity stage. However, only 40 athletes completed this stage. Two athletes were excluded because they only completed the questionnaire once and refused to continue. At the surveillance stage, only 46 athletes had completed this stage, for a reason; two athletes entered the national team, one athlete withdrew from the Cibubur Youth Athletes Sports Center and one athlete was excluded because he did not fill out the questionnaire for 2 consecutive weeks.

Measurement

In the first and second stages of the research, the Questionnaire will be given online every Saturday morning and the deadline for completion is 14.00 on Sunday. Athletes will be reminded 4 times, namely; on Friday afternoon, Saturday afternoon, Saturday evening, and Sunday afternoon. All participants who reported injuries and illnesses were confirmed and recorded by the medical team. The medical team does not know the athlete's questionnaire answers.

Statistics

The validity test was carried out using the Pearson Correlation test and reliability was assessed from the Cronbach's α and intraclass correlation coefficient (ICC) values, with the expected Cronbach's α value being more than 0.6 and the intraclass correlation coefficient (ICC) value being more than 0.8 (Koo & Li, 2016). All reports from the surveillance stage were used to assess the application of the questionnaire. The application will be evaluated using the receiver operating characteristic curve (ROC) and the area under the ROC curve (AUC). Therefore, an AUC of 0.81 to 0.9 is considered excellent, and greater than 0.9 is considered outstanding in distinguishing injured from uninjured, and ill or not ill subjects. Data were prepared and analyzed using the SPSS 18 series.

This research has passed an ethical process review from the Ethical Committee of the Faculty of Medicine, University of Indonesia – Cipto Mangunkusumo Hospital (approval no: KET-840/UN2.F1/ETIK/PPM.00.02/2022).

Result

There were no problems in the translation process questionnaire from English to Bahasa in both the forward and backward translation. However, all suggestions from linguists and the research team were carefully considered, so there were slight changes, such as: adding the words practice or competition to the second question that assesses performance. The characteristics of athletes who completed the validity stage are shown in Table 1.

Table 1. Characteristics of athletes in the adaptation and validation stages

Variable	N	%
Gender:		
Male	23	57.5
Female	17	42.5
Type of sports:		
Pencak silat	6	15
Athletics	5	12.5
Archery	4	10
Sepak takraw	4	10
Pétanque ball	3	7.5
Taekwondo	3	7.5
Basketball	2	5
Handball	2	5
Volleyball	2	5
Cricket	2	5
Karate	2	5
Synchronized swimming	2	5
Wushu	2	5
Water polo	1	2.5

The validation test was carried out from the results of 40 athletes with a range of 13-18 years, using the Pearson Correlation Test between the score of each answer compared to the total severity score. The r value from the Pearson Correlation Test was compared with the r table value seen via $df=N-2$, so the r table $=0.264$. If the calculated r result is greater than the able r value, then the OSTRC Bahasa version questionnaire is considered valid. The results of the Pearson

Correlation Test are shown in Table 2.

Reliability was assessed by test-retest carried out within 7-day intervals, the entire questionnaire showed very good test-retest reliability values, with an intraclass correlation coefficient value; health problems 0.905 (95% CI; 0.846-0.945), shoulder region 0.94 (95% CI; 0.902-0.965), knee region 0.933 (95% CI; 0.891-0.961) and ankle region 0.840 (95% CI; 0.740-0.908). The internal consistency (IC) of the OSTRC question-

Table 2. Item-total correlation and effect of removing items on internal consistency

OSTRC Bahasa Version	Item-total correlation (r)*	Cronbach's α if the item deleted
Health problems:		
Question 1 (Participation)	0.915	0.862
Question 2 (Modification)	0.881	0.893
Question 3 (Performance)	0.931	0.846
Question 4 (Symptoms)	0.829	0.908
Shoulder region:		
Question 1 (Participation)	0.989	0.88
Question 2 (Modification)	0.677	1
Question 3 (Performance)	0.989	0.88
Question 4 (Symptoms)	0.989	0.88
Knee region:		
Question 1 (Participation)	0.984	0.879
Question 2 (Modification)	0.899	0.938
Question 3 (Performance)	0.628	0.938
Question 4 (Symptoms)	0.739	0.875
Ankle region:		
Question 1 (Participation)	0.984	0.645
Question 2 (Modification)	0.899	0.736
Question 3 (Performance)	0.628	0.875
Question 4 (Symptoms)	0.739	0.853

*Pearson correlation test result

naire on injury and health problems Bahasa version was also determined from 40 athletes. The results showed that the OSTRC injury questionnaire in the Bahasa version had very good internal consistency for the three regions; shoulder, knee, and ankle regions. Cronbach's α of the OSTRC injury questionnaire for the shoulder, knee, and ankle regions were 0.940, 0.933, and 0.840 respectively. Meanwhile, the Bahasa version of the OSTRC health problems questionnaire also had a very good internal consistency with Cronbach's α 0.905. At this stage, testing was also carried out by assessing Cronbach's α if one of the question items was removed and if changes in Cronbach's α were still within the acceptable range for both language versions of the OSTRC questionnaire. If the item was removed for the knee region questionnaire, there were 2 items (question number 2 and 3), and the ankle region also had 2 items (question

number 3 and 4) which slightly changed Cronbach's α greater than Cronbach's α of all items. This result showed that no item disproportionately influences the overall assessment reliability. The results of Cronbach's α calculations are shown in Table 2. Overall, the Bahasa version of the OSTRC questionnaire, which assesses health problems, shoulder region, knee region, and ankle region, had very good validity and reliability.

In the second stage, the final version of the OSTRC Bahasa version was applied for surveillance for 8 weeks. Athletes who took part in the surveillance stage were different from athletes who took part in the adaptation and validation stages. The athlete characteristics who completed the second stage; age range 14-18 years, length of time practicing sports 8.45 ± 2.61 years, all the athletes in the regular training program, and other characteristics are described in Table 3.

Table 3. Characteristics of athletes who finish the surveillance stage

Variable	N	%
Gender:		
Female	27	58.7
Male	19	41.3
Type of sports:		
Badminton	13	28
Pencak silat	7	15.2
Taekwondo	6	13
Swimming	6	13
Athletics	5	11
Archery	5	11
Weightlifting	3	6.6
Gymnastic	1	2.2

Table 4. Characteristics of questionnaire answers

Questions	Health Problem		Injury	
	N	%	N	%
Question 1 (Participation):				
Answer A (score 0)	307	84.57	307	81.2
Answer B (score 8)	42	11.58	59	16.6
Answer C (score 17)	8	2.20	5	1.3
Answer D (score 25)	6	1.66	7	1.9
Question 2 (Modification):				
Answer A (score 0)	335	92.28	338	89.4
Answer B (score 8)	16	4.41	28	7.4
Answer C (score 17)	7	1.93	4	1.1
Answer D (score 25)	5	1.38	8	2.1
Question 3 (Performance):				
Answer A (score 0)	325	89.54	320	84.7
Answer B (score 8)	25	6.89	40	10.6
Answer C (score 17)	6	1.66	11	2.9
Answer D (score 25)	7	1.93	7	1.9
Question 4 (Symptoms):				
Answer A (score 0)	311	85.68	309	81.7
Answer B (score 8)	39	10.75	50	13.2
Answer C (score 17)	11	3.04	17	4.5
Answer D (score 25)	2	0.56	2	0.5

During 8 weeks of surveillance, 36 (78.2%) athletes filled out the questionnaire 8 times, and 10 (21.8%) athletes filled out the questionnaire 7 times. The level of adherence every week was between 91.66-100%, with average adherence in the surveillance period reaching 97.32%. During the surveillance period, there were 378 reports from the injury questionnaire, consisting of 307 reports of no injuries and 71 reports of injuries. Examinations carried out by the medical team found 69 confirmed reports of injury and 2 reports of no injury. The median injury severity score reached score 32, with a score range of 16 to 100. Meanwhile, for health problems, 363 reports were found, consisting of 307 reports of no health problems and 56 reports of having health problems. A confirmation examination of 56 reports of health problems found 42 reports confirmed as having illness and 14

confirmed had no illness. The median severity score of illness reached score 32, with a score range of 8 to 100. Mann-Whitney test showed a statistically significant difference in severity score between athletes with injury and non-injury as well as severity score between athletes with illness and no illness ($p < 0.001$). An overview of the questionnaire answers is presented in Table 4.

For the application in the surveillance stage, the AUC value of the Bahasa version of the OSLO health problems questionnaire score was 0.973 (95% CI; 0.956-0.989) with $p < 0.0001$ and the AUC value of the Bahasa version of the OSLO injury questionnaire score was 0.996 (95% CI; 0.990- 1) with $p < 0.0001$. The Receiver Operating Characteristic (ROC) curve and Area Under Curve (AUC) for the Bahasa Version of the OSTRC Questionnaire are shown in Figure 1.

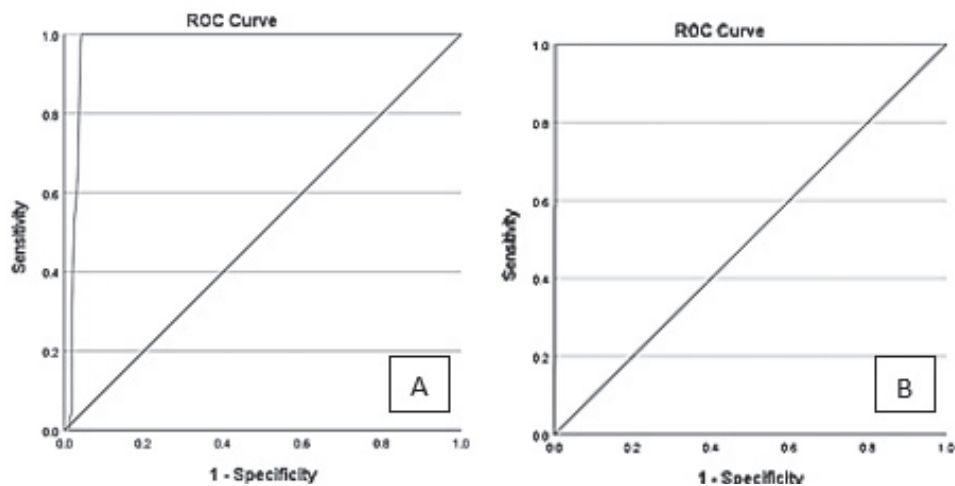


FIGURE 1. The ROC and Area AUC for the Bahasa Questionnaire: A; health problems, B; Injury

The application of the OSTRC Injury and Health Problems Questionnaire Bahasa version also evaluated the cut-off value of the severity score in the illness and injury group. The cut-off value of the health problems questionnaire was 12, which had

a sensitivity of 97.6% and a specificity of 96% and the cut-off value of the injury questionnaire was 8, with a sensitivity of 100% and a specificity of 99.4%. Details of the curve intersection points are shown in Figure 2.

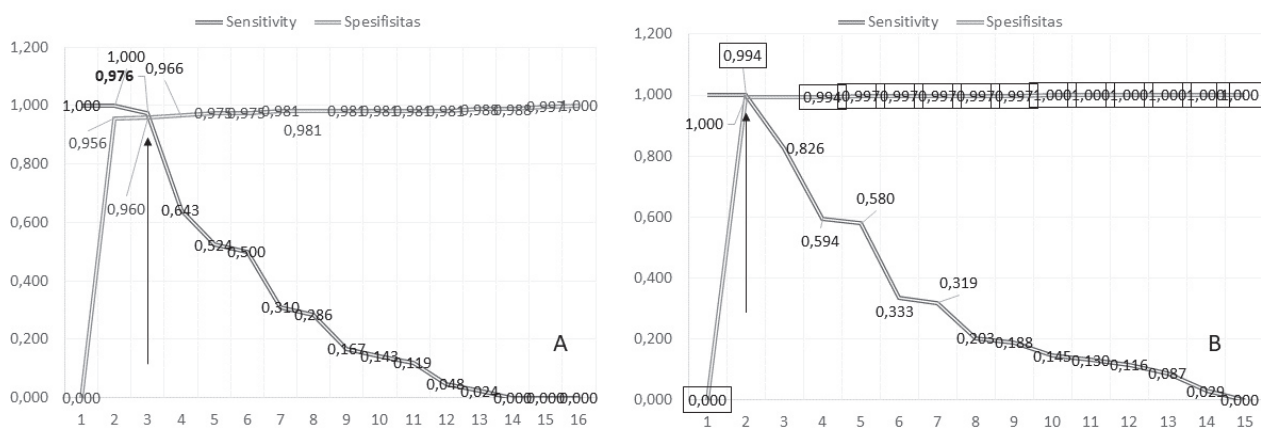


FIGURE 2. ROC cut-off point curve for A: health problems questionnaires and B: injury questionnaires

Discussion

The Bahasa version of the OSTRC Health Problems and Injuries Questionnaire has good validity and reliability in identifying health problems and injuries. This assessment is seen from two things, namely: (1) Validity and reliability test and (2) Sensitivity and specificity. The validity of this study was assessed by the r -value from the Pearson Correlation Test

which reached: (1) health problems ($r = 0.829$ to 0.931), shoulder region ($r = 0.677$ to 0.989), knee region ($r = 0.628$ to 0.984), and ankle region ($r = 0.628$ to 0.984). The reliability of the questionnaire assessed using Cronbach's Alpha reached the range of 0.840 to 0.940, and also the intraclass correlation coefficient reached the range of 0.840 to 0.940. These results are in accordance with Julie Gallagher et al, regarding self-reported

measurement for injury and illness for athletes which must have a Cronbach's Alpha between 0.7-0.95, as well as assessing reliability based on intra-class correlation coefficient (ICC) ≥ 0.7 (Gallagher et al., 2017). The grouping of injury areas into three regions was also carried out based on research from Kaewkul et al and research from Nagano et al (Kaewkul et al., 2021; Mashimo et al., 2020), and according to Ben Clarsen et al as the developer of the initial version of the OSLO questionnaire stated that the validity and reliability values of the OSLO questionnaire would not have a significant difference between the divisions of injury areas (Clarsen et al., 2013, 2014). The addition of open questions regarding the location of injury or illness is very necessary, this is because each sport has different characteristics of injury and illness locations (Clarsen et al., 2013, 2014). The time for filling out the questionnaire every 7 days in this study is still within the appropriate time range, based on Streiner et al regarding the health means measurement scale, which revealed that periodic reporting of subjective measurement results must be carried out with a minimum duration of 3 days to a maximum of 14 days, this aims to minimize the occurrence of recall bias (Streiner et al., 2015).

The Bahasa version of OSTRC has good performance in separating athletes who experience health problems with a sensitivity of 97.6% and a specificity of 96% and also separates injured athletes with a sensitivity of 100% and a specificity of 99.4% based on ROC and AUC. From the literature review, only a study from (Kaewkul et al., 2021) also assessed the sensitivity and specificity of OSTRC after being adapted to the Thai version, and the results of the (Kaewkul et al., 2021) study

also had high sensitivity and specificity in separating health problems and injuries. Clarsen et al in their original method, recommend preventing direct comparison of data between the results of the OSTRC questionnaire, this is because there are bound to be differences in the data collection process and operational definitions of injury or illness. The sensitivity and specificity of this study cannot be compared with other studies (Clarsen et al., 2013, 2014). The strengths of this study are (1) The first research to validate the OSTRC in the Bahasa version, (2) the Bahasa version of the questionnaire only consists of 4 questions, which makes it easier for athletes to report their health conditions periodically, (3) the Bahasa version of the questionnaire is easy to adapt to various sports without changing the questionnaire's ability to screen for health conditions. However, the Indonesian version of the OSTRC Questionnaire is a new option for regular monitoring of health problems and injuries in athletes. In the future, it is important to determine methods and operational definitions of illness and injury comprehensively, this is to obtain continuous data.

Conclusion

Based on the results of this research, it was found that the validity and reliability of the Bahasa version of the OSTRC questionnaire were very good. Apart from that, the Indonesian version of OSTRC can screen for athletes' health conditions, which can be seen through the ability to separate injured or ill athletes through the severity score. If the severity score is ≥ 8 , it is an early indicator of the possibility that the athlete will experience health problems (injury or illness).

Acknowledgments

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Conflict of interest

The authors declare no conflict of interest. OSTRC Bahasa versions are available from the corresponding author on request.

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ORIGINAL SCIENTIFIC PAPER

Analysis of Sociodemographic– and Sport-Factors as Correlates of Doping Tendency in Professional Handball Players

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Abstract

Factors associated with doping in sports are frequently studied, but sport specific, gender-stratified investigations are rare. This study aimed to evaluate sociodemographic and sport factors associated with doping tendency (DT) in professional handball players. The participants were handball professionals from Croatia and Bosnia and Herzegovina ($n=173$; 22.12 ± 3.11 years of age, 64 females and 109 males) who were tested on sociodemographic variables (gender, age, education), sport factors (experience in sport, achievement at the junior and senior level), and doping factors (personal opinion on the problem of doping, doping knowledge, and personal DT). Logistic regressions with sociodemographic- and sport-factors as predictors were calculated for binarized outcome (positive DT vs. negative DT). A greater likelihood of having a positive DT was found for males than for females ($OR=1.60$, 95% CI: 1.16-2.12) and for those who achieved success at the junior level ($OR=1.21$, 95% CI: 1.02-1.95). Among females, positive DT increased with experience in handball ($OR=1.31$, 95% CI: 1.11-1.55). Male players who achieved better competitive/sport results at a younger age were more prone to doping ($OR=1.50$, 95% CI: 1.21-1.83). While a greater tendency toward doping in males might be expected due to sociocultural factors, the gender-specific associations indicated specific factors that must be noted in the development of anti-doping strategies in this sport.

Keywords: performance enhancement, predictors, logistic regression, association, athletes

Introduction

Doping in sports (doping) refers to the use of prohibited substances or methods by athletes to gain an unfair advantage in competitions. These substances can include performance-enhancing drugs, such as anabolic steroids, stimulants, hormones, and other illicit drugs. Doping also encompasses methods such as blood doping (increasing the blood's ability to carry oxygen) and gene doping (altering genetic material to enhance performance) (McLean, Naughton, Kerherve, & Salmon, 2023). Doping is currently considered one of the most important problems in sports for several important reasons. First, doping enhances athletic performance beyond natural abilities, giving dopers an unfair advantage over clean athletes.

Second, many doping substances have serious health risks and side effects, and athletes jeopardize their well-being by using these substances. Third, doping goes against the spirit of fair play and violates the rules and regulations of most sports organizations. This is directly connected with (fourth) ethical concerns since doping raises questions related to cheating, honesty, and integrity in sports. Finally, it is clear that doping directly and indirectly damages the reputation of sports, leading to a loss of public trust and interest. Therefore, efforts to prevent doping in sports are crucial to maintain the integrity of sports (Özkan et al., 2020; Varfolomeeva, Kozyreva, & Beresneva, 2023).

The World Anti-Doping Agency (WADA) plays a crucial



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role in the global fight against doping in sports (Houlihan, Vidar Hanstad, Loland, & Waddington, 2019). Apart from the well-known duties of WADA (i.e., setting standards, forming a prohibited list of substances and methods, and anti-doping testing), one of the key duties of WADA is education and research (Deng, Guo, Wang, Huang, & Chen, 2022). Specifically, WADA promotes anti-doping education and prevention programs for athletes, coaches, and sports organizations to raise awareness about the dangers and consequences of doping. Additionally, WADA supports and funds scientific research to improve anti-doping methods, detection techniques, and understanding of doping trends in sports. Therefore, it is clear that WADA not only defines rules and regulations but also actively proclaims the necessity of an evidence-based approach in the global fight against doping in sports. One of the best-known and widely accepted practices is identifying factors associated with doping behavior in (certain) sports (Rodek, Idrizovic, Zenic, Perasovic, & Kondric, 2013; Sajber, Maric, Rodek, Sekulic, & Liposek, 2019). The idea is to control factors associated with a positive likelihood of doping and to identify factors associated with a negative tendency toward doping in athletes and supportive personnel (coaches, physicians, etc.) (Rodek et al., 2013).

A range of factors have been identified as correlates of doping behavior and doping tendency (DT) in sports. For example, Lazuras et al. highlighted the role of attitudes, normative beliefs, situational temptation, and behavioral control in predicting doping intentions (Lazuras, Barkoukis, Rodafinos, & Tzorbatzoudis, 2010). Boardley et al. explored the role of moral disengagement, with the former identifying specific mechanisms such as moral justification and displacement of responsibility and the latter developing and validating measures of team-based efficacy beliefs and moral disengagement (Boardley, Grix, & Harkin, 2015). Morente-Sánchez et al. emphasized the need for educational programs to combat doping (Morente-Sánchez & Zabala, 2013), while Kirby et al. provided game-theoretic and qualitative analyses of the economic and psychological factors influencing doping decisions, respectively (Kirby, Moran, & Guerin, 2014).

However, most of the studies reported the correlates of DT in mixed samples of athletes involved in individual and team sports and/or did not perform gender-stratified analysis. In one of the rare studies where only athletes involved in team sports were observed, the authors found that male athletes, those with better junior-level results, and those who regularly consumed dietary supplements were more likely to dope (Sekulic et al., 2016). However, the abovementioned study clearly revealed sport-specific templates of DT and differences among sports (the authors observed four Olympic team sports) and highlighted the necessity of further sport-specific, and gender-specific analyses.

Doping is currently considered one of the most important problems in sports, and sport and scientific authorities intensively seek the most efficient solutions in the global fight against doping. Among other approaches, this includes systematic studies performed with the aim of elucidating the factors associated with DP in sports (Rodek, Sekulic, & Pasalic, 2009; Sajber et al., 2019; Versic, Uljevic, & Pelivan, 2022). However, studies investigating specific sports are rare, and we have found no study which examined factors associated with DT, specifically in handball (team-handball) athletes. Therefore, the aim of this study was to evaluate the sociode-

mographic and sport-related factors associated with DT in professional handball players. Initially, we hypothesized that different factors would be correlated with DT in female and male professional handball players.

Methods

Participants

The participants in this study were professional handball players from Croatia and Bosnia and Herzegovina ($n=173$; 22.12 ± 3.11 years of age, 64 females and 109 males). At the time of testing, all participants were members of teams competing at the highest level of national competitions. Handball teams were invited to participate in the study after the study was approved by the Ethics Committee of the Faculty of Kinesiology, University of Split (protocol code 2181-205-02-05-14-004; 17 June 2014). Participants were informed that the study was anonymous, that there was no intention to connect responses to a specific person, that they could leave some questions or that the whole questionnaire unanswered.

Variables

Variables were collected by a previously validated questionnaire and consisted of sociodemographic factors, sport factors and doping-related factors (Rodek et al., 2013; Sajber et al., 2019).

Sociodemographic factors included queries on age (in years), gender (male – female), and education (elementary school, high school, university/college student, university/college degree). Sports factors included questions on experience in handball (in years), the highest competitive achievement at the junior level (local competitions, national competitions, national-level achievement-medal, national team member), and at the senior level (local competitions, national competitions, national-level achievement-medal, national team member).

Doping-related factors included questions on self-perceived knowledge about doping in sport (poor, average, good), personal opinion on the main problem of doping in sports (doping is mainly a health hazard, doping is mainly a problem of fair-play, doping is equally health hazard and a fair-play issue, I don't think doping is a problem), and opinions on personal DT (I will never use doping, I don't know, I will use doping if it will help me with no health hazard, I will use doping). For the purpose of the later logistic regression calculations (please see Statistics for details), DT was categorized into "Negative DT" (first answer) and "Neutral and positive DT" (remaining answers).

Online internet platform was used for testing all the players. A link for the questionnaire was sent to all players, and participants directly connected to the online platform where they answered the questionnaire on the QSU. All players received the link at approximately the same time of day and were informed that the questionnaire would be closed within 24 hours. The used platform was preprogrammed to allow a single connection from one IP address to prevent multiple answers from same participant.

Statistics

Since the Kolmogorov-Smirnov test identified all variables except age and experience in sport as nonparametric, descriptive statistics included calculations of frequencies

and percentages (means and standard deviations for age). Differences between males and females were calculated with the Mann-Whitney test (for ordinal variables), Chi-square test (for nominal variables), and independent samples t-test (for parametric variables).

Spearman's rank order correlation was calculated to determine the associations between the study variables when appropriate. The associations between sociodemographic and sport factors as predictors of DT were determined by performing logistic regressions. Specifically, DT was dichotomized (please see previously for categorization) and observed as a criterion, with "Negative DT" coded as 1 and "Neutral and positive DT" coded as 2. Odds ratios (ORs) and 95% confidence intervals (CIs) were reported.

Results

Male and female players were of the same age (22.03 ± 4.43 and 22.28 ± 3.12 years for males and females, respectively, t test = 0.38, $p=0.69$) and had similar experience in handball (8.17 ± 4.32 and 7.71 ± 3.15 years for males and females, respectively; t test = 0.73, $p=0.46$).

Table 1 presents descriptive statistics for the study variables and differences between genders. Compared with their male peers, females were better educated (Mann-Whitney test = 2.68, $p=0.01$) and self-declared poorer doping knowledge (Mann-Whitney test = 2.38, $p=0.01$). Positive DT was more common in males than in females (Mann-Whitney test = 2.41, $p=0.01$).

The correlations between the study variables are presented in Table 2. Apart from some logical correlations (i.e., age

Table 1. Descriptive statistics and differences between genders in study variables

	Total		Males		Females		Gender differences	
	F	%	F	%	F	%	Test value	p
Education MW								
Elementary school	0	0	0	0	0	0		
High school	119	68.8	38	80.1	34	53.1		
College/University student	39	22.5	15	14.1	24	37.5		
College/University level	11	6.4	5	4.7	6	9.4	2.68	0.01
Junior level achievement MW								
Local Competitions	0	0.0	0	0.0	0	0.0		
National Competitions	108	62.4	70	64.2	38	59.4		
National level achievement	61	35.3	35	32.1	26	40.6		
National team member	2	1.2	2	1.8	0	0.0	0.68	0.49
Senior level achievement MW								
Local Competitions	0	0.0	0	0.0	0	0.0		
National Competitions	30	17.3	23	21.1	7	10.9		
National level achievement	110	63.6	63	57.8	47	73.4		
National team member	33	19.1	23	21.1	10	15.6	0.47	0.63
Doping knowledge MW								
Poor	98	56.6	69	63.3	29	45.3		
Average	37	21.4	21	19.3	16	25.0		
Good	38	22.0	19	17.4	19	29.7	2.38	0.01
Main problem of doping Chi square								
Doping is health-hazard	105	60.7	58	53.2	47	73.4		
Doping is against fair-play	42	24.3	29	26.6	13	20.3		
Both, health-hazard and against fair-play	9	5.2	7	6.4	2	3.1		
Doping is not the problem	12	6.9	10	9.2	2	3.1	6.18	0.10
Doping tendency MW								
Positive	20	11.6	12	11.0	8	12.5		
Somewhat positive	25	14.5	23	21.1	2	3.1		
Neutral	25	14.5	24	22.0	13	20.3		
Negative	93	53.8	52	47.7	41	64.1	2.41	0.01

Legend: MW denotes variables where differences were established by Mann Whitney test; Chi square denotes variables where differences were established by Chi square test; for detailed explanation of variables please see Variables subsection

is significantly associated with experience in handball, while experience is associated with achievement), several significant correlations deserve attention. Among men, DT was associated with greater junior-level achievement (Spearman's $R=0.49$,

$p<0.01$). Additionally, among females, DT was significantly associated with experience (Spearman's $R=0.61$, $p<0.05$) and age (Spearman's $R=0.32$, $p<0.05$).

The results of the logistic regressions calculated sepa-

Table 2. Spearman's rank order correlations between study variables (* denotes significant correlations)

	Age	Experience	Education level	Junior level achievement	Senior level achievement	Doping knowledge
TOTAL SAMPLE						
Age	-					
Experience	0.68*	-				
Education level	0.18*	0.06	-			
Junior level achievement	-0.10	-0.23*	-0.27*	-		
Senior level achievement	0.36*	0.35*	0.19*	0.24*	-	
Doping knowledge	-0.01	-0.05	0.06	0.12	0.04	-
Doping tendency	0.07	0.13	0.03	0.36*	0.02	-0.01
MALES						
Age	-					
Experience	0.70*	-				
Education level	0.03	0.11	-			
Junior level achievement	-0.01	-0.22*	-0.41*	-		
Senior level achievement	0.44*	0.43*	0.12	0.24*	-	
Doping knowledge	-0.18	-0.19*	-0.11	0.15	-0.05	-
Doping tendency	0.04	-0.05	0.00	0.49*	0.04	-0.02
FEMALES						
Age	-					
Experience	0.62*	-				
Education level	0.31*	0.08	-			
Junior level achievement	-0.27*	-0.24	-0.17	-		
Senior level achievement	0.10	0.13	0.26*	0.26*	-	
Doping knowledge	0.33*	0.27*	0.26*	0.03	0.20	-
Doping tendency	0.32*	0.61*	0.09	0.21	0.02	0.02

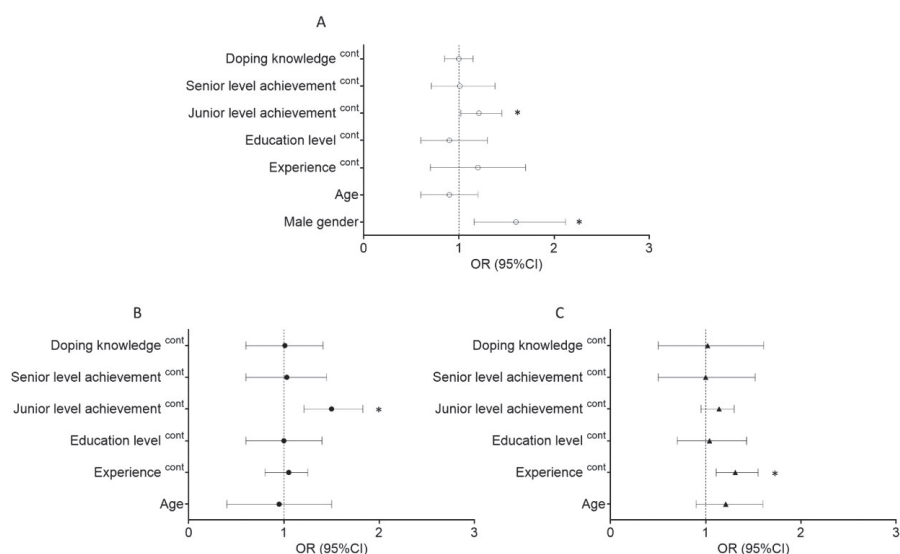


FIGURE 1. Results of logistic regression between study variables and doping tendency outcome in the total sample (A), in males (B), and in females (C); * denotes significant associations at $p<0.05$; cont denotes variables observed as continuous for the purpose of logistic regression calculation

rately for the total sample and gender stratified are presented in Figure 1. For the total sample, male gender was a factor associated with an increased risk for neutral/positive DTs (OR=1.60, 95% CI: 1.16-2.12). Additionally, higher likelihood for neutral/positive DT was found for players who were more successful at the junior level (OR=1.21, 95% CI: 1.02-1.95) (Figure 1A). For males, a greater likelihood of neutral/positive DT was found for players who achieved competitive success as juniors (OR=1.50, 95% CI: 1.21-1.83) (Figure 1B). Among females, those with more experience in handball were more prone to having neutral/positive DTs (OR=1.31, 95% CI: 1.11-1.55) (Figure 1C).

Discussion

Evidently, males are more prone to doping than females are. This is generally not a novel finding, but to the best of our knowledge, this is the first study in which this topic has been specifically addressed for handball sport (Sekulic et al., 2016). In general, the issue of why males are more prone to doping than females is complex and multifaceted, and several factors contribute to this disparity and can generally be categorized as (i) sociocultural influences, (ii) psychological factors, or (iii) physiological characteristics.

Sociocultural expectations and influences are known to be important determinants of any human behavior (Reynolds, Haycraft, & Plateau, 2022). In regard to sports, men are more pressured than women are to perform at high levels and to display physical capacities. There is no doubt that the region where the sample was drawn from (i.e., southeastern Europe) follows the same template. Consequently, such societal pressure can drive males to look for any kind of performance enhancement, including doping, simply to meet these expectations. This is additionally aggravated by the fact that men typically participate in competitive sports at higher rates than women (Gilic, Ostojic, Corluka, Volaric, & Sekulic, 2020; Sekulic et al., 2021). This puts men at increased risk of exposure to competitive environments and can result in a greater tendency toward doping. Finally, there is no doubt that doping as a phenomenon is not as stigmatized in men as it is in women. Consequently, male athletes (including the handball players studied here) probably experience greater peer pressure to use doping than their female peers involved in the same sport.

In regard to the psychological factors that can influence the likelihood of doping in males, the most important factor is likely gender differences in risk-taking behavior. Indeed, studies have repeatedly confirmed that men are generally more prone to risk-taking behavior than women are (Byrnes, Miller, & Schafer, 1999). This is confirmed in various circumstances, and sport-related behaviors are no exception (Woodman et al., 2013). Doping usage is known to be connected to various risks (i.e., health hazards and the danger of being caught and proclaimed as cheater). As a result, the decision to use doping can be seen as a form of risk-taking, being naturally more associated with males than with females. The previously described mechanism is typically amplified by typical male characteristics, aggression and competitiveness. In brief, higher levels of aggression and competitiveness among men may contribute to a greater likelihood of using doping to achieve superior performance.

In addition to sociocultural and psychological factors, some physiological differences between males and females deserve attention. Testosterone levels are positively correlat-

ed with greater muscle mass and strength, both of which are typical for males. The desire to enhance these traits, even by doping, might be stronger in men. Finally, metabolic differences between genders also contribute to a greater likelihood of doping among males. It seems that certain doping substances are more likely to be effective in males, while usage is related to “fewer risks” in males than in females. Specifically, one of the widely known side effects of anabolic-steroid usage is masculinization (Sjöqvist, Garle, & Rane, 2008). Naturally, this side effect presents bigger problem in females than in males, resulting in certain barriers to doping among female handball players.

More experienced female players were more prone to doping. Again, this finding can be supported by several factors. While some of these factors are relatively biased by certain sociocultural characteristics of the region where the sample was drawn from, others are probably more generalizable globally. Additionally, although we found a correlation between experience in handball and DT solely in females, there is a certain possibility that the following discussion is transferable to males as well.

As athletes age, they naturally experience a decline in physical performance due to factors such as reduced muscle mass, slower recovery times, and decreased stamina (Gries & Trappe, 2022). Doping can be seen as a way to counteract these age-related declines, being more accepted in older and more experienced players than in younger ones. Additionally, athletes who are involved in sports for a longer time are more vulnerable in terms of being injured, while recovery from intensive training and exhaustive competitions is slower (Bourogianis, Hatzimanouil, Semanltianou, Georgiadis, & Sykaras, 2023; Hatzimanouil, Skandalis, Terzidis, Papasoulis, & Mavromatis, 2022). Doping drugs were originally developed as medicinal treatments for the treatment of injuries and illness. Therefore, if athletes experience such “benefits” from drugs (and this is more likely to occur in more experienced players), they will be more prone to doping as well. While specific mechanisms are characteristic for both males and females, the gender-specific association is understandable because of the previously discussed factors of a higher risk for doping in males (please see the previous discussion for more details).

Experienced athletes generally feel more pressure to maintain their competitive and performance levels. They try to stay competitive, and it is hard to accept that their achievement cannot be at the same level as when they are younger. Taken together, these factors can lead them to consider doping as a solution that can help them. While some of them are professionals and sport assure them financial income, as retirement approaches, they might resort to doping as a last chance to achieve their career goals and secure finances. In explaining differences between males and females with regard to these explanations, it must be highlighted that female players earn less in sports than males do (Mogaji, Badejo, Charles, & Millisits, 2021). Consequently, they might be more interested in prolonging their careers to earn additional money, resulting in a specified correlation solely among them and not among their male colleagues.

This is not the first study in which competitive achievement at a younger age was found to be correlated with DT in athletes. This correlation was also highlighted in a previous study on team sport athletes (Sekulic et al., 2016). However, this is one of the first studies in which such an association was

found for athletes involved in one specific sport, meaning that eventual sport-type bias cannot be identified as a factor of influence. Additionally, this is found only in male players, which clearly highlights gender-specific correlation.

High competitive/sport achievement at a junior age is naturally associated with high career expectations (Whitehead, Andrée, & Lee, 2004). In other words, if players achieve success at a young age, they consequently expect to be successful even at the senior level. However, this rarely occurs since competition at a younger age is limited to a very narrow age span (one or two years). At the senior level, rivals vary from 10-15 years, making competition much more challenging. As a result, the frustration of those who were successful at the junior level but not successful at the senior level is understandable. These athletes often face immense pressure to perform better than they are objectively able. This pressure can lead to the temptation to use doping to meet those expectations.

The social factor of external expectations could also contribute to a greater likelihood of doping in players, particularly those who achieved high success in the early phases of their career (Petróczi & Aidman, 2008). High expectations from coaches, sponsors, fans, and even families can increase the pressure on athletes to perform well. This external pressure can make doping a viable option for meeting these expectations, especially if success at the senior level is not achieved. This is logically connected with fear of failure, or the fear of not living up to one's own or others' expectations. This can drive athletes to seek any means necessary to succeed, including doping.

Finally, one typical sport factor and its potential influence on this correlation is particularly interesting. The volume and intensity of training load at the senior level are much greater than those at the junior level (Murray, 2017). This is particularly evident in handball due to its contact nature. Namely, at the very beginning of their senior-level career, young athletes must compete against much stronger rivals, who are not on-

ly better conditioned but also familiar with higher demands of the senior-level sport. Over the first few years of their senior-level career, most young players suffer injuries and/or at least experience dramatic psychophysiological exhaustion. As a result, they are challenged to seek any kind of help, including doping.

Limitations and strengths

The most important limitation of this study is that DT was self-reported by players. Therefore, there is a certain possibility for social desirability bias. However, we believe that the strict anonymity of the survey decreased this possibility. On the other hand, the fact that we studied a relatively large number of players competing at the highest competitive level, in a region where handball is a highly popular sport and both male and female teams are among the most successful in the world, is an important strength of the study.

Conclusion

The results revealed gender-specific correlates of DT in handball players. Therefore, our initial study hypothesis can be accepted. The established associations indicated specific factors that must be noted in the development of anti-doping strategies in this sport.

Since experience in sports is a risk factor for DT in females, special attention should be given to female players who are involved in handball for a longer time. Most likely, the anticipated end of one's career and pressure to maintain one's competitive and performance level are the most important factors contributing to this association.

Success at younger ages was shown to be a risk factor for positive DT in male players. Overall, some career-specific factors (i.e., high expectations) and sport-specific factors (i.e., exponential increase in training and competition demands at the senior level) should be studied in the future as circumstantial and contextual factors of identified associations.

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Conflict of interest

The authors declare that there are no acknowledgments.

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ORIGINAL SCIENTIFIC PAPER

The Impact of Service Quality on Self-Confidence of Wheelchair Athletes: An Analysis of Family Support as Moderator

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Abstract

This study aims to analyze further the effect of service quality on the confidence of wheelchair athletes. The family support is referred to as a moderator variable. This quantitative research type used the Moderated Regression Analysis (MRA) method. Sampling was carried out at a special open tournament for wheelchair athletes. The total number of athletes was 122 (basketball =30, tennis =16, badminton =22, archery =20, fencing =16, table tennis =18), with characteristics as follows: aged 25.36 ± 2.4 years, and had 11.37 ± 1.2 years of coaching rate. The instrument was a questionnaire which had been validated through the focus group discussion (FGD) with experts, namely sports training lecturers and wheelchair trainers. The initial results were typically distributed data based on the Kolmogorov-Smirnov test 0.200. The second regression test analysis without moderation was 0.003, and $R^2=0.466$, meaning there was an effect between service quality and self-confidence. The third moderation analysis was 0.000, and $R^2=0.583$. The family support variable moderated positively and significantly between service quality and self-confidence. In conclusion, the wheelchair athletes' self-confidence is determined in one way, namely developing and improving good service quality. With the support of parents, wheelchair athletes feel supported by emotional, nutritional, instrumental, appreciation, and informative aspects. Thus, good service quality and family support can increase wheelchair athletes' self-confidence, and hopefully it impacts their achievement.

Keywords: athlete disabilities, para-athlete, wheelchair

Introduction

Self-confidence is a part of psychological elements related to the belief in an individual's ability to do a task or job. Success in a task or job can be interpreted as achievement (Flores Ferro et al., 2023). Everyone, including an athlete, can achieve achievements. The results of the study have been published that self-confidence can influence athletes to produce high performance in competition (Astuti et al., 2023; Jun et al., 2024; Lochbaum et al., 2022). However, training and increasing self-confidence should be done not only from the level of training or experience in participating in competitions. It must also be seen from the condition of the athlete. One of the conditions of the athlete is the wheelchair athlete (Berardi et

al., 2018; Mortenson et al., 2022).

Individuals with disabilities that require wheelchairs generally experience low self-confidence (Foley et al., 2020; Gutiérrez-García et al., 2023). Based on the evidence in the existing environment, a lack of self-confidence is caused by the feeling that they do not have the expertise. Being an athlete is a speciality for an athlete with a wheelchair disability. By becoming an athlete, it is hoped that individuals with disabilities can raise their self-esteem. Self-confidence of athletes with a disability is generally determined by many factors, both internal and external factors. Internal factors consist of motivation, unstable mood and emotions, anxiety, and concentration (Huenullán et al., 2023; Palencia & Gallón, 2022; Požerienė et



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al., 2018; Youngson et al., 2023). Meanwhile, external factors include family support, organization, trainer competence, and training facilities (Balan & Mujea, 2022; Ballas et al., 2022; Sudarko et al., 2023; Tusiime & Odiango, 2022).

However, few scientific publications have examined the factors that influence the confidence level of wheelchair athletes (Graikinis-Evaggelinos et al., 2019). Furthermore, to improve the quality of life of athletes with disabilities, service quality will undoubtedly be indispensable (Kunene, 2023). In theory, based on the results of previous studies, it is explained that good service quality will affect the level of athlete satisfaction (Cuesta-Valiño et al., 2023; Günel & Duyan, 2020; Nugroho et al., 2023). Satisfaction is desirable for every athlete with a disability because satisfaction is part of improving the quality of life for individuals with disabilities. Research studies further explain that to obtain satisfaction in people with disabilities, one way of physical activity is exercising (Biagini et al., 2022; Zheng et al., 2023). Thus, based on the results of those studies, there is an assumption that the quality of service at sports clubs can provide a good quality of life for athletes with disabilities. The few findings encourage the researchers to explore more deeply the relationship among service quality, wheelchair athletes, and self-confidence.

The results of other scientific reports also showed that the confidence level of male wheelchair-fencing athletes was better than that of women in pre-competition (Peron & Elsner, 2020). These results are characterized by a tense body, a sense of worry about appearance, and the body feeling stiff when doing movements. Wheelchair basketball also shows that shooting is the most critical performance in wheelchair basketball athletes. However, from the analysis of this study, it is explained that the confidence level of a wheelchair basketball athlete plays a vital role in a shot (Ali & Al-Oudah, 2022; Ceruso et al., 2022; Hernandez-Beltran et al., 2023). Another study compares the confidence levels of wheelchair tennis athletes and wheelchair badminton athletes. The results showed no difference in the confidence level between the two (Abdullah et al., 2021). However, the advice from the study requires any factors that affect the level of self-confidence of both. In addition, the results of studies that affect the quality of life of wheelchair athletes, namely fencing, basketball, tennis, rugby, handball and athletics, show that self-confidence is part

of the psychological factors that make them able to improve the quality of life through disability sports clubs (Calheiros et al., 2021; Clemente et al., 2019, Côté-Leclerc et al., 2017; Rengifo Cruz et al., 2023).

This study aims to analyze further the effect of service quality on self-confidence of wheelchair athletes, namely, basketball, tennis, badminton, archery, fencing, and table tennis. Then, this study also involves family support as a moderator variable because there is a suspicion from the results of previous studies that family support for athletes with disabilities affects the level of athlete psychology (Alanazi, 2023; Coates & Howe, 2023). Family support is shown by emotional closeness, appreciation of processes and results, meeting the needs for training and matches, and participating in sports (Rumahpasal et al., 2020). Furthermore, family support is also shown by paying attention to nutrition, finding solutions when problems occur, and establishing good relationships with coaches (Mandan et al., 2024). Therefore, the published results of this study can provide an evaluation for organizational administrators, officials, sports club facilities providers, coaches, and athletes' families to be able to pay attention to athletes with wheelchair disabilities. Getting attention through improving the quality of services and family support will undoubtedly directly impact their quality of life.

Research Method

Research Design and Procedure

This quantitative research type used the Moderated Regression Analysis (MRA) method (Figure 1). Quality of service is an independent variable, and self-confidence is a dependent variable. Then, the moderation variable aims to test how family support can strengthen or weaken the service quality and self-confidence variables. Data collection was carried out during the wheelchair-open tournament event. However, the types of wheelchair sports involved in the open tournament were carried out at different times and places, so it takes four months (from November 2023 to February 2024). Then, the data was collected using a questionnaire sheet containing statement items. Only researchers and coaches are authorized to provide instructions and accompany the athlete when the athlete fills out the questionnaire sheet. The data collection time on the questionnaire was carried out two to one day before the athlete competed.

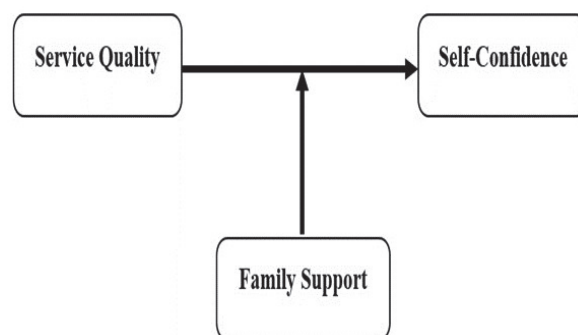


FIGURE 1. Research Design with Moderated Regression Analysis

Study Participants

This study was conducted by total sampling at the special open tournament championships for athletes with wheelchair disabilities. Thus, the research involved a total of 122 wheelchair athletes consisting of 30 basketball athletes, 16 tennis

athletes, 22 badminton athletes, 20 archery athletes, 16 fencing athletes, and 18 table tennis athletes. The characteristics were 29.98 ± 2.4 years of age and 11.37 ± 1.2 years of training. The competition experience is 26 athletes who have participated in competitions at the international and national levels,

44 athletes at the national level, and 52 athletes at the regional and provincial levels. The samples that will be involved in this research are selected voluntarily and if they successfully complete the questionnaire, they will then collect the results of the questionnaire to the researcher. This research has been conducted in accordance with Helsinki Declaration and approved by the author's institution, the coach accompanying the athlete, and the competition organizer.

Research Instrument

Instrument preparation was done by reviewing literature and scientific books with keywords such as athletes with disabilities, athletic achievements, training achievements, parental support, and self-confidence (Yusoff et al., 2021). Then, modifications are made according to the needs and circumstances of the facts in the field. Furthermore, the FGD (focus group discussion) was carried out to test the instrument's validity by involving several experts, namely sports coaching lecturers, teaching lecturers for athletes with disabilities, and disability coaches who have national coach licenses. The scale used was 1 to 4 (1 "very poor", 2 "poor", 3 "good", and 4 "very good").

The results that can be identified are service quality variables (independent variables) which are tangible, empathy, reliability, responsive, assurance (Juita et al., 2024; Prabowo et al., 2024; Prayoga et al., 2024). Confidence variables (dependent variables) are optimistic, independent, sporty, not worried, and adaptable (Abdullah et al., 2021; Berardi et al.,

2018; Djaba et al., 2024; Lochbaum et al., 2022). The moderation variable, namely family support, consists of emotional, nutritional, instrumental, appreciation, and informative factors (Mandan et al., 2024).

Statistical Analysis

The first step of this research analysis was to conduct a normality test as a condition before the regression test. The second step was to test the regression by displaying the coefficient of determination (R Square) and the results of the multicollinearity test (tolerance and VIF values). The third step was the same as the second step, combining the quality of service (independent variable) with the support of family members (moderation variable). Statistical analysis of this study used SPSS 26 (IBM, Armonk, NY, US). The significance level in this regression test is 5% ($p < 0.05$).

Results

Normality Test

Before the regression analysis with moderation, the second stage is the normality test of the normality test based on the Kolmogorov-Smirnov value due to a sample of more than 50 wheelchair athletes.

Based on the results of the normality test in table 1, the Kolmogorov-Smirnov values for service quality, self-confidence and parental support show significant values ($p > 0.05$), meaning the data is normally distributed.

Table 1. Normality (Kolmogorov-Smirnov) Test Results

		Service Quality	Self-Confidence	Family Support
N		122	122	122
Normal Parameters	Mean	95.76	119.58	96.03
	Std. Deviation	26.510	46.093	28.472
	Absolute	0.162	0.255	0.174
Most Extreme Differences	Positive	0.161	0.254	0.183
	Negative	-0.109	-0.212	-0.113
Test Statistic		0.162	0.255	0.176
Asymp. Sig. (2-tailed)		0.082	0.077	0.081

Regression Test without Moderation

The third stage was regression analysis without using moderation variables. Thus, the analysis focused on the quality of service and the self-confidence of wheelchair athletes.

Based on the results of unmoderated regression (Table 2),

it shows a sig value of 0.003 or ($p < 0.05$), which means that there is a relationship between performance in training and the self-confidence of wheelchair athletes in basketball, tennis, badminton, archery, fencing, and table tennis. Then, from these results, there were no symptoms of multicollinearity.

Table 2. Regression Test Results without Moderation

		Unstandardized Coefficients		Standardized Coefficients		Collinearity Statistics		
Model		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	66.682	13.205		5.050	0.000		
	Service Quality	0.354	0.116	0.358	3.042	0.003	0.846	1.053

a. Dependent Variable: Self-Confidence

Table 3. R Square Value Results without Parental Support; Model Summary

Change Statistics									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	df1	df2	Sig. F Change
1	0.638	0.466	0.453	4.458	0.466	9.758	1	121	0.001

a. Predictors: (Constant), Service Quality

Then, the R square value in Table 3 shows 0.466, which means that the value of service quality on the confidence of wheelchair athletes in basketball, tennis, badminton, archery, fencing, and table tennis 46.6%. Meanwhile, 87.2% were influenced by other variables that this study cannot explain.

Regression Test Using Moderation Variable (moderated regression analysis)

The fourth stage was regression analysis using moderation variables (moderated regression analysis). Thus, the analysis

focused on testing family support as a moderation variable and whether it can affect service quality and self-confidence variables.

Based on the regression results using moderation (Table 4) shows a sig value of 0.000 or ($p < 0.05$), there is a relationship between the quality of service and the confidence of wheelchair athletes in basketball, tennis, badminton, archery, fencing, table tennis through family support relationship as moderator. Then, from these results, there were no symptoms of multicollinearity.

Table 4. Regression Test Results using Moderation; Coefficients

		Unstandardized Coefficients		Standardized Coefficients		Collinearity Statistics	
	Model	B	Std. Error	Beta	t	Sig.	Tolerance VIF
	(Constant)	67.241	10.359		6.491	0.000	
1	Service Quality	0.567	0.097	0.574	5.840	0.000	0.880 1.136
	Service Quality*Family Support	0.604	0.101	0.625	6.356	0.000	0.873 1.214

a. Dependent Variable: Self-Confidence

Then, the R square value in Table 5 shows 0.583, which means that the value of the quality of parental service and support as a moderator of the confidence of wheelchair athletes

in basketball, tennis, badminton, archery, fencing, and table tennis 58.3%. Meanwhile, 87.2% were influenced by other variables that this study cannot explain.

Table 5. R Square Value Results; Model Summary

Change Statistics									
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	df1	df2	Sig. F Change
1	0.764a	0.583	0.572	1.928	0.583	28.719	2	120	0.000

a. Predictors: (Constant), Service Quality*Family Support, Service Quality

Discussion

The results revealed a significant positive effect of service quality on wheelchair athlete's self-confidence, which was 0.003, with a coefficient of determination of 0.466. The quality of service required by wheelchair athletes consists of five factors: tangible, empathy, reliability, responsiveness, and assurance. The results of the five factors on the quality of service have also been carried out in non-disabled sports, such as individual and group sports (Nugroho et al., 2023; Prabowo et al., 2024; Prayoga et al., 2024). In wheelchair sports, the quality of tangible services needed by athletes is in the form of organizational, official, and coach involvement in paying attention to athletes during and outside the training schedule. In addition, it is not only involvement but also handling the needs or problems of athletes quickly. Thus, actual involvement also affects other factors, such as empathy, reliability, and responsiveness. Forms of involvement from the organization include providing adequate training facilities, monthly allowance, and housing. Trainers and officials must have a straightforward training program, help wheelchair athletes meet training equipment, and pay attention to nutrition through assistance with supplements if needed. In the assurance factor, organization, trainer, and official are also responsible for health, ensuring safe training places, protecting from injury, and guaranteeing accidents outside of training time. The results of this study are in line with the results of previous studies that say, in principle, the quality of service is the fulfillment of all the needs of athletes, such as facilities, training equipment, assistance in the form

of material, and moral support (Nugroho et al., 2023). Then, a friendly and polite service, quick in handling problems, and able to provide solutions when athletes experience problems, is the hope of every athlete (Juita et al., 2024).

Proper quality of service will affect athlete's satisfaction. The results of this study on wheelchair sports showed that athletes were satisfied with the service performance of organizations, coaches, and officials. It shows that athletes are always disciplined in training, feel happy and enthusiastic, do not experience complaints, and maintain good relationships by communicating with officials and coaches during or outside the training schedule. The indication highlights the results of previous studies on an athlete's satisfaction, which can indirectly increase self-confidence (Berardi et al., 2021). Other research findings also revealed that life satisfaction can affect the confidence of wheelchair athletes during competition (Abdullah et al., 2021). Based on some wheelchair athletes' disclosures, they assume that the quality of service is a direct support to athletes. Support provided by organizations, coaches, officials, and other support teams can positively affect the psychological condition of athletes, one of which is confidence. This reason has been made clear in previous research studies that athletes who feel valued and cared for tend to perceive their abilities better. It can increase their achievement motivation (Anderson et al., 2023; Hanh et al., 2023; Huenullán et al., 2023).

Then, the regression analysis results using moderation, namely family support, can affect the confidence level of wheelchair athletes in basketball, tennis court, badminton, archery,

fencing, and table tennis 0.000 with a determination coefficient of 0.583. The regression analysis results with moderation are also based on their statement that family involvement, namely support in the form of emotional, nutritional, instrumental, appreciation, and informative, is significant in influencing self-confidence. They also assess the role of the family and are very aware of the needs of a wheelchair athlete. It gives athletes more attention and certainly gets recognition from the family (Richard et al., 2022). Previous qualitative research studies have explained, family support can also help find the meaning of life by becoming an athlete after suffering because it is looked down upon by others (Nopiyanto et al., 2022). Then, the results of qualitative research from deaf athletes explained that parents help athletes win gold medals in national Paralympic championships even though the form of support is often moral support during matches (Mustamin et al., 2021). From the results of previous studies, it is clear that parental support can affect athlete performance, one of which is self-confidence. Increased self-confidence in athletes can suppress anxiety, and athletes become confident in their skills from the results of previous training performances. Hence, the opportunity for lively success in competition is even more fantastic (Hidayat et al., 2023; Lochbaum et al., 2022; Waluyo et al., 2022).

Then, the combination of quality of service and family involvement is indispensable in improving the self-confidence of wheelchair athletes. Further results in this study showed wheelchair athletes revealed that organizations, coaches, and officials always ask about the condition of athlete through the athlete's family. On the other hand, the athlete's family also provides information to organization, coach, and official about daily activities outside of training, including sleep time, nutrition, and the athlete's psychological state. It will be more beneficial for the coach to prepare an effective training program according to the athlete's condition. In addition, it will be easier for coaches to know athlete's personality characteristics so that training programs and materials will be easily conveyed (Mandan et al., 2024). Open communication among athletes, families, organizations, coaches and officials can help identify the needs and expectations of each party (Azhari et al., 2023). By supporting and understanding each other, athletes can feel more accepted, cared for and supported, boosting their confidence. In addition, coaches and athlete families can share information about training goals, athlete development, and achievement targets with the athlete's family. In contrast, the athlete's family can provide insight into the unique needs and

preferences of wheelchair athletes outside of the training environment (Orr et al., 2021). Thus, family support can positively strengthen the impact of service quality on self-confidence in wheelchair athletes.

One of the main factors of a disability athlete's self-confidence is determined by a good quality of service. Then, parental support is able to significantly moderate the relationship between service quality and the self-confidence of wheelchair athletes. The limitation of this study was that the number of samples was considered insufficient because not all wheelchair athletes were willing to fill out this research questionnaire. In addition, wheelchair competence in Indonesia is also minimal. It is due to the interest and motivation of individuals with disabilities to participate as athletes. It is expected that the publication of this study will be a good step for all disability sports and pay attention to the quality of services to improve performance in matches through increasing self-confidence. In addition, the athlete's family continues encouraging individuals with special needs to improve their quality of life. One of them is becoming a paralympic athlete.

Conclusion

Based on the results of this study, it can be concluded that the quality of service can increase the confidence of wheelchair athletes before competing. Then, the variables of parental support could positively and significantly influence the performance in training and the confidence of wheelchair athletes. These results showed the effect of service quality on self-confidence without parental support variables 0.003 and ($R^2=0.466$). In contrast, parental support variables could moderate performance in training on self-confidence 0.000 and ($R^2=0.583$). There needs to be a severe response regarding facilities to support athlete training performance because facilities are vital in service quality. However, it should be noted that if the facilities are exemplary, wheelchair athletes feel cared for. Of course, the most important thing is to train more actively and discipline them. In addition, cooperation between organizations, officials, coaches and family support is essential to pay more attention to athletes. The recommendation from the results of this research study is the involvement of a government agency to foster, develop, and facilitate all individuals with special needs. Further research studies should pay more attention to the number of samples involved and how to collect data, of course, with a simple research design which can produce accurate data.

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Conflict of Interest

The researchers have no conflicts both with the researchers and the results of other studies.

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ORIGINAL SCIENTIFIC PAPER

The Correlation between Body Composition and Physical Fitness Parameters in High Theological School Students

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Abstract

Previous research findings support the claim that physical activity (PA) and maturation result in changes in physical fitness (PF) and body composition (BC) parameters. Fitness assessment tests are especially important for monitoring the health of adolescents. The purpose of this study was to find the correlation between BC and PF parameters in students enrolled in High Theological School in Niš. Thirty-seven male students living in boarding schools were included in the participant sample. The average age of the participants was 17.48 ± 1.48 years. The body composition analyser Omron BF 511 (Omron, Tokyo, Japan), was used to measure the BC values. A modified Eufitmos testing protocol was determined to evaluate PF parameters such as: Progressive aerobic cardiovascular endurance run (Pacer), 1-Mile run (1MR), Push-ups (PU), Handgrip (HG), Standing broad jump (SBJ), 20-Meter run (20mR), Back-saver sit and reach (BSR), and Body mass index (BMI). The results showed that the weight status parameters were highly correlated with each other. Regarding PF parameters, they correlated moderately with each other. The correlation analysis between BC and PF parameters showed that BMI, Fat%, and Muscle% achieved moderate correlations with PACER, VO₂max, PU, SBJ, and 20mR, while BW had a significant correlation only with HG and PU. Also, BH had a moderate correlation with VO₂max, PACER, SBJ, and HG. The findings indicate a correlation between body composition and physical fitness parameters among theology students. However, a suggestion for further research is to conduct multiple regression analysis, which would reveal the relationship between the overall body composition composite and the fitness of the students, rather than just individually determined parameters.

Keywords: *physical fitness, anthropometric characteristics, adolescents, Eufitmos testing protocol*

Introduction

As an important health indicator, physical fitness (PF) is a set of measures that demonstrate how effectively the body can carry out exercises and physical activities (PA) (Oja & Pixsöt, 2022). PA is defined by the World Health Organization (WHO, 2020) as any movement of the body caused by the activation of skeletal muscles, including the expenditure of energy and being able to be performed at different intensities. PF is a term used to describe an assortment of characteristics that people have or develop throughout their lives that are associated with their capacity for exercise and PA. It indicates an integrated measurement of the body systems and func-

tions that are engaged in their performance (Kolb et al., 2021; Landry & Driscoll, 2012; Ortega, Ruiz, Castillo, & Sjöström, 2008; WHO, 2020). An assortment of motor abilities, including muscle strength and endurance, movement speed, agility, joint flexibility, coordination, and balance, are included in this concept. A few authors also incorporate body composition (BC) as another aspect of PF (Ortega, Ruiz, & Castillo, 2013). Anthropometric characteristics, including body mass index (BMI) and BC, have an impact on the PF parameters that are associated with health (Hyska, Mersini, Mone, & Burazeri, 2014; Rakic, Bozic-Krstic, & Pavlica, 2011).

The findings of earlier research support the idea that changes



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in BC and PF indices are also caused by the maturing process (Malina, 2005). Country-specific epidemiological data on BC, the prevalence of overweight or underweight, and PF parameters are essential for creating public health policies that work and apply to practice for PA (Ried-Larsen, Grøntved, Kristensen, Froberg, & Andersen, 2015). Body fluids, bones, muscle, and adipose tissue are the basic components of BC. The distribution of muscle and fat mass within the body can be assessed and understood with the aid of this information (Heymsfield & Waki, 1991). PA is the most effective strategy to prevent overweight and obesity in children and adolescents (Hills, Okely, & Baur, 2010).

Standardized test batteries can be used to reliably measure PF parameters in children and adolescents. There are currently over fifteen test groups available worldwide to evaluate PF parameters (Cvejic, Pejovic, & Ostojic, 2013). The test batteries that are most frequently used are Unifittest, FitnessGram, and Eurofit (Bruggeman et al., 2020; Mekota & Kovar, 1995; Pluim & Gard, 2018). The goal of the European Fitness Monitoring System (EUFITMOS) is to create a system across Europe for measuring adolescents' physical fitness. The authors found an existing test battery used in the EUFITMOS project to assess physical fitness based on a review of all prior research (Marques et al., 2021). The tests that were chosen were: Progressive aerobic cardiovascular endurance run (Pacer), 1-Mile run (1MR), Push-ups (PU), Handgrip (HG), Standing broad jump (SBJ), 20-meter run (20mR), Back-saver sit and reach (BSR), and Body mass index (BMI).

Although some studies have examined the relationship between BC and PF among students (Ivanyshyn et al., 2021; Ortega et al., 2007; Yohannes, Östenberg, & Alricsson, 2020), they have used only one parameter of weight status in relation to certain PF parameters. Additionally, to our knowledge, there are no studies that have examined this relationship among theology students. In this regard, our study includes multiple BC parameters, as well as a complete battery of PF tests to assess the fitness status of the participants. Therefore, the study aims to explore the correlation between physical fitness parameters and body composition in adolescents participating in the High Theological School.

Methods

Sample Participants

Thirty-seven male students from the High Theological School in Niš, Serbia made up the sample of participants. The average age of the male participants was 17.48 ± 1.48 years. The exclusion criteria were: the participants' or parents' lack of consent to participate in the research; missing one measurement session; lower extremity injuries excluding physical activity; poor health; diagnosed chronic or noncommunicable diseases; incorrectly performed physical fitness tests; and abandonment from conducting tests during the research. Every adult participant has read the informed consent, understood the aim of the research, given written consent to take part, and is free to terminate participation at any moment without facing negative consequences. Consent was signed by parents or legal guardians for underage participants. Participants had to fulfill certain criteria to be included in the research: they had to be in good general health and not have any long-term medical conditions that might interfere with their ability to participate in all anthropometric and motor assessments at the scheduled times. The study took place at the Faculty of Sports and Physical Education in Niš. The research was conducted in accordance with the Declaration of Helsinki, and the protocol

received approval from the Ethics Committee of the Faculty (Approval Document No. 04-504/2).

Study procedure

The researchers received instructions on how to conduct physical fitness tests correctly, code data, and maintain test results confidential. They prepared all the required equipment before the measurements started. The study was carried out in the Faculty of Sport and Physical Education hall between 2 and 4 p.m. The researchers consciously and voluntarily decided to take part in the study. The study's goal and the confidentiality of the results were explained to participants, their parents, the school administration, and the teachers. Teachers and the school administration were always consulted while organizing the planned measurements.

Measurements

Body composition

Trained evaluators conducted anthropometric measurements. All participants received instructions to avoid food or liquids for three hours before the measures, starting on the day of the appointment. The Martin Anthropometer was used to measure participants' body height (BH) with 0.5 cm precision. The Omron BF 511 device (Omron, Tokyo, Japan), which offered information on body weight (BW), percentage of body fat (Fat%), and percentage of muscle (Muscle%), was used to measure body composition parameters. Body Mass Index (BMI) is calculated using the following formula: $BMI = BW(kg)/BH(m)^2$.

Physical fitness parameters

A modified EUFITMOS protocol (Marques et al., 2021) for evaluating PF parameters was chosen for this research, taking into account the participant sample. Pacer, HG, PU, SBJ, 20mR, BSR, and BMI were the tests that were selected. EUFITMOS was proposed by authors from six different countries, creating a test battery from valid motor tests that cover the entire physical fitness of a person. Thus, a reliable and valid test for measuring cardiorespiratory fitness is the Pacer (Meredith & Welk, 2010). The maximal oxygen consumption (VO_{2max}) is then calculated from the lap count that was recorded (Saint-Maurice, Welk, Finn & Kaj, 2015). To evaluate resistance and strength in the upper body, a PU test is used (Baumgartner, Oh, Chung & Hales, 2002). The valid measure of maximum isometric strength is HG (Wind, Takken, Helders, & Engelbert, 2010). According to España-Romero et al. (2010), HG has acceptable reliability despite differences in dynamometer brands. A valid field-based test to assess lower-body muscular power is the SBJ (Fernandez-Santos, Ruiz, Cohen, Gonzalez-Montesinos & Castro-Piñero, 2015). For measuring linear speed, 20mR is used. BSR has low validity for lumbar flexibility but moderate validity for hamstring flexibility (Mayorga-Vega, Merino-Marban & Vician, 2014).

Statistical data analysis

To determine whether the data distribution was normal, the Kolmogorov-Smirnov test (K-S test) was used. The data are presented as mean values with a standard deviation (SD). To determine the relationship between BC variables and PF, the Pearson correlation coefficient was used with correlation intensity ranging from 0.01 to 0.39 (weak), 0.4 to 0.69 (moderate), and 0.7 to 1.0 (strong). For the statistical analysis, SPSS software version 21 (IBM, Chicago, IL, USA) was used, with a significance level of $p=0.05$.

Results

The results of the descriptive statistics used to measure the characteristics of BC are shown in Table 1. The average BH is 179.35 ± 7.36 cm, and BW is 83.70 ± 17.92 kg. The BMI average was very high (26.05 ± 5.52). The results of the descriptive statistics for the assessment of PF measures are also included in the same table.

The K-S test indicated that all variables have a normal distribution at a significance level of $p < 0.05$. Therefore, the Pearson correlation coefficient was used to determine the associations.

When it comes to anthropometric characteristics (table 2), all parameters were highly intercorrelated (.797-.971), except for BH, which had a low correlation with other anthropometric parameters.

Table 1. Descriptive Statistics Results

Variables	Mean	SD	Min	Max	Skew	Kurt	K-S	p
BH	179.35	7.36	161.00	192.00	-.210	-.181	.615	.844
BW	83.70	17.92	61.00	132.30	.788	.001	.829	.498
BMI	26.05	5.52	18.20	41.30	.737	.153	.730	.661
Fat%	22.90	9.45	5.80	39.30	-.021	-1.018	.558	.914
Muscle%	38.31	4.81	29.70	46.60	-.001	-.968	.618	.840
Pacer	4.81	1.81	2	9	.239	-.730	1.010	.259
VO2max	30.42	6.19	20.40	44.90	.250	-.559	.672	.757
PU	12.86	8.54	0	38	1.020	1.197	.855	.458
HG	408.57	84.02	228	566	.194	-.676	.794	.554
SBJ	181.00	26.69	118	238	-.106	-.336	.644	.801
20mR	3.80	.311	3.25	4.45	.430	-.418	.480	.975
Back saver	26.65	9.47	0	47	-.474	1.323	.727	.666

Note. BH-Body height; BW-Body weight; BMI-Body mass index; Fat%-Percentage of body fat; Muscle%-percentage of muscle; Pacer-Progressive aerobic cardiovascular endurance run; VO2max-maximal oxygen consumption; PU-Resistance and strength in the upper body; HG-Maximum isometric strength; SBJ-Lower-body muscular power; 20mR-Linear speed test; BSR-Back-saver sit and reach

Table 2. Correlations Between Body Composition Variables And Physical Fitness

Variables	BH	BW	BMI	Fat%	Muscle%	Pacer	VO ₂ max	PU	HG	SBJ	20mR
BW	.220										
BMI	-.177	.919**									
Fat%	-.300	.797**	.923**								
Muscle%	.225	-.810**	-.904**	-.971**							
Pacer	.402*	-.296	-.458**	-.571**	.549**						
VO ₂ max	.397*	-.308	-.470**	-.580**	.548**	.990**					
PU	.269	-.477**	-.584**	-.687**	.688**	.693**	.713**				
HGP	.636**	.361*	.101	-.023	.027	.312	.315	.076			
SBJ	.564**	.002	-.226	-.411*	.399*	.668**	.666**	.501**	.588**		
20mR	-.309	.215	.355*	.483**	-.482**	-.680**	-.670**	-.453**	-.385*	-.649**	
BSR	.259	.141	.020	-.006	-.044	.087	.076	.015	.288	.339*	-.257

Note: BH-Body height; BW-Body weight; BMI-Body mass index; Fat%-Percentage of body fat; Muscle%-percentage of muscle; Pacer-Progressive aerobic cardiovascular endurance run; VO2max-maximal oxygen consumption; PU-Resistance and strength in the upper body; HG-Maximum isometric strength; SBJ-lower-body muscular power; 20mR-linear speed test; BSR-Back-saver sit and reach; *-Correlation is significant at the 0.05 level (2-tailed); **-Correlation is significant at the 0.01 level (2-tailed).

The correlation analysis between physical fitness parameters showed that PACER and VO2 max were highly correlated with each other, as well as with the tests SBJ, 20mR, and PU. There was a moderate correlation between PU, HG, SBJ, and 20mR, while BSR had a weak correlation with the PF tests.

Regarding the relationship between body composition parameters and physical fitness (table 2), BH achieved a moderate correlation with VO2max, PACER, SBJ, and HG, with weak correlations for the rest. BW had a significant correlation only with HG and PU. While BMI, Fat%, and Muscle% achieved moderate correlations with PACER, VO2max, PU, SBJ, and 20mR.

Discussion

This is the first study of its kind that has been carried out among Serbian theological high school students, as far as we are informed. Main findings of the study are: a) All anthropometric parameters, except for BH, were highly intercorrelated; b) The correlation analysis between physical fitness parameters showed that PACER and VO2max were highly correlated with each other, as well as with the tests SBJ, 20mR, and PU. There was a moderate correlation between PU, HG, SBJ, and 20mR; c) BMI, Fat%, and Muscle% achieved moderate correlations with PACER, VO2max, PU, SBJ, and 20mR, while BW had a significant correlation only with HG and PU. BH had a moderate correlation with VO2max, PACER, SBJ, and HG; d) The students had high average BMI values of over 25 and are classified as overweight.

Since there isn't much information in the literature about using the Eufitmos test battery on a sample of adolescents, the outcomes were compared to test batteries that are more widely used, including the Fitnessgram, EuroFit, or their modifications. Eufitmos is regarded as valid, reliable, relevant, health-related, and specially designed for adolescents (Marques et al., 2021). Our outcomes can be contrasted with those of adolescent participants from seven European countries who completed the Eufitmos assessment battery and submitted information for 2022 (Marques et al., 2024). The methodology used in our research, has been applied in several studies (Marques et al., 2021; Marques et al., 2024; Masanovic et al., 2020).

In this study, a high mutual correlation between anthropometric parameters was established, consistent with previous studies that examined the correlation, especially of weight status parameters, which showed high intercorrelation (Katanic et al., 2023; Kavak, Pilmane, & Kazoka, 2014; Ranasinghe et al., 2013). This is expected since these parameters are used to assess weight status and obesity prevalence in adolescents (Kavak et al., 2014).

When it comes to the interrelationship of PF parameters, our results show a moderate mutual correlation. These findings are in line with previous findings that have generally confirmed a moderate correlation between fitness parameters (Batez et al., 2021). This is somewhat understandable since the given parameters are selected to assess the same component of PF.

Our research findings are consistent with evidence from other studies showing that PF is impacted by BW (Huotari, Nupponen, Laakso, & Kujala, 2009; Johansson Brissman, Morinder, Westerståhl & Marcus, 2020). As evidenced by trends in China (Bi, Zhang, Gu, Song, & Cai, 2020), Sweden (Westerstahl, Barnekow-Bergkvist, Hedberg, & Jansson, 2003),

and New Zealand (Albon, Hamlin, & Ross, 2008), increased BW also related to higher values of BMI. Moreover, our study's outcomes align with those of Yohannes et al. (2020), which found a relationship between a sample of Swedish adolescents' PF measures and their BMI. Mak et al. (2010) suggest that results should be interpreted carefully because lifting a larger BMI may need more energy in people who are obese or have extra BW. BMI and Fat% negatively correlate with parameters PACER and VO2max, which are direct indicators of cardiorespiratory endurance. These findings are consistent with previous studies (Ivanyshyn et al., 2021; Ortega et al., 2007), which also found a negative association between BMI, BW, and Fat% with measured cardiorespiratory fitness.

This understanding supports previous research (Chen, Cui, Zhang, & Peng, 2020; Huang & Malina, 2010) and our study's findings, indicating adolescents with normal weights typically have greater levels of PF than those who are overweight or obese. Some authors have evaluated PF parameters related to strength and flexibility (Chen et al., 2020; Joshi, Bryan & Howat, 2012; Zar, Karan, & Ahmadi, 2017). Consistent with our research findings, these studies have also determined that individuals of normal weight perform better in endurance tests as well as strength and flexibility tests. In adolescents between the ages of 12 and 18, a nonlinear association was found between BMI and the outcomes of PF tests in the study by authors Dewi, Rimawati and Purbodjati (2021). They confirmed that obese or underweight participants performed worse on PF tests than participants with normal nutritional state. Conversely, in healthy adolescents, developing more muscle mass has a significant impact on improving PF. Furthermore, these results generally correspond with the results of our study. A study conducted in our region (Leskošek, Strel & Kovač, 2007) looked at the morphological characteristics and motor abilities of students from different high schools who were 16, 17, and 18 years old. The authors concluded that most variables get better with age. Similar outcomes were seen in the study conducted by Glavač et al. (2015), which looked into variations in students' morphological characteristics and motor abilities over four years at a military high school. Although morphological parameters, or parameters of weight status, are generally observed for assessing obesity (Kavak et al., 2014), preference should be given to body composition analysis using precise measuring devices with defined measurement protocols for these purposes (Đorđević et al., 2024).

These results indicate the importance of promoting physical activity among students to increase their aerobic conditioning and physical fitness (Oforka et al., 2023). When compared to the recommended weekly level of 60 minutes of moderate to vigorous PA per day, five days a week, the level of PA among children and adolescents is not sufficient in many countries (WHO, 2020). Students in high theological schools require PA even when physical education PE sessions are not feasible to organize in practice. This is especially due to the specific conditions of boarding life. The Serbian church community has been significantly engaged in the matter of reforming theological education throughout the last two decades of the 20th century and the beginning of the 21st (Мићин, 2009; Сандо, 2012). Consequently, any prospective reform in the future must endorse the effort to include regular PE as a subject in high theological schools (Bogdanović, 2019). Studies of this nature aim to influence policies to promote physical activity and healthy lifestyles among children and adolescents (Peulić,

Katanic, Jovanović, & Bjelica, (2024). Therefore, this should be just the first step in highlighting the significance of physical fitness for the population's health, calling for collective involvement in taking necessary further steps in that direction.

Practical implications suggest that these findings can be useful in designing appropriate interventions to curb the high rates of overweight and obesity and to improve PF test outcomes among adolescents, especially those who lack access to organized physical activity and live in tightly controlled environments.

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Conflict of interest

The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Enhancing Elementary School Students' Visual Perceptual Skills: A Comparative Study of Oculus Quest 2 Exergames and Conventional Activities

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Abstract

The purpose of this study was to investigate potential differences between two exercise training programs among elementary school students: one based on Oculus Quest 2 exergames and the other on conventional activities, focusing on the Visual Pursuit (VP) skills of visual tracking and selective attention. The study comprised two cohorts: the Oculus Quest (OQ) group and the Typical Activity (TA) group. Engaged a total of forty (n=40) students, specifically selected from the fifth and sixth grades of two public elementary schools situated in the northern region of Greece. The participants, aged 10 to 12 years, consisted of 17 boys and 23 girls. Evaluation of the children's VP abilities occurred through a Visual Pursuit Test administered prior to the interventions (pre-test), after the intervention sessions (post-test), and one month after the interventions (1-month retention test). This program spanned a duration of 6 weeks, with sessions held twice per week, each session lasting 30 minutes. Two-way analyses of variance with repeated measures, were conducted to determine effect of training program groups (TA, OQ) and measurements (pre-test, post-test, 1-month retention test) across time on VP performance. Analysis of the data illustrated that the post-test and 1-month retention test VP scores were remarkably greater than pre-test VP scores in both groups. In conclusion, the findings indicate that both utilizing Oculus Quest 2 VR and conventional activities as interventions show promise in enhancing the VP abilities of elementary school children. This suggests that VR technology, alongside typical methods, could be a valuable tool in Physical Education settings for improving cognitive skills among young learners.

Keywords: visual pursuit, virtual reality, elementary education, perceptual abilities, children

Introduction

Visual perceptual skills play a pivotal role in achieving success across a myriad of sports endeavors. Whether it's striking a baseball, blocking a soccer ball, or sinking a basketball shot, the execution of precise and intricate motor actions heavily relies on the faculty of vision. This intricate interplay between perception and action stands as a cornerstone for excellence in physical education and sports, with the visual senses serving as the fundamental catalyst (Klostermann & Mann, 2019).

The process of transducing light into neural signals relies on the intricate mechanisms of optics and retinal sampling to achieve heightened spatial acuity (Hejtmancik & Shiels, 2015). Moreover, the dynamic nature of ocular movements enables athletes to sustain focus amidst swift bodily motions and changes in the field of view (Kozik & Enns, 2021).

Following transduction, visual information traverses through the intricate pathways of the visual system to the cerebral cortex, where it amalgamates to form three-dimen-



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sional representations (Mahabadi & Al Khalili, 2023). These representations undergo a complex interplay with attentional mechanisms, cognitive control, memory, and motor systems, facilitating seamless perception and action. This symbiotic relationship ultimately culminates in the awe-inspiring displays of athleticism witnessed on the field (Kozik & Enns, 2021).

Different sports necessitate distinct visual aptitudes. For instance, the visual demands placed on a tennis player vastly differ from those placed on a hockey goaltender. A tennis player must swiftly anticipate the trajectory of a fast-moving ball to execute a precise shot, while a hockey goaltender must vigilantly track the puck amidst the flurry of activity in the rink. The ramifications of success or failure in these tasks extend far beyond the realm of sports, encompassing profound physical and psychological repercussions.

Past research endeavors have underscored the divergence in visual skills required across various sports disciplines. Studies examining Olympic athletes and participants in interceptive versus strategic sports have illuminated the nuanced intricacies of visual acuity and perceptual processing (Burris, Liu & Appelbaum, 2020; Laby, Kirschen & Pantall, 2011). The burgeoning field of "sports vision" posits that heightened visual abilities underpin superior athletic performance. Previous investigations have elucidated disparities in visual acuity, contrast sensitivity, and tracking abilities among athletes at different levels of proficiency (Uchida, Kudoh, Higuchi, Honda & Kanosue, 2013; Williams, Ford, Eccles, Ward & Library, 2011). Meta-analyses of physical education and sports expertise literature have further corroborated these findings, elucidating the superior perceptual prowess and attentional processing exhibited by high-achieving athletes in comparison to their counterparts (Mann, Williams, Ward & Janelle, 2007; Voss, Kramer, Basak, Prakash & Roberts, 2010).

Early methods of training for visual skills in physical education and sports primarily consisted of analog drills that placed significant demands on oculomotor skills. Trainees were tasked with rapidly adjusting visual focus and alignment, as well as tracking moving or spatially separated targets through saccadic and smooth pursuit eye movements. Despite initial enthusiasm, empirical studies and consensus reviews have raised doubts about the efficacy of such analog training techniques (Smeeton et al., 2013).

However, recent years have witnessed the emergence of a diverse array of new technologies aimed at revolutionizing sports visual skills training. Digital innovations, including stroboscopic eyewear and mobile tablet-based devices, now facilitate training within the natural context of sporting activities (Appelbaum & Erickson, 2018; Poltavski, Biberdorf, & Poltavski, 2021). Augmented and virtual reality simulations further expand the possibilities by offering realistic and immersive environments for sports-specific training (Akbaş et al., 2019).

The effectiveness of these modern approaches, as outlined in the "Modified Perceptual Training Framework," hinges on various factors. These include the specific perceptual functions targeted for training, the alignment of training stimuli with desired competitive skills, and the congruence between elicited responses and competitive demands. Thus, successful sports visual skills training strategies entail a nuanced interplay of multiple factors within a program that is both feasible

and manageable for athletes who contend with demanding schedules (Hadlow, Panchuk, Mann, Portus, & Abernethy, 2018).

Specifically, Tirp, Steingröver, Wattie, Baker & Schorer (2015) examined the transferability of throwing accuracy and quiet eye duration between virtual and real learning environments, specifically in dart throwing. Participants were divided into virtual training, real training, and control groups. They completed pre- and post-tests on both real and virtual dartboards. The real training group showed superior throwing accuracy compared to the control group, while throwing accuracy was better in the virtual group. Both training groups improved between tests, with the control group performing the worst. Quiet eye duration increased significantly between tests, particularly in the virtual group. Overall, our results indicate the effectiveness of both training methods and suggest a degree of transferability between tasks.

Rauter et al. (2013) examined how well technical skills learned on a scull rowing simulator could transfer to actual sculling on water. Two groups, each consisting of four recreational rowers, were involved, with one group training on water and the other using the simulator. Both groups had four training sessions across two weeks under the guidance of the same licensed rowing instructor. Both groups demonstrated enhanced performance on water. In conclusion, realistic simulator training facilitated skill gains similar to real-world training, suggesting potential for further motor learning enhancement with augmented feedback.

Finally, Gray (2017) examined the transfer of training from virtual to real-world baseball batting among athletes with intermediate experience levels. The study compared a group undergoing adaptive training in a virtual environment with groups undergoing repetitive batting practice, where balls of consistent speed and trajectory were hit, both in the real world and virtual environment. In the adaptive training group, factors like pitch speed and spin would adjust based on performance, either increasing with success or decreasing with failure. Gray (2017) discovered that adaptive training led to significantly greater enhancements in real-world performance compared to both repetitive practice groups.

However, within the domain of sports visual skills, there exists a notable gap in research regarding the reliability and efficacy of Virtual Reality (VR) approaches. Despite advancements in VR technology, studies investigating its utility in enhancing perceptual motor skills performance remain scarce. This underscores the necessity for further exploration and evaluation of VR applications within the realm of sports visual skills to better understand their potential benefits and limitations. Such research endeavors could provide valuable insights into optimizing training methodologies and improving perceptual motor skills performance across various sports disciplines. Therefore, the purpose of this study was to investigate potential differences in the Visual Pursuit (VP) skills of visual tracking and selective attention between two exercise training programs among elementary school students: one based on Oculus Quest 2 exergames and the other on conventional activities. The study comprised two cohorts: the Oculus Quest (OQ) group and the Typical Activity (TA) group. Evaluation of the children's VP abilities occurred through a Visual Pursuit Test administered prior to the interventions (pre-test), after the intervention sessions (post-test), and one month after the interventions (1-month retention test). Specifically, the re-

search addressed the following hypotheses: (H1) There will be no significant difference in VP performance measures between the two groups of children (TA, OQ) at the pre-test stage, and (H2) children in both experimental groups (TA, OQ) will demonstrate improvement in their VP abilities, which will be sustained over time. These findings could inform the development of guidelines regarding the utilization of such devices among children.

Methods

Participants

This study engaged a total of forty ($n=40$) students, specifically selected from the fifth and sixth grades of two public elementary schools situated in the northern region of Greece. Their ages ranged between 10 and 12 years old, with an average age of 10.35 years and a standard deviation of 0.71. Among these participants, there were 17 boys (constituting 42.5% of the sample) and 23 girls (comprising 57.5%).

The sampling approach employed in this research was self-selected sampling, where individuals volunteered to participate. Subsequently, the participants were randomly divided into two distinct groups, each consisting of 16 children: the TA group, comprising 5 boys and 15 girls, and the OG group, comprising 12 boys and 8 girls.

Before the assignment to groups, a screening process was conducted for children whose parents had expressed interest in their involvement. This screening aimed to confirm the participants' willingness to participate and ensure they met the study's requirements. These requirements were clarified to the children, and they were assessed against specific inclusion and exclusion criteria.

The inclusion criteria specified that participants must be between the ages of 10 and 12 years at the commencement of the study, possess proficiency in using exergames, and commit to attending all intervention program sessions. Conversely, the exclusion criterion encompassed the presence of any clinically severe illness or disorder that would impede the participant's ability to engage in the intervention program effectively.

Furthermore, prior to the voluntary participation of each child in the study, informed consent was diligently obtained from their respective parents or legal guardians. This comprehensive process ensured that all ethical considerations were meticulously addressed before the commencement of the research.

The Ethics Committee of the Faculty of Physical Education and Sport Sciences at Democritus University of Thrace approved (decision number: 39936/277; date: 29.03.2023) that this experimental study complied with all ethical guidelines for research involving human subjects, as stipulated by the Helsinki Declaration (World Medical Association, 2023).

Procedures

Prior to the data collection, meticulous efforts were undertaken to ensure full compliance with ethical standards and procedures. Comprehensive parent permission forms and child assent forms were distributed and collected, affirming participants' voluntary consent to take part in the study. These documents outlined the nature of the research, including its objectives, procedures, and potential risks and benefits, providing parents and children with a clear understanding of their roles and responsibilities.

Upon securing all necessary permissions and assents,

the data collection phase began with the administration of pre-tests. These pre-tests were conducted diligently between January 23rd and February 3rd, allowing for the initial assessment of participants' baseline visual pursuit skills prior to any intervention.

Following the completion of pre-testing, a rigorous random assignment process was implemented to ensure the formation of two distinct experimental groups: the TA group and the OQ group. This random assignment method helped to minimize biases and ensure the comparability of the two groups, thereby enhancing the internal validity of the study.

Both experimental groups participated in a structured visual perceptual (VP) training program focused on enhancing visual tracking abilities. The program lasted 6 weeks, with sessions held twice a week for 30 minutes each.

As the 6-week intervention period drew to a close on March 17th, participants underwent post-test assessments to evaluate the immediate effects of the VP training program on their visual pursuit abilities. These post-test assessments provided valuable insights into the efficacy of the intervention in promoting skill acquisition and improvement.

Furthermore, to assess the durability of acquired skills over time, retention test assessments were scheduled one month after the conclusion of the intervention period. This extended assessment timeline allowed researchers to examine the retention and transferability of skills acquired during the VP training program, providing critical insights into the long-term impact of the intervention on participants' visual pursuit abilities.

In summary, the meticulously structured timeline of the study, encompassing pre-tests, intervention sessions, post-tests, and retention tests, facilitated a comprehensive examination of the effectiveness and sustainability of the VP training program in enhancing participants' visual pursuit skills.

Experimental treatments

As mentioned before, both experimental groups embarked on a structured VP training program, meticulously designed to target and enhance visual tracking abilities. The carefully structured nature of the VP training program aimed to optimize learning outcomes while accommodating the participants' schedules and minimizing potential fatigue or burnout.

The VP training program officially launched on February 6th, marking the beginning of an intensive and immersive learning experience for participants. Over the ensuing weeks, participants engaged in a series of engaging and interactive activities tailored to enhance their visual pursuit skills, leveraging both traditional activities (e.g., change directions, changing reactions, you pass, badminton, dodge ball, etc.) and innovative Oculus Quest exergames (e.g., REAKT performance trainer, eleven table tennis, beat saber, holofit, etc.). The traditional activities were designed based on established physical education practices (e.g., Logan, Robinson, Wilson & Lucas, 2012; Payne & Isaacs, 2011), while the Oculus Quest exergames were selected for their proven effectiveness in enhancing visual and motor skills in previous research (e.g., Amprasi, Vernadakis, Zetou & Antoniou, 2022; Hashemi, Khodaverdi & Zamani, 2022).

A typical lesson for the intervention groups began with children transitioning from their classrooms to engage in Visual Pursuit (VP) abilities instruction. The session started

with warm-up activities and games designed to prepare the children for the tasks ahead, with each child having their own space for these activities, which lasted 2 to 3 minutes. The children were then randomly divided into two VP abilities stations, where they participated in specific VP games. Each half of the class spent 12 minutes on one VP game before switching to the other station, allowing each child to spend a total of 24 minutes actively participating in two distinct VP games. The session concluded with closing activities, including 2 to 3 minutes of stretching exercises. During this time, the importance of critical elements for the development of VP abilities was highlighted through interactive discussions, encouraging the children to engage and reflect on the key concepts of the session.

Measures

The present study utilized the Visual Pursuit test, a component of the Vienna Test System developed by Schuhfried GmbH in Austria. Widely recognized in psychological diagnostics (Schmid, Sauter, Stepansky, Lobentanz & Zeitlhofer, 2005), this computerized assessment tool is specifically designed to gauge concentrated targeted perception and selective attention within the visual domain. Consequently, successful performance on this test necessitates both selective and sustained attention abilities.

Comprising a short form of 40 items, the test presents participants with arrays of nine intricately intertwined dark lines leading to nine distinct endpoints against a light background computer screen. Each item begins with a marked starting point on one of the lines, prompting participants to visually track this line to its corresponding endpoint as swiftly as possible. Participants are required to respond by pressing one of nine number buttons on a response panel within a four-second time frame before the screen disappears. Any items not responded to correctly or within the designated time limit are marked as incorrect responses.

Automated scoring of participant performance considered the number of correct responses and the mean Reaction Time (RT) for correct answers. The mean Reaction Time for correct answers (in seconds) was standardized, with all norms provided as percentile ranks. Scores on the Visual Pursuit test ranged from 0 to 24, indicating below-average visual perception; scores from 25 to 75 showed average visual perception; and scores from 76 to 100 represented above-average visual perception. A higher score signifies better selective attention and orientation performance. Each test run

typically lasts approximately 6 minutes per participant. The Visual Pursuit test's reliability and validity have been well-established through extensive research efforts. For example, a study conducted by Biehl, Wagner, Karner & Bernd (2005) reported high internal consistency for the short form of the test, with Cronbach's alpha coefficients reaching .92, indicating excellent reliability.

Statistical analysis

The research design utilized in this study followed a pre-test/post-test control group structure with a 1-month retention test, wherein participants were assigned to groups through randomization. This random assignment process was executed using computer-generated random student numbers, which determined the allocation of students to respective groups. Prior to data analysis, thorough screening was conducted to ensure adherence to statistical assumptions, revealing no detected violations (Green & Salkind, 2013).

To evaluate the impact of the training programs and the progression of measurements over time on Visual Pursuit (VP) performance, a two-way analysis of variance (ANOVA) with repeated measures was employed. The dependent variable under scrutiny was VP test scores. The within-individual factors included training program groups (TA, OQ) and time points (pre-test, post-test, 1-month retention test). Significance of mean differences across time points was determined at the 0.05 alpha level. Additionally, effect size calculations were conducted using the partial eta-squared statistic (partial η^2) to gauge the practical significance of the findings, with interpretation based on Cohen's guidelines: 0.01=small, 0.06=medium, and 0.14=large (Cohen, 1988).

Results

An independent samples t-test was conducted to evaluate the hypothesis that both experimental groups (TA and OQ) would not differ significantly on measure of VP performance at pre-test. Indeed, there were no significant initial differences between the two intervention groups in the mean VP scores, $t(38)=1.12$, $p=.27$.

Two-way analysis of variance (ANOVA) with repeated measures was conducted to evaluate the Hypothesis II (that children in both the TA and the OQ groups would improve and retain their VP abilities). As demonstrated in what follows, this hypothesis was corroborated.

Table 1. Results of two-way analysis of variance (ANOVA) with repeated measures.

Factors' effects	Sum of Squares	df	Mean Square	F	p	η^2p
Time	4241.0	2	2120.48	31.944	<0.001	0.457
Time * Group	14.1	2	7.06	0.106	0.899	0.003
Group	267	1	267	2.24	0.143	0.056

A significant main effect was noted for Time, $F(2, 38)=31.94$, $p<0.001$, partial $\eta^2=0.457$, while the Training programs x Time interaction effect was not significant, $F(2, 38)=0.106$, $p=0.899$, partial $\eta^2=0.003$. The univariate test associated with the Group's main effect was also not significant, $F(1, 38)=2.24$, $p=0.143$, partial $\eta^2=0.056$.

Pairwise comparisons using t-test with a Bonferroni adjustment were used to analyzing the main effect of Time. The

results revealed significant mean differences in VP scores between pre-test and post-test ($MD=-13.12$; 95% CI: 62.2 to 78.8, $p<0.001$) and between pre-test and 1-month retention test ($MD=11.13$; 95% CI: 62.2 to 76.6, $p<0.001$) in both experimental groups (TA, OQ). As shown in Figure 1, the post-test and 1-month retention test VP scores were remarkably greater than pre-test VP scores for both experimental groups.

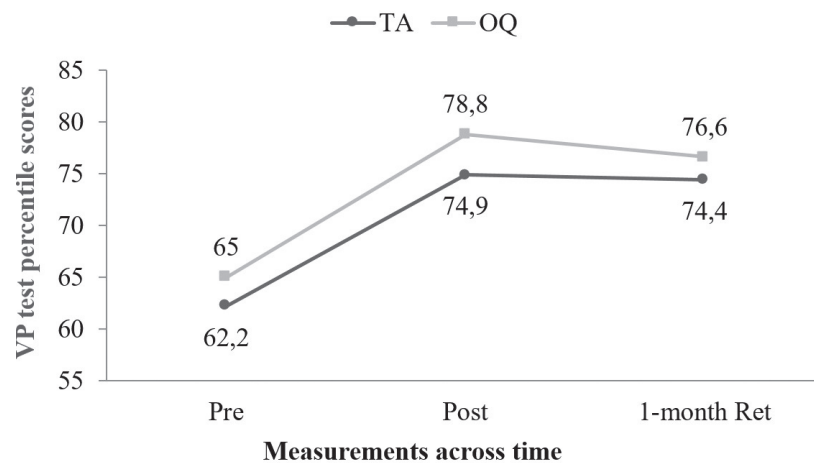


FIGURE 1. VP performance of the three groups on all measurements across time

Discussion

In recent years, there has been a growing body of research focusing on the physical and cognitive advantages of visual pursuit (VP) exercises for children (Dalziell, Booth, Boyle & Mutrie, 2019). VP exercises involve engaging multiple body parts and/or manipulating objects to accomplish goal-directed actions, necessitating precise timing, temporal, and spatial estimation, which in turn requires perceptual and high-level cognitive information processing (Pesce, 2012). For instance, activities such as kicking, throwing, catching, dribbling, volleying, or striking a ball exemplify coordinative exercises, where children must coordinate visual input with manual actions.

Considering the existing body of knowledge, our research sought to explore potential differences between two Visual Pursuit (VP) exercise training programs for elementary school students: one utilizing Oculus Quest 2 exergames and the other involving conventional activities. In order to achieve this goal, researchers established specific hypotheses to direct our investigation. The following sections detail the findings and conclusions derived from our study.

The first research hypothesis, which posited that there would be no significant differences in Visual Pursuit (VP) skills between the two experimental groups during the pre-test phase, was substantiated by our findings. This outcome can be interpreted in several ways. One possibility is that the VP tasks administered during the pre-test phase were inherently familiar and accessible to all participants, regardless of their group assignment. Consequently, the lack of significant differences in VP skills at the pre-test stage highlights the importance of considering contextual factors and task familiarity when interpreting research findings in educational settings.

The confirmation of the second research hypothesis further strengthens the validity of our study. It suggested that both the OQ and TA groups experienced improvements in their VP skills and were able to maintain these improvements over time. This finding was supported by the significant increase observed in the VP scores at both the post-test and the 1-month retention test, compared to the pre-test scores, across both experimental groups.

These outcomes are consistent with previous research findings that emphasize the effectiveness of early instruction in visual perceptual skills in enhancing children's perceptual development (Amprasi et al., 2022; Hashemi et al., 2022). The

positive outcomes observed in our study align with the existing body of literature, providing further evidence of the beneficial effects of targeted interventions aimed at improving VP skills among elementary school children.

The results of the post-test were anticipated, given the documented effectiveness of visual perceptual skill instruction in facilitating immediate changes in children's perceptual abilities (Logan et al., 2012; Payne & Isaacs, 2011). However, the findings from the 1-month retention test are particularly noteworthy, as they demonstrate that participants in both the OQ and TA groups were able to sustain their improved VP skills over the course of one month following the completion of the intervention. This suggests that the interventions were not only effective in the short term but also led to enduring improvements in VP skill development.

These retention results are significant in the context of visual perceptual skill development literature, as they provide evidence of long-term retention, which is less commonly reported in existing studies (Hashemi, Khodaverdi & Zamani, 2022; Robinson & Goodway, 2009). The retention test results highlight the distinction between immediate perceptual learning observed in post-test assessments and the more enduring effects of perceptual skill development demonstrated in retention tests. This differentiation is crucial for understanding how initial gains in perceptual skills can be solidified and maintained over time, a topic that warrants further exploration in future research.

Moreover, the effectiveness of the Oculus Quest-based intervention in enhancing students' perceptual abilities can be attributed to several factors. The immersive and interactive nature of Oculus Quest exergames likely played a critical role in this regard. Previous research has shown that VR environments can enhance cognitive and motor skills by engaging multiple sensory modalities and providing real-time feedback, which promotes deeper learning and skill acquisition (Brown et al., 2020; Grosprêtre, Marcel-Millet, Eon & Wollesen, 2023). VR environments have been found to facilitate experiential learning, which is critical for the development of perceptual skills (Kolb, 1984; Merchant et al., 2014).

The adaptive and personalized nature of Oculus Quest technology allowed for tailored learning experiences that cater to individual learning styles and preferences. This adaptability is consistent with findings from previous studies that highlight the benefits of adaptive learning technologies in

optimizing educational outcomes (Clark & Feldon, 2014; Johnson & Mayer, 2009). Personalized learning experiences have been shown to significantly enhance engagement and motivation, which are essential for sustained learning and skill retention (Schunk & Zimmerman, 2012; Wang, Woo, Quek, Yang & Liu, 2014).

Furthermore, the active engagement and intrinsic motivation elicited by the VR exergames likely contributed to the sustained improvements observed in VP skills. Research indicates that interactive and engaging learning environments can lead to higher levels of motivation and, consequently, better learning outcomes (Deci & Ryan, 2000; Ryan & Deci, 2018). The motivational aspects of VR exergames, which combine elements of gamification and immersive experiences, have been shown to enhance both cognitive and physical engagement in learning activities (Hamari & Keronen, 2017; Lister, West, Cannon, Sax, & Brodegard, 2014).

The traditional activities, which were based on established physical education practices, also proved effective in enhancing VP skills. These activities, such as those involving changes in direction, reaction exercises, and sports like badminton and dodge ball, have been shown to improve motor and perceptual skills through repetitive practice and feedback (Gabbard, 2012; Vernadakis et al., 2012). These activities require children to process visual information quickly and accurately, enhancing their visual perceptual skills and selective attention.

Research has consistently demonstrated that physical activities involving coordinated movement and perceptual tasks can significantly improve visual perceptual skills in children (Payne & Isaacs, 2011). For example, activities that require children to track moving objects, such as catching or striking a ball, help develop their ability to anticipate and respond to visual stimuli (Logan et al., 2012). These tasks not only improve visual tracking but also enhance hand-eye coordination, spatial awareness, and reaction times, all of which are critical components of visual perceptual skills (Robinson & Goodway, 2009).

Moreover, traditional activities that incorporate dynamic and varied movements can stimulate cognitive processes related to attention, memory, and executive functions. For instance, research by Alesi, Bianco, Luppina, Palma & Pepi (2016) demonstrated that participation in complex physical activities, such as a football exercise program, which demand rapid decision-making and problem-solving, can effectively improve cognitive flexibility and executive control in children. This is supported by findings from Best (2010), who noted that physical activities requiring rapid shifts in attention and control, such as those found in sports and team games, can improve cognitive functions related to attention and inhibitory control.

Thus, it appears that the findings of this study highlight the potential of both traditional activities and the innovative Oculus Quest educational games to enhance and maintain visual perceptual skills in elementary school children. Traditional activities, which have long been valued for their role in cognitive and sensory development, continue to demonstrate their effectiveness in fostering visual perception. Meanwhile, the integration of Oculus Quest educational games introduces a modern, interactive approach that not only engages children through immersive experiences but also aligns with the technological advancements and digital fluency of today's youth.

Together, these methods provide a comprehensive strategy to support the visual perceptual development of elementary school students, offering a balanced blend of conventional and cutting-edge educational tools.

Limitation

Upon assessing the findings of the current research, it's imperative to acknowledge several limitations. Firstly, one significant challenge was the practicality of conducting scientific investigations within a live school environment. The testing conditions were inherently non-clinical, subject to regular fluctuations, which may have impacted the consistency of data collection.

Furthermore, it's important to recognize that the results reported in this study were based on the utilization of four specific interactive gaming software platforms (REAKT performance trainer, Eleven Table Tennis, Beat Saber, & Holofit). This specificity introduces a potential limitation, as different gaming software covering alternative games and exercises may yield varied outcomes.

Lastly, the interventions targeting Visual Pursuit (VP) skills were implemented exclusively in two public elementary schools located in northern Greece. The characteristics of these specific schools may have influenced the results, thereby limiting the generalizability of findings to broader populations.

Despite these acknowledged limitations, it is hoped that this study contributes to the existing body of knowledge surrounding Physical Education practices, particularly regarding the utilization of exergames to enhance children's perceptual development. By identifying these constraints, future research endeavors can aim to address and overcome these challenges, further advancing our understanding of the potential benefits and limitations associated with exergaming in educational contexts.

Conclusion

This study represents a significant contribution to the field of educational research and physical education by exploring the efficacy of VP skill interventions using Oculus Quest 2 exergames among elementary school students. The findings of this research highlight several key insights.

Firstly, the results confirm the effectiveness of both Oculus Quest-based interventions and conventional activity-based interventions in improving VP skills among elementary school children. The significant improvements observed in VP scores at both post-test and 1-month retention test sessions underscore the potential of these interventions to enhance perceptual abilities over time.

Secondly, the study identifies various factors contributing to the success of the interventions. The immersive and interactive nature of the Oculus Quest 2 environment engages multiple sensory modalities, facilitating experiential learning and skill acquisition. Additionally, the personalized and adaptive nature of Oculus Quest 2 technology enables tailored learning experiences, catering to individual needs and preferences.

Moving forward, future research could explore the long-term effects of Oculus Quest-based interventions on perceptual development and academic performance among elementary school students. Additionally, investigating the optimal duration and frequency of exergaming interventions, as well as their potential effects on other aspects of children's health

and well-being, could further enhance our understanding of the role of exergaming in education.

Overall, this research contributes valuable insights into the potential of exergaming interventions to improve perceptual development among elementary school students. By elucidat-

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Conflict of interest

The authors declare that there are no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Effect of Different Instructions on Learning Proprioceptive Neuromuscular Facilitation Skills among Physiotherapy Students

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Abstract

This study aims to assess the effect of different instructions on learning proprioceptive neuromuscular facilitation (PNF) skills among physiotherapy students. Twenty-five (n=25) first-year undergraduate physiotherapy students were randomly assigned into two groups. The first group (n=12) was a live demonstration group watching the instructor performing the technique, while the second group (n=13) was a video demonstration group watching a pre-recorded video by the same instructor. Both groups were shown identical eight (8) patterns of PNF skills. After watching the skill, the students in both groups were asked to perform in an isolation room using all the techniques shown to measure their psychomotor skills using the standard Objective Structured Clinical Examination form. The test revealed an insignificant difference in the total performance score between live and video demonstrations of PNF skills among physiotherapy students. Live demonstration group (Mean=59.50±25.69) and video demonstration group (Mean=75.00±13.71), U=48.5, z=-1.605, p>0.05, r=0.321. The finding of this study indicates that live and video demonstration groups are on par in performing the PNF skills. Although video demonstration may reduce time teaching and at the same time ensure the students learn motor skills accordingly. The current study suggests that video demonstration should be employed as an adjunct to pre-, during, and post-live demonstration to augment knowledge retention. This finding can also be a reference for policymakers or higher institutions to decide to change the existing policy for new norms after the break-out of COVID-19 as there is no difference in students' performance between these two groups.

Keywords: Covid-19, proprioceptive neuromuscular facilitation, live instructions, video instructions

Introduction

Proprioceptive Neuromuscular Facilitation (PNF) is part of the treatment given by the physiotherapist to the patient to promote or hasten the response of the neuromuscular mechanism through stimulation of the proprioceptors, and it is aimed at effect motor learning from a lasting response of the neuromuscular mechanism (Chaturvedi et al., 2017; Smedes, Heidmann, Schäfer, Fischer, & Stępień, 2016). Sagittal, frontal, and transverse planes and movement diagonally in line with muscle fiber arrangement were the patterns involved. There were three types of contraction combined in PNF which are concentric, isometric, and eccentric contraction and at the

same time, auditory, visual, and cutaneous also been stimulated to improve neuromuscular facilitation (Nunes & Martins e Silva, 2016).

PNF is known for improving the flexibility and strength of the muscles, reducing pain and stiffness, and improving trunk stability. According to Lim, Nam and Jung (2014) PNF stretching is being used widely in athletic training, and it serves to avoid injuries. Borges, Medeiros, Minotto and Lima (2018) claim athletes who sustain hamstring tightness can implement PNF stretching to enhance hamstring flexibility due to it is effectiveness in improving the flexibility of the muscles. Besides improving flexibility and strength of the muscles Birinci,



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Razak Ozdinciler, Altun and Kural (2019) claim an individual with a golfer's elbow (elbow tendinitis) can benefit from PNF stretching by alleviating pain, enhancing range of motion, and reducing stiffness. However, studies by Papadimitriou, Loupos, Tsalis, and Manou (2017) on PNF among swimmers didn't produce a positive or negative effect in their results (i.e. 10 10-10-second kick trial & hip curl and extension).

PNF requires a complex task and requires many demonstrations due to mapping ample input space to a large output space, for example, from camera input to the desired joint movement (Manschitz, Gienger, Kober, & Peters, 2018). Farsi, Bahmanbegloo, Abdoli, and Ghorbani (2016) stated that coaches, instructors, and physical educators used live demonstrations widely when teaching new motor skills. According to White, Rodger, and Tang (2016), skill is a stable and reliable link between the perception of humans and the environmental constraint to perform the task, which is compatible across persistent performances of the action and can be adapted to changes in task constraints flexibly. While this task is trained repeatedly, it can enhance spatial and temporal, which control the accuracy of movements (Bonassi et al., 2020).

Face-to-face is a well-recognized form of learning; however, it is necessary to prepare for distance learning in higher institutions due to unexpected events in the future such as COVID-19 (Arifiati et al., 2020). The COVID-19 pandemic has put forth a great challenge for higher institutions with the remarkable responsibility of reconsidering delivering excellent quality education when implementing social distancing policies by the government to prevent further spreading the virus. As a result, all higher institutions carried out distance learning to overcome the constraints in delivering the lessons and continuing the academic schedule (Amir et al., 2020; Zutshi, 2020). Various approaches to e-learning are being used to make sure the continuousness of the higher education learning process does not stop, including Zoom, Google Meet, Google Classroom, Telegram, and many more platforms (Kapasia et al., 2020).

Even though live demonstration are well known for delivering the best outcome in motor learning skills for the past several decades, Seals, Gustowski, Kominski, and Li (2016) claim a live demonstration can be replaced by a video demonstration to give more time on assessment and feedback in the practical session. There were several advantages of video demonstration compared to live demonstration. Brockfeld, Müller, and de Laffolie (2018) mention that one of the advantages was video demonstration could be repeated at any time and place while the student does not have to attend lecture class. Besides that, the university improves the learning process through e-learning. In addition, the learning effects of live and video demonstrations are equal, and this has been confirmed in other fields (Azeem, Ali, Shaukat & Arfan ul Haq, 2018; Thulakumar, Jayasinghe, Rasnayaka, Jayasinghe, & Abeysundara, 2018).

Nowadays demand for delivering the best approach to education in higher institutions puts great pressure on lecturers teach motor learning skills in physiotherapy programs. With the constraint of live demonstration teaching hours and increasing demand in clinical settings and real-world situations, the students need to prepare themselves to overcome this limitation, especially in the pandemic situation of COVID-19 (Stone, Cooke, & Mitchell, 2020). Video demonstrations are pre-recorded demonstrations of the motor skills that need to be learned and are part of the solution that is thought can fill the gap in delivering and transferring the knowledge

among the students. Therefore, further research and evaluation of transferability of the motor learning skills through watching video demonstrations need to be done compared to live demonstrations which are already known to deliver the knowledge (Stone et al., 2020).

Nevertheless, to our knowledge, the effectiveness of video demonstration compared with a live demonstration in acquiring hands-on skills among physiotherapy students is still limited in the previous literature. PNF exercises are among the complex exercises that need to be learned by the student to conduct the exercise to the client. Information technologies are alternative to fill the gap or lack in lab activity on campus. Therefore, the purpose of this study was to compare live instruction with video-based instruction methods for learning proprioceptive neuromuscular facilitation skills among physiotherapy students.

Material and methods

Participants

The study population was Physiotherapy students in a local university. The sampling technique that was used in this study was purposive sampling. Only semester one student were selected due to only this group had not learned or knew anything about PNF while the other Physiotherapy students in higher semesters had learned PNF. The students participated voluntarily in this study and was assigned into two groups: live demonstration group ($n=12$, mean age 19.67 ± 0.89 years old) and video demonstration group ($n=13$, mean age 19.69 ± 1.11 years old). This study was approved by the Research Ethics Committee, Research Management Centre, Universiti Teknologi MARA (REC/09/2021(MR/762)).

Procedures

During the live demonstration, the researcher performed the eight patterns of PNF skills, and the participants could only watch while disallowed to interrupt or ask any questions in that session. It took only 20 minutes to perform all eight PNF patterns (Diagonal 1 Flexion Upper Limb, Diagonal 1 Extension Upper Limb, Diagonal 2 Flexion Upper Limb, Diagonal 2 Extension Upper Limb, Diagonal 1 Flexion Lower Limb, Diagonal 1 Extension Lower Limb, Diagonal 2 Flexion Lower Limb, and Diagonal 2 Extension Lower Limb). For the video demonstration, the researcher did a pre-recorded video on eight patterns of PNF skills (same movement as in the live demonstration group), and participants only watched the video through the LCD provided by the researcher while disallowed to interrupt or ask any question in that session. It took only 20 minutes to watch all the PNF patterns. The equipment used to record the demonstration in this study was a handphone camera model OPPO F11 (Oppo, Hong Kong), while during the video demonstration laptop brand Lenovo (Chengdu, China) was used to play the video. The measurement of PNF skills was taken in one trial during the post-test. A researcher with more than 10 years of experience teaching the Physiotherapy Program conducted the assessments using the Objective Structured Clinical Examination Form. The participants were graded one at a time using the same human model. All eight patterns of PNF skills were assessed for each participant at one time by the researcher. The participant read the instructions given and followed by performing the PNF skills for each eight pattern involving four patterns for the upper limb and four patterns for the lower limb. The participants started by intro-

ducing themselves and then they explained the procedure to the model. After that, the participant is required to move the model's limb passively three times followed by asking the model to perform the movement pattern three times while the participant resists the movement as shown in the demonstration.

Data Analysis

IBM SPSS Statistics for Windows Version 21.0 (IBM Corp., Armonk, NY, USA) was used for statistical analysis. The data was checked using Shapiro-Wilk Test to identify normal distribution. Mann-Whitney U test was carried out to determine the significant level of the Objective Structured Clinical Examination (PNF Movement) results due to the data are generally not normal distributed when using Shapiro-Wilk Test ($p < 0.05$). The significant level of this study was set at $p < 0.05$ and the effect size (r) of < 0.3 (small effect), 0.3 to 0.5 (medium) and > 0.5 (large).

Results

The performance score of PNF Diagonal 1 Flexion Upper Limb pattern among physiotherapy students (Table 1) by live demonstration (Mean=11.17±1.80) and video demonstration group (Mean=11.69±1.03), $U=61.5$, $z=-0.924$, $p=0.355$, $r=0.185$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 1 Extension Upper Limb pattern among physiotherapy students by live demonstration (Mean=9.75±4.86) and video demonstration group (Mean=10.92±1.04), $U=73.5$, $z=-0.249$, $p=0.803$, $r=0.050$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Flexion Upper Limb pattern among physiotherapy students by live demonstration (Mean=6.08±5.55) and video demonstration group (Mean=8.23±4.02), $U=61.0$, $z=-0.0944$, $p=0.345$, $r=0.0190$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Extension Upper Limb pattern among physiotherapy students by live demonstration (Mean=6.17±5.52) and video demonstration group (Mean=8.00±4.81), $U=65.5$, $z=-0.704$, $p=0.482$, $r=0.141$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 1 Flexion Lower Limb pattern among physiotherapy students by live demonstration (Mean=7.25±4.58) and video demonstration group (Mean=10.15±3.29), $U=35.5$, $z=-2.344$, $p=0.019$, $r=0.459$ (there was a significant difference between the two groups). The performance score of PNF Diagonal 1 Extension Lower Limb pattern among physiotherapy students by live demonstration (Mean=6.92±5.20) and video demonstration group (Mean=9.62±3.20), $U=59.5$, $z=-1.027$, $p=0.305$, $r=0.205$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Flexion Lower Limb pattern among physiotherapy students by live demonstration (Mean=5.50±4.70) and video demonstration group (Mean=8.62±3.36), $U=48.0$, $z=-1.654$, $p=0.098$, $r=0.331$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Extension Lower Limb pattern among physiotherapy students by live demonstration (Mean=6.67±5.42) and video demonstration group (Mean=7.77±3.85), $U=71.5$, $z=-0.358$, $p=0.720$, $r=0.072$ (there was no significant difference between the two groups).

Table 1. The results obtained by live and video demonstrations group for each PNF patterns

PNF Movements	Live		Video		U	z	p	r
	Mean	SD	Mean	SD				
Diagonal 1 Flexion Upper Limb	11.17	1.80	11.69	1.03	61.5	-0.924	0.355	0.185
Diagonal 1 Extension Upper Limb	9.75	4.86	10.92	1.04	73.5	-0.249	0.803	0.050
Diagonal 2 Flexion Upper Limb	6.08	5.55	8.23	4.02	61.0	-0.0944	0.345	0.0190
Diagonal 2 Extension Upper Limb	6.17	5.52	8.00	4.81	65.5	-0.704	0.048	0.141
Diagonal 1 Flexion Lower Limb	7.25	4.58	10.15	3.29	35.5	-2.344	0.019	0.459
Diagonal 1 Extension Lower Limb	6.92	5.20	9.62	3.20	59.5	-1.027	0.305	0.205
Diagonal 2 Flexion Lower Limb	5.50	4.70	8.62	3.36	48.0	-1.654	0.098	0.331
Diagonal 2 Extension Lower Limb	6.67	5.42	7.77	3.85	71.5	-0.358	0.720	0.072

Abbreviations: U - Mann-Whitney U Test; p - significant level; z - z score; r - effect size.

The performance score of PNF Diagonal 2 Extension Upper Limb pattern among physiotherapy students by live demonstration (Mean=6.17±5.52) and video demonstration group (Mean=8.00±4.81), $U=65.5$, $z=-0.704$, $p=0.482$, $r=0.141$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 1 Flexion Lower Limb pattern among physiotherapy students by live demonstration (Mean=7.25±4.58) and video demonstration group (Mean=10.15±3.29), $U=35.5$, $z=-2.344$, $p=0.019$, $r=0.469$ (there was a significant difference between the two groups). The performance score of PNF Diagonal 1 Extension Lower Limb pattern among physiotherapy students by live demonstration (Mean=6.92±5.20) and video demonstration group (Mean=9.62±3.20), $U=59.5$, $z=-1.027$, $p=0.305$, $r=0.205$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Flexion

Lower Limb pattern among physiotherapy students by live demonstration (Mean=5.50±4.70) and video demonstration group (Mean=8.62±3.36), $U=48.0$, $z=-1.654$, $p=0.098$, $r=0.331$ (there was no significant difference between the two groups). The performance score of PNF Diagonal 2 Extension Lower Limb pattern among physiotherapy students by live demonstration (Mean=6.67±5.42) and video demonstration group (Mean=7.77±3.85), $U=71.5$, $z=-0.358$, $p=0.720$, $r=0.072$ (there was no significant difference between the two groups).

The test revealed insignificant difference (Table 2) in the total performance score between live and video demonstration on PNF skills among physiotherapy students by live demonstration group (Mean=59.50±25.69) and video demonstration group (Mean=75.00±13.71), $U=48.5$, $z=-1.605$, $p=0.109$, $r=0.321$ (there was no significant difference between the two groups).

Table 2. Mean (SD) for score performance for the total mark of all the patterns

PNF Movements	Live		Video		U	z	p	r
	Mean	SD	Mean	SD				
Total Performance Score	59.50	25.69	75.00	13.71	48.5	-1.605	0.109	0.321

Abbreviations: U - Mann-Whitney U Test; p - significant level; z - z score; r - effect size.

Discussion

The purpose of this study was to determine the effect of different instructions on learning proprioceptive neuromuscular facilitation (PNF) skills among physiotherapy students.

In this study, the participants were given 20 minutes each to learn the PNF technique either via live or video demonstrations. It was hypothesized that there is no significant difference in total performance score between live and video demonstra-

tion of PNF skills among physiotherapy students, and it was proved that the result was the same. This finding is consistent with some recent studies (Alqahtani et al., 2015; Azeem et al., 2018; Gorucu-Coskuner, Atik & Taner, 2020). The causes of no significant differences between live and video demonstration group in this study maybe due to experience by the participants and the time constraints when participants receiving live instructions. The content of delivering instructions for live and video demonstrations in this study are similar, thus the participants can follow the instructions and remembers the flow the movements easily in order to perform the PNF movements. In addition, all participants in this study were in the first year, and they don't any experience yet in performing the movement task. However, in the previous study by Ramlogan, Raman and Sweet (2013) and Gorucu-Coskuber, Atik and Taner (2020), the participants in their study were in the third and fifth year, and have some basic experience in performing the skill tasks, that produced significant differences between live and video instructions groups.

During live instructions, participants have to shift their eyes many times to follow the cues by the instructor in order to look for important points when learning the PNF movement patterns. Preston et al. (2012) claimed the weaknesses of live demonstration as current students of physiotherapy program were taught through the live demonstration. After a live demonstration, the lecturer will allow students to practice and get feedback in a limited time during class. The students who wholly depend solely on memory or simple notes taken during the class will be inaccurate in practicing the skills acquired via live demonstration. Alqahtani et al. (2015) also agreed with Preston et al. (2012) by revealing that live demonstration had disadvantages: lack of manpower to demonstrate, low ratio of lecturer to students, and time constraint to demonstrate. In

addition, despite there were no significant differences between the live and video demonstration groups, participants in the video group may see better on the technique as they can bring the device closer to their eyes, or can focus on the image as compared to the video demonstration where in reality it will be with every-one crowding around (Alqahtani et al, 2015).

As this study found no significant differences between live and video demonstrations, the usage of these methods are advisable for skill learning task in physiotherapy, sports and clinical settings. Gorucu-Coskuber, Atik and Taner (2020) states that the participants in the study preferred both method of instructions in order to achieve highest effect in the learning the skills task. In addition, the use of both methods also should be considered as the learning style of the students are not the same (Alqahtani et al, 2015). Future studies also should look at whether combining live and video demonstrations can produce highest learning effect and compared to live or video demonstrations only.

Conclusion

The finding of this study indicates that live and video demonstration groups are on par in performing the PNF skills. This finding can also be a reference for policymakers or higher institutions to make necessary changes to the existing policy in the education sector after the breakout of COVID-19 as there is no difference in students' performance between these two groups. The video demonstration is equally as practical as a live demonstration. To improve students' learning experiences and match multiple intelligences, both methods should be considered in teaching undergraduate physiotherapy students. Future studies should investigate the effect of different feedback methods on learning PNF skills among physiotherapy students in the universities.

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Conflict of interest

The authors declare that there are no conflict of interest.

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Guidelines for Authors

Revised September 2019

*** Please use the bookmark function to navigate within the guidelines. ***

When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

1. UNIFORM REQUIREMENTS

1.1. Overview

The *Sport Mont* (SM) applies the Creative Commons Attribution (CC BY) license to articles and other works it publishes.

The submission with SM is free of charge but author(s) has to pay additional 215 euros per accepted manuscript to cover publication costs. If the manuscript contains graphics in color, note that printing in color is charged additionally.

SM adopts a double-blind approach for peer reviewing in which the reviewer's name is always concealed from the submitting authors as well as the author(s)'s name from the selected reviewers.

SM honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

Manuscripts must be provided either in standard UK or US English language. English standards should be consistent throughout the manuscripts accordingly.

Format the manuscript in A4 paper size; margins are 1 inch or 2.5 cm all around.

Type the whole manuscript double-spaced, justified alignment.

Use Times New Roman font, size eleven (11) point.

Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

Include line numbers (continuous) for the convenience of the reviewers.

Apart from chapter headings and sub-headings avoid any kind of formatting in the main text of the manuscripts.

1.2. Type & Length

SM publishes following types of papers:

Original scientific papers are the results of empirically- or theoretically-based scientific research, which employ scientific methods, and which report experimental or observational aspects of sports science and medicine, such as all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side. Descriptive analyses or data inferences should include rigorous methodological structure as well as sound theory. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒ Open Submissions

☒ Indexed

☒ Peer Reviewed

Original scientific papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

Review papers should provide concise in-depth reviews of both established and new areas, based on a critical examination

of the literature, analyzing the various approaches to a specific topic in all aspects of sports science and medicine, such as all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Review papers should be:

- Up to 6000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
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- Maximum number of references is 100.

Editorials are written or commissioned by the editors, but suggestions for possible topics and authors are welcome. It could be peer reviewed by two reviewers who may be external or by the Editorial Board.

☐Open Submissions

☒Indexed

☒Peer Reviewed

Editorials should be:

- Up to 1000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 10.

Short reports of experimental work, new methods, or a preliminary report can be accepted as two page papers. Your manuscript should include the following sections: Introduction, Methods, Results, and Discussion.

☒Open Submissions

☒Indexed

☒Peer Reviewed

Short reports should be:

- Up to 1500 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Peer review - fair review provides authors who feel their paper has been unfairly rejected (at any journal) the opportunity to share reviewer comments, explain their concerns, and have their paper reviewed for possible publication in SM.

☒Open Submissions

☒Indexed

☐Peer Reviewed

Peer review - fair review should be:

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- A structured abstract of less than 250 words;
- Maximum number of references is 15.

Invited papers and award papers include invited papers from authors with outstanding scientific credentials. Nomination of invited authors is at the discretion of the SM editorial board. SM also publishes award papers selected by the scientific committee of the International Scientific Conference on Transformation Processes in Sport.

☐Open Submissions

☒Indexed

☐Peer Reviewed

Invited papers and award papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
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- Maximum number of references is 30;
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SM only accepts electronic submission to the e-mail of the Journal Office: **sportmont@ucg.ac.me**.

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- A manuscript prepared according to the Guidelines for the Authors;
- A signed form that states the study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere, that states that all of the authors are in agreement with submission of the manuscript to SM, and that, for studies that use animal or human individuals, authors must include information regarding their institution's ethics committee, and which identifies the official approval number;
- A signed form that there is no conflict of interest.

Name the files according to the family name of the first author. Authors submitting revised versions of the manuscript can use the identification number of their manuscript as provided by the Journal Office. *See example:*

- ✓ FAMILY NAME-manuscript.doc – (main manuscript file)
- ✓ FAMILY NAME-statement.PDF – (authorship statement)
- ✓ FAMILY NAME-declaration.PDF – (declaration of potential conflict of interest)
- ✓ FAMILY NAME-fig1.tiff – (Figure 1)

1.4. Peer Review Process

A manuscript submitted for publication will be submitted to the review process as long as it fits the following criteria:

- The study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere;
- All persons listed as authors approved its submission to SM;
- Any person cited as a source of personal communication has approved the quote;
- The opinions expressed by the authors are their exclusive responsibility;
- The author signs a formal statement that the submitted manuscript complies with the directions and guidelines of SM.

The editors-in-chief and associate editors will make a preliminary analysis regarding the appropriateness, quality, originality and written style/grammar of the submitted manuscript. The editors reserve the right to request additional information, corrections, and guideline compliance before they submit the manuscript to the ad-hoc review process.

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1.6. After Acceptance

After the manuscript has been accepted, authors will receive a PDF version of the manuscripts for authorization, as it should look in printed version of SM. Authors should carefully check for omissions. Reporting errors after this point will not be possible and the Editorial Board will not be eligible for them.

Should there be any errors, authors should report them to the Office e-mail address **sportmont@ucg.ac.me**. If there are not any errors authors should also write a short e-mail stating that they agree with the received version.

1.7. Code of Conduct Ethics Committee of Publications



SM is hosting the Code of Conduct Ethics Committee of Publications of the **COPE** (the Committee on Publication Ethics), which provides a forum for publishers and Editors of scientific journals to discuss issues relating to the integrity of the work submitted to or published in their journals.

2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. See example:

Body Composition of Elite Soccer Players from Montenegro

Original Scientific Paper

Elite Soccer Players from Montenegro

Dusko Bjelica¹

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Dusko Bjelica

University of Montenegro

Faculty for Sport and Physical Education

Narodne omladine bb, 81400 Niksic, Montenegro

E-mail: sportmont@t-com.me

Word count: 2,946

Abstract word count: 236

Number of Tables: 3

Number of Figures: 0

2.1.1. Title

Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: “and”, “or”, “between” etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

2.1.2. Type of publication

Authors should suggest the type of their submission.

2.1.3. Running head

Short running title should not exceed 50 characters including spaces.

2.1.4. Authors

The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

In case all authors belong to the same institution affiliation numbering is not needed.

Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

2.1.6. Corresponding author

Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

2.1.7. Manuscript information

All authors are required to provide word count (excluding title page, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References), the Abstract word count, the number of Tables, and the number of Figures.

2.2. Abstract

The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

Since abstract is independent part of your paper, all abbreviations used in the abstract should also be explained in it. If an abbreviation is used, the term should always be first written in full with the abbreviation in parentheses immediately after it. Abstract should not have any special headings (e.g., Aim, Results...).

Authors should provide up to six key words that capture the main topics of the article. Terms from the Medical Subject Headings (MeSH) list of Index Medicus are recommended to be used.

Key words should be placed on the second page of the manuscript right below the abstract, written in italic. Separate each key word by a comma (and a space). Do not put a full stop after the last key word. See example:

Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

2.3. Main Chapters

Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements

(optional), Conflict of Interest (optional). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

2.3.1. Headings

Main chapter headings: written in bold and in Title Case. *See example:*

- ✓ **Methods**

Sub-headings: written in italic and in normal sentence case. Do not put a full stop or any other sign at the end of the title. Do not create more than one level of sub-heading. *See example:*

- ✓ *Table position of the research football team*

2.3.2 Ethics

When reporting experiments on human subjects, there must be a declaration of Ethics compliance. Inclusion of a statement such as follow in Methods section will be understood by the Editor as authors' affirmation of compliance: "This study was approved in advance by [name of committee and/or its institutional sponsor]. Each participant voluntarily provided written informed consent before participating." Authors that fail to submit an Ethics statement will be asked to resubmit the manuscripts, which may delay publication.

2.3.3 Statistics reporting

SM encourages authors to report precise p-values. When possible, quantify findings and present them with appropriate indicators of measurement error or uncertainty (such as confidence intervals). Use normal text (i.e., non-capitalized, non-italic) for statistical term "p".

2.3.4. 'Acknowledgements' and 'Conflict of Interest' (optional)

All contributors who do not meet the criteria for authorship should be listed in the 'Acknowledgements' section. If applicable, in 'Conflict of Interest' section, authors must clearly disclose any grants, financial or material supports, or any sort of technical assistances from an institution, organization, group or an individual that might be perceived as leading to a conflict of interest.

2.4. References

References should be placed on a new page after the standard title written in upper and lower case letters, bold.

All information needed for each type of must be present as specified in guidelines. Authors are solely responsible for accuracy of each reference. Use authoritative source for information such as Web of Science, Medline, or PubMed to check the validity of citations.

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SM adheres to the American Psychological Association 6th Edition reference style. Check "American Psychological Association. (2009). Concise rules of APA style. American Psychological Association." to ensure the manuscripts conform to this reference style. Authors using EndNote® to organize the references must convert the citations and bibliography to plain text before submission.

2.4.2. Examples for Reference citations

One work by one author

- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

Works by two authors

- ✓ Duffield and Marino (2007) studied
- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.

- ✓ First citation: Bangsbo, Iaia, and Krstrup (2008) stated that
- ✓ Subsequent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year

- ✓ Krstrup et al. (2003) studied
- ✓ In one study (Krstrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)

- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

- Nepocatych, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine*, 6(2), 57-65. doi: 10.26773/mjssm.2017.09.008
- Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology*, 100(6), 727-735. doi: 10.1007/s00421-007-0468-x
- Krstrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise*, 35(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

- Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies*, 50(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

- Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia*, 4. Retrieved November 15, 2005, from <http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html>

Conference paper:

- Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from <http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf>

Encyclopedia entry (print, with author):

- Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

- Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from <http://www.bartleby.com/65/et/ethnolog.html>

Thesis and dissertation:

- Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport*. Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

- Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

- Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from *Routledge Handbook of Applied Sport Psychology* ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

- Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See example:*

- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. ^{a,b,c}), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript ^a.

- ✓ ^aOne participant was diagnosed with heat illness and n = 19.^bn = 20.

Probability notes provide the reader with the results of the tests for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

- ✓ *P<0.05, †p<0.01.

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In the text, tables should be cited as full words. *See example:*

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓in our results (Tables 1, 2 and 5)... (citing more tables at once)

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Figures and figure legends should be completely intelligible without reference to the text.

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Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written below the figure, in sentence case. *See example:*

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, * $p > 0.01$, † $p > 0.05$.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See example:*

- ✓ Figure 1
- × figure 1
- × Figure 1.
- ✓ ...exhibit greater variance than the year before (Figure 2). Therefore...
- ✓ ...as shown in Figures 1 and 3. (citing more figures at once)
- ✓ ...result has shown (Figures 1-3) that... (citing more figures at once)
- ✓ ...in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See example:*

- ✓ Figure 1a
- ✓ ...in Figures 1a and b we can...
- ✓ ...data represent (Figures 1a-d)...

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

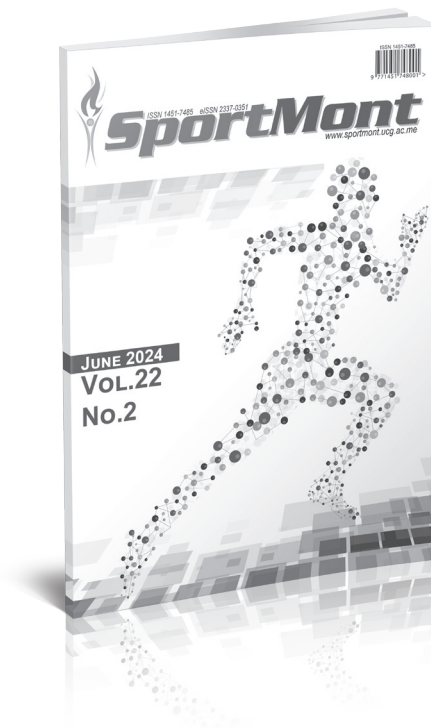
Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10°	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 °	× 10kg	× 12 : 2	× .056
Signs should be placed immediately preceding the relevant number.				
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age		
× 45 ± 3.4	× p < 0.01	× males > 30 years of age		

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. See example:

✓ First time appearing: *musculus biceps brachii*
Abbreviated: *m. biceps brachii*



ISSN 1451-7485

Sport Mont Journal (SMJ) is a print (ISSN 1451-7485) and electronic scientific journal (eISSN 2337-0351) aims to present easy access to the scientific knowledge for sport-conscious individuals using contemporary methods. The purpose is to minimize the problems like the delays in publishing process of the articles or to acquire previous issues by drawing advantage from electronic medium. Hence, it provides:

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- Fast publication time;
- Peer review by expert, practicing researchers;
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- Community-based dialogue on articles;
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SMJ is published three times a year, in February, June and October of each year. SMJ publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

SMJ covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

Prospective authors should submit manuscripts for consideration in Microsoft Word-compatible format. For more complete descriptions and submission instructions, please access the Guidelines for Authors pages at the SMJ website: <http://www.sportmont.ucg.ac.me/?sekcija=page&p=51>. Contributors are urged to read SMJ's guidelines for the authors carefully before submitting manuscripts. Manuscripts submissions should be sent in electronic format to sportmont@ucg.ac.me or contact following Editors:

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MONTENEGRIN JOURNAL OF SPORTS SCIENCE AND MEDICINE



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CALL FOR CONTRIBUTIONS

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Prospective authors should submit manuscripts for consideration in Microsoft Word-compatible format. For more complete descriptions and submission instructions, please access the Guidelines for Authors pages at the MJSSM website: <http://www.mjssm.me/?sekcija=page&p=51>. Contributors are urged to read MJSSM's guidelines for the authors carefully before submitting manuscripts. Manuscripts submissions should be sent in electronic format to office@mjssm.me or contact following Editors:

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Spring issue – March 2025



MONTENEGRIN SPORTS ACADEMY

Founded in 2003 in Podgorica (Montenegro), the Montenegrin Sports Academy (MSA) is a sports scientific society dedicated to the collection, generation and dissemination of scientific knowledge at the Montenegrin level and beyond.

The Montenegrin Sports Academy (MSA) is the leading association of sports scientists at the Montenegrin level, which maintains extensive co-operation with the corresponding associations from abroad. The purpose of the MSA is the promotion of science and research, with special attention to sports science across Montenegro and beyond. Its topics include motivation, attitudes, values and responses, adaptation, performance and health aspects of people engaged in physical activity and the relation of physical activity and lifestyle to health, prevention and aging. These topics are investigated on an interdisciplinary basis and they bring together scientists from all areas of sports science, such as adapted physical activity, biochemistry, biomechanics, chronic disease and exercise, coaching and performance, doping, education, engineering

and technology, environmental physiology, ethics, exercise and health, exercise, lifestyle and fitness, gender in sports, growth and development, human performance and aging, management and sports law, molecular biology and genetics, motor control and learning, muscle mechanics and neuromuscular control, muscle metabolism and hemodynamics, nutrition and exercise, overtraining, physiology, physiotherapy, rehabilitation, sports history, sports medicine, sports pedagogy, sports philosophy, sports psychology, sports sociology, training and testing.

The MSA is a non-profit organization. It supports Montenegrin institutions, such as the Ministry of Education and Sports, the Ministry of Science and the Montenegrin Olympic Committee, by offering scientific advice and assistance for carrying out coordinated national and European research projects defined by these bodies. In addition, the MSA serves as the most important Montenegrin and regional network of sports scientists from all relevant subdisciplines.

The main scientific event organized by the Montenegrin Sports Academy (MSA) is the annual conference held in the first week of April.

Annual conferences have been organized since the inauguration of the MSA in 2003. Today the MSA conference ranks among the leading sports scientific congresses in the Western Balkans. The conference comprises a range of invited lecturers, oral and poster presentations from multi- and mono-disciplinary areas, as well as various types of workshops. The MSA conference is attended by national, regional and international sports scientists with academic careers. The MSA conference now welcomes up to 200 participants from all over the world.

It is our great pleasure to announce the upcoming 22th Annual Scientific Conference of Montenegrin Sports Academy "Sport, Physical Activity and Health: Contemporary Perspectives" to be held in Dubrovnik, Croatia, from 3 to 6 April, 2025. It is planned to be once again organized by the Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, University of Montenegro and other international partner institutions (specified in the partner section).

The conference is focused on very current topics from all areas of sports science and sports medicine including physiology and sports medicine, social sciences and humanities, biomechanics and neuromuscular (see Abstract Submission page for more information).

We do believe that the topics offered to our conference participants will serve as a useful forum for the presentation of the latest research, as well as both for the theoretical and applied insight into the field of sports science and sports medicine disciplines.





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Sports Science and Medicine Journals from Montenegrin Sports Academy

We have expanded the quality of our journals considerably over the past years and can now claim to be the market leader in terms of breadth of coverage.

As we continue to increase the quality of our publications across the field, we hope that you will continue to regard MSA journals as authoritative and stimulating sources for your research. We would be delighted to receive your comments and suggestions, mostly due to the reason your proposals are always welcome.

Look Inside!



Sport Mont Journal

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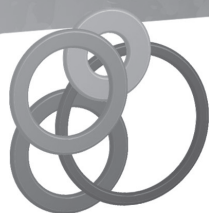
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